

Focus Sensitivity in Mabilia and Yoruboid

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Abstract

This dissertation investigates the syntax-semantics properties of association with focus sensitive particles in three West African languages: Kasem, Kusaal, and Yorùbá. Specifically, I investigate three focus sensitive particles that have been classified as involving conventional association with focus (Beaver and Clark, 2008). These are *only*, *also*, and *even*. These particles are focus sensitive because they interact in different ways with the alternatives evoked by the focus constituent. In the three languages, these focus sensitive particles can either be realized ad focally or adverbially, each realization having a distinct morphological form. I propose a uniform analysis for the different associative properties of these particles by arguing for a Propositional Approach. This approach is based on the assumption that the interpretation of each of these focus sensitive particles is always licensed by a propositional operator, which must occupy a propositional position in the syntax. This is especially the case when we are dealing with ad focal association with focus. The minimal propositional node in this case is vP, and the adverbial focus sensitive particles may overtly realize these propositional operators. However, I show that while the covert operators have flexible distribution in the clausal spine, the adverbial focus sensitive particles are restricted in their distribution due to language-specific structural/morphological constraints. The consequence of such a proposal is that we are able to account for a wide range of structural and semantic possibilities that these focus sensitive particles present in these languages. On the other hand, I argue against a Quantifier Approach, which assumes that ad focal particles undergo type-shifting and composes with its focus associate to form a single quantifier, which then undergoes quantifier-raising at LF to a propositional position. I argue that this approach is deficient in many regards, and unlike the Propositional Approach, cannot account for all the syntax-semantic properties of these particles. To support my claim, I provide morphological, syntactic, and semantic evidence in these languages. Furthermore, contrary to the claim in the literature that, with the Propositional Approach, the ad focal exclusive particles are just agreement morphemes that are semantically vacuous, I argue that these particles perform distinct and dedicated functions. They parameterize the evoked alternatives' scale used for different exclusive scalar interpretations. The study then provides structural analyses for these focus sensitive particles in relation to their focus associates. The motivation for this study is due to the lack of adequate investigation on the phenomenon of association with focus among the African languages. This study, therefore, provides both empirical and theoretical contributions to association with focus phenomenon, and the theory of focus generally.

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List of Abbreviations

1	first person
2	second person
3	third person
ACC	accusative
CONJ	conjunction
CQ	current question
DEF	definite
DISJ	disjunct
FOC	focus
FUT	future
FV	final vowel
HAB	habitual
IMM	immediate
IPFV	imperfective
LF	logical form
LOC	locative
NEG	negation
PFV	perfective
PLA	principle of lexical association
POSS	possessive
PROG	progressive
PST	past
QP	quantifier phrase
QUD	question under discussion
SEC	secundative
SG	singular
SS	surface structure

Chapter 1

Introduction

This dissertation investigates the syntax-semantic properties of three focus sensitive particles (*only*, *also*, and *even*) in three West African languages: Kasem, Kusaal and Yorùbá. The three focus sensitive particles have been classified as involving conventional association with their focus associates. This means that (a) they directly reference focus in their semantics, i.e, focus sensitivity is built into their lexical meaning, and therefore (b) they require their focus associate to be in their c-command domain. Depending on the constituent that is in focus, these particles engage the alternatives that is evoked by the focus. This is exemplified in (1), with the meanings contributed by the particles given in (2).

- (1) Adam {only/even/also} gave A SHOE to Sam.
- (2)
 - a. Adam did not give anything else to Sam.
 - b. That Adam gave something else to Sam is more likely than a shoe.
 - c. In addition to a shoe, Adam gave something else to Sam.

Although extensive cross-linguistic studies have been carried out on the nature and properties of these particles, African languages have not received so much attention in this regard. Apart from Grubic (2015), and a few other studies such as Carstens and Zeller (2020); Hartmann and Zimmermann (2008), among others, there has been very little that has been done with respect to association with focus among the African languages, as far as I know. This is surprising considering the rich use of morphology in the area of information structure among the African languages. This is evident in the development and advancement of the cartographic framework (Rizzi, 1997), which African languages have consolidated in many regards (see Aboh, 1998, 2004a; Hartmann and Zimmermann, 2012; Amaechi and Georgi, 2019; Issah and Smith, 2020, among many others). Therefore, the phenomenon of association with focus is worth investigating in these languages. One of the aims of this dissertation is to further fill this gap. Specifically, I investigate the properties of *only*, *also*, and *even* in Kasem, Kusaal and Yorùbá. The goal is to describe the behaviour of these particles in these languages, and to propose an analysis for the observed properties.

In order to accomplish this goal, this dissertation investigates specific syntax-semantic properties of these particles in the chosen languages. These properties include how to reconcile the syntactic distribution of these particles with their semantic properties. For example, while these particles are considered to be propositional operators, i.e., they compose with a proposition, they can syntactically occur in different positions within the clause. They may occur adverbially, which causes association from a distance with

the focus constituent, as in (3-a). On the other hand, they may also occur adjacent to the focus constituent, as in (3-b). While the position of *only* in (3-a) corresponds with its propositional semantic property, the position of the same particle in (3-b) does not correspond to this semantic requirement. While various analyses have been proposed to account for such an asymmetry, there is no consensus yet.

- (3) a. Adam *only* gave Sam A SHOE.
b. Adam gave Sam *only* A SHOE.

In this dissertation, I investigate such cases, and I argue for a Propositional Approach, which emphasizes that out of the two possible positions of these focus sensitive particles, as in (3), the interpretation is always licensed by a high operator, which may be realized as the adverbial particles. In cases where the adfocal particles are used, a covert operator is assumed, and evidence for such an element is provided. One of these evidence comes from particle doubling cases, where both the adverbial and the adfocal particles co-occur, and produce a single (concord) exclusive/additive/scalar-additive meaning.¹ An example from Yorùbá is given in (4). This sentence means that Tolù ate nothing other than yam. The presence of an adverbial exclusive particle, which co-occurs with an adfocal exclusive particle supports the presence of a clausal exclusive operator.² The same result applies to additive and scalar-additive particles as well.

(4) **Yorùbá exclusive doubling**

Tolù **kàn** jẹ́ [iṣù]_F **nìkan**.
Tolù only eat yam only
Int. ‘Tolù only ate only YAM.’

However, this leaves us with the question of the nature and/or function of the adfocal particles in such situations. Previous studies have argued that the adfocal focus sensitive particles are semantically vacuous elements. In other words, they mark concord (cf. Lee, 2005; Quek and Hirsch, 2017; Hirsch, 2019, 2022; Sun, 2021; Branan and Erlewine, 2023; Yip, 2025, a.o.). However, in this study, I will argue to the contrary. I will show, based on empirical evidence from the three languages, that on a closer look, the adfocal particles, in particular the adfocal exclusive particles, are not mere concord markers. They carry not-at-issue function of ranking the alternatives that are evoked by the focus associate. On the other hand, the adverbial exclusive particles are responsible for the at-issue exclusion of the alternatives that are informationally stronger than the prejacent (i.e., the sentence without the focus sensitive particle) or that are ranked higher than the prejacent.³

Another syntax-semantic property of the focus-sensitive particles investigated is the phenomenon of Backward Association with Focus. Consider the examples in (5), from Erlewine (2014a, p.129). The claim in the literature is that, in English, while *even* can associate with fronted materials, *only* cannot (cf. Jackendoff, 1972). This has been ar-

¹ It is important to make a terminological note here. In many instances in this dissertation, it may seem that the terms *doubling* and *concord* are used interchangeably. However, while I use the former to simply refer to the co-occurrence of the particles, I use the latter to refer to a syntax-semantic relation, where the co-occurrence of the particles produces a single interpretive meaning, similar to what we know with negative concord phenomena (cf. Zeijlstra, 2004a, 2022, a.o.).

² Note that such a sentence is not felicitous in English, if we assume that the sentence has a single focus constituent, which is the object in this case.

³ In this study, I assume that the set of alternatives that the focus sensitive particles interact with consists of full propositions, and not individual words. I will come back to this.

gued to be related to the semantic contributions that each focus sensitive particle makes. Erlewine (2014a,b) argues that while *only* requires a strict structural compliance of its prejacent in order to assert its truth-condition, *even* (and *also*) does not have such a requirement. They allow for some structural flexibilities. He argued that what seems to be a backward association with *even* is actually ‘illusory’. That is, *even* does not associate backwards. Rather it associates with a lower copy of the moved constituent.

- (5) a. [Mary]_F John *even* saw ___ at the party.
 b. *[Mary]_F John *only* saw ___ at the party.

In this dissertation, I will provide evidence for exclusive backward association in Kasem, Kusaal and Yorùbá. I show that in these languages, the adverbial exclusive particles associate backward/leftward with a fronted focus constituent. The analysis that I propose is based on reconstruction. I argue that the fronted focus constituent reconstructs to its canonical position at LF, and it is at this point that the interpretation is derived. For the other two focus sensitive particles, *also* and *even*, I follow Erlewine’s analysis that they do not require the reconstruction of their focus associate. What is required is that the fronted focus constituent should originate from their c-command domain. These particles will then associate with the lower copy of the fronted constituent. This is based on the assumption that this lower copy still has a focus feature.

In summary, the dissertation provides a uniform Propositional account to the syntax-semantic properties of association with focus in languages (belonging to a group) which have largely been understudied. The study, therefore, provides both empirical and theoretical contributions to association with focus phenomenon.

1.1 Chapters’ summary

Including this introduction, the dissertation has eight chapters in total. I will briefly summarize what each chapter contains.

The remaining part of this chapter involves discussions on the relevant areas of the individual languages investigated in this study. This includes a general background of the languages, and discussions on the basic syntactic structures on the languages.

Chapter 2 of this dissertation will introduce the basic theoretical framework that I assume in this study. The chapter summarizes the relevant aspects of earlier studies, that form the basic background for the dissertation. The topics discussed includes the notion of focus, association of focus and the Quasi, Free, and Conventional (QFC) model, question under discussion and approaches to focus association.

In **Chapter 3**, I describe the focus realization strategies in Kasem, Kusal and Yorùbá respectively. I argue that all three languages employ both in-situ focus marking and ex-situ focus marking strategies. However, while Kasem and Kusaal mark both in-situ and ex-situ focus morphologically, Yorùbá only uses focus morphology for ex-situ focus marking. In addition, I show that all three languages have instances of (anti-)pied-piping. In pied-piping, the logical/semantic focus is contained within the morphologically marked constituent, while in anti-pied-piping, it is the morphologically marked constituent that is contained within the logical/semantic focus. For all these cases, I propose syntactic analyses for them.

Chapter 4 describes the exclusive focus sensitive particle in the three languages. I show that all three languages have both adverbial and adfocal exclusive focus sensitive

particles, which have distinct forms. This is unlike English, where the same form is used for both adfocal and adverbial uses of *only*. Importantly, I show that unlike English, these languages allow for exclusive doubling or concord, where both the adverbial and the adfocal exclusive particles co-occur to derive a single exclusive meaning. This is not possible in English, as in: #Adam only saw only Sam_F. Furthermore, in this chapter, I argue against the claim that adfocal exclusive particles are vacuous concord items. This chapter lays the foundation for the proposal and analysis of exclusive focus sensitive association in the next chapter.

Chapter 5 introduces the approach, which this dissertation is develops. I argue for a uniform account, which analyzes both adfocal and adverbial use of exclusives. I argue that a Propositional Approach uniformly accounts for the syntax-semantic properties of exclusives. I introduce the alternative Quantifier Approach, and argue that it does not account for the different possibilities and properties of *only* in these languages. In addition, I argue against the claim in the literature that adfocal exclusive particles are semantically inert or concord items under the propositional approach. Using empirical evidence, I show that adfocal exclusive particles parameterize the relevant scale of the set of alternatives in accordance with the different scalar interpretations associated with exclusives. Lastly, in this chapter I show that all three languages allow for Backward Association with focus, which is not allowed in English. I provide an analysis for this by assuming reconstruction of the moved constituents at LF.

In **Chapter 6** of this dissertation, I discuss the additive focus sensitive particle. Again, all three languages have both adverbial additives and adfocal additives. I argue for a propositional analysis here as well, i.e., the additive interpretation comes from a high covert additive operator, which may be overtly realized by the adverbial additive particles. I provide evidence for additive backward association as well. However, unlike the exclusive backward association cases, I argue that additive backward association does not require reconstruction. All that is required is for the fronted constituent to have originated within the scope of the additive operator. I assume that the additive operator associates with the lower copy of the fronted constituent instead. Therefore, additive backward association with focus is a mere illusion. The reason for this is based on the semantic contribution of additives, which does not affect the truth-condition of their assertion, unlike *only* which does.

Chapter 7 discusses scalar-additive association with focus in these languages. I show that for scalar-additives, only Yorùbá has both adverbial and adfocal scalar-additives. Kasem and Kusaal only have adverbial scalar-additives. Nonetheless, I argue for a propositional approach for scalar-additive as well. Similar to *only* and *also*, scalar-additives also allow for backward association. I extended the analysis, which I assumed for additive backward association, to the scalar-additive cases as well. Furthermore, in this chapter, I also discussed the notion of conventional association, and argue that the three focus sensitive particles that are investigated in this dissertation involve conventional association, which supports Beaver and Clark's (2008) classification of these particles in English.

The last **Chapter 8** provides a conclusion to the dissertation, and highlights areas for future research.

1.2 Relevant backgrounds to the languages

The three languages investigated in this dissertation are Kasem, Kusaal and Yorùbá. Although the three languages belong to the Niger-Congo languages family, Kasem and

Kusaal belong to the Mabilia (formerly Gur) group, and Yorùbá belongs to the Yoruboid group. In this section, I will discuss aspects of these languages that are relevant for the discussion in this dissertation. Therefore, this section is not meant to describe the languages in detail because this is not the aim of the dissertation.

1.2.1 Kasem

Kasem is a Mabilia language spoken predominantly in the Upper East region of Northern Ghana and some parts of Burkina Faso. In Ghana, Kasem speakers are found in northern Ghanaian communities such as Navrongo, Chiana, and Paga. In Burkina Faso, they are found in Kampala, Tiebele, and Pa (cf. Taluah, 2020). The language is spoken by approximately 400,000 people (called Kasenas) according to Bodomo (2020) (cf. Jeckel et al., 2024). Kasem has three dialects namely, Navrongo Kasem 1 (NK1), Navrongo Kasem 2 (NK2), and Navrongo Kasem 3 (NK3). This dissertation is based on NK1, which is also the one taught in schools according to my language consultants. For some studies on Kasem, see Callow (1965a, 1968); Hewer (1976); Halle (1978); Haas (1986); Awedoba (1996); Taluah (2020) among others. Most of these studies are on the phonology of Kasem. The syntax of the language is still very much understudied.

Kasem is an SVO language, as shown with the example in (6-b), which answers a question requiring an all new information. The language has independent tense morphemes preceding the verb. The tense morpheme *yaa* below is used to mark hesternal past, i.e., events that happened yesterday.⁴

- (6) a. What happened?
 b. Adam *yaa* go chworo.
 Adam PST kill.PFV fowl
 ‘Adam slaughtered a fowl.’

Negation is marked based on aspects in the language. If the aspect is perfective, then *mo* is used, as in (7-b). However, if the aspect is imperfective, the negative morpheme is *ba*, as in the progressive construction in (8-b).

- (7) a. Adam *di* mumuna.
 Adam eat rice
 ‘Adam ate rice.’
 b. Adam **wo** *di* mumuna.
 Adam NEG eat rice
 ‘Adam did not eat rice.’
- (8) a. Adam *wora* *(o) *di* mumuna.
 Adam PROG 3SG eat rice
 ‘Adam is eating rice.’
 b. Adam **ba** *wora* *(o) *di* mumuna.
 Adam NEG PROG 3SG eat rice
 Adam is not eating rice.’

Progressive constructions in Kasem have unique structures. For example, the subjects of the sentences in (8) have pronoun copies following the morpheme glossed as PROG

⁴ There are other past tense morphemes: *pa’a* = immediate past (i.e., events that happened today); *deem* = events that happened last year or a few years ago; *deém* = remote past (i.e., events that happened long time ago).

here. Tentative analysis of this clause points to a bi-clausal structure for the language. For example, consider the examples in (9). Generally, PP modifiers in Kasem occur clause-finally, and cannot occupy a pre-verbal position, as shown in a simple perfective clause in (9-a). However, in a progressive construction, the PP-modifier can occupy either a position just below *wora*, or clause-finally, see (9-b). The possibility of having the PP-modifier following *wora* suggests that we are dealing with a bi-clausal structure. See Aremu and Apebuga (2025) for some discussions on the nature of the progressive clauses and the composition of *wora* in Kasem. The analysis provided there is that of a copy raising structure. But I will not go into the details here.⁵

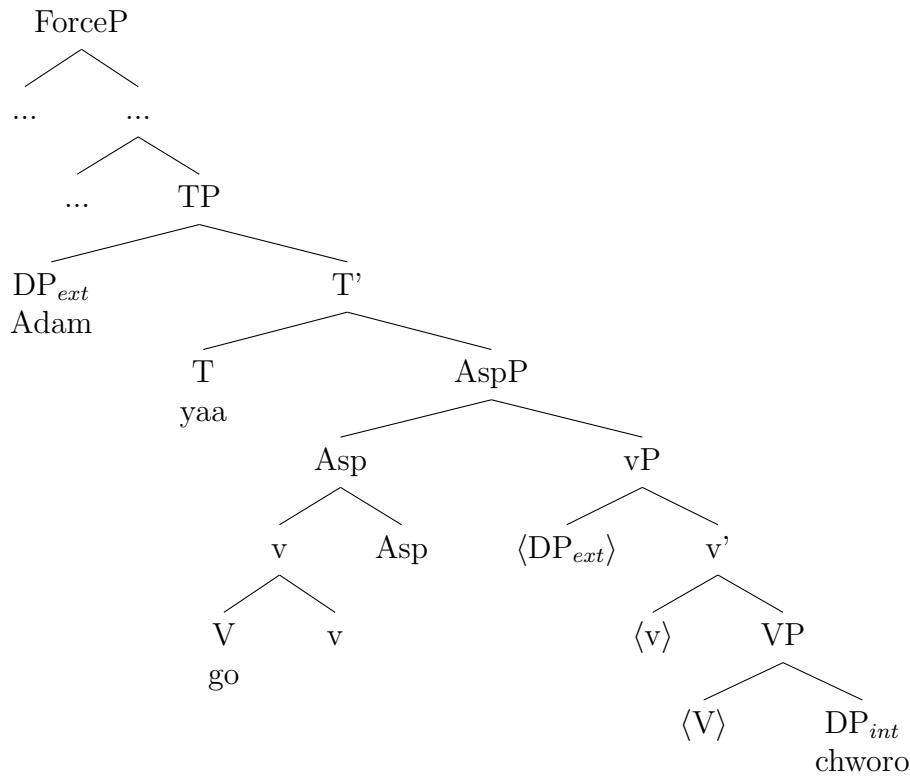
- (9) a. Adam (*[**de foone**]) tōŋē *([**de foone**]).
Adam with fear work.PFV with fear
‘Adam worked with fear.’
b. Adam **wora** ([**de foone**]) o di wodiū kom ([**de foone**]).
Adam PROG with fear 3SG eat food DEF with fear
(i) ‘Adam is eating the food with fear.’
(ii) Lit. ‘Adam is there with fear eating the food.’

Based on the facts presented above for Kasem, we can propose that there is V to v movement in the language. I also assume that there is V+v to Asp movement. The progressive sentence, as in (8-a), has a complex structure, and so cannot serve as a basic structure for Kasem. Adopting the sentence in (6-b), repeated in (10) below, I propose the structure in (11) for Kasem’s simple declarative clause. After the internal argument is merged with V, V moves to v, whose Spec is occupied by the external argument subject. Then, there is V+v to Asp movement. The T head is filled with an independent past tense morpheme, so no V+v+Asp to T movement. Afterwards, the subject argument moves from Spec,vP to Spec,TP for EPP requirement. The left periphery of the clause (abbreviated as [...]) hosts information structural projections, such as the Focus Phrase and the Topic Phrase. The focus position would become relevant in subsequent chapters of this dissertation.

- (10) Adam yaa go chworo.
Adam PST kill.PFV fowl
‘Adam slaughtered a fowl.’

⁵ Kasem has more structures that are similar to that of the progressive clauses. As we will see in Chapters 4, 5, and 7, the adverbial exclusive particle *weeni* (‘only’), the modal auxiliary *wae* (‘may’), and the adverbial scalar-additive particle *ta-mape* (‘even’) all have structures involving a lower copy of the subject. Since these structures are not entirely important for the present study, I will not dwell too much on them because a comprehensive investigation is required, which I leave for a future occasion.

(11) **A simple sentence structure of Kasem**



Another property of Kasem is its noun class system which is clearly seen in its determiner system. According to Awedoba (1996, p.9), Kasem has nine noun classes which are distributed under five grammatical gender classification (cf. Callow, 1965b). The table in 1.1 summarizes the nominal gender and class system of Kasem as marked on the definite determiners (adapted from Awedoba, 1996, p.9).

Gender	Singular		Plural	
	Nominal Class	Determiners	Nominal Class	Determiners
1	I	wom	II	bam
2	III	dem	IV	yam
3	V	kam	VI	sem
4	VII	kom	VIII	tem
5	VII	kom	IX	dem

Table 1.1: Kasem Nominal Gender and Classes

The five grammatical gender classifications are summarized in (12), and are marked as suffixes on the nouns belonging to each group.

- (12)
- Gender 1: mostly human nouns.
 - Gender 2: fruits, body parts, abstract nouns, etc.
 - Gender 3: young plants, small mammals, insects, diminutive or pejorative attributes, etc.
 - Gender 4: trees, animals, illnesses, etc.
 - Gender 5: domestic livestock

(Awedoba, 1996, p.10)

The noun class system in Kasem, especially the Table 1.1, was included in the discussion here because of the varying definite determiners that might be noticed in the course of the discussion in this dissertation.

1.2.2 Kusaal

Kusaal is also a Mabilia language spoken in Ghana, Burkina Faso, and Togo by approximately 500,000 people (Bodomo, 2020). Kusaal has two dialects namely, Agole and Toende. This dissertation is based on the Agole dialect. According to my language consultants, Agole is the dialect that is taught in schools, and is seen to be the standard variety. For some studies on the grammar of Kusaal, including discussions on information structure, see Abubakari (2018, 2019); Musah (2018); Eddyshaw (2025). As a tonal language, Kusaal has three tones: high (´), low (`), and mid (˘) (cf. Musah, 2018). Our study here does not depend much on the tones, and as stated by Musah (2018, p.54), “(the) implications of tone on morpho-syntax are noteworthy *in the aspectual system* of the language.” Since our focus here is not on the aspectual marking system of the language, I do not dwell on tone nor include tone in the data. This is done in order to avoid inconsistency.

The simple syntactic structure in Kusaal is SVO, as shown in (13-b). The direct object follows the verb, and tense is marked as an independent morpheme preceding the verb. The past tense morpheme can change depending on how far past is the event. For instance, if the event happened yesterday, then *sa* is used (instead of *pa’a*), see (13-c).⁶

- (13) a. What happened?
 b. Adam pa’a kɔdig nua la.
 Adam IMM.PST kill fowl DEF
 ‘Adam slaughtered the fowl (earlier today).’
 c. Adam sa kɔdig nua la.
 Adam HEST.PST kill fowl DEF
 ‘Adam slaughtered the fowl (yesterday).’

In addition, Kusaal has verbal complexes, in the form of suffixes, which mark aspects, especially the imperfective aspects. The habitual marker in Kusaal is the suffix *-Vd*. This vowel of the suffix changes depending on the final vowel of the root verb, see (14-a). If the verb ends with a consonant, the V of the suffix is realized with a default high vowel /i/, see (14-b). Lastly, the habitual suffix has an allomorph *-t*, which usually occur when the root verb ends with a high vowel, as in (14-c).

- (14) a. Adam sɔ-ud.
 Adam bath-HAB
 ‘Adam baths.’
 b. Adam tɔmm-id.
 Adam work-HAB
 ‘Adam works.’
 c. Adam di-t mui.
 Adam eat-HAB rice
 ‘Adam eats rice.’

⁶ Other past tense morphemes include: *daa* = two days ago but less than a year, and *da* = a year and beyond (remote past) (Abubakari, 2018).

In a progressive construction, the suffix *nɛ* is used, as in (15).

- (15) Adam di-*t-nɛ* mui.
 Adam eat-HAB-IPFV rice
 ‘Adam is eating rice.’

The form *nɛ* is homophonous with different grammatical functions. So, it is important for us here to distinguish its functions. Apart from marking imperfective aspect, *nɛ* performs, at least, two other functions in the language: (a) it is used to mark in-situ focus constituents, and (b) it is used for conjoining two DPs in the language. For the in-situ focus function, consider the sentences in (16) and (17). The in-situ focus marker *nɛ* precedes the in-situ direct object focus in (16-b). This position might be argued to be similar to that of the imperfective *nɛ*. However, if we consider the in-situ direct object focus marking in (17-b), then the function of *nɛ* as an in-situ focus marker becomes obvious. I will discuss in-situ focus in Kusaal extensively in Chapter 3 (see also Abubakari (2018), Ch. 3, and Musah (2018), Ch. 2).

- (16) a. Awini tis anɔɔn ligidi la?
 Awini give who money DEF
 Lit. ‘Awini gave who the money?’
 b. Awini tis **nɛ** [Sam]_F ligidi la.
 Awini give FOC Sam money
 ‘Awini gave SAM the money.’
- (17) a. Awini tis Sam bɔ?
 Awini give Sam what
 Lit. ‘Awini gave Sam what?’
 b. Awini tis Sam **nɛ** [ligidi la]_F.
 Awini give Sam FOC money DEF
 Awini gave Sam THE MONEY.

The other function of *nɛ* is as a DP coordinator. It can only conjoin DPs, and not VPs or TPs. Consider the following example which is a title of a story from Musah (2018). *Nɛ* conjoins the three DPs in (18).

- (18) A-sumbul **nɛ** A-zaɲkɔar **nɛ** gɛla yɛla
 Mr.-Hare CONJ Mr.-Wolf CONJ eggs matter
 ‘Mr. Hare and Mr. Wolf and the story of the eggs’

(Musah, 2018, p.277)

There is another coordinator *ka* that is used for clausal coordination in the language. The following example in (19) is from a story documented by Musah (2018).

- (19) Pu’a la yu’un kpɛ dɔɔg-in la, **ka** dau la pibigi o
 Woman DEF then enter room-LOC DEF CONJ man DEF remove 3SG.POSS
 zupibig la.
 hat DEF
 ‘The woman then went into the room, and the man removed his hat.’

(Musah, 2018, p.274)

Interestingly, *ka* is also used to mark ex-situ focus constituents. Both the fronted

wh-phrase and the fronted focus answer in (20) are followed by *ka*. We will see more of this in Chapter 3 where I discuss focus constructions in the languages under investigation.

- (20) a. Bɔ **ka** Awini tis Sam?
 what FOC Awini give Sam
 ‘What did Awini give to Sam?’
 b. [Ligidi la]_F **ka** Awini tis Sam.
 money DEF FOC Awini give Sam
 ‘It is the money that Awini gave to Sam.’

Another interesting aspect of the language which is also relevant for us here is negation. Kusaal has distinct negative morphemes that are dependent on the aspect of the clause. This is similar to what we saw in Kasem above. In the examples in (21), while *pɔ* is used in a perfective clause, *kɔ* is used in an imperfective clause. See Musah (2018) for further discussion.

- (21) a. Adam **pɔ** di mui.
 Adam NEG eat.PFV rice
 ‘Adam did not eat rice.’
 b. Adam **kɔ** di mui.
 Adam NEG.FUT eat rice
 ‘Adam will not eat rice.’

At this point, we can now propose a syntactic structure for a simple declarative clause in Kusaal. Using the sentence in (13-b), repeated in (22) below, I propose the structure in (23) for a simple declarative clause in Kusaal. The structure is built up to TP. I assume that, apart from the V to v movement, there is also the V+v to Asp movement as we have seen with the imperfective clauses in (14) and (15) above.⁷ The T head is occupied by the tense morpheme, and the subject moves from its externally-merged position in Spec,vP to Spec,TP for EPP requirement. Above the TP is where information structural notions like focus are interpreted and hosted (especially in ex-situ focus cases).

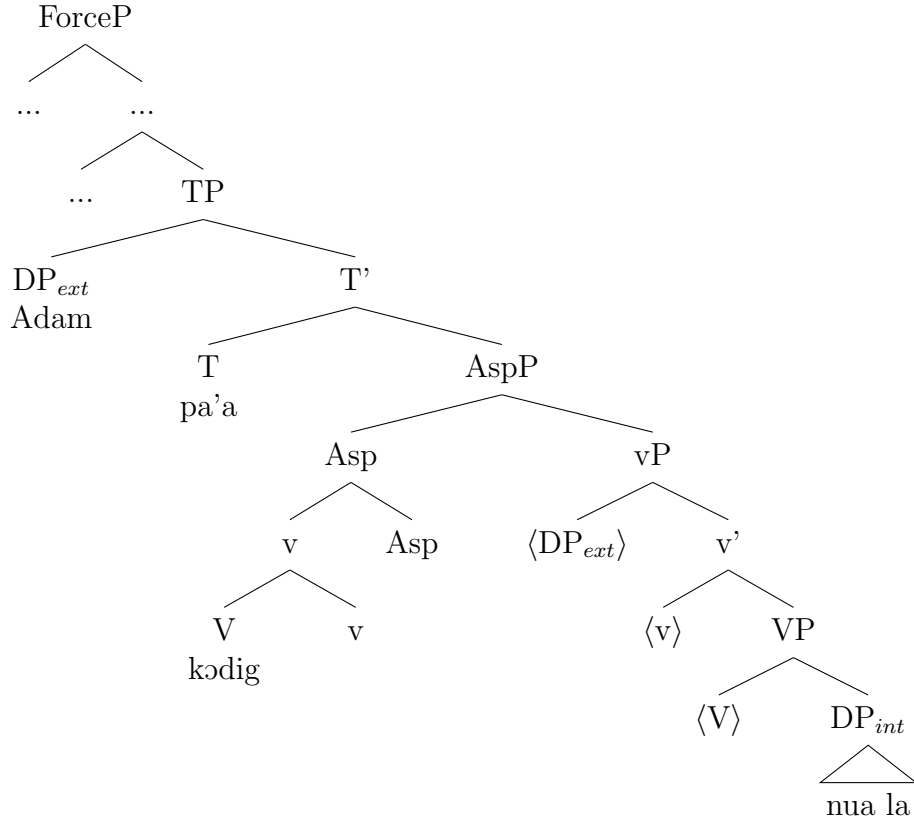
- (22) Adam pa’a kɔdig nua la.
 Adam IMM.PST kill fowl DEF
 ‘Adam slaughtered the fowl (earlier today).’

⁷ In fact, Abubakari (2018) claimed that perfective clauses in Kusaal are not always without an overt marker. There is *-ya*, which she claimed marks perfective aspect only in intransitive clauses, see (i). The status of this *-ya* marking, and why it is only used in intransitive clauses, is still unclear.

- (i) a. Biig la di-**ya**.
 child DEF eat-PFV
 ‘The child has eaten.’
 b. *Biig la di-**ya** diib la.
 child DEF eat-PFV food DEF
 ‘The child has eaten the food.’

(from Abubakari, 2018, p.85, ex.3.113)

(23) **A simple sentence structure of Kusaal**



Similar to Kasem, Kusaal also has noun classes. It has 11 noun classes which are grouped based on the semantic affinity that they share. Nouns are grouped under the following classes: human, kinship and honorary terms, places, animals, body parts, liquids, mass objects, abstract objects, fruits/food, and so on. The members of each classes are distinguishable based on their distinct suffixes. The classification of these classes overlaps and has an intricate system. I will not discuss these classes because it is not significant to our discussion in this study, but see Abubakari (2018), Ch.3 and Musah (2018), Ch.3 for a comprehensive list and discussion of these classes.

1.2.3 Yorùbá

Yorùbá belongs to the Yoruboid group of languages of the Niger-Congo language family. It is spoken mainly in South-Western Nigeria, some parts of Benin Republic, and Togo. The speakers of Yorùbá are about 47 million (Wikipedia). It is one of the three national languages in Nigeria (others are Hausa and Igbo). According to Ethnologue, Yorùbá has an institutional status because it is well-documented and studied in and outside its home community. There are grammars which have been developed for the language, such as Bámgbóṣé (1967); Awobuluyi (1978). There are many dialects of Yorùbá whose classification is regional-based. However, there are still debates on the actual number of dialects of the language. For some of the studies and discussion on the dialects of Yorùbá, see, a.o., Adetugbo (1967); Akinkugbe (1976); Adeniyi (2010); Akintoye (2020), and their references. For this study, I use what is considered as the Standard dialect of the language by majority of the speakers. This is the Ọ̀yọ́ dialect.⁸

⁸ I want to put a note here that there has always been a controversy among the Yorùbás on what is the standard. However, the Ọ̀yọ́ dialect has always been considered the standard variety by the

Yorùbá is a tonal language with three tones: high (´), low (`), and mid (ˉ). While the high and low tones are marked in the orthography of the language, the mid tone is usually unmarked. Since the phonology and the tonal system will not play any significant role here, I will not go into it in detail. For some studies on the phonology of Yorùbá, see, among many others, Oyelaran (1970); Courtenay (1971); La Velle (1974); Akinlabi (1985); Connell and Ladd (1990); Laniran (1988, 1992); Akinlabi and Liberman (2001); Laniran and Clements (2003).

The syntax of Yorùbá has also been widely studied. The basic word order in the language is SVO, with tense, aspect and modal (TAM) morphemes occupying distinct positions between the subject and the verb. Consider the sentence in (24-b), which is an answer to the question in (24-a). The simple structure is Subject >> Verb >> Object.

- (24) a. Kí ni ó sẹ̀lẹ̀?
 what FOC EXPL happen
 ‘What happened?’
 b. Tolú jẹ̀ ìrẹ̀sì.
 Tolú eat rice
 ‘Tolú ate rice.’

Although the past or present tenses are not morphologically marked in Yorùbá, future tense is marked. So, this supports the presence of a T head, as in the example in (25-a). Perfective and progressive aspects are also marked as free morphemes preceding the verb, as in (25-b) and (25-c) respectively. Example (25-d) represents a canonical negative sentence in the language.

- (25) a. Tolú **máa/yóò** jẹ̀ ìrẹ̀sì.
 Tolú FUT eat rice
 ‘Tolú will eat rice.’
 b. Tolú **tí** jẹ̀ ìrẹ̀sì.
 Tolú PFV eat rice
 ‘Tolú has eaten rice.’
 c. Tolú **n** jẹ̀ ìrẹ̀sì.
 Tolú IPFV eat rice
 ‘Tolú is eating rice.’
 d. Tolú **o** jẹ̀ ìrẹ̀sì.
 Tolú NEG eat rice
 ‘Tolú did not eat rice.’

In a negative sentence, the negative marker precedes the aspectual marker when they co-occur, see (26).⁹ Based on all the examples given, it is obvious that Yorùbá does not make use of verbal morphology (complexes) such as suffixes or prefixes.

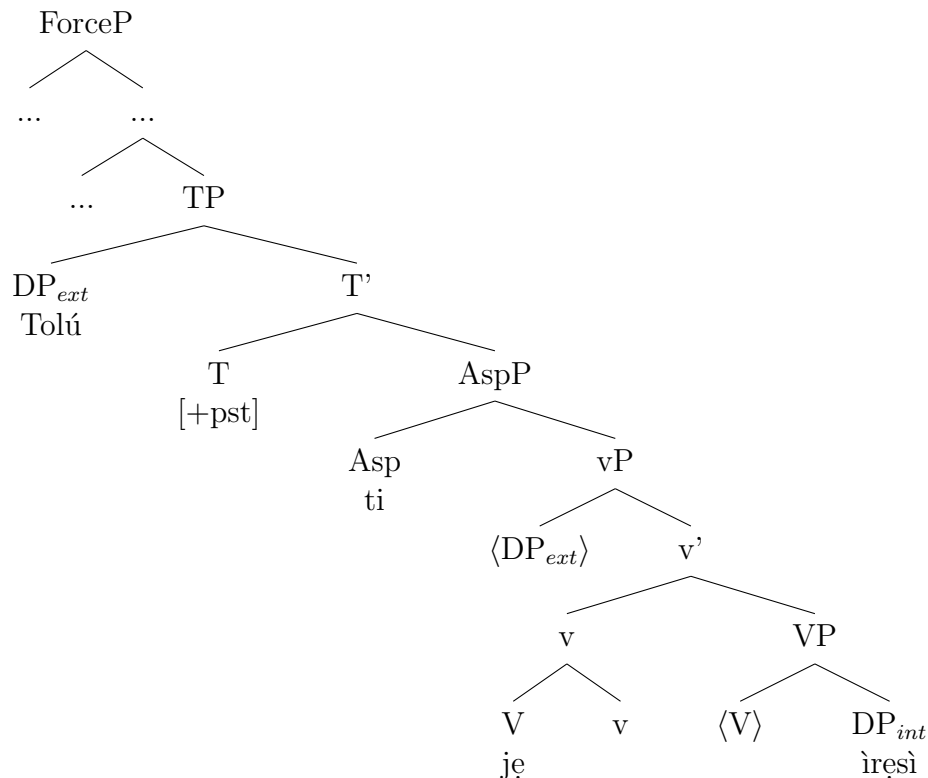
- (26) Tolú **ò tî** jẹ̀ ìrẹ̀sì.
 Tolú NEG PFV eat rice
 ‘Tolú has not eaten rice.’

majority, and it is taught in schools and used in writing materials, including the Yorùbá Holy Bible (Bíbẹ̀lì Mímọ̀).

⁹ There seems to be a phonological reflex on the perfective marker in the presence of negation. Compare (25-b) with (26). The vowel of the perfective morpheme is lengthened, and the tone changes. At this point, I do not have an answer as to why this is the case.

The structure of a simple clause (with perfective aspect, cf. (25-b)) that I assume for Yorùbá is given in (27). The internal argument merges with the lexical verb, and then there is V to v movement. However, there is no V+v to Asp/T movement in the language, as the language employs independent morphemes which occupy Asp and T (see (25)) respectively. The external argument (i.e., the subject) merges at Spec,vP, and moves from there to Spec,TP for EPP requirement on T. The position above TP hosts information structural notions, such as focus (FocP) and topics (TopP), hence the use of [...] above TP, and below ForceP. These projections are only present only when they are needed (cf Rizzi, 1997).

(27) **A simple sentence structure of Yorùbá**



In summary, we have seen that the three languages have an SVO word order. One point of variation among the languages is that while the Mabilia languages, Kasem and Kusaal, have v to Asp movement, Yorùbá does not. Aspects are independently marked in Yorùbá.

1.2.4 Data collection

Unless stated otherwise, the data used in this dissertation are from original fieldwork and elicitation sessions. The Kasem and Kusaal data were first collected during my fieldwork in Ghana in August, 2022, and then from many remote elicitation sessions with the language consultants. As a native speaker, the Yorùbá data are based on my introspective judgments, with occasional confirmations from at least five Yorùbá speakers. The data collection methods employed include the use of prepared questionnaire and structured interview sessions. In addition, I consulted other sources for data, such as the Kusaal dictionary, Bible, and grammar book, and the Kasem dictionary and Bible. Some of the data for this study, and more on other Mabilia languages, can be found on <https://mabilia-vp.com>. One last point is that while the Yorùbá data are consistently

tone-marked, the Kusaal and Kasem data are not tone-marked. This is due to a lot of inconsistencies in the orthography of these languages. Moreover, as far as I know, tone does not seem to play a role in the current investigation. Of course, this is not in any way to relegate the importance of tones, which is in fact an integral property of most African languages. For the recorded pronunciation of some of the data collected, see the Mabilia database: <https://mabilia-vp.com>.

1.3 Why these languages?

The choice of the languages under study was made deliberately in order to provide a broader coverage of association with focus phenomenon among the (West) African languages. The three languages represent two distinct language groups of the Niger-Congo language family. Although both Kasem and Kusaal are Mabilia languages, they have a few differences as well. For instance, while Kusaal uses distinct morphological focus markers for the in-situ and the ex-situ focus strategies (*nɛ* and *ka*), Kasem uses the same morphological focus marker *mo* for both focus strategies. Yorùbá, on the other hand, only uses focus morphology in ex-situ focus. In-situ focus are unmarked. Importantly, investigating languages from two distinct language groups allow us to make certain generalizations that would otherwise not be founded. Similarly, as I will show, these languages offer rich morphology with respect to the phenomenon under investigation. The aim is to also provide a window for further investigation on association with focus phenomenon, within and across the language groups of the Niger-Congo language family.

Chapter 2

Association with Focus: A theoretical framework

In this chapter, I discuss the basic foundation and development of focus and association with focus, which are relevant to the present dissertation. This also forms the framework that this dissertation is built upon. Section 2.1 presents an introduction to the notion of focus, its semantics, its prosody, and its morphosyntax. Section 2.2 begins with a general overview of association with focus, and then discusses Beaver and Clark’s (2008) Quasi, Free & Conventional Model (QFC-Model), with a special attention on the Conventional association with focus particles *only*, *also*, and *even*. Section 2.3 summarizes the different approaches that have been proposed for the syntax-semantic interface of association with focus. Lastly, section 2.4 summarizes the chapter.

As for the notations employed throughout this dissertation, I put the focus constituent in the target languages inside an F-marked square bracket $[XP]_F$ (cf. Jackendoff, 1972), while the English translation that corresponds to the focus XP would be in SMALL CAPS.¹ Focus markers and focus sensitive particles are in **boldface**. Except otherwise mentioned or footnoted, I follow the Leipzig glossing rule for interlinear glossing.

2.1 Focus and its realizations

2.1.1 The meaning and prosody of focus

There are quite a number of reasons why human language is able to emphasize a smaller part of a clause or even the whole clause itself. This may be due to the need to present new information in a discourse, or even to contrast or correct supposedly wrong information. There may be other reasons, of course, but whatever may be the reason, they all involve information structuring or packaging (Chafe, 1976; Krifka, 2008).² A semantically-oriented

¹ Note that the use of small caps does not indicate the precise location of the focus pitch accent in the English translation. Thus, both the accented word/syllable and their de-accented sub-constituents, e.g. articles will all be in small caps.

² Typically, focus is said to introduce *new information* into the discourse which is different from the *old* or *given* information (Rochemont, 1986). However, this definition is constantly debated because not only is it pragmatic, but it also does not capture the fact that focus can involve an old or a given information (Ladd, 1980, 2008; Zubizarreta, 2016). An example of such type of focus is the *Second Occurrence Focus* or *De-accented focus* which is both given and can be indicated by the presence of a focus-sensitive particle like *only* (Partee, 1999; Baumann, 2016; Beaver and Clark, 2008). For instance, in example (i-a), the focus is a new information, and it is accented. However, in (i-b),

definition is given by Krifka (2008), where “focus indicates the presence of alternatives that are relevant for the interpretation of linguistic expressions” (Krifka, 2008, 247).³ Krifka’s definition takes a cue from Rooth’s (1985; 1992) *alternative semantics* approach to focus.⁴ In alternative semantics, a focus constituent has, in addition to its ordinary semantic value $\llbracket XP \rrbracket^0$ (assigned to every syntactic node), a focus semantic value $\llbracket XP \rrbracket^f$. This focus semantic value is the set of alternatives that is evoked by the focus constituent; the F-marked constituent according to Jackendoff (1972).

Marking a constituent as focus does not only place discourse prominence on the constituent, it also entails that there is at least one other alternative constituent. For instance, in example (1) below, the focus is on the direct object of the clause (1-b), and it evokes a set of alternative propositions where the focus is substituted with contextually-relevant alternatives such as Adam bought a shoe, Adam bought a truck, Adam bought a bike, etc.

- (1) a. What did Adam buy?
b. Adam bought [a CAR]_F.
c. #[Adam]_F bought a car.

The potential alternatives are not only context-dependent, but also (usually) share the same grammatical function and/or lexical category as the focus.⁵ Thus, we cannot have constituents such as *blue*, *small*, *eat*, *laugh*, *yesterday*, and so on as alternatives to the focus under discussion. Similarly, example (1-c) is infelicitous with regard to the context created (the *wh*-question) because the subject is focused instead of the object; the grammatical function differs. (1-c) will be felicitous if the question is changed to *Who bought a car?* This type of focus is called a *narrow focus* (Lambrecht, 1994). We will make use of Krifka’s (2008) definition in this dissertation because (i) it is more general, and (ii) it fits well with the behaviour of the focus-sensitive particles discussed in this dissertation: the interpretation of the particles is sensitive to the alternatives introduced by the focus constituent.

The size of the focus can be bigger than what we just observed. The whole VP (2) and even the entire clause (3) can be focused. In a language like English, what is common to all the cases is the placement of the *focal stress* on the most deeply embedded argument, which is usually the object.

the same focus *vegetable*, which is already given in the previous example, is repeated as the focus associate of the focus-sensitive particle *only*. Since such particles are known to associate with focus, thus, the second occurrence of the focus *vegetable* is de-accented, or has a secondary prominence according to some experimental studies (see a.o. Beaver et al., 2007; Beaver and Clark, 2008; Féry and Ishihara, 2009; Howell, 2010; Biring, 2015).

- (i) a. Everyone knew that Mary only eats [VEgetables]_{FOF}.
b. If even [PAUL]_{FOF} knew that Mary only eats [vegetables]_{SOF}, then he should have suggested a different restaurant.
(Adapted from Baumann (2016, p.483) citing Partee (1999); FOF=First Occurrence Focus, SOF=Second Occurrence Focus)

³ See Krifka (2008), section 3.2, for a classification of the different nature of the alternatives relevant for interpretation- *expression focus* versus *denotation focus*.

⁴ The term *alternative semantics* was introduced by von Stechow (1989) (cf. Rooth, 1992, fn.1).

⁵ See Schwarzschild (1999), however, for a discussion on some of the problems that arise with regard to the semantic type of the alternatives.

- (2) a. What did Adam do?
b. Adam [bought a CAR]_F.
- (3) a. What happened?
b. [Adam bought a CAR]_F.

There are many studies that discuss the position of the focus stress in relation to the larger focused constituent. In the simplest case, the word which possesses the focal stress corresponds to the focus, as in (1-b) above. However, this is not always the case. A focus constituent (for example, a VP or a CP/TP) may contain the word bearing the focal stress. This relationship between information structure (and focus in particular) and prosody has received much attention starting with earlier works such as Bolinger (1958, 1972); Chomsky (1971); Jackendoff (1972). There have been many more works on the theories of focus prosody with some variations among them (see a.o. Selkirk, 1984, 1995; Rochemont, 1986; Truckenbrodt, 1995; Zubizarreta, 1998; Schwarzschild, 1999; Reinhart, 2006; Assmann et al., 2023). These studies, according to Arregi (2016), can be classified into two separate accounts: the *Default prosody* account and the *F-projection* account. Briefly, in the default prosody account, there is an indirect relationship between prosody and focus. The latter is taken to be a result of the general prosodic property of the language, where the default prosody is independent of the focus marking. For instance, in the default prosody of a clause, the VP is believed to be metrically stronger than the subject. Thus, the focal stress is most likely to be on a constituent embedded in the VP: the object (Assmann et al., 2023). On the other hand, the F-projection account takes a more direct approach to focus marking and prosody. The sentence prosody is said to be dependent on the focus, with cross-linguistic differences based on focus projection rules and pragmatic constraints (cf. Arregi, 2016; Assmann et al., 2023).

To elucidate these processes, I provide Selkirk’s (1995) focus assignment rules below.⁶

- (4) *Basic Focus Rule:*
An accented word is F-marked.
- (5) *Focus Projection Rules:*
 - a. *Head Projection:*
F-marking of the head of a phrase licenses the F-marking of the phrase.
 - b. *Argument Projection:*
F-marking of an internal argument of a head licenses the F-marking of the head.

(Selkirk, 1995, p.555)

There are also semantic-pragmatic principles of F-marking that distinguishes focus (new) elements from given elements (6).

- (6) a. The focus of a sentence is an unembedded F-marked constituent (i.e., an F-marked constituent not dominated by any other F-marked constituent).
b. An embedded F-marked constituent is new (i.e. not given).
c. A non-F-marked constituent is given.

(Selkirk, 1995, pp.555-556)

⁶ The focus assignment rules pay emphasis on argument structure (consider the rule in (5-b) especially) which is not the case with the default prosody account (cf. Jackendoff, 1972).

Here is how it works. In (7-a), the direct object is accented, and thus F-marked, based on the rule in (4). Since there is no other F-marked constituent dominating the direct object, it is, therefore, the focus of the sentence, based on the interpretive principle in (6-a). Next, the argument projection rule in (5-b) allows the F-marking on the direct object to license the F-marking on the verb. F-marking on the verb then licenses F-marking of the entire VP, as in the head projection rule in (5-a). This is exemplified in (7-b) with the VP now the focus in line with (6-a). Based on the assumption that the verb heads the sentence, F-marking of the verb can license the F-marking of the entire sentence (cf. Head projection (5-a)). This would result in the sentence being focused, (7-c). In each of these cases, any constituent that is neither F-marked nor licensed is assumed to be given following the interpretive rule in (6-c).

- (7) a. [TP John [VP saw MARY_F]]
 b. [TP John [VP saw MARY_F]_F]
 c. [TP John [VP saw MARY_F]_F]_F

Subject focus, on the hand, is accented (and thus F-marked) independently and does not project since it is not an internal argument (8).⁷

- (8) [TP JOHN_F [VP saw Mary]]

2.1.2 Focus in morphosyntactic focus marking languages

Apart from prosody, cross-linguistic studies have shown that languages also mark focus morphologically and/or syntactically. This type of focus marking is dominant among the African languages.⁸ For instance, in some African languages, the focus morpheme is only present when the focus constituent is fronted to the left periphery, while there are those that mark focus morphologically with or without movement. In the Yorùbá (Benue-Congo) example in (9), the focus constituent is obligatorily followed by the focus marker *ni* when the former is fronted (9-b). However, when the focus constituent is in an in-situ position (i.e. its canonical base-generated position), the focus marker cannot be present (9-c).

- (9) a. Kí **ni** Adé rà?
 what FOC Adé buy
 ‘What did Adé buy?’
 b. Iṣu ***(ni)** Adé rà.
 yam FOC Adé buy
 ‘Adé bought A YAM.’
 c. Adé rà iṣu ***(ni)**.
 yam FOC Adé buy
 ‘Adé bought A YAM.’

A language like Likpakpaanl (a Mabia language spoken in Northern Ghana), on the other hand, obligatorily marks focus, morphologically, both when the focus is realized

⁷ See Schwarzschild (1999); Wagner (2005); Biring (2006); Breen et al. (2010); Arregi (2016) a.o., for a consideration of the wrong predictions made by the F-projection approach and the arguments supporting the view that focus projection depends on the default prominence of a sentence.

⁸ Due to the wide range of studies in this regard, we cannot do justice to all of them here. But see Assmann et al. (2023) for a recent overview of the placement of different focus markers from different African languages.

clause-internally (10-b), and when it is fronted to the left periphery (10-c).⁹ In addition to É. Kiss’ (1998) proposal that contrastive focus is syntactically marked by focus fronting, focus fronting can also result in some other discourse related interpretations like existence presupposition and mirativity (cf. Cruschina, 2009, 2012, 2019; Grubic et al., 2019; Zimmermann and Kouankem, 2024).

- (10) a. Stephen gà tér ɲmà dìn?
Stephen FUT help who today
‘Who will Stephen help today?’
b. Stephen gà tér Enoch *(lè) dìn.
Stephen FUT help Enoch FOC today
‘Stephen will help ENOCH today.’
c. Enoch *(lè) Stephen gà tér dìn.
Enoch FOC Stephen FUT help today
‘Stephen will help ENOCH today.’

In addition to the strategies above, it is also possible to have instances where different focus sizes are realized with the focus marker occurring in the same surface position in the clause. An example of such is found in Kasem. The two *wh*-questions in (11-a) and (11-b) can be answered with (11-c). The linear order of the sentence in (11-c) remains the same regardless of whether it is the VP or the direct object that is focused.

- (11) a. Bɛ **mo** Adam kea?
what FOC Adam do
‘What did Adam do?’
b. Bɛ **mo** Adam go-a?
what FOC Adam slaughter-DISJ
‘What did Adam slaughter?’
c. Adam [go [chwóró kóm]_(F)]_(F) **mo**.
Adam slaughter fowl DEF FOC
‘Adam [SLAUGHTERED [THE FOWL]].’

At a first glance, one might assume that the focus projection rules discussed at the end of the last subsection (§2.1.1) can also account for the observed behaviour in morphosyntactic focus marking languages. This would mean that the focus marker *mo* morphologically marks the object, and then this marking projects upward unto the verb and even the VP, according to the focus projection rules in (5) above. However, this raises a few questions and/or challenges. First, can we equate morphological focus marking to prosodic focus marking? Secondly, many morphosyntactic focus marking languages have no in-situ morphological marker (see Hartmann and Zimmermann (2007): Hausa; Duah (2015): Akan; Hartmann and Zimmermann (2012): Bura, etc.). For example, if we consider the earlier Yorùbá example repeated below (12) (given as (9) above), the in-situ focus is morphologically unmarked (12-c). This is problematic for focus projection which requires a focus exponent for focus projection to proceed.

- (12) a. Kí **ni** Adé rà?
what FOC Adé buy
‘What did Adé buy?’

⁹ The data is from Samuel Acheampong (pers. comm.).

- b. Iṣu *(**ni**) Adé rà.
yam FOC Adé buy
'Adé bought A YAM.'
- c. Adé rà iṣu (***ni**).
Adé buy yam FOC
'Adé bought A YAM.'

Thirdly, when we consider more cross-linguistic data, focus projection fails to account for the different strategies of focalization in morphosyntactic focus marking language. For example, in Buli (another Maba language), the focus sentence in (13-b) can be used to correct the proposition in (13-a). Notice that here the focus marker *ká* is not linearly adjacent to the focus. It occurs at the periphery of the DP containing the focus. It remains to be seen how focus projection would account for such cases because nothing is projecting upward.

- (13) a. The boy is riding a blue bicycle.
b. Aaya, nidoa-bini a do **ká** kutug-wusum [sogluk]_F.
no male-small.DEF IPFV ride FOC iron-horse dark
'No, the boy is riding a [black] bicycle.'

(adapted and modified from Assmann et al., 2023, p.1365)

Another problem that morphosyntactic focus marking languages pose to the focus projection approach is with respect to cases where complex constituents like VP and TP are focally marked directly. In such cases, there is no projection whatsoever from a sub-constituent as predicted by the focus projection approach. This is exemplified in Gùrùntùm (a Chadic language spoken in Northern Nigeria) example given below (from Hartmann and Zimmermann, 2009, p.1359). The morphological marking is consistently on the whole TP. Thus, no focus projection has taken place.

- (14) [TP Tí vún lúurìn nvùrì]_F-à.
3SG wash clothes yesterday-FOC
'She washed clothes yesterday.'

These issues, coupled with the fact that no attempt has been made to compare the focus projection theory proposed for the prosodic focus marking languages like English and the morphosyntactic focus marking languages, have led to a proposal by Assmann et al. (2023).

Assmann et al.'s (2023) "Towards a theory of morphosyntactic focus marking"

In a recent study, Assmann, Büring, Jordanoska and Prüller (2023) did a detailed study of six morphosyntactic focus marking languages, and developed a theory of morphosyntactic focus marking which captures what they called *focus syncretism*: although the marking of different focus sizes (V(P), object or TP) may seem to involve the same form, there is no focus projection.¹⁰ In their approach focus marking can directly mark a complex constituent which may be the logical focus or may directly embed the logical focus as the case may be. This is in contrast to the focus projection approach which requires that a sub-constituent of a phrase has to be focally marked before it projects to

¹⁰ Although they did a detailed study of 6 languages, their survey covered over 20 morphosyntactic focus marking languages.

the phrase itself through the head of the phrase. The focus exponent in this approach is not necessarily a word, but can be a VP or even a TP.

Take for instance a language like Gùrùntùm. The focus marker *á* can exclusively mark subject focus when it precedes the constituent (15-b). However, when *á* occurs between the verb and the direct object as in (16), it can mark not only V focus, but also object, and even VP focus. The sentence in (16) is a felicitous answer to questions which require the three foci respectively (*What is he doing to the seeds?*, *What is he doing?*, and *What is he gathering?*). So this structure is *syncretic* by corresponding to three possible semantic foci: V, VP & Object.

- (15) a. ‘Who is chewing colanut?’
 b. **Á** fúrmáyò bà wúm kwálingálá.
 FOC [Fulani]_F PROG chew colanut
 ‘THE FULANI is chewing colanut.’
- (16) Tí bà [[ròmb]_F-**á** [gwéi]_F]_F.
 3SG PROG gather-FOC seeds.
 ‘He is [[GATHERING] [THE SEEDS]].’

Assmann et al. (2023, pp.1352-3) citing Hartmann and Zimmermann (2009)

Thus, going by Assmann et al.’s (2023) proposal, the *biggest* pragmatically focused constituent that a syncretic structure produces is focally marked (regardless of the surface position of the morphological marker, e.g., (16)). This is the case in Gùrùntùm where the biggest possible focus constituent is focally marked, which is the VP in this case. Their analysis seems to dissociate *focal marking* from *morphological placement*, although the latter appears to feed the former (see Assmann et al., 2023, section 4). Thus, in the Gùrùntùm example in (16), while the focus morpheme suffixes to the verb, it focally marks the VP, which is the broadest possible pragmatic focus that the given construction can have.

This view of the morphosyntactic focus marking languages, according to Assmann et al. (2023), raises at least two observations in comparison to what we already know with the focus projection theory. First, one might say that this theory is just an alternative way of presenting what has already been discussed in the focus projection theories; which is the idea that a nuclear accent on the object constituent can project up to both the VP level and the clausal level (but not the subject or the verb) through focus projection rules (Selkirk, 1995; Schwarzschild, 1999). Secondly, how do we possibly exclude cases where distinct focus markers mark distinct focus constituents or focus sizes? These observations are attended to below.

In order to distinguish their novel theory of morphosyntactic focus marking languages from the mainstream focus projection theory, Assmann et al. (2023, p.1350) propose two basic constraints which guide the type of focus marking in morphosyntactic focus languages (17).

- (17) a. **Direct marking/No projection:** Each focal marking marks exactly one constituent as FOCAL; crucially, this constituent may be non-terminal and in fact as big as an entire clause.
 b. **Blocking:** One has to choose the most specific focus marker that is pragmatically appropriate.

The first constraint excludes the possibility of focus projection. A focal marking that

is syncretic (marking more than one focus sizes) marks one and only one constituent, which may be the biggest constituent. Also, “a focal marking can never realize a focus that is bigger than the focally marked constituent” (Assmann et al., 2023, p.1354). Here, there is a difference between formal focal marking and pragmatic focus marking. While a constituent like the VP can be focally marked, the pragmatic focus may be on the object of the verb; a sub-constituent within the VP. This is reminiscent of the Gùrùntùm case in (16) above where the VP is argued to be focally marked, but its sub-parts, V and object, can be focused as well. Subject focus (15), on the other hand, is both the focally marked constituent as well as the pragmatic focus. Thus, all focally marked constituents can be a pragmatic focus, but not all pragmatic foci can be focally marked.¹¹ So “broad foci do not project from narrow foci by specialized projection rules, they are directly licensed by morphological focal marking” (Assmann et al., 2023, p.1355).

The second constraint ensures that focus syncretism is blocked when there is a specific focus marker that marks a pragmatic focus. This is dependent on the set of morphological focus markings that are present in a given language. Take for instance, clausal focus in Gùrùntùm (18), where the structure below cannot be used for either VP or subject foci because both already have specialized focus marking as in (15-b) & (16) above. Since we say that only the broadest possible focus is focally marked, while its sub-constituent can be pragmatic focus, one would predict that the focus structure in (18) (given as (14) above) might be used not only for clausal focus, but also for any of its sub-constituents. However, this is not the case because the sub-constituents (subject and V(P)/Object) are marked distinctively. This is captured by the blocking effect, whereby if a language has more specific focus markers for sub-constituents of a (focally marked) bigger constituent, the former marking is preferred over being syncretic with the bigger constituent.

- (18) Tí vúm lúurìn nvùrì-à.
 3SG wash clothes yesterday-FOC
 ‘[She washed clothes yesterday].’

The blocking effect is comparable to Schwarzschild’s (1999) *AvoidF* constraint which minimizes f-marking as much as possible. So, if a sub-constituent of a complex phrase can be independently marked, there is no need to focally mark the complex phrase again.¹²

¹¹ Additionally, Assmann et al. (2023, p.1375) (citing Fominyam and Šimík (2017)) discussed data from Awing (a Grassfield Bantu language spoken in Cameroon) that shows focus syncretism of object, V, VP, subject and TP foci. Although Awing has morphological focus markers which are used for contrastive focus or exhaustive focus, in new information focus, the markers are often absent. So the sentence in (ii) is a syncretic answer to all the *wh*-questions in (i).

- (i) a. What did Alombah cook?
 b. What did Alombah do with maize?
 c. What did Alombah do?
 d. Who cooked maize?
 e. What happened?
- (ii) Alombah a-pe’-náŋnə ŋgəsáŋə.
 Alombah SM-PST-cook maize
 ‘Alombah cooked maize.’

¹² See Beaver and Clark (2008) ch. 2 & 3 for a revision of Schwarzschild’s (1999) focus projection constraints due to some empirical challenges, which the latter cannot account for.

It however remains unclear how such a theory would address the interaction between focus fronting and the blocking effect discussed above. In languages where both in-situ and ex-situ focus realizations are possible, the open question is “which of the realizations does blocking apply to?”¹³ A similar question arises when we consider the morphological marking asymmetry that exist between local subject focus and non-subject focus or non-local subject focus. In some of the Mabia languages, the form of the focus marker changes depending on whether it is the local subject that is focused or the non-subjects and non-local subject. This is discussed in the next chapter for the focus realization strategies of Kusaal (see ch. 3, subsection 3.3).

2.1.3 The QUD model and focus

Before we close this subsection, there is a need to discuss the approach upon which the focus realization, that is discussed in this dissertation, is built. This is necessary because focus and its salient alternatives form the basis for the subsequent discussion and analyses in this study.

In the preceding section we saw how focus is realized from questions. In fact, both have a lot in common, including their semantics. For instance, focus realization involves a *background* which is the part that is unmarked for focus; no prosodic prominence as in English or focus morpheme or movement as in the African languages discussed above. Thus, the background of the sentence with a focus in (19-a) is (19-b).

- (19) a. Adam bought [a CAR]_F.
b. Adam bought *x*.

Additionally, the earlier discussions show that each focus seems to answer a question, whether explicitly or implicitly. So, the answer is congruent to a *Question Under Discussion* (QUD henceforth) (cf. Roberts, 1996; Büring, 2003; Beaver and Clark, 2008, a.o.). Analysing the relationship between such question-answer pairs has led to different approaches. The most commonly discussed is the Hamblin (1976) approach to question semantics. According to Hamblin, the meaning of a question like (20-a) is the set of *possible* answers (20-b) (Beaver and Clark, 2008).¹⁴

- (20) a. [What did Adam buy?]
b. {Adam bought a car, Adam bought a bicycle, Adam bought a plane, ...}

What we just saw is reminiscent of something we have discussed earlier, i.e., Rooth’s (1985; 1992) alternative semantics approach to focus. In fact, Rooth’s alternative semantics of focus is similar to Hamblin’s alternative semantics of questions. Although both approaches may appear to be the same on the surface, they are however different both in their basic assumptions and domain of application. In Hamblin’s approach to question semantics, questions denote the alternative set of propositions, which are possible answers to the question, as in (20-b) above.¹⁵ In Rooth’s focus semantics approach on the other

¹³ There are a few other open questions raised by Assmann et al. (see especially Assmann et al., 2023, sec. 7 & 8).

¹⁴ This is different from a similar proposal by Karttunen (1977) who sees question meaning as just a set of true answers. See also Groenendijk and Stokhof (1984) for a denotation of question as not only a set of (true) possible answers, but a set of *exhaustive* possible answers (Beaver and Clark, 2008).

¹⁵ Later, Karttunen (1977, p.10) proposes that “questions denote the set of propositions expressed by

hand (as discussed above), in addition to an ordinary value, focus has a semantic value which is a set of alternative propositions which is only different in the sub-constituent that is in focus (in the case of narrow focus). Rooth’s alternative semantics goes further to allow the conjunction of alternatives in a set of propositions. Thus, not only {Adam bought a car, Adam bought a bicycle, Adam bought a plane...} is a set of alternatives, but also {Adam bought a car and a bicycle, Adam bought a bicycle and a plane, Adam bought a plane and a car, Adam bought a car, a bicycle and a plane...}. In addition, Rooth’s alternative semantics is used to model other focus-related effects like focusing adverbs (association with focus), contrast, scalar implicature, and the question-answer congruence. So “in each case, the story is that a semantic or pragmatic rule somehow uses the focus semantic value” (Rooth, 1992, p.77). Hamblin’s theory is only concerned with the alternative propositions of *wh*-questions. But generally speaking, both approaches rely on computing alternative sets for their theory. Now, let’s go back to how focus relates to questions.

A focus constituent is part of a discourse and does not occur in a vacuum. Moreover, discourse is dynamic, and so is the information shared. In fact, every assertion is seen as a development of the shared *common ground* (which is a set of propositions taken as true) between interlocutors (Stalnaker, 1974, 1978). The length and breath of the discourse is determined by the commitment of the interlocutors (Hamblin, 1971). The idea is that at the initial stage, the common ground is empty. As the conversation proceeds, the common ground is updated by adding new information that is judged to be true by the participants. In more technical terms, Stalnaker (1974, 1978) proposes that a discourse involves sets of possible worlds which are intersected in the course of the discourse until the context set is reduced to the minimum: the actual world.

Roberts (1996, 2004), on the other hand, ties the management of the common ground in the discourse to “sequences of questions” (Beaver and Clark, 2008, p.34). Unlike Stalnaker’s (ibid.) intersection of possible worlds, Robert’s (ibid.) strategy for reducing the set of possible worlds is a sequence of questions. Here is how it works. Assume a scale of possible questions that are relevant to the common ground. On this scale, there are questions from less specific (or more general) to more specific ones. Any question that is attended to (attempted) at any specific point in the discourse is called the *Current Question* (CQ henceforth) (Roberts, 1996, 2004; Beaver and Clark, 2008). So, as the discourse proceeds, and the common ground is updated, successive CQs emerge (which were once part of the *Open Questions* according to Beaver and Clark (2008)) in the form of propositions after the answer to a preceding CQ has become an old information. The CQ rule is given below (21).

(21) **Current Question Rule (CQR)** (Beaver and Clark, 2008, p.36):

The Current Question must contain at least one true alternative, and contain multiple alternatives which are not resolved as true or false in the common ground.

Influenced by Hamblin’s semantics, CQR requires that out of the set of possible alternative propositions, at least one proposition must be the true answer to the CQ.

Furthermore, there is a lot of intricacies going on within the discourse. For instance, certain assertions can be accommodated in the common ground. The only condition is

there *true* answers instead of the set of propositions expressed by their *possible* answers.” However, Karttunen admits that he has no concrete argument against Hamblin’s original proposal, and believes that Hamblin’s proposal could also be made to do the job his own proposal does. Therefore, I maintain Hamblin’s question semantics for this dissertation.

that every assertion or contribution to the discourse must be relevant, in what Beaver and Clark (2008, p.37) call the *Discourse Principle*.

(22) **Discourse Principle:**

Utterances should be maximally relevant to the CQ.

Not only accommodation, but *conversational implicature* is also employed in order to keep the discourse running. Recall that it was mentioned at the beginning of this section, that the (open) questions that form the basis for the assertions in the discourse are not only explicit, but can also be implicit. In fact, in a natural language setting, most of the discourse is centered around implicit questions, which can be implied by previous knowledge or context (cf. Grice and White, 1961; Grice, 1975, 1978, 1989; Beaver and Clark, 2008).

In conclusion, this section has summarized some of the different focus marking strategies found recorded in the literature. I have also briefly introduced the QUD model which seems to unify both alternative semantics of focus and alternative semantics of questions (cf. Beck, 2006; Beaver and Clark, 2008). The conclusion is that not only semantics, but also pragmatics (à la Roberts, 1996, 2004) is pivotal in the discussion of focus and focus sensitivity. I will now turn to focus sensitivity in the next section.

2.2 Focus sensitivity and association

In the last section, we saw how focus indicates the presence of alternatives. In this section, I will discuss how these focus-induced alternatives are quantified in different ways for the interpretation of certain particles. There are a number of expressions which are conventionally sensitive to focus. Consider the example in (23). In (23-a), the presupposition is that apart from the gold medal won by the athlete, he/she must have won at least one other medal, maybe a bronze medal for example. The shift of the focal stress (from the adjective to the verb) in example (23-b), on the other hand, presupposes that in addition to winning, the athlete did something else to the gold medal, like touching the medal. The source of the differences in meaning lies with the expression *also*. Such expressions are called *Focus Sensitive Particles*.¹⁶

- (23) a. The athlete *also* won a [gold]_F medal.
 b. The athlete *also* [won]_F a gold medal.

In both examples, the ordinary semantic value of the propositions is their truth-condition without the contribution of the focus sensitive particle (i.e., there is an athlete who won a gold medal in the real world). This is the *prejacent*; which is the meaning of the proposition before the introduction and influence of a focus sensitive particle. Which constituent is F-marked is dependent on the current question (Beaver and Clark, 2008). Thus, the prejacent or assertion of both sentences are the same, as in (24) and (25) below. However, due to the different placement of focus in both sentences, different alternative sets are produced. The focus sensitive particle *also* then quantifies over these alternative sets which result in different presuppositions in (24) and (25).

¹⁶ At least for *only* and *even*, Fischer (1968) happens to be one of the earliest references to have linked these particles to focus (see also Chomsky, 1971; Jackendoff, 1972; Jacobs, 1983; König, 1991; Beaver and Clark, 2008, among others).

- (24) Focus sentence: The athlete *also* won a [gold]_F medal.
Prejacent & Assertion: The athlete won a gold medal.
Presupposition: The athlete won another medal.
- (25) Focus sentence: The athlete *also* [won]_F a gold medal.
Prejacent & Assertion: The athlete won a gold medal.
Presupposition: The athlete did something else to the gold medal.

The phenomenon we have been discussing is called *Association with Focus*.¹⁷ The term *Association with Focus* (AwF henceforth) was first used by Jackendoff (1972) for certain linguistic expressions whose interpretation are sensitive to phonological prominence. Such expressions include *negation*, *comparatives*, *superlatives*, *sentential connectives*, *modals*, *quantificational adverbs*, *quantificational determiners*, and so on (see also Rooth, 1992). That this group of expressions are indeed AwF expressions have been a source of debate for many years. Not until a few years ago, many analyses (in-)directly classify all of them as AwF expressions: expressions whose meaning are derived from the focus in a structure (Rooth, 1985, 1992; König, 1991; Beaver and Clark, 2008). Not until Beaver and Clark (2008) was a proper classification of these expressions done. Beaver and Clark (2008) argues that on closer scrutiny, it is not all focus sensitive expressions that are indeed sensitive to focus as far as grammatical focus is concerned. Following Tancredi (1990), a focus sensitive particle needs to c-command its focus associate (another name for the focus constituent in the context of AwF) (26). However, as we will see in the next section, this is not always the case.

- (26) **Principle of Lexical Association (PLA):**
 An operator like *only* must be associated with a lexical constituent in its c-command domain.

(Tancredi, 1990, p.30)

The violation of the PLA and some other behavioural differences among the focus sensitive particles led to the proposal of a three-way classification of these expressions by Beaver and Clark (2008). The following are the classifications (QFC-model for short).

- (27) a. Quasi association particles
 b. Free association particles
 c. Conventional association particles

2.2.1 Beaver and Clark's (2008) QFC-Model

Under the QFC-Model, all the expressions that were initially classified as focus-sensitive, do not have a uniform behaviour toward the focus associate and the set of alternatives that it indicates. In this section, I discuss briefly each of the classifications, and point out the areas where they differ.

Conventional association

Focus particles such as *only*, *also* and *even* are said to be conventional association operators because: (a) they are lexically encoded for focus sensitivity, and (b) their semantics

¹⁷ The particles have also been called *Alternative Sensitive Particles*, and the phenomenon, *Association with Alternatives* (Hartmann and Zimmermann, 2008; Grubic, 2015)

make direct reference to the set of alternatives of the focus constituent.¹⁸ In contrast to quasi and free association with focus particles, Beaver and Clark (2008, p.68) propose that conventional association with focus particles have a grammatical dependency on the CQ. This relationship is based on the idea introduced earlier that a focus constituent is naturally an answer to (an explicit or implicit) CQ. It is now the set of alternatives invoked by the focus that the conventional association particles control.¹⁹ Consider the examples in (28) in which the additive focus particle *also* associates with the indirect object in (28-a) and the direct object in (28-b). Having *also* in sentence (28-a) gives the interpretation that in addition to *Sam*, there is at least one other alternative (could be more) in the set of alternatives invoked by the focus $[Sam]_F$. Thus, the focus constituent cannot be the sole alternative who got a banana from Adam. In (28-b), the focus particle has the same effect. The only difference is with respect to the constituent in focus. So, in addition to *a banana*, Adam definitely gave Sam something else, e.g., an apple.

- (28) a. Adam *also* gave $[Sam]_F$ a banana.
b. Adam *also* gave Sam $[a\ banana]_F$.

Another conventional association with focus particle is *even*. The semantics of *even* is *scalar-additive* in nature. It involves a scale of the set of alternatives of the focus constituent where the alternatives are higher on the scale of the contextually-salient alternatives. Assuming that the set of alternatives of the focus in (29-a), include $\{Sam, John, Peter, Simon, \dots\}$, the presence of *even* in the sentence implies that it is more likely for Adam to give a banana to any of the other alternatives than it is to give a banana to Sam. In other words, the focus *Sam* ranks higher among the unlikely (noteworthy) people that Adam would give a banana to. The same goes for (29-b). The fact that Adam gave Sam a banana seems less likely than other contextually-salient alternatives (cf. Karttunen and Karttunen, 1977; Karttunen and Peters, 1979; König, 1991). Usually, the meaning expressed in the context of *even* is that of a surprise or unexpectedness.²⁰

- (29) a. Adam *even* gave $[Sam]_F$ a banana.
b. Adam *even* gave Sam $[a\ banana]_F$.

Unlike the additive and the scalar-additive particles, the exclusive particle *only* excludes all other alternatives apart from the constituent in focus. For example, the presence of the exclusive particle in (30-a) presupposes that Adam did not give a banana to any other individual aside Sam. Thus, the whole proposition *p* is true such that any other alternative proposition *p'* to *p* is false. Likewise, the prejacent in (30-b) is the only true proposition which answers the QUD. Any other alternative proposition like $\{Adam\ gave\ Sam\ an\ apple, \dots\ a\ mango, \dots\ an\ orange, \text{ and so on}\}$ are false based on the exclusive interpretation.

- (30) a. Adam *only* gave $[Sam]_F$ a banana.
b. Adam *only* gave Sam $[a\ banana]_F$.

So far, the discussion on the exclusive particle *only* has been on its *non-scalar* property

¹⁸ I use the terms ‘focus (sensitive) particles’ and ‘focus (sensitive) operators’ interchangeably. Both of them refer to the same notion. The latter may be used in order to fit certain contexts, however.

¹⁹ This is in contrast with the probabilistic view of Roberts (1996) and von Stechow (1994) where the focus particles “may be anaphoric on the current question or the discourse topic” (Beaver and Clark, 2008, p.42).

²⁰ This is very similar to mirativity (cf. Cruschina, 2009, 2012, 2019)

or use. It has however been argued in the literature that the exclusive particle also has *scalar* uses (see a.o. Jacobs, 1983; König, 1991; Beaver and Clark, 2008; Greenberg, 2019, 2022). Here, the domain of *only*'s quantification is a scale which consist of a set of salient alternatives. Both the role of *only* on the set of alternatives on a scale, and its evaluative property have been sources of debate for a while (see Jacobs, 1983; König, 1991; Beaver and Clark, 2008). Considering the example in (31-a), the presence of *only* presupposes that the excluded set of alternatives are ranked higher than the focus associate itself. But based on the scalar uses of *only*, (31-b) is odd since being a pro is assumed to be the highest level of expertise that can be attained. However, if we assume a non-scalar use of *only*, (31-b) may mean that out of all the possible conditions to be at the bootcamp, Adam only met the condition of being a pro when he joined.

- (31) a. Adam was *only* [a novice]_F when he joined the bootcamp.
 b. Adam was *only* [a pro]_F when he joined the bootcamp.

In a direct contrast to instances like (31-a), examples such as (32-a) show a reverse context where the excluded alternatives seem to be ranked lower than the focus associate. This becomes even more obvious if there is a continuation as in (32-b). While the first continuation fits well with the previous sentence because any other excluded alternative is ranked lower, the second continuation is bad because the focus associate ranks higher than every other excluded alternatives.

- (32) a. *Only* [a professor]_F can supervise a doctoral dissertation.
 = Alternatives: [an associate professor can ..., an assistant professor can ...]
 b. *Only* [an army]_F can stop the ongoing riot. (Anything less cannot do/*anything more can do)
 = Alternatives: [the police can..., the vigilante can ..., etc.]

However, the reverse scale effect in (32) is only superficial according to König (1991), who assumes that the excluded contextually-salient set of alternatives still ranks higher than the focus associate on the scale. Consider the following examples from König (1991, p.101). According to König, the evaluation of the scale of the salient set of alternatives in sentences such as (33) remains the same as in the original scalar evaluation by *only* given earlier. The only difference is that (33) can be accounted for based on what Fauconnier (1975b, 1979) called *pragmatic entailment* as opposed to entailment in the regular sense of the word (König, 1991).

- (33) a. *Only* a [miracle]_F can save us (i.e. nothing short of a miracle can save us).
 b. *Only* a [radical]_F change will save our economy.

What this means is that the (a) sentences in (34) and (35) 'pragmatically entail' the (b) sentences (adapted from König, 1991, p.102).

- (34) a. Ordinary measures can save us.
 b. A miracle can save us.
 (35) a. A cosmetic operation can save our economy.
 b. A radical operation can save our economy.

Other exclusive particles such as *solely*, *simply*, *but*, and *alone* are compatible with both scalar and non-scalar interpretations. However, while *purely* and *exclusively* are reported to be strictly non-scalar (i.e. they do not refer to a scale in their quantification) (36),

merely and *just* on the other hand are strictly scalar (37) (König, 1991; Beaver and Clark, 2008; Coppock and Beaver, 2014).²¹

(36) Adam was *exclusively/purely* [a novice]_F when he joined the bootcamp.

(37) Adam was *merely/just* [a novice]_F when he joined the bootcamp.

The apparent questions that arise with respect to the different uses of *only* is whether (a) *only* is polysemous- having both scalar and non-scalar meaning- or (b) *only* is always scalar and evaluative or (c) *only*'s semantics is canonically vague with respect to (non-)scalar interpretation, leaving the discourse to determine the (non-)scalar meaning. For discussions of these questions, see Jacobs (1983); König (1991); Beaver and Clark (2008); Greenberg (2022); Coppock and Beaver (2014), among others.²² I will return to an elaborate discussion on the notion of exclusive scalarity in Chapters 4 and 5, where it will become important in the languages under investigation.

Free association

The second degree of association, according to Beaver and Clark (2008), consists of expressions that are not constrained in their association with focus. In other words, they are not conventionally dependent on focus per say, nor are they grammaticalized for focus dependency. They include quantificational adverbs, quantificational determiners, generics, superlatives, verbs of desire, emotive factive verbs, and counterfactuals (Beaver and Clark, 2008). For example, in each of the sentences in (38), with their implications quoted below, the quantificational adverb *always* involves two arguments: the *restrictor argument* and the *scope argument*. While the scope argument is the immediate clause containing it, the restrictor is the created context which might be explicit or implicit. When the restrictor argument of the operator is explicit, it becomes a free variable *x* which equals to a set of events where the use of *always* (with respect to the scope argument) is true. In (38-b) for example, it is not the case that Adam never gives Sam anything else other than a banana. It only means that based on the salient set of events in which *always* quantifies over, Adam always gives Sam A BANANA. Thus, each of the examples in (38) are felicitous in their respective discourse contexts (Beaver and Clark, 2008).

(38) a. Adam *always* gives [Sam]_F a banana.

“Whenever Adam gives someone a banana, that person is Sam.”

b. Adam *always* gives Sam [a banana]_F.

“Whenever Adam gives something to Sam, that has to be a banana.”

In summary, the truth-conditions of the propositions in which these operators appear are dependent on the domain of their quantification which involve or embed focus, thus explaining why they have been described as focus-sensitive in earlier works with the lexical (conventional) association with focus (Beaver and Clark, 2008).

Quasi association

The term *quasi* simply means to have a resemblance or likeness to something. There-

²¹ The concept of scale and scalarity is more complex than can be outlined here. The scale that is used by focus sensitive particles is only a part of the whole concept. There are other areas within the sentence which utilizes scales. For example, the sentence *The water is hot* entails that the water is warm, and thereby involves a scale of the temperature of the water. For a more comprehensive discussion on scalar phenomena, see among others Horn (1972); Fauconnier (1975a, 1979); Gazdar (1979); König (1991, ch.3)

²² See, especially, Greenberg (2022) for a good study on the similarities and differences between *even* and *only* with respect to their scalar interpretation and evaluation.

fore, although quasi-operators may seem to behave like conventional focus sensitive operators, they are non-veridical operators (i.e. they do not entail the truth of the proposition that they modify). According to Beaver and Clark (2008), “the interaction between focus and the quasi operators produces cancelable implicatures rather than indefeasible truth conditional effects” (which we find with conventional operators). These operators include: negation; sentential connectives like *either ... or*; possibility modals like *perhaps, might*; verbs of appearance like *seem*; and belief operators like *think* (Beaver and Clark, 2008). Although these operators appear to be focus-sensitive, this sensitivity is not due to a direct semantic dependency on the alternative of the evoked set of alternatives of the focus. Instead, it arises from general pragmatic principles, e.g. the QUD.

Take for instance the negative operator in the sentences below (39). In (39-a), the focus (a banana) is excluded from a possible set of things that Adam could have given to Sam. In (39-b) on the other hand, it is the indirect object- Sam- that is excluded from the possible individuals who might have received a banana from Adam.

- (39) a. Adam did *not* give [a banana]_F to Sam.
 b. Adam did *not* give a banana [to Sam]_F.

Although it may seem like the negative operator’s impact is detectable with respect to what the focus is in the sentence, this impact is rather indirect. This is shown in the questions that both sentences can answer. Recall that the Rooth-Hamblin alternative semantic approach allows a set of alternatives to be a set of individuals and/or the conjunction of individuals which answer a CQ under discussion. With this in mind, (39-a), for example, can be a direct answer to (40-a) as well as a partial answer considering the set of alternatives. Adam might as well not have given Sam something else (e.g. an avocado) or even anything; except if we consider (39-a) to be exhaustive (see Beaver and Clark, 2008, pp.45-49 for a discussion on how different intonation contours can constrain the appropriateness of some answers to some CQs.). Example (39-a) is also a possible answer to the CQ in (40-b). Similarly, (39-b) is a possible answer to both questions in (41). Thus, the noticeable inference that we get between the negative operator and the focus is more or less as a result of the fact that the argument (i.e. the prejacent) of the operator corresponds to the CQ. It is this CQ that comes with a requirement of containing at least one true alternative in its set of alternatives.

- (40) a. What didn’t Adam give to Sam?
 b. What did Adam give to Sam?
- (41) a. To whom did Adam not give a banana?
 b. To whom did Adam give a banana?

The point here is that the interpretation of quasi-operators like negation can change with explicit changes in CQ. More so, negation as a non-veridical operator only operates on the proposition in its scope, which happens to include a focus constituent.

2.2.2 Testing the conventional dependency of focus sensitive particles

So far, we have maintained the idea that focus sensitive particles do not behave the same way, leading to the advancement of the QFC model by Beaver and Clark (2008). The motivation for the QFC model is also evident on the basis of the behaviours of the different

focus particles under certain conditions. Consider the following examples with *always* and *only* under ellipsis.

- (42) a. Kim *always* SALUTES because Sandy *always* does ~~salute~~.
 [can mean: ‘Kim salutes at every ceremony because Sandy salutes at every ceremony’]
 b. *Kim *only* SALUTES because Sandy *only* does ~~salute~~.
 [cannot mean: ‘Kim salutes (and does nothing else) because Sandy salutes (and does nothing else).’]
 (adapted from Beaver and Clark, 2008, p.177)

While *always* can associate with elided materials (42-a), *only* cannot (42-b). Bassi et al. (2022) call this the ‘Beaver & Clark’s Constraint’ (43). (42-b) is ungrammatical because the focus associate of *only* in the second clause has been elided.

- (43) **Beaver & Clark’s Constraint** (formulated by Bassi et al., 2022, p.817):
Only and its associated Foc cannot be separated by a node targeted for ellipsis.

Another noticeable interaction, according to Beaver and Clark (2008) is with extracted materials. Observe the *wh*-movement constructions in (44) & (45). When the associated *wh*-phrase of *always* is fronted, the sentence can have the interpretation where the quantificational adverb associates with the fronted element (44-b). *Only* on the other hand, can only associate with the materials in its scope (45-c) & (45-d) but not the fronted material (45-b).

- (44) a. What does Adam *always* buy ___ at the market?
 b. → ‘What is x such that whenever Adam goes to the market he buys x?’
 (45) a. What does Adam *only* buy ___ at the market?
 b. → *‘What is x such that whenever Adam goes to the market he buys x and nothing else?’
 c. → ‘What is x such that whenever Adam goes to the market he only buys x and does nothing else to x in addition (like eat or sell it)?’
 d. → ‘What is x such that Adam only buys x at the market, and no where else?’

The Principle of Lexical Association given above in (26), repeated below, predicts that all focus sensitive particles must c-command their focus associate. But the examples show that while *only* seems to obey the PLA by failing to associate with an extracted material, *always* violates the PLA because it is able to associate with an extracted material.

- (46) **Principle of Lexical Association (PLA):**
 An operator like *only* must be associated with a lexical constituent in its c-command domain.

(Tancredi, 1990, p.30)

Other evidence that support the call for a sub-categorization of the focus sensitive particles include their interactions with reduced materials (i.e. second occurrence focus), monotonicity, and presupposition. See Beaver and Clark (2008, ch. 6, 7, & 8) for a detailed discussion on these interactions.

2.2.3 Some exceptions and the need for further investigation

Not only do the quasi, free and conventional focus sensitive particles behave differently, but cross-linguistic studies have also shown that even the so-called conventional association particles- *only*, *also*, and *even*- do not always associate conventionally in a few studied languages so far (see a.o. Zimmermann, 2015, 2006; Hartmann and Zimmermann, 2008; Grubic and Zimmermann, 2011; Duah, 2015). In some of these languages, ex-situ focus is morphological marked in the presence of an exclusive particle, but the focus marker must be absent in the presence of additive/scalar-additive particles. First, consider the behaviour of the exclusive particle and the additive particle in the following examples from Akan (Duah, 2015, p.14 & fn.7).²³ While the focus marker *nà* is optionally present with the exclusive particle (47), it is obligatorily absent with the additive particle (48).

- (47) a. Did Kwame and Kwaku go home?
 b. Kwame **ńkóáá** (nà) ɔ-kɔɔ-ɔ fíé.
 Kwame only FOC 3SG-go-DISJ home
 ‘Only KWAME went home/KWAME went home alone.’
- (48) a. Who else ate the food?
 b. Frema **ńsó** (*nà) dì-ì àdùàné nó bì.
 Frema also FOC eat-COMPL food DET some
 ‘FREMA also ate the food.’

Another contrast within the class of Beaver and Clark’s (2008) conventional association particles comes from Ngamo, a West Chadic language spoken in Northern Nigeria (Grubic and Zimmermann, 2011, p.298). In Ngamo, *-i* is a background marker according to Grubic and Zimmermann (2011); Grubic (2015). It separates the backgrounded part of a sentence from the focused constituent. The subject in (49) is obligatorily focused (indicated by the presence of the background marker) when associating with the exclusive particle *yak*. The exclusive particle associates with an inverted subject, dislocated from its canonical subject position, according to (Grubic and Zimmermann, 2011).

- (49) a. Sàlko bànò -ì **yak** [Kulè]_F.
 build-PFV house BM only Kule
 ‘Only KULE built a house.’
 b. **#(Yak)** [Shuwa]_F (**yak**) sàlko bànò (**yàk’i**).
 only Shuwa only build-PFV house only
 (intended:) ‘Only SHUWA built a house.’

On the other hand, the additive particle *ke* and the scalar-additive particle *har* cannot associate with an inverted subject that follows the background marker (50). Association with (scalar)-additive particles is only possible if the subject remains in the canonical pre-verbal position without the background marker (51).²⁴

²³ Note that what I glossed as the focus marker here was glossed as a particle in Duah (2015). But it is indeed what many speakers of Akan believe to mark the focus. It behaves as such even in Duah’s paper. The same morpheme is also used in *wh*-questions.

²⁴ Note also the free occurrences of the focus sensitive particles which suggest their adverbial nature. Only the exclusive particle can be adfocal (Grubic and Zimmermann, 2011).

- (50) a. #Salko bano -i **ke** [Kulè]_F.
 build-PFV house BM also Kule
 (intended) ‘KULE also built a house.’
 b. #Salko bano -i **har** [Kulè]_F.
 build-PFV house BM even Kule
 (intended): ‘Even KULE built a house.’
- (51) a. **Kè/Har** [Kulè]_F sàlko bànò.
 also/even Kule build-PFV house
 b. [Kulè]_F **kè/har** sàlko bànò.
 Kule also/even build-PFV house
 c. [Kulè]_F sàlko bànò **kè’ê/hàr’î**.
 Kule build-PFV house also/even
 ‘KULE built a house, too.’

The last case is from Bura, a Central-Chadic (Afro-Asiatic) language spoken in North-eastern Nigeria (Hartmann and Zimmermann, 2008, p.202, 208). We find the same result as in the other two preceding languages in Bura. The focus associate of the exclusive particle *daci* must be morphologically marked, regardless of the position of the particle (52).²⁵ In contrast, the additive particle *ma* must only associate with a morphologically unmarked focus, as in (53-b).²⁶

- (52) a. Mtaku **daci** *(an) liha Biu.
 Mtaku only FOC go Biu
 ‘Only MTAKU went to Biu.’
 b. Mtaku *(an) liha Biu **daci**.
 Mtaku FOC go Biu only
 ‘Only MTAKU went to Biu.’
- (53) a. Ladi **ma** thlika whada ni.
 Ladi too plant peanut DEF
 ‘LADI, too, grew peanuts.’
 b. Ladi *(an) thlika whada **ma**.
 Ladi FOC plant peanut too
 ‘LADI, too, grew peanuts.’

These notable differences in morphological focus marking within the class of Beaver & Clark’s conventional association operators call for a cross-linguistic research. If *only*,

²⁵ Similar to Akan, the focus marker was glossed as PRT in the original text.

²⁶ It is however unclear whether the subject focus in (53) is in an ex-situ or in-situ position. If ex-situ subject focus always requires the focus marker as indicated by the following example from Hartmann and Zimmermann (2008, p.199), then it would be interesting to know whether the example in (53-a) can occur with the focus marker or not. Nevertheless, the different behaviour between the exclusive particle and the additive particle is clear.

- (i) a. Wa an tira ri?
 who FOC leave Q
 ‘Who left?’
 b. Ladi *(an) tira.
 L. FOC leave
 ‘LADI left.’

also and *even* are conventional focus operators, then going by PLA (see (46) above), they require focus in their scope. We have seen that in languages like English, focus is marked with prosody. However, in the languages discussed above, and the languages under the current study, focus is marked morphosyntactically. We expect to always find a marked focus (depending on the strategy employed by a given language) in the scope of conventional focus-sensitive operators. It is in this spirit, as well as a response to a call for cross-linguistic research by Beaver and Clark (2008, p.181) on this area, that this study will also investigate whether these operators (*only*, *also*, *even*) are indeed conventional operators in these languages or not. This is one of the objectives of this study, and would be discussed in Chapter 7, specifically Section 7.5.

Before ending this section, I want to point out Jackendoff's (1972) counterexample to the c-command requirement by *even* but not *only* in English. In the case of subject focus, and fronted focus generally, *even*, but not *only*, is said to be able to associate with a focus associate outside of its scope. (54-a) is ungrammatical considering PLA because the subject focus is outside the scope of *only*. (54-b) on the other hand, is grammatical, and yet it violates the PLA.

- (54) a. *[Adam]_F *only* ate a banana.
 b. [Adam]_F *even* ate a banana.

Other fronted constituents produce the same result. (55) is a case of a direct object fronting. Such leftward or backward association cases have been a puzzle for a long time. The question is that why is it possible for *even* (and *also*) to associate with an element that is not in its c-command domain at the surface syntax, and *only* cannot?

- (55) a. *[A banana]_F, Adam *only* ate ___ yesterday.
 b. [A banana]_F, Adam *even* ate ___ yesterday.

A few studies have proposed analysis for this puzzle. Two of such studies are by Erlewine (2014a,b), which we now turn to.

2.2.4 Erlewine's (2014a; 2014b) leftward focus association

In a paper titled "Explaining leftward focus association with 'even' but not 'only'", and his thesis on "Movement out of Focus", Erlewine (2014a,b) proposed an analysis for the asymmetric behavior of the focus particles in examples such as (54) and (55) above.

Adopting the copy theory of movement, Erlewine (ibid.) argues that the leftward focus associate of *even*, the subject in (56-b), has an unpronounced lower copy in the scope of *even*. Thus, what looks like a backward association with *even* is a mere illusion (Erlewine, 2014a). However, the same is not true for VP-*only* (56-a).²⁷ According to

²⁷ Erlewine argues for a copy theory of movement for VP-*even* association with leftward materials with data that shows the possibility of scope ambiguity in a sentence involving VP-*even*, universal quantifier, and negation. Considering the sentence in (i) (from Erlewine, 2014a, p.133), the focus subject *student* with a universal quantifier can take surface or inverse scope with the negative element. The fact that *even* can associate with the subject, regardless of the scope, points to the fact that it does not force the reconstruction of the subject. If it does, then scope ambiguity should not be possible. This therefore excludes a reconstruction analysis according to Erlewine.

- (i) Every [student]_F didn't *even* come to the party.
 a. $\forall > \text{Neg}$: No student came.

Erlewine (2014a), this is due to the different semantic contributions of each particle. “The crucial difference is that *even*’s semantic contribution is a projective inference that does not modify the assertion, whereas *only*’s semantic contribution modifies the assertion” (Erlewine, 2014a, p.129).

- (56) a. $*[_{TP} [Adam]_F \text{ only } [_{vP} Adam [_v \text{ ate a banana}]]]$.
 b. $[_{TP} [Adam]_F \text{ even } [_{vP} Adam [_v \text{ ate a banana}]]]$.

As shown in section 2.2.1, *even*’s scalar inference does not affect the truth-conditions of its prejacent. The set of salient alternatives, which is the focus-semantic value of the focus associate, is only used for *even*’s scalar inference. But *only*’s semantics affects the truth-conditions of its prejacent. So backward association with *only* is ungrammatical because the semantics of *only* asserts the exclusion of alternatives (Erlewine, 2014a). Note that what is presented here is only on the surface compared to the detailed argument presented by Erlewine (2014a,b). I will return to this discussion in Chapters 6 and 7.

So far, I have discussed the notion of focus and association with focus which involves the three different degrees of association. The QFC-model of sub-categorization of these particles is needed since not all of the particles are indeed directly associating to a focus constituent. Only the conventional association particles do. Even within the class of conventional association particles, we saw that there could still be minor differences.

2.3 Syntax-semantic approaches to association with focus

There have been several attempts to analyze the syntax-semantic properties of the conventional association of focus-sensitive particles. Previous analysis can be grouped into at least two approaches: (a) Quantifier approach, and (b) Propositional approach (see, a.o., Chomsky, 1976; Rooth, 1985; Kayne, 1998; Büring and Hartmann, 2001; Bayer, 1996, 2018, 2020; Lee, 2005; Wagner, 2006; Hole, 2017; Quek and Hirsch, 2017; Hirsch, 2022; Sun, 2021; Bassi et al., 2022; Branan and Erlewine, 2023; Yip, 2025). Most of the works in this area have been on the exclusive particle *only*. Other conventional focus sensitive particles like *even* and *also* have not received as much attention.²⁸ So, in the next two subsections, I will summarize the two approaches based on the exclusive particle *only*. In English, *only* can occupy different syntactic positions in the clausal structure to express the same meaning (57). It can occupy the pre-verbal position (57-a), which has been called adverbial-*only* or sentential-*only*, or adjoins directly to the focus constituent in the clause. The latter case has variously been called adnominal-*only*, adfocal-*only* or constituent-*only* (cf. Büring and Hartmann, 2001; Hirsch, 2017, 2022; Sun, 2021; Bassi et al., 2022).²⁹ The structures for the adjunction sites of *only* for the examples in (57) are given in (58). The structures result in the same interpretation at LF, which leads to a form-meaning mismatch situation.

-
- b. Neg > $\forall$: Not every student came, but some may have.

²⁸ However, see Hole (2017) and Sun (2021) for studies on not only *only*, but also *even* in Vietnamese and Chinese using the propositional approach for association. See also Branan and Erlewine (2023) for a treatment of all of them with the particle theory association.

²⁹ I will use the term ‘adfocal-*only*’ in this dissertation since it covers not only the nominal elements which are focused, but also non-nominal constituents such as PP focus or even adjunct focus.

- (57) a. Adam *only* drank [water]_F.
b. Adam drank *only* [water]_F.
- (58) a. [_{TP} Adam [_{vP} **only** [_{vP} ⟨Adam⟩ drank [_{DP} water]_F]]]
b. [_{TP} Adam [_{vP} ⟨Adam⟩ drank [_{DP} **only** [_{DP} water]_F]]]

The need for a syntax-semantic analysis of *only* becomes necessary when the semantic interpretation and property of *only* is considered. *Only* is a propositional operator which excludes any other alternative proposition apart from the one with its focus associate (59) (Horn, 1969; Rooth, 1985).

$$(59) \quad \llbracket \mathbf{only} \rrbracket^{ALT} = \lambda p_{\langle st \rangle} . \lambda w : p(w) . \forall p' \in ALT[p'(w) \rightarrow p \subseteq p']$$

So while adverbial-*only* patterns well with *only*'s propositional property because it scopes over a proposition, the vP, adfocal-*only* doesn't. In formal semantics, the semantic type of a DP represents the DP as an individual or as a quantifier. Thus, a DP is not a proposition that *only* can compose with based on (59). This raises the question of how to reconcile the syntactic property of *only* with its semantics, considering adfocal-*only* examples such as (58-b).

A solution was to assume two different meanings for *only* (Rooth, 1985, 1992; Beaver and Clark, 2008). One is the original meaning, which is already given in (59), where *only* composes with a proposition. The other meaning, which was suggested to address the adfocal-*only* cases, forms the basis for the Quantifier Approach which we now turn to.

2.3.1 The Quantifier approach

Under, the quantifier approach, it is believed that the focus sensitive particle *only* type-shifts and composes with its focus associate in order to form a single quantifier phrase (60), which then raises to a scope-taking position in the clause; usually above vP, as in (61) (Rooth, 1985, 1992; Bayer, 1996; Jacobson, 2014). So based on the semantic type of what it composes with, *only* has multiple interpretations.

$$(60) \quad \llbracket \mathbf{only}_Q \rrbracket^{ALT} = \lambda Q_{\langle est, st \rangle} . \lambda f_{\langle est \rangle} . \llbracket \mathbf{only} \rrbracket(Q(f))$$

(Hirsch, 2022, p.2)

$$(61) \quad \text{LF: } [_{TP} \text{ Adam } [_{QP} \mathbf{only} [_{DP} \text{ water}]_F] [_{vP} \langle \text{Adam} \rangle \text{ drank } \langle [_{QP} \mathbf{only} [_{DP} \text{ water}]_F \rangle]]]]$$

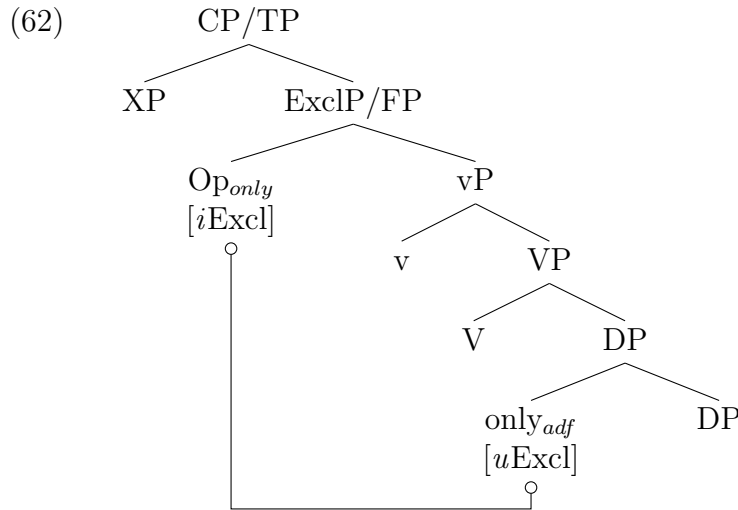
The implication of this approach is that the adfocal *only* remains the locus of exclusive semantics, only that it type-shifts from being a propositional operator to being able to compose with DPs, as in (60). This accounts for the semantic requirement. In the syntax, the adfocal *only* and the DP it composes with together quantifier-raises to a propositional position (at LF), as in (61). This approach predicts that *only* and its focus associate cannot be dissociated. Thus, even in contexts where *only* takes scope, both it and its focus associate would have to scope together. I will return to this approach in more detail in Chapter 5.

2.3.2 The Propositional approach

The Propositional Approach (PA henceforth) simply maintains that both the adverbial-*only* and the adfocal-*only* cases can be accounted for without necessarily resorting to

a different semantics for *only*. In other words, *only*'s original (propositional) semantics can account for all the cases of *only* association. We only need to accommodate a few assumptions to make this work. Different terminologies have been used for a few analyses and/or proposals that can be classified under PA: *bipartite analysis* (Bayer, 1996; Lee, 2005; Hole, 2017; Sun, 2021), *concord analysis* (Hirsch, 2017, 2019, 2022), and more recently, *operator-particle approach* (Branan and Erlewine, 2023). All the accounts have one thing in common: although the adfocal-*only* may be syntactically low, its interpretation depends on a syntactically high position in the clause.

A major mechanism employed by the propositional approach to adfocal *only* is *agreement* (see Quek and Hirsch, 2017; Hole, 2017; Hirsch, 2022; Sun, 2021; Bassi et al., 2022; Branan and Erlewine, 2023, among others). Under this assumption, the exclusive interpretation (feature) comes from an exclusive null operator [Op] which occupies a high position (head of an Exclusive Phrase) in the clausal spine, usually adjoined to vP, as in (62) or higher. The exclusive null Op then establishes agreement with the lower adfocal particle, which adjoins to the focus associate to form an extended DP projection. Thus, the adfocal-*only* is only seen as an agreement morpheme, which is often assumed to be semantically inert (but see Section 5.4 of Chapter 5 for a different proposal). Its interpretation comes from the null Op via Agree.

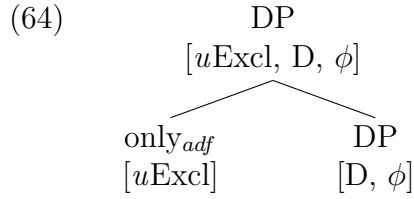


Therefore, it is the exclusive null Op or $\llbracket \text{ONLY} \rrbracket$ as it is commonly represented in the literature, that carries the exclusive semantics given in (59) above, repeated below. This means that ONLY takes two arguments, which are the proposition and an individual. The individual in this case is the focus associate. So, the assertion is that the proposition is true iff there is no other proposition containing the focus which answers the current question in a given world of reference. The assumption is that the exclusive Op/ONLY does not have a fixed position. That is, it can be below or above the TP, as long as its quantificational scope property is satisfied (cf. Lee, 2005).

$$(63) \quad \llbracket \text{ONLY} \rrbracket^{ALT} = \lambda p_{\langle st \rangle} . \lambda w : p(w) . \forall p' \in ALT[p'(w) \rightarrow p \subseteq p']$$

Since the exclusive null Op and the adfocal-*only* are linked via agreement, one question that might arise is how then does the association with the focused constituent takes place? A straightforward answer, following Branan and Erlewine (2023), is that the extended DP is a projection which contains the features of both the exclusive particle and the focused DP. In other words, the formal features of both the adfocal-*only* and the focus associate

project to the extended DP (64) (Branan and Erlewine, 2023). Therefore, agreement with adfocal-*only* is agreement with the whole extended DP.³⁰



Interestingly, there is an explicit support for such an agreement operation in Lavukaleve (one of the four Central Solomons languages) according to Branan and Erlewine (2023) (see also Mursell, 2021). In Lavukaleve, focus particles reflect ϕ agreement with the focus constituent. In (65-a) and (65-b), the focus particles show ϕ agreement with the subject focus and the object focus respectively. Furthermore, there is a postverbal particle which also reflects the ϕ agreement of the focus constituent. Branan and Erlewine (2023) proposes that this postverbal particle corresponds to the exclusive null Op that we have discussed so far.³¹

- (65) Lavukaleve: Branan and Erlewine (2023, p.625) citing Terrill (2003, p.277)
- a. [Aira la]_F **feo** fo'sal na aua **heo**.
 woman(f) ART.SG.F PRT.3SG.F fish(m) ART.SG.M ate.AGR OP.3SG.F
 'THE WOMAN ate a fish.'
- b. Aira la [fo'sal na]_F **fin** oum **hin**.
 woman(f) ART.SG.F fish(m) ART.SG.M PRT.3SG.M ate.AGR OP.3SG.M
 'The woman ate A FISH.'

In addition to agreement between the exclusive Op and the adfocal-*only*, the latter (can) still undergo focus movement in languages like Mandarin Chinese (Sun, 2021) and Korean (Lee, 2005) where scrambling occurs. The languages investigated here also involve focus movement, and this applies to them as well, as I will show in this dissertation.

2.3.3 Arguments in support of the Propositional Approach

There has been cross-linguistic support for the propositional/bipartite approach to *only* in languages like Korean (Lee, 2005), Mandarin Chinese (Erlewine, 2010; Sun, 2021; Yip, 2025), Vietnamese (Erlewine, 2017), Cantonese (Lee, 2019; Yip, 2025), German (Bayer, 1996, 2018, 2020), and Dutch (Barbiers, 2010). Unlike English (cf. (66)), most of these languages allow the co-occurrence of both the adverbial *only* and the adfocal *only* (*only*-concord), which results in a single *only*-reading.³² Consider the Vietnamese (67) and

³⁰ For similar proposal, see Bayer and Obenauer (2011); Bayer (2018, 2020) who see the exclusive focus particle as a *syncategorematic* element (instead of an adverb).

³¹ See also Mursell (2021) on the interaction between information-structural features with ϕ -features in languages like Tagalog, Swahili, and Lavukaleve.

³² Bayer (2020, p.64-65) however gave at least three examples in which he claims are rare cases of *only*-doubling in English.

- (i) a. They are solid options for the wireless speaker fan, even if they **only** support **only** the original AirPlay.
 b. The stakes have never been higher as he **only** has **only** 48 hours to find someone to take care of his young daughter.
 c. He doesn't love me at all; he **only** thinks **only** of himself.

the Cantonese (68) examples below. Apart from the singleton cases for adverbial-*only* and the adfocal-*only*, both can be doubled in these languages (cf. (67-c) & (68-c)). The truth-condition remains the same as in the individual-occurrence cases. This results in a form-meaning mismatch. Therefore, it has been argued that the adverbial-*only* position is the same as the position of the exclusive Op that we discussed above.

(66) *John **only** ate **only** [banana]_F.

(67) **Vietnamese:** (Erlewine, 2017, p.331)

- a. Nam mua **moi** [cuon sách]_F.
Nam buy only CL book
'Nam bought only THE BOOK.'
- b. Nam **chi** mua [cuon sách]_F.
Nam only buy CL book
'Nam only bought THE BOOK.'
- c. Nam **chi** mua **moi** [cuon sách]_F.
Nam only buy only CL book
'Nam only bought THE BOOK.'

(68) **Cantonese:** (Yip, 2025)

- a. Aaming maai-zo [joengjuk]_F bei Aafan **zaa**.
Ming buy-PFV lamb to Fan only
'Ming bought only Fan LAMB (but not beef or pork).'
- b. Aaming **zinghai** maai-zo [joengjuk]_F bei Aafan.
Ming only buy-PFV lamb to Fan
'Ming only bought Fan LAMB (but not beef or pork).'
- c. Aaming **zinghai** maai-zo [joengjuk]_F bei Aafan **zaa**.
Ming only buy-PFV lamb to Fan only
'Ming only bought Fan LAMB (but not beef or pork).'

The morphological doubling is not the only evidence for PA. Another evidence comes from the scope ambiguity that is found in the presence of adfocal-*only* in relation with certain elements like modal and embedding predicate (see Taglicht, 1984; Rooth, 1985; Lee, 2005; Hirsch, 2022, among others). The sentence in (69) (with adfocal-*only*) is ambiguous with respect to the scope of *only*. It can scope below the embedding verb *require* (69-a) or above it (69-b). The argument here is that based on the scope (c-command) requirement involving the exclusive particle, the exclusive interpretation resides somewhere higher than the adfocal-*only* which allows for the different scopes that are observed. Thus, the different position of the exclusive Op is shown in (70). There is, however, no scope ambiguity with the adverbial-*only* (71). It is assumed that the adverbial-*only* and the exclusive Op are in complementary distribution.

(69) She is required to eat **only** [fruits]_F.

- a. [require > only]: She is required to eat nothing else.
- b. [only > require]: The only requirement is for her to eat fruits (she can eat other things in addition).

(70) [_{TP} She_i is [(**Op**) required [_{TP} t_i to (**Op**) [_{vP} eat **only** [fruits]_F]]]]

- (71) a. She is required to **only** eat [fruits]_F. [require > only; *only > require]
 b. She is **only** required to eat [fruits]_F. [*require > only; only > require]

The scope ambiguity cases can also be analysed with the quantifier approach. We only need to assume that depending on the scope relation, the quantifier phrase, comprising of the adfocal-*only* and the focus associate, can raise to a scope position below or above the verb *require* (72) in order to account for the scope ambiguity (Hirsch, 2019, 2022).

- (72) a. LF: [_{TP} She_i is [required [_{TP} t_i to [_{QP} **only** [fruits]_F]_k [_{vP} eat t_k]]]]
 b. LF: [_{TP} She_i is [[_{QP} **only** [fruits]_F]_k required [_{TP} t_i to [_{vP} eat t_k]]]]

However, things become more interesting in a split scope context according to Hirsch (2019, 2022). Consider the following sentence in (73) with a quantified DP. While the interpreted *only* scopes above the verb *require*, the focused quantifier DP scopes below it. There is a split in the scope of the exclusive particle and the quantified DP. The possibility for a split in the scope of the exclusive particle and the quantified DP is unexpected if the quantifier approach is the only approach. This is because the quantifier approach predicts that the adfocal-*only* + the DP must scope together since they form a single quantifier phrase (QP). Thus, it does not predict split-scope (Hirsch, 2022).

- (73) She is required to eat **only** [two fruits]_F. [only > require > two fruits]
Meaning: x is *only* required to eat two fruits (not specific).

PA on the other hand, can directly explain the split-scope, assuming that the adfocal-*only* is semantically inert. Since the exclusive Op can be high, it scopes above *require* while the quantified DP raises above the embedded verb but scope below *require* as in (74).³³

- (74) LF: [_{TP} She_i is [**Op** [required [_{TP} t_i to [_{QP} [two fruits]_F]_k [_{vP} eat t_k]]]]]]

2.4 Summary of the chapter

In this chapter, I have discussed the basic notions of focus (section 2.1) and association with focus (section 2.2) which are relevant for the present study. I showed that focus realization, as an answer to a current question, presents a well-founded way for realizing not only the focus constituent, but also the contextually-salient set of alternatives. This is the Rooth-Hamblin semantics by Beaver and Clark (2008). Of particular importance is the fact that these alternatives are evaluated by the focus sensitive particles (*only*, *also* & *even*) that are of concern in this dissertation. The choice of these focus sensitive particles is influenced by the behavioural differences which are found among the expressions that were previously classified as focus sensitive. I discussed that Beaver and Clark's (2008) QFC-model to focus sensitive particles have come under some scrutiny from cross-linguistic studies that show that *only*, *also*, and *even* do not always behave the same way in the presence of their focus associate. In other words, morphosyntactic focus marking languages appear to morphologically mark focus in the presence of *only*, but not *also* and *even* (Hartmann and Zimmermann, 2008; Grubic and Zimmermann, 2011; Duah, 2015), which leaves us with the question: why the asymmetry in focus marking?. One of the aims of this dissertation is to contribute to this ongoing discussion. Finally, in section 2.3,

³³ For other evidence in support of PA, see Bassi et al. (2022) on ellipsis, and Yip (2025) on minimality and locality effects.

I presented the approaches for mapping the syntax-semantic properties of focus sensitive particles, making a case for the propositional/bipartite/concord approach, which plays an important role in my proposal for association with focus phenomena in the Mabia and Yoruboid languages studied in this dissertation.

Chapter 3

Focus realization in Kasem, Kusaal and Yorùbá

3.1 Introduction

In the last chapter, we saw that focus indicates the presence of alternatives and that languages employ different focus marking strategies, ranging from the use of prosody, to the use of morphology and/or syntax. In this chapter, I investigate the focus realization strategies employed by the three languages under the current study: Kasem, Kusaal and Yorùbá. As is common to many (West) African languages, we will see that (a) these languages employ morphosyntactic focus marking strategies, but (b) still show some differences with respect to the focus marking patterns. In each of the sections, I will first describe the focus constructions in the languages, and then provide syntactic analyses for the focus realization patterns and variation within each language. Thus, this chapter will serve as a background to subsequent discussions on the association with focus phenomena in these languages.

As a strategy to elicit focus, I will employ the Question-Answer pair that was introduced in chapter 2. This strategy is preferred among other strategies because not only is it efficient, but it is also simple to use when eliciting data on focus with language speakers.

Here is how the chapter is structured. Section 3.2 starts with a description of the focus construction in Kasem and then ends with a proposed syntactic analysis. Similarly, section 3.3 introduces the focus system in Kusaal (especially a morphological marking asymmetry between local subject focus and the others), and provides an analysis for the observed focus marking asymmetry. Section 3.4 describes the focus realization strategies in Yorùbá, taking into consideration the earlier studies, and then contains a proposed analysis. Finally, in section 3.6, I give a typological summary of the chapter, which focuses on the focus realization strategies in the languages, and the syntactic placement of the focus markers and focus constituents.

3.2 Focus realization in Kasem

The focus marking strategies in Kasem follow from well-known strategies among the West African languages (cf. Fiedler et al., 2010; Kalinowski, 2015). These strategies include the use of a dedicated morphological focus marker, focus fronting, and clefts. In Kasem, a constituent can be focused in an in-situ position with a dedicated focus marker, or can

be fronted to the left periphery of the clause. Interestingly, focus marking in Kasem also involves the phenomena of pied-piping and anti-pied-piping in focus constructions. Each of these strategies is discussed below.¹ After the general discussion, I will propose a structural analysis for the focus marking strategies in the language.

3.2.1 In-situ focus in Kasem

Focused constituents in Kasem can remain in situ, without movement. In such cases, the focused constituent remain in their canonical argument position, and are usually identified by a focus context (*wh*-question, for instance) and the presence of the morphological focus particle **mo** in Kasem. Usually, the focus particle adjoins directly to the logical focus if it is any of the following: DP, PP, and adjuncts.

The example in (1-b) is an in-situ object focus which answers the *wh*-question in (1-a). The focus particle *mo* immediately follows the focus object thereby forming a constituent with the latter. Notice that the *wh*-question shows the same marking with *mo* in the in-situ position as well, as in (1-a).

- (1) a. Adam go bε ***(mo)?**
 Adam slaughter what FOC
 Lit.‘Adam slaughtered what?’
 b. Adam go [chworɔ]_F ***(mo)**.
 Adam slaughter fowl FOC
 ‘Adam slaughtered A FOWL.’

The constituency of the focus marker *mo* and the focused constituent is even more evident when we consider the focus construction in ditransitive constructions such as (2) and (3). While the *mo* follows the direct object in (2), it follows the indirect object in (3). These sentences also exemplify in-situ focusing of direct and indirect objects in ditransitive constructions in Kasem.

- (2) a. Adam pε bε ***(mo)** John?
 Adam give what FOC John
 Lit.‘Adam gave what to John?’
 b. Adam pε [tɔnɔ]_F ***(mo)** John.
 Adam give cup FOC John
 ‘Adam gave a CUP to John.’
 (3) a. Adam pε kɔpe wɔ ***(mo)?**
 Adam give cup who FOC
 Lit.‘Adam gave the cup to who?’
 b. Adam pε tɔnɔ [John]_F ***(mo)**.
 Adam give cup John FOC
 ‘Adam gave a cup to JOHN.’

Adjunct focus can also be realized in an in-situ position. The adjunct *wh*-question in (4-a) is in-situ, and it can be answered with the in-situ adjunct focus construction in (4-b).

¹ I will put the focus constituent in an F-marked square bracket [XP]_F, and use **boldface** for the morphological focus marker. In the English translation, I will use SMALL CAPS for the focus constituent.

- (4) a. Adam go chworo kom maŋ-kɔ **mo**?
 Adam slaughter fowl DEF time-which FOC
 ‘When did Adam slaughter the fowl?’
 b. Adam go chworo [diim]_F **mo**.
 Adam kill fowl yesterday FOC
 ‘Adam slaughtered a fowl YESTERDAY.’

3.2.2 Ex-situ focus

In Kasem, the focus marker *mo* is still present in ex-situ focus. The only difference with in-situ focus is that in the ex-situ focus construction, the focused constituent is dislocated to the left periphery of the clause.

Ex-situ non-subject focus

Apart from their in-situ realization, non-subject focus can be fronted to the left periphery. Consider the question-answer pair in (5) which shows that both object *wh*-question and focused answer can be realized ex-situ in Kasem.

- (5) a. Bɛ *(**mo**) Adam go-a ____?
 what FOC Adam slaughter-DISJ
 ‘What did Adam slaughter?’
 b. [Chworo]_F *(**mo**) Adam go-a ____.
 fowl FOC Adam slaughter-DISJ
 ‘Adam slaughtered A FOWL.’

In ditransitive constructions, the direct object or the indirect object can be fronted to an ex-situ position. The example in (6) shows this.

- (6) a. Bɛ **mo** Adam pɛ ____ John?
 what FOC Adam give John
 ‘What did Adam give to John?’
 b. [Tɔnɔ]_F *(**mo**) Adam pɛ ____ John.
 cup FOC Adam give John
 ‘Adam gave a CUP to John.’
 (7) a. Wɔ **mo** Adam pɛ kɔpɛ ____?
 who FOC Adam give cup
 ‘To whom did Adam give the cup?’
 b. [John]_F *(**mo**) Adam pɛ tɔnɔ ____.
 John FOC Adam give cup
 ‘Adam gave a cup to JOHN.’

Adjunct focus can also be realized in an ex-situ position. Both the ex-situ adjunct *wh*-question and the focused answer are given in (8-a) and (8-b) below.

- (8) a. Maŋ-kɔ **mo** Adam go chworo kom ____?
 time-which FOC Adam slaughter fowl DEF
 ‘When did Adam slaughter the fowl?’
 b. [Diim]_F **mo** Adam go chworo kom ____.
 yesterday FOC Adam slaughter fowl DEF
 ‘Adam slaughtered the fowl YESTERDAY.’

Ex-situ Subject Focus:

In Kasem, subject focus is always realized in an ex-situ position. In other words, in-situ subject focus does not exist. An example of subject focus in Kasem is given in (209-b) below. The focus marker *mo* follows the subject of the clause, and is assumed to be adjoined to it. Both the focused constituent and *mo* form a constituent, as I have argued for non-subject foci above.

- (9) a. Wɔ **mo** go chworɔ kom?
 who FOC kill fowl DEF
 ‘Who slaughtered the fowl?’
 b. [Adam]_F **mo** go chworɔ kom.
 Adam FOC kill fowl DEF
 ‘ADAM slaughtered the fowl.’

Based on the assumption that *mo* forms a constituent with the focused constituent, and the data in (209-b), it is hard to tell whether the focused subject is in an in-situ position (i.e., Spec,TP) or an ex-situ position, somewhere in the left periphery of the clause. However, I assume that local subject focus in Kasem is realized in an ex-situ position which is different from the canonical subject position, i.e. Spec,TP. One evidence for this comes from its behaviour with respect to scope. While non-focused subjects allow inverse scope in Kasem, as in (10-a), focused subjects do not (10-b). If a DP subject is focused, we get surface scope. If local subject focus occupies the same position as a non-focus DP subject, then the different scopal behaviour is surprising. But if the focused subject occupies a separate position, then the difference in scopal behaviour is less surprising.

- (10) a. Nɔɔn-wɔdwɔŋi nɛ nɔɔno maama. [$\exists > \forall$; $\forall > \exists$]
 person-one see person all
 ‘Someone saw everyone.’ (presence of ambiguity: inverse scope)
 (i) There exists an individual *x* such that for all *y*, *x* saw *y*.
 (ii) For all *y*, there exists an individual *x* such that *x* saw *y*.
 b. [Nɔɔn-wɔdwɔŋi]_F **mo** nɛ nɔɔno maama. [$\exists > \forall$; * $\forall > \exists$]
 person-one FOC see person all
 ‘SOMEONE saw everyone.’ (no inverse scope)
 (i) There exists an individual *x* such that for all *y*, *x* saw *y*.
 (ii) *For all *y*, there exists an individual *x* such that *x* saw *y*.

Similarly, the scopal behaviour of local subject focus (with a universal quantifier *all*) in relation to negation is another argument for an ex-situ position for local subject focus. While a quantified non-focused local subject allows for an inverse scope in the presence of negation (11-a), its focus counterpart does not; it is scope-rigid (11-b). In other words, (11-a) is ambiguous between the interpretations given in its (i) & (ii). (11-b) on the other hand, can only have the surface scope interpretation where the universal quantifier scopes over negation.

- (11) a. Bia bam maama wo di pea. [$\neg > \forall; \forall > \neg$]
 children DET all NEG eat yam
 ‘All the children did not eat yam.’
 (i) It is not the case that all the children ate yam (meaning some did not eat while some might have).
 (ii) It is the case that none of the children ate yam.
- b. [Bia bam maama]_F **mo** wo di pea. [* $\neg > \forall; \forall > \neg$]
 children DET all FOC NEG eat yam
 ‘All the children did not eat yam.’
 (i) *It is not the case that all the children ate yam (meaning some did not eat while some might have).
 (ii) It is the case that none of the children ate yam.

I wish to mention here that the test with negation in (11) seems to be stronger than the one with the existential and universal quantifiers in (10). This is because of the nature of negative operators. Unlike the existential/universal quantifier, sentential negation generally takes scope in its pronounced position. So, any ambiguity that may be seen in a negative context would have to be as a result of the reconstruction of the quantifier to its VP-internal position for narrow scope. Without reconstruction, the quantifier takes a wide scope. This means that the local subject does not reconstruct when it is in focus, hence the absence of an inverse scope. While I am aware that there could be other potential reasons for the scope disambiguation of the subject focus (e.g., licensing by congruent question), a structural explanation is what I presented here (cf. Bošković, 2024).²

The idea that local subject focus or *wh* does not occupy the same position as the canonical subject is not new. This has already been argued for by Bošković (2024, and its references) for English and some other languages. For instance, consider the anti-agreement phenomenon in Kinande (12). The agreement on the verb changes depending on whether the subject is in the canonical in-situ position, or has undergone local *wh*-fronting (compare (12-a) with (12-c)). If the agreement marking for in-situ subject is used in a local *wh*-subject fronting context, the sentence becomes ungrammatical (12-b). This is an argument for different positions for local subject focus and non-focused subject (cf. Bošković, 2024, and its reference for more).

- (12) **Kinande** (adapted from Bošković (2024, p.10) citing Schneider-Zioga (1995))
- a. Kambale **a**.langira Marya.
 Kambale AGR.saw Mary
 ‘Kambebe saw Mary.’
- b. *Iyondi yo **a**.langira Marya.
 who C AGR.saw Mary
 ‘Who saw Mary’
- c. Iyondi yo **u**.langira Marya.
 who C ANTI-AGR.saw Mary
 ‘Who saw Mary?’

Non-local subject focus involves a resumptive pronoun in the embedded subject position when the subject is dislocated to the left periphery of the matrix clause (13-b). However, if the focused subject is realized within the embedded clause periphery, the

² The discussion on the ex-situ nature of local subject focus is not peculiar to Kasem alone. As would be seen in §3.3, Kusaal also displays a similar property. The same is available in Yorùbá as well.

resumptive pronoun cannot be present (13-c), which is typical of local subject focus in the language.³

- (13) a. Wɔ **mo** Peter we **o** go chworɔ kom.
 who FOC Peter say 3SG kill fowl DEF
 ‘Who did Peter say slaughtered the fowl?’
 b. [Adam]_F **mo** Peter we **o** go chworɔ kom.
 Adam FOC Peter say RP kill fowl DEF
 ‘Peter said that ADAM slaughtered the fowl.’
 c. Peter we [Adam]_F **mo** (***o**) go chworɔ kom.
 Peter say Adam FOC RP kill fowl DEF
 ‘Peter said that ADAM slaughtered the fowl.’

Cleft structure: In addition to ex-situ focus, Kasem employs a cleft construction for focusing similar to the English *it*-cleft (14). The expletive *ko* (‘it’) is followed by a copula *ye*, and then the focus phrase, which is immediately followed by the focus marker *mo*. This results in an exhaustive focus reading, which is typical of cleft constructions in English.

- (14) a. Bɛ *(**mo**) Adam go-a?
 what FOC Adam slaughter-DISJ
 ‘What did Adam slaughter?’
 b. Ko ye [chworɔ]_F **mo** Adam go-a.
 3SG COP fowl FOC Adam slaughter-DISJ
 ‘It is A FOWL that Adam slaughtered.’

Before ending this section, it is important to mention here that the focus realization strategies (i.e. in-situ versus ex-situ focus realization) are optionally used by the speakers of the language. Although it may seem as though there is no preference at first, but on a closer look, ex-situ focus seems to involve exhaustive focus and/or mirative focus.⁴ I will not go into the details here, as I leave it for a detailed research in the future.

In the next section, we will discuss cases of focus marking mismatches where the logical focus does not correspond to the focus marked constituent.

3.2.3 (Anti)-pied-piping in Kasem

In the previous sections, we saw that the focus particle *mo* adjoins directly to the logical focus. However, the focus particle does not always adjoin directly to the logical focus. To my knowledge, there are at least four instances where the *mo*-adjunction site does not correspond to the logical focus. In other words, there is a mismatch between *mo*’s adjunction site and the logical focus or the answer to the current question under discussion. These instances include verb focus, DP-internal focus, verb phrase focus, and clausal focus. Such mismatches have been recorded for some other languages such as Hausa (Hartmann and Zimmermann, 2007), and German and Czech (Fanselow and Lenertová, 2011). For an overview of such logical focus (morphosyntactic) marking mismatches in about 53 languages, see Branen and Erlewine (2023).

The mismatches between focus marking and logical focus are of two types. The first involves the focus marking of a phrase which contains the logical focus. This is called *pied-*

³ See Section 3.5 for a discussion on the resumptive pronoun in the embedded subject position in Kasem.

⁴ See, for example, Abubakari’s (2019) discussion for Kusaal.

*pip*ing. The second type involves focus marking a sub-constituent/part of the logical focus. This is called *anti-pied-piping* (Fanselow and Lenertová, 2011; Branen and Erlewine, 2023).⁵ This will become explicit as we discuss them in turn below.

Pied-piping:

Out of the four instances of logical focus versus focus marking mismatches mentioned above, only DP-internal focus and verb focus involve pied-piping. The other two, VP focus and clausal/TP focus, involve anti-pied-piping. I discuss pied-piping in DP-internal focus and verb focus below.

DP-internal focus and pied-piping:

In addition to the focus of a simple DP, sub-constituents inside the DP can also be focused. This is DP-internal focus. For example, the possessor inside the DP is in focus in (15-b). However, the focus particle follows the entire DP and not the possessor itself. This is a case of focus pied-piping because the adjunction site of the focus marker *mo* contains the logical focus which is the possessor.

- (15) a. Adam go wɔ chwóró **mo**
 Adam slaughter.PFV who fowl FOC
 ‘Whose fowl did Adam slaughter?’
 b. Adam go [_{DP} [Peter]_F (***mo**) chwóró] **mo**.
 Adam slaughter Peter FOC fowl FOC
 ‘Adam slaughtered Peter’s fowl.’

Although the widely known cases of pied-piping introduced by Ross (1967, §4.3) describe instances of movement where a focus expression takes more than itself when it is fronted, as in (16), Branen and Erlewine (2023) also use the term ‘pied-piping’ to include instances like (15-b) above, where the focus particle adjoins to a phrase that contains the *logical focus* (_F) without movement.

- (16) [[_{DP} [Peter]_F (***mo**) chworó] **mo**] Adam go-a ____.
 Peter FOC fowl FOC Adam slaughter-DISJ
 ‘Adam slaughtered Peter’s fowl.’

Verb Focus and pied-piping:

Another instance of pied-piping involves the verb focus. In order to focus the verb, the focus particle follows the entire VP in a transitive sentence, as in (17-b). The sentence is ungrammatical if the focus particle follows the verb in the presence of a direct object, as in (17-c). The focus particle may seem to follow the verb in an intransitive construction such as (18). However, this is a case of an adjunction to the whole VP, and not an adjunction to the verb itself.

- (17) a. What did Adam do to the fowl?
 b. Adam [go]_F chworó kom **mo** dīm.
 Adam slaughter fowl DEF FOC yesterday
 ‘Adam SLAUGHTERED the fowl yesterday.’
 c. *Adam [go]_F **mo** chworó kom dīm.
 Adam slaughter FOC fowl DEF yesterday
 ‘Adam SLAUGHTERED the fowl yesterday.’

⁵ Anti-pied-piping cases have also been called *discontinuous focus*, or *partial focus movement* (see Hartmann and Zimmermann, 2007, §5.2).

- (18) Adam [toŋe]_F **mo**.
 Adam work FOC
 ‘Adam WORKED.’

Anti-pied-piping:

Both VP focus and clausal focus involve anti-pied-piping. Unlike the pied-piping cases, where the logical focus is contained inside the morphological focus marked constituent, in anti-pied-piping, it is the focus marked constituent that is contained inside the logical focus.

VP Focus and Anti-pied-piping:

The congruent answer to the current question in (19-a) produces a VP focus which can involve an intransitive construction, as in (19-b), or a transitive construction, as in (19-c).

- (19) a. Bε **mo** Adam kea dīim?
 what FOC Adam do yesterday
 ‘What did Adam do yesterday?’
 b. Adam [toŋe]_F **mo** dīim.
 Adam work FOC yesterday
 ‘Adam WORKED yesterday.’
 c. Adam [go chworō]_F **mo** dīim.
 Adam kill fowl FOC yesterday
 ‘Adam SLAUGHTERED A FOWL yesterday.’

The VP focus answers above may not seem to involve anti-pied-piping because it may be the case that *mo* directly adjoins to the whole VP. Consider the simple structural representations in (20), where the logical focus is the whole vP, but the focus particle adjoins to the object DP instead of the vP/VP. As noted by Branam and Erlewine (2023), while it is easier to observe VP focus anti-pied-piping in OV languages, as in (20-a) (because the verb is the last element in the phrase), it is more difficult to observe the same in VO languages because it can be argued that the focus particle adjoins to the VP and not the object (20-b). If this is the case, then there is no anti-pied-piping in VP focus in Kasem which is a VO language. However, I present empirical data that shows that there is indeed VP focus anti-pied-piping in Kasem.

- (20) a. [_{vP} [[DP]**Prt**] v]_F
 b. [_{vP} v [[DP](**Prt**)] (**Prt**)]_F

The first evidence for VP focus anti-pied-piping comes from transitive constructions where the direct object plus the focus particle *mo* can optionally be fronted to the left periphery, as in (21-c). Such ‘anti-pied-piping by movement’ cases were also reported by Branam and Erlewine (2023) for languages such as Japanese, Hindi-Urdu, Bangla, Hungarian, and so on (cf. Aoyagi, 1998; Kenesei, 1998; Kotani, 2009; Dash and Datta, 2022). If the focus particle does not adjoin to the direct object in VP focus, then it is surprising why (21-c) is as much a felicitous answer as (21-b) to the question in (21-a).

- (21) a. Bε mo Adam ke diim?
 what FOC Adam do yesterday
 ‘What did Adam do yesterday?’

- b. Adam [_{VP} go [_{DP} chworo **mo**]]_F diim.
Adam kill fowl FOC yesterday
'Adam SLAUGHTERED A FOWL yesterday.'
- c. [_{DP} Chworo **mo**]_i Adam [go t_i]_F diim.
fowl FOC Adam kill yesterday
'Adam SLAUGHTERED A FOWL yesterday.'

Another evidence for VP focus anti-pied-piping comes from the ditransitive constructions in (22). The focus particle adjoins to the direct object in (22-b), and can be fronted in (22-c). However, the focus particle cannot adjoin to the indirect object in a VP focus context. Both in-situ and ex-situ cases (cf. (22-d) & (22-e)) of indirect object adjunction can only have a narrow indirect object focus reading. This appears to suggest that the focus particle *mo* adjoins to the leftmost sub-constituent of the logical focus, the VP in this case. I will return to this property in the next section.

- (22)
- a. Bε mo Adam kea?
what FOC Adam do
'What did Adam do?'
 - b. Adam [pε tənɔ **mo** John]_F.
Adam give cup FOC John
'Adam GAVE A CUP TO JOHN.'
 - c. Tənɔ **mo** Adam [pε ___ John]_F.
cup FOC Adam give John
'Adam gave a cup to John.'
 - d. #Adam [pε tənɔ John **mo**]_F. *only answers an indirect object question*
Adam give cup John FOC
'Adam gave a cup to John.'
 - e. #John **mo** Adam [pε tənɔ ___]_F. *only answers an indirect object question*
John FOC Adam give cup
'Adam gave a cup to John.'

There is an important thing to note based on the data. If the focus particle really adjoins to the entire VP, then (22-d) should be a felicitous answer since the focus particle appears in a clause-final position. However, we have seen that this is not the case.⁶

TP Focus and Anti-pied-piping:

Another case of anti-pied-piping comes from TP/clausal focus. To answer a question like (23-a), the focus particle has to follow the subject as in (23-b). (23-c) is an infelicitous answer to the question *what happened?* The felicitous structure in (23-b) can be used for subject focus as well.

⁶ I will show in Section 3.2.4 that the anti-pied-piping cases follow from what Branan and Erlewine (2023) called the *leftmost requirement*. This is a requirement that constrains the focus particle to adjoin to the leftmost sub-constituent of the logical focus which involves anti-pied-piping. This is why it is the direct object, and not the indirect object, that gets to host *mo*. This also explains why the adjunction of the focus particle to the indirect object is infelicitous with the current question.

- (23) a. Bε mó kèá?
 what FOC do
 ‘What happened?’
 b. [_{TP} [Adam **mó**] go chworo]_F.
 Adam FOC kill fowl
 ‘Adam slaughtered a fowl.’
 c. #Adam go chworo **mó**.
 Adam kill fowl FOC
 ‘Adam slaughtered a fowl.’

It is worth mentioning that several accounts have shown similar focus realization patterns in some other languages. For instance, Fanselow and Lenertová (2011) showed that in German and Czech, a constituent that is part of a logical or semantic focus can be fronted to the left periphery (‘Subpart of Focus Fronting’ (SFF) in their terms), posing a challenge to the theory which argues that information structure is marked in the syntax. In each of the German and Czech examples in (24), the fronted constituent is the direct object. But this is only a sub-constituent of the logical focus considering the current questions under discussion. The questions require either a VP focus or a clausal focus answer. The fact that (24-a) and (24-b) can be used to answer both questions in the two languages challenges theories which directly tie syntactic movement to information structural features, according to Fanselow and Lenertová (2011). If focus movement is triggered by agreement between the semantic/logical focus and a left peripheral head (Foc) (cf. Rizzi, 1997), then the cases in (24) are a mystery as the fronted constituents are not the logical focus, but only a sub-part of it (see also Hartmann and Zimmermann, 2007).

- (24) What did you do?, What’s new?/What happened?
 a. [Einen HAsen]_i habe ich t_i gefangen. (German)
 ART.ACC rabbit have I caught
 ‘I caught a rabbit.’
 b. [ZAJíce]_i jsem chytil t_i. (Czech)
 rabbit.ACC AUX.1SG caught.SG.MS
 ‘I caught a rabbit.’

(adapted & modified from Fanselow and Lenertová, 2011, p.174)

In summary, we have seen that there are cases of pied-piping and anti-pied-piping in Kasem. In addition, we saw that there are certain surface strings that are ambiguous in their focus structure. For example, adjoining *mo* to the direct object can be felicitous for VP focus (anti-pied-piping) as well as object focus itself. Similarly, adjoining *mo* to the subject of the clause can both felicitously answer a TP focus as well as a subject focus. Therefore, the question of how to recognize the logical focus in such ambiguous marking cases arises. The simple solution is for us to depend on the current question under discussion (pragmatics) which helps in revealing the constituent that is the logical focus. This forms part of my analysis for focus construction in Kasem in the next section.

In the next section, I propose an analysis for the different focus marking strategies described in Kasem above. I start by proposing the adjunction structure for *mo* and the focused constituent. Then, I argue for the presence of a FocP in the left periphery in both in-situ and ex-situ focus cases. In in-situ focus, the head of Foc establishes agreement with the focused constituent. However, in ex-situ focus, the focused constituent is fronted

to the Spec of FocP in the left left periphery (à la Rizzi, 1997, and many others). This includes subject focus which we have seen to always be ex-situ. It occupies a position that is distinct from the normal subject position as I have argued above. For anti-pied-piping cases, I propose an analysis which makes use of not only syntax but also pragmatics. I will not provide an analysis for the cleft strategy because it does not play much of a role in our overall analysis here.

3.2.4 Towards an analysis for focalization in Kasem

Based on the data discussed so far, there are a few things that we must take note of. First is the fact that the focus particle **mo** is obligatory in both the ex-situ and the in-situ contexts (see (5) and (1) repeated below as (25) for simplicity). As a result, the focus particle forms a constituent with the focused DP, especially in the one-to-one focus marking cases. Secondly, we have also seen cases of mismatches between the morphologically or morphosyntactically marked constituent and the logical focus, i.e. pied-ping and anti-pied-piping, in the language.

- (25) [What did Adam slaughter?]
- a. Adam go [chworɔ]_F *(**mo**).
Adam slaughter fowl FOC
‘Adam slaughtered A FOWL.’
 - b. [Chworɔ]_F *(**mo**) Adam go-a.
fowl FOC Adam slaughter-DISJ
‘Adam slaughtered A FOWL.’

Thus, any analysis that will be proposed must capture the following properties of Kasem’s focus system:

- (26)
- a. The constituent nature of the focus marker *mo* in relation to its host;
 - b. The direct (one-to-one) focus marking, pied-piping and anti-pied-piping cases;
 - c. The movement of this constituent to a sentence-initial position in the left periphery of the clause (ex-situ focus);

Beginning with the first property, I assume that the focus marker *mo* forms a constituent with the focus phrase. This assumption is made on the basis of two grounds. The first is with respect to the placement of *mo* in the data already presented above. In the in-situ focus case for example, we see that *mo* does not seem to have a fixed position in the clausal spine. Instead, it always follows the focused constituent. For instance, the (b) examples in (27)-(29) show different focused constituents in a ditransitive construction which are marked appropriately by *mo*. In other words, each of the answer to the current questions is distinguished by the distinct placement of *mo*. If we take adjunct focus for example, assuming a fixed position for *mo* (e.g. the internal vP periphery) would predict or require the movement of both the direct object and the indirect object to a higher position in order to get the linearization correctly.⁷ Such movement steps would be unmotivated and unnecessary.

⁷ See Kayne (1998); Jayaseelan (2001); Belletti (2004, 2009); Collins and Essizewa (2007) for instances of clause-internal/low focus projections.

- (27) a. What did Adam give to John yesterday?
 b. Adam pɛ tɔnɔ *(**mo**) John dīim.
 Adam give cup FOC John yesterday
 ‘Adam gave A CUP to John yesterday.’
- (28) a. To whom did Adam give a cup yesterday?
 b. Adam pɛ tɔnɔ John *(**mo**) dīim.
 Adam give cup John FOC yesterday
 ‘Adam gave a cup TO JOHN yesterday.’
- (29) a. When did Adam give a cup to John?
 b. Adam pɛ tɔnɔ John dīim *(**mo**).
 Adam give cup John yesterday FOC
 ‘Adam gave a cup to John YESTERDAY.’

Secondly, *mo* does not only occur in the post-verbal position. As we have seen, it can also occur in the pre-verbal position in the case of the subject focus (30), and even ex-situ focus in general (31). Thus, the structure in (32) represents the adjunction of the focus marker to the focus phrase.

- (30) [Adam]_F **mo** go chworɔ kom.
 Adam FOC kill fowl DEF
 ‘ADAM slaughtered the fowl.’
- (31) [Chworɔ]_F **mo** Adam go diim.
 fowl FOC Adam kill yesterday
 ‘Adam slaughtered A FOWL yesterday.’
- (32) **Focus particle adjunction**
- DP

DP

chworɔ

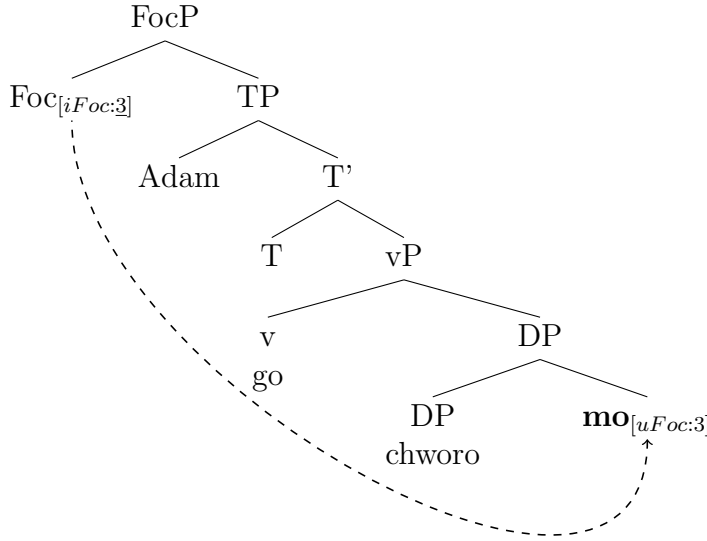
mo

For the *in-situ direct focus*, I assume that there is a high null focus head that licenses the focus constituent, and that *mo* morphologically realizes the focus feature on the focus exponent. By direct focus, we mean instances where *mo* directly adjoins to the logical or semantic focus (e.g. an answer to a *wh*-question). Taking the in-situ object focus in (33) as a case study, the structure is given in (34). The null focus head is interpretable but unvalued and probes downward in order to get valued by the focus marker. It is the result of the feature valuation that is seen in the structure.⁸

- (33) a. What did Adam slaughter?
 b. Adam go [chworɔ]_F **mo**.
 Adam kill fowl FOC
 ‘Adam slaughtered A FOWL.’

⁸ The use of numbering (as in [3]) is just an index for feature valuation. Basically, it shows the instances of a specific feature, here focus feature (cf. Pesetsky and Torrego, 2007). When the index is underlined, it shows that the feature was initially unvalued, but then gets valued after probing for, and agreeing with a suitable goal. So, the idea here is based on the traditional approach to Agree where an unvalued feature acts as a probe (Chomsky, 2000, 2001; Pesetsky and Torrego, 2007).

(34) **In-situ Focus: Direct adjunction**



The presence of a higher licensing focus head is supported by the behaviour of *mo* with respect to DP-internal focus constructions. Consider example (15) above, repeated in (35) below.

- (35) a. Wɔ chworo **mo** Adam go-a?
 who fowl FOC Adam slaughter-DISJ
 ‘Whose fowl did Adam slaughter?’
 b. Adam go [_{DP} [Peter]_F (***mo**) chworo] **mo**.
 Adam slaughter Peter FOC fowl FOC
 ‘Adam slaughtered Peter’s fowl.’

The focus morpheme can only adjoin to the whole complex DP even though the logical focus is only a sub-part, i.e. the possessor *Peter*. If *mo* is all that there is, then it is surprising why it cannot immediately follow the possessor in this case. But if we assume a higher licensing head that must Agree with *mo*, then we can explain why *mo* is in the DP boundary in (35-b). This is because agreement is subject to locality constraints (Chomsky, 2000, 2001, 2008; Béjar and Rezac, 2009, i.a.). The path between *mo* and the corresponding Foc head cannot cross a phase boundary, assuming the DP to be a phase in this case (cf. Svenonius, 2004; Bošković, 2005; Citko, 2014). Thus, given the *Phase Impenetrability Condition* (PIC), the domain of a phase is not accessible to operations outside of the given phase (Chomsky, 2000). The domain here is the complement of the DP phase head. This is why *mo* does not adjoin directly to *Peter* in (35-b). However, the edge of the phase head is accessible to external operations. The edge includes its specifier (and any adjoined phrase) and the head. Therefore, the logical focus is marked at the edge of its phase due to locality.⁹

This is further confirmed by cases of two levels of embedding of the DP-internal focus. Following the question in (36-a), the focus answer in (36-b) is only grammatical if the focus marker adjoins to the highest DP layer. Multiple embeddings produce similar result.

⁹ There are other recorded Agree phenomena that pattern similarly to *mo*-adjunction with respect to strict locality. Two of such cases are the Panoan Switch Reference (Baker and Camargo Souza, 2020) and Cuzco Quechuan object agreement (Lefebvre and Muysken, 1988).

- (36) a. Wɔ̌ nakɔ̌ chworo **mo** Adam go-a?
 who sister fowl FOC Adam kill-DISJ
 ‘Whose sister’s fowl did Adam slaughter?’
 b. Adam go [Peter (***mo**) nakɔ̌ (***mo**) chworo] *(**mo**) .
 Adam kill Peter FOC sister FOC fowl FOC
 ‘Adam slaughtered PETER’S sister’s fowl.’

The same locality restriction is seen with the focus marking of an element within a relative clause. Relative clauses in Kasem (and a few other Mabilia languages) end with a relative determiner *to* which seems to mark the boundary of the relative clause (37). If a constituent within the relative clause is focalized, the focus marker does not follow the focused constituent, but adjoins to the boundary of the relative clause; after the relative determiner (cf. (38-b)). Summarily, what this implies is that the placement of the focus particle is governed by locality constraints.

- (37) A nɛ baaro wolo na go naao **to**.
 1SG see man REL PFV kill cow REL.DET
 ‘I saw the man who slaughtered a cow.’
 (38) a. N nɛ baaro wolo na go chworo to na?
 2SG see man REL PFV kill fowl REL.DET Q
 ‘Did you see the man who slaughtered a fowl?’
 b. Aawo, a nɛ baaro wolo na go naao (***mo**) to **mo**.
 no 1SG see man REL PFV kill cow FOC REL.DET FOC
 ‘No, I saw the man who slaughtered A COW.’

There is one challenge that must be addressed, taking into consideration the locality and PIC explanation given above. Even though *mo* adjoins to the edge of the DP phase, for example, how do we explain the fact that there is another phase, the vP, which c-commands the DP? With the current assumption of the PIC, the prediction is that agreement is blocked because of the presence of another phase, the little vP. This is based on the strong PIC in (39). The strong PIC disallows any operation into a phase once it is built. So, the complement of H, which is the v head in our case, is not accessible once HP (vP in our case) is built.

- (39) **Strong PIC** (Chomsky, 2000, 2001)
 The domain of a phase head H is not accessible to operations outside HP; only H and its edge are accessible.

However, for our analysis here, the version of PIC that we adopt must be loosened in order for the high focus head to be able to access the focused constituent. Fortunately, Chomsky (2001) proposes a loosened form of the PIC. This is the *weak PIC*, and it is stated in (40). With the weak PIC, there is a longer accessibility window for syntactic operations (such as agreement and movement) to take place. The domain (complement) of a phase head H is accessible as long as a higher phase (ZP) has not been merged.

- (40) **Weak PIC** (Chomsky, 2001, p.14, ex.(11))
 The domain of H is not accessible to operations at ZP; only H and its edge are accessible to such operations.

The implication of the weak PIC is that the complement of a phase remains accessible

until the next phase head is merged. In our case, the (edge of) DP complement of the vP (= H) phase remains accessible until the next phase CP (= ZP) is merged (41). Therefore, since Foc^o is part of the C-system (Rizzi's (1997) split-CP hypothesis), its agreement with *mo* falls within the coverage of the weak PIC.¹⁰

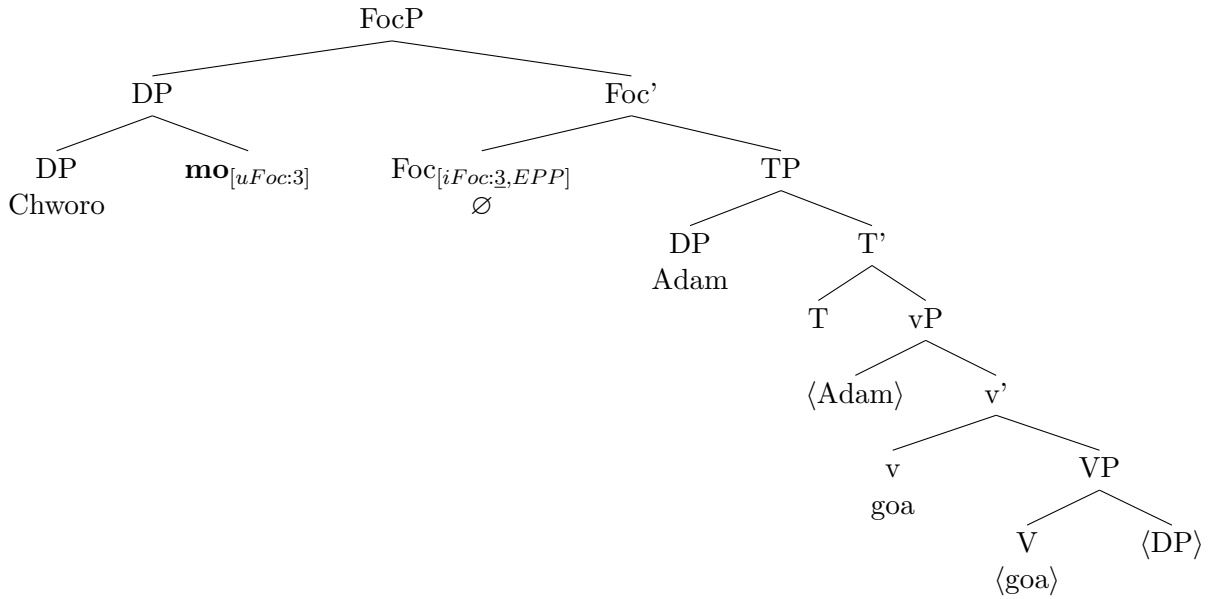
$$(41) \quad [_{CP} \dots \text{Foc} \dots [_{vP} v [_{DP} \dots] \text{mo}]]$$

Ex-situ Focus analysis

Moving on to ex-situ focus, I simply assume that the focus constituent moves to Spec,FocP to satisfy an EPP requirement on Foc. Using the ex-situ object focus in (42-b) as a reference, we can derive the structure in (43).

- (42) a. Bε *(**mo**) Adam go-a?
 what FOC Adam slaughter-DISJ
 ‘What did Adam slaughter?’
 b. [Chworo]_F *(**mo**) Adam go-a.
 fowl FOC Adam slaughter-DISJ
 ‘Adam slaughtered A FOWL.’

(43) Kasem ex-situ focus structure



After the derivation is built up to the TP level, the FocP is projected above it. However, unlike the in-situ derivation given above (cf. (34)), the focus head in the ex-situ strategy is specified with an EPP which licenses the movement of the focus. Here is how the ex-situ focus derivation goes. First, the unvalued focus head probes its c-command domain for a suitable goal. It finds the object phrase which has a valued feature (exponed as *mo*), and Agrees with the latter. After agreement has taken place, the EPP on Foc requires the agreement goal to be fronted to its Spec. So, basically, agreement feeds movement. In addition, the fronted focus does not have to be an object DP only, a PP or even an adjunct can also be fronted (see (8) above).

The idea is that although the focus head in the left periphery is consistently null,

¹⁰ While the strong PIC is more restrictive and stricter with respect to locality and access (Chomsky, 2000), the weak PIC is less restrictive and grants more access to operations across phase boundaries (Chomsky, 2001).

its feature specification or combination differs depending on the focus strategy that is employed, i.e. in-situ versus ex-situ strategies (44).¹¹

(44) **Feature specification based on focus strategy**

- a. In-situ focus = $\text{Foc}_{[\text{iFoc:}__]}$
- b. Ex-situ focus = $\text{Foc}_{[\text{iFoc:}__, \text{EPP}]}$

Anti-pied-piping analysis in Kasem

So far, we have provided analyses for (a) the constituent property of the focus marker *mo*, and (b) cases of direct focus marking, i.e. the direct morphological marking of the logical or semantic focus. But we have also seen the locality restriction that comes with *mo* marking and the agreement with a high null Foc^0 in the left periphery of the clause. In the process, we have touched some parts of the second property of Kasem focus system listed in (26) above.

We now turn to the anti-pied-piping focus marking cases. While the analysis that I am going to give for anti-pied-piping is not entirely different from the proposal so far, however, we would need to make some additional assumptions in order to explain the ‘non-correspondence’ between the morphological marking of focus and the logical focus itself in anti-pied-piping cases. Recall that in the anti-pied-piping cases, the focus marker *mo* adjoins to a sub-part of the logical focus, as in the sentence focus example below (given as example (23) above). Although the logical focus is the whole clause (45-b) (going by the *wh*-question in (45-a)), the morphological marking is on the subject. (45-c) is infelicitous or incongruent as an answer to the current question. It can answer a direct object *wh*-question (What did Adam slaughter?), for example. Before I propose my analysis for the anti-pied-piping cases, I will briefly discuss the views and claims from two previous studies on anti-pied-piping in other languages. The aim is to show the similarities they have with my analysis for anti-pied-piping, as well as highlight the areas where they differ. The two studies are Fanselow and Lenertová (2011) and Branen and Erlewine (2023).

- (45)
- a. Bε **mo** kea?
what FOC do
‘What happened?’
 - b. $[_{TP} [\text{Adam } \mathbf{mo}] \text{ go } \text{chworo}]_F$.
Adam FOC kill fowl
‘Adam slaughtered a fowl.’
 - c. #Adam go chworo **mo**.
Adam kill fowl FOC
‘Adam slaughtered a fowl.’

Fanselow and Lenertová (2011): They presented data from German and Czech to argue for a *formal fronting* approach to focus (Frey, 2005). In other words, they claim that anti-pied-piping cases are clear evidence that focus fronting is not directly tied to

¹¹ There is a general question for such an optional movement, which is: how does the focus head know whether a focus constituent should move or not? This is a question that does not have an absolute answer, and is usually overlooked in the literature. A potential answer to this question, at least for Kasem, is that focus fronting is linked to some semantics/pragmatic interpretations. Usually, ex-situ focus in Kasem is associated with exhaustive presupposition or identificational focus. Another use of focus fronting is with mirative focus (see Cruschina, 2009, 2012, 2019; Bianchi et al., 2016; Cruschina and Bianchi, 2021, a.o., on mirative focus fronting). So, focus constituents move to Spec,FocP for interpretive purposes, and the focus head is specified with an EPP to trigger such movement.

information structure, contrary to what has been proposed in the literature, especially the cartographic tradition (Rizzi, 1997; Aboh, 2004b, 2007). In the cartographic tradition, information structure, focus fronting to be precise, is triggered by the need to satisfy a focus criterion in the left periphery via Spec-Head agreement (Rizzi, 1997, i.a.). However, Fanselow & Lenertová’s analysis removes information structure from the syntax. Some of the data presented by Fanselow & Lenertová had already been presented at the end of the previous section. For example, (46) (as (24) above) shows that in German and Czech, the fronted constituents are only sub-parts of the logical foci that answer any of the current (wh-)questions (Fanselow and Lenertová, 2011, p.174).

- (46) What did you do?, What’s new?/What happened?
- a. [Einen HAsen]_i habe ich t_i gefangen. (German)
 ART.ACC rabbit have I caught
 ‘I caught a rabbit.’
 - b. [ZAJíce]_i jsem chytil t_i. (Czech)
 rabbit.ACC AUX.1SG caught.SG.MS
 ‘I caught a rabbit.’

Fanselow & Lenertová linked these instances of fronting to a formal feature or an unspecific edge feature on C. Due to this feature, the latter attracts the sub-part of the logical focus to its specifier (Spec,CP). This raises the question of whether there is any restriction on what can or cannot be fronted. Fanselow & Lenertová shows that while accented sub-constituents can cause an intervention effect (i.e. block sub-part focus fronting), unaccented constituents like (weak) pronouns and indefinites do not. By accent, we mean structural accent here, which might be different from focus (nuclear) accent which is usually on a focus exponent (cf. Selkirk, 1995; Kratzer and Selkirk, 2007; Truckenbrodt, 2007).¹² In the idiomatic expression in (47), while the sub-part focus fronting retains the idiomatic reading in (47-b), the reading is lost when a lower accented phrase is fronted, crossing a higher accented phrase (47-c) (Fanselow and Lenertová, 2011, p.179).¹³

- (47) What happened?/What did he do?
- a. Er hat die FLInte ins KORN geworfen. (German)
 he has the gun into-the grain thrown
 ‘He has given up.’
 - b. [Die FLInte]_i hat er t_i ins KORN geworfen.
 - c. #[Ins KORN]_i hat er die Flinte t_i geworfen.

On the other hand, unstressed or weak pronouns (48-a) and indefinites (48-b) do not constitute blocks for sub-part focus fronting (Fanselow and Lenertová, 2011, p.180).

- (48) What’s new?
- a. GULáš_i jsem uvářila t_i. (Czech)
 goulash AUX.1SG cooked.SG.FM
 ‘I cooked goulash.’

¹² Selkirk (1995), Kratzer and Selkirk (2007), Fanselow and Lenertová (2011), among others, argue for an obligatory assignment of accents to all non-given sub-constituents of a clause.

¹³ According to Fanselow and Lenertová (2011), cases like (47-a) show that the focus exponent (bearing the primary nuclear accent) in German can be different from the XP that undergoes sub-part focus fronting.

- b. KARla_i někdo hledal t_i.
 Karel.ACC somebody.nom looked-for.SG.MS
 ‘Somebody was looking for Karel.’

Refining Fox and Pesetsky’s (2005) theory of cyclic linearization, Fanselow & Lenertová assume that the fronted sub-constituent is determined by accenting constituents immediately at the point of external merge (at linearization). It is the absence of such an accent that leads to given elements, which in turn do not qualify for formal fronting. However, given elements may later be qualified for fronting if they are accented in contrastive topic constructions or multiple focus contexts. The emphasis here is that “... accented XPs are linearized before they move, so they cannot cross each other.” Thus, an accented phrase can only move across an unaccented phrase.¹⁴

Branan and Erlewine (2023): Although Fanselow & Lenertová assume that “it is always the leftmost accented XP that undergoes SFF movement”, it is Branan and Erlewine (2023) who formalize such a requirement. Branan and Erlewine’s anti-pied-piping paper covers many more languages (over 60 languages), and show that not only prosodic focus marking languages, but also morphological and/or morphosyntactic focus marking languages attest anti-pied-piping. The Japanese examples in (49) display anti-pied-piping in predicate focus (49-a) and sentence focus (49-b) respectively. In each cases, the focus sensitive particle *mo* adjoins to a sub-part of the logical focus. The data is from Branan and Erlewine (2023, p.634-635) based on Ohno (2003) and Aoyagi (1998, 2006). This is a case of anti-pied-piping by particle placement.

- (49) a. Taro-ga [tako]-mo tabe-ta.
 Taro-NOM octopus-also eat-PST
 ‘Taro also [ate octopus]_F.’
 b. [Taro]-mo piano-o hii-ta.
 Taro-also piano-ACC play-PST
 ‘[Taro played piano]_F, too.’

Branan and Erlewine propose an operator-particle analysis to capture not only anti-pied-piping cases, but also pied-piping cases. This is similar to what we propose above for the direct focus marking cases in Kasem. They argue for the presence of a high null operator that agrees with the particle that adjoins to the sub-constituent of the logical focus, as in (50) below (where ‘MSF’ stands for MORPHOSYNTACTIC RESPONSES TO FOCUS).

- (50) [CP/TP Op [XP YP_{MSF} ZP]_F]
 ↑_{Agree}

In order to formalize the placement of the particle, Branan and Erlewine suggested that particle placement has access to the same information that prosodic languages use to determine structural stress. They refer to this information as an abstract bullet sign • which leads to the assignment rule in (51). This means that each constituent that is embedded within a phase is assigned a •. Therefore, the choice of the element that takes MSF is formalized as the particle placement condition in (52). The phrase that host MSF is mostly left-aligned within the logical focus in most of the languages investigated

¹⁴ Fanselow and Lenertová (2011) argue against approaches such as remnant movement and scattered deletion that might be adopted in order to enforce the feature checking of focus.

by Branán and Erlewine (2023). There were a few exceptions, however, including the pied-piping instances (e.g., DP-internal focus) discussed above. These exceptions plus the pied-piping facts are accommodated by the other parts of the particle placement condition, which allows for ‘preferably ●-left-alignment or overlapping’ with the logical focus.

(51) **●-assignment:**

At phasal Spell-out, assign a ● to each phrase that is not a part of the extended projection that contains the phase head.

(52) **Particle Placement:**

During phasal spell-out, Late Adjoin the particle to a phrase that is ●-left-aligned/is preferably ●-left-aligned/overlaps with the logical focus.

In summary, both studies presented data that show that a sub-part of a logical focus can be overtly marked, whether syntactically, as in sub-part focus fronting in German and Czech, or morphologically (morphosyntactically) as presented by Branán and Erlewine (2023) in many languages. Secondly, the choice of the constituent that is marked (by movement or morphology) is dependent on the linearization of accented phrases (Fanselow and Lenertová, 2011), and/or the left-most condition of the particle placement (Branán and Erlewine, 2023). Lastly, Fanselow and Lenertová’s claim that information structure should be separated from syntax cannot be generalized, especially considering morphosyntactic focus marking languages like Kasem (including Kusaal and Yorùbá as we would see soon). Such claim might get support from prosodic focus marking languages that they presented because of the in-situ focus marking with accent in such languages. However, morphosyntactic focus marking languages support a relationship between information structure and syntax.

However, there are a few open issues from these earlier analyses. First, in anti-pied-piping cases, how do we know what the logical focus is (seeing that the sub-part focus fronting (Fanselow and Lenertová, 2011) and the particle placement (Branán and Erlewine, 2023) is to a sub-part of the logical focus)? Secondly, Branán and Erlewine’s particle placement condition raises a question about late adjunction. Although they explained the version of late adjunction that they adopt, but generally, late adjunction is assumed to occur after the completion of the phase. In other words, late-adjoined elements should not be accessible to syntax in a cyclic spell-out approach. Also, late-adjoined elements are often seen as optional elements which do not affect core syntax.¹⁵

My proposal: My proposal for anti-pied-piping in Kasem takes insights from the previous analyses above. However, I make the following additional assumptions for the observed patterns in Kasem.

- (53) a. *Mo* prefers to adjoin to [-V] constituents, i.e., constituents that are not part of the verbal extended projection (e.g., DPs and PPs).
 b. A modified version of the particle placement requirement, which depends on the weak PIC, is proposed.
 c. Direct focus fronting and anti-pied-piping fronting in Kasem are motivated differently. While the former is motivated by pragmatic reasons, the latter

¹⁵ Stepanov (2001), for instance, argued that constructions involving raising across an experiencer (e.g., John seems to Mary to be smart) does not affect the minimal link condition (subject-to-subject raising) because the latter is an adjunct.

involves formal fronting in the spirit of Fanselow and Lenertová (2011).

- d. The syntax of Kasem’s focus anti-pied-piping (and focus in general) is not independent of pragmatics. The context, in the form of the current question under discussion, plays a crucial role in the recognition of the logical focus.

To begin with, the two instances of anti-pied-piping in Kasem are repeated below: VP focus (54) and TP focus (55). In both cases, it is the leftmost sub-constituent of the logical focus that hosts the morphological marking with *mo*. The fronting of other sub-constituents, like the adverb in (54-d) (in VP focus anti-pied-piping) and the direct object in (55-c) (in TP focus anti-pied-piping), cannot be felicitous answers to the questions in (54-a) and (55-a) respectively. They are only compatible with narrow focus on the adverb and the direct object respectively.

- (54) a. Bε mo Adam ke diim?
 what FOC Adam do yesterday
 ‘What did Adam do yesterday?’
 b. Adam [_{VP} go [_{DP} chworo **mo**] diim]_F.
 Adam kill fowl FOC yesterday
 ‘Adam SLAUGHTERED A FOWL YESTERDAY.’
 c. [_{DP} Chworo **mo**]_i Adam [_{VP} go t_i diim]_F.
 fowl FOC Adam kill yesterday
 ‘Adam SLAUGHTERED A FOWL YESTERDAY.’
 d. [_{Adv} Diim **mo**]_i Adam [_{VP} go chworo t_i]_F.
 yesterday FOC Adam kill fowl
 (i) ‘Adam slaughtered a fowl YESTERDAY.’
 (ii) *‘Adam SLAUGHTERED A FOWL YESTERDAY.’
- (55) a. Bε **mo** kea?
 what FOC do
 ‘What happened?’
 b. [_{TP} [_{DP} Adam **mo**] go chworo]_F.
 Adam FOC kill fowl
 ‘ADAM SLAUGHTERED A FOWL.’
 c. [_{DP} Chworo **mo**]_i [_{TP} Adam go t_i]_F.
 fowl FOC Adam kill
 (i) ‘Adam slaughtered A FOWL.’
 (ii) *‘ADAM SLAUGHTERED A FOWL.’

For the particle placement (left-most) requirement, Branau and Erlewine (2023) show certain constituents “are skipped for the evaluation of ‘leftmost’”. These include indefinites, given elements, and/or less informative elements (see also Fanselow and Lenertová, 2011, for a similar observation).

Interestingly, unlike the cases discussed by Branau and Erlewine (2023), in Kasem, the particle placement does not skip indefinites if they are the leftmost constituent.¹⁶ This is the case in both VP focus (56) and TP focus (57). This shows that the leftmost requirement seems to be stricter in Kasem.

¹⁶ This might be due to the absence of prosodic influence in focus marking in Kasem, as indefinites are said to involve secondary prominence (cf. Selkirk, 1984, 1995; Krifka, 2008).

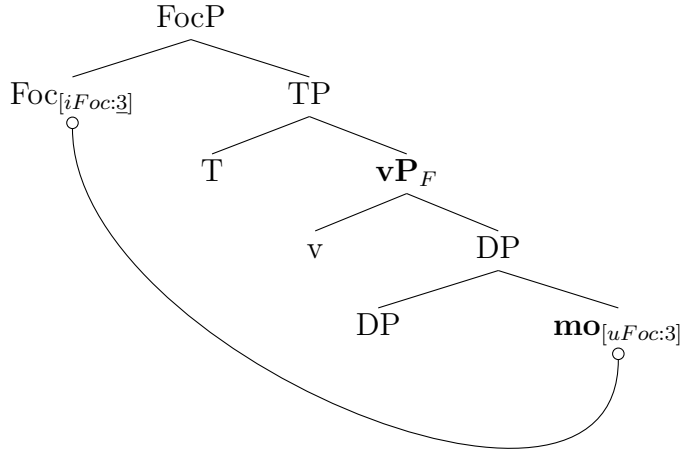
- (56) a. Bε **mo** Adam ke diim?
 what FOC Adam do yesterday
 ‘What did Adam do yesterday?’
 b. Adam [_{VP} go [_{DP} woŋo **mo**]] diim.
 Adam kill something FOC yesterday
 ‘Adam SLAUGHTERED SOMETHING YESTERDAY.’
 c. [_{DP} Woŋo **mo**]_i Adam [_{VP} go t_i] diim.
 something FOC Adam kill yesterday
 ‘Adam SLAUGHTERED SOMETHING YESTERDAY.’
- (57) a. Bε **mo** kea?
 what FOC do
 ‘What happened?’
 b. [_{TP} [_{DP} Nɔɔno **mo**] ŋɔge lu-siseiŋa].
 someone FOC steal metal-horse
 ‘SOMEONE STOLE A BICYCLE.’

Furthermore, in an intransitive TP focus construction, particle adjunction is not sensitive to whether the subject is unergative (58-b) or unaccusative (58-c). In as much as the subject is the leftmost sub-constituent at the point of the merging of the TP, *mo* adjoins to it.

- (58) a. Bε **mo** kea?
 what FOC happen
 ‘What happened?’
 b. [[Adam **mo**] saŋe].
 Adam FOC cook
 ‘ADAM COOKED.’
 c. [[Adam **mo**] to-a].
 Adam FOC fall-DISJ
 ‘ADAM FELL.’

The structure that I propose for anti-pied-piping (of VP focus) (59) is similar to the one proposed for the direct focus marking on the logical focus in (34) above. There is a high null *Foc*^o which licenses *mo*-adjunction via Agree. *Foc*^o is interpretable but unvalued. Thus, it probes and gets valued by *mo*, which is valued but uninterpretable. It is important that the interpretability comes from a higher head due to semantic consideration. The logical focus has to be in the scope of the licensor. This is the first requirement in order to be able to account for anti-pied-piping.

(59) **In-situ VP Focus: Anti-pied-piping structure**



There is an important property of *mo* that must be further discussed as mentioned in (53-a) above. Based on the focus data presented so far, one conclusion can be reached: *mo* does not adjoin to verbal elements in the presence of non-verbal elements. This means that while *mo* easily adjoins to DPs and PPs directly, it does not seem to adjoin to VP and TP. The latter categories have one thing in common. They are [+V] (à la Grimshaw, 2000, 2005; Biberauer and Sheehan, 2012; Sheehan et al., 2017, a.o.). They are part of the extended verbal projection. Thus, we can have the following formulation for *mo*-adjunction (60).

(60) **Mo-adjunction property**
Mo adjoins to [-V] elements.

However, on a closer look, there is an instance where *mo* would have to adjoin to the VP. This is with intransitive constructions. Consider the question and answer pair in (61). The focus here is the VP, but in the absence of a direct object, *mo* adjoins to the VP *as a last resort*.

- (61) a. Bε **mo** Adam kea?
 what FOC Adam do
 ‘What did Adam do?’
 b. Adam [toɽe]_F **mo**.
 Adam worked FOC
 ‘Adam WORKED.’

Considering the data above, we would need to modify the adjunction property of *mo* in (60) as (62). The consequence of (62) is that *mo* always looks out for [-V] elements, specifically DPs.

(62) **Modified Mo-adjunction property**
Mo adjoins to [+V] elements only in the absence of [-V] elements within the logical focus (in the scope of Foc).

Therefore, the assumption is that in (VP or TP) anti-pied-piping, after the completion of either the VP or the TP, *mo* (with a [-V] property) does not adjoin to either of them in the presence of a DP. In the next step, *mo* adjoins to a [-V] element of the logical focus. But in the absence of a DP, it adjoins to VP as a last resort. This is peculiar to the

VP because it can be intransitive. TP focus will always have a subject DP, whether it is unergative or unaccusative (cf. (58) above). So there is a host for *mo*-adjunction for TP anti-pied-piping. Next, we turn to an endeavour to propose a constraint that will account for the placement of *mo* in anti-pied-piping cases.

The formulation in (62) by itself would allow any [-V] element to host *mo* in anti-pied-piping. For instance, in TP focus anti-pied-piping, *mo* should be able to adjoin to the direct object, as in (63-b) (repeated from (55) above). But we know that this is not the case (63-b)-ii. Only *mo* adjunction to the subject is felicitous (63-c). Therefore, we need to constrain our particle placement for anti-pied-piping cases.

- (63) a. What happened?
 b. $[_{DP} \text{ Chworo } \mathbf{mo}]_i \text{ } [_{TP} \text{ Adam go } t_i]_F$.
 fowl FOC Adam kill
 (i) ‘Adam slaughtered A FOWL.’
 (ii) *‘ADAM SLAUGHTERED A FOWL.’
 c. $[_{TP} [_{DP} \text{ Adam } \mathbf{mo}] \text{ go } \text{chworo}]_F$.
 Adam FOC kill fowl
 ‘ADAM SLAUGHTERED A FOWL.’

As discussed above, for Fanselow and Lenertová (2011) and Branan and Erlewine (2023), prosody or accent placement plays a vital role in their leftmost requirement. In contrast, we have seen that prosody does not play any role in our case. It is rather the word order alone that seems to play a role with respect to particle placement.

If *mo* adjoins to the leftmost constituent in a VP focus anti-pied-piping, the direct object in this case, we are faced with two challenges. The first challenge borders on how to capture this requirement, and the second borders on how to agree into the vP phase. These challenges can be combined into one question: how do we capture the leftmost requirement in such a way that the locality requirement of Agree is met? In order to solve these challenges, I propose a version of the particle placement requirement which is built upon the weak PIC (65). Consider the weak PIC version given in (40) above, repeated below as (64) for ease of referencing. With the weak PIC, the complement of a phase head only becomes inaccessible at the merge of the next phase head (cf. Müller, 2004; Richards, 2004, 2007a,b, 2011). It delays spell-out and allows Agree to take place with the complement of a phase. Therefore, agreement between Foc^o and *mo* that is inside vP is possible, but *mo* inside another phase that is embedded in vP is not. With this in place, we can now make sense of the particle placement requirement in (65). Since spell-out of a lower phase domain is delayed until the merging of the next phase, *mo* is late-adjoined to the leftmost non-verbal projection of the logical focus.¹⁷

- (64) **Weak PIC** (Chomsky, 2001, p.14, ex.(11))
 The domain of H is not accessible to operations at ZP; only H and its edge are accessible to such operations.
- (65) **Particle Placement Requirement (with weak PIC)**
 Before merging a higher phase, late-adjoin the focus particle to the left-most non-verbal maximal projection (phrase) of a logical focus.

To make this clearer, let’s compare the idea here with that of Branan and Erlewine

¹⁷ The requirement in (65) does not only cover anti-pied-piping cases. It is more general, i.e. it also covers cases where *mo* adjoins directly to the logical focus, like object focus for example.

(2023) that was introduced above. Branán and Erlewine’s idea of particle placement is tied to the timing of particle adjunction which is conditioned by stress placement. Since stress placement is related to the PF interface, and it feeds particle placement, the particle adjunction in their analysis has to take into consideration the interface with phonology. Thus, the timing for particle adjunction for them is at least during spell-out. For our analysis here however, the timing for the particle placement is conditioned by access to word order information. We already saw that the focus system in Kasem does not make use of prosody but morphosyntax. The timing of *mo*-particle placement therefore is conditioned by word order. This is further supported by the cases of weak elements like indefinites which could not host the particles in Branán and Erlewine’s (2023) survey due to lack of primary stress (cf. Fanselow and Lenertová, 2011). However, as we saw in examples (56) and (57) above, weak indefinites are not exempted from particle placement in Kasem. This indicates that word order, and not stress assignment, is the pivot for particle adjunction in Kasem. This is one of the areas that distinguish our proposal here from that of Branán and Erlewine (2023) and Fanselow and Lenertová (2011).

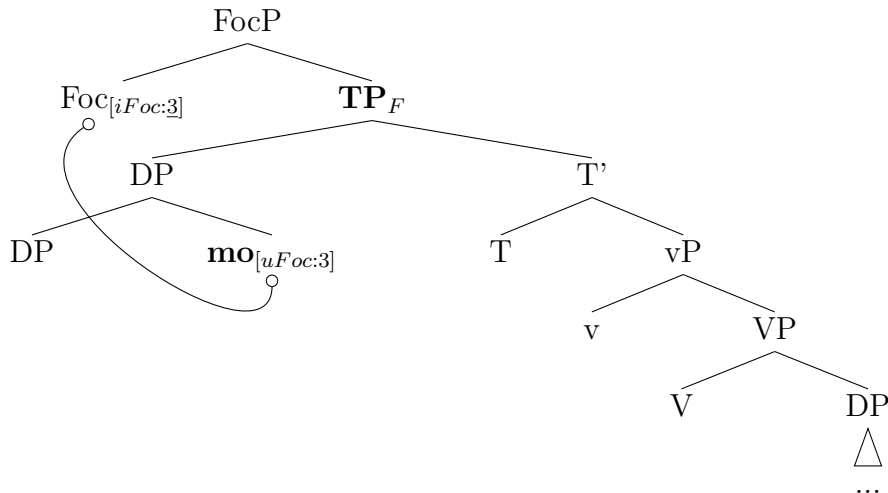
The particle placement requirement in (65) constrains the adjunction of *mo* in anti-pied-piping cases to the left-most non-verbal maximal projection of the logical focus. Thus, in VP anti-pied-piping, it does not matter whether it is the big VP or the little vP that we assume as the logical focus. *Mo* late-adjoins to the left-most non-verbal constituent. Recall the *mo*-adjunction property in (62) which prioritizes adjunction to a non-verbal constituent. Since word order is always available during the derivation, the notion of *late-adjunction* adopted here is one which can access the word order of a logical focus that involves anti-pied-piping.

TP anti-pied-piping can also be accounted for by the current proposal. Based on the word order within the TP, the subject happens to be the leftmost constituent. Recall that the data in (58) already shows that this is regardless of whether the subject is unaccusative or unergative. What matters is that at the completion of the TP, the subject occupies a leftmost position. So for instance, in an unaccusative construction, even though the subject started out in the thematic object position, it ends up in the canonical subject position. The particle placement requirement therefore requires that it hosts *mo* in TP anti-pied-piping.

In (66), we have the structure for TP anti-pied-piping. The agreeing focus head is just above the TP, and are both within the CP phase. So Foc licences *mo* and has the logical focus (TP) in its scope.¹⁸

¹⁸ Note that the structure in (66) takes for granted the vP-subject internal hypothesis, as the assumption is that the subject is externally-merged in Spec,vP before moving to Spec,TP.

(66) **TP Anti-pied-piping structure**



The next point is how to reconcile the size of the logical focus and morphological focus marking, especially in anti-pied-piping cases. This is answered by the assumption in (53-d). I assume that the current question under discussion (pragmatics) plays an important role in anti-pied-piping cases. This is captured by the discourse principle of Beaver and Clark (2008, p.37) that was introduced in chapter 2 (repeated in (67) below). It is important for utterances to be relevant to the current question. In addition to the discourse principle, Beaver and Clark propose another principle which leans on Rooth's alternative semantics to focus and Hamblin's alternative semantics to question (the Rooth-Hamblin approach that we introduced in chapter 2). This is called the Focus Principle (68).¹⁹ The two principles connect the focus answer to CQ. In our case, the logical VP/TP focus is not given if we consider a question like *What did Adam do?/What happened?*²⁰ Thus, the new information in the declarative answer is the focus that evokes alternatives (of the CQ). Although the morphological marking is realized on a sub-part of the logical focus, the logical focus is interpreted at LF through the help of the CQ.

(67) **Discourse Principle** (Beaver and Clark, 2008, p.37):

Utterances should be maximally relevant to the CQ.

(68) **Focus Principle** (Beaver and Clark, 2008, p.37):

Some part of a declarative utterance should evoke a set of alternatives containing all the Rooth-Hamblin alternatives of the CQ.

To some extent, this is similar to what we see in prosodically focus marking languages, e.g. English. In English, the focus exponent in a VP focus is the direct object (69). This is accounted for by Selkirk's (1984; 1995) *Focus Projection (Rule)* which allows the F-marking (i.e. accenting) of an internal argument x of a phrase YP to project and license the F-marking of YP. While there is a similarity between the focus projection in (69) and VP anti-pied-piping in Kasem, it is obvious that the same is not true for TP anti-pied-

¹⁹ Beaver and Clark (2008) claim that their focus principle is more relaxed in congruency than Roberts's 1996 account. For example, while the sentence [I think [MARY]_F laughed] can answer the question [Who laughed?] in Beaver and Clark's account, the same sentence can only answer the question [Who do you think laughed?] in Robert's account. There is a strict requirement for congruency in Robert's account; the declarative sentence (that embeds the focus) must be congruent to the CQ completely (Beaver and Clark, 2008, p.37).

²⁰ A focus can be given, as in being mentioned in the background. But it can still be contrastively focused or a second occurrence focus (cf. Partee, 1999; Baumann, 2016, among others)

pip-ing, whereby the subject is morphologically marked and not the direct object. When the TP is the focus in English, the focus exponent is still the object (70) as this would be the rightmost content word in a phrase which typically receives the main stress going by Chomsky and Halle’s 1968 *Nuclear Stress Rule* (cf. Chomsky and Halle, 1968; Arregi, 2016).²¹ Thus, unlike TP anti-pied-piping, subject focus does not usually “project” to TP in English. A nuclear stress on the subject usually translates into a subject focus that answers a question like *Who slaughtered a fowl?*, and not *What happened?*

- (69) a. What did Adam do?
b. Adam [_{vP} slaughtered a FOWL_F]_F.
- (70) a. What happened?
b. [_{TP} Adam [_{vP} slaughtered a FOWL_F]_F]_F.

In summary, the focus marking strategies in Kasem has been the subject of discussion in this section. After describing the focus marking patterns that are present in the language, I have proposed distinct analyses for direct focus marking (i.e., cases where the morphological marking is on the logical focus itself) and anti-pied-piping cases (morphological marking of a sub-part of the logical focus) which is tied to the presence of a licensing focus head in the left periphery of the clause. This focus head licenses both the in-situ and the ex-situ realization of focus. The only difference is that in the latter case, there is an EPP requirement that drives the fronting of the focus constituent to Spec,FocP.

In the next section, I introduce the focus system in Kusaal and propose an analysis for it. The analysis is very similar to what I just presented for Kasem. Kusaal, however, provides further morphological support to some of the assumptions made in this section.

3.3 Focus realization in Kusaal

Although the realization of focus in Kusaal is similar to what we saw in Kasem, there are, however, noticeable differences. Kusaal allows for focus realization both in the in-situ position and the ex-situ position. Just like Kasem, Kusaal also marks in-situ focus morphologically: with the particle *nɛ*. But there is a difference in the order of the focus particle. Unlike Kasem, Kusaal’s in-situ focus particle precedes the focus constituent: [*nɛ* XP_F]. Another difference between the two languages is that ex-situ focus in Kusaal uses a different focus marker *ka*. Before going into the description and analysis of the data, I want to briefly discuss previous descriptions of focus in the language.

Focus realization strategies in Kusaal have been discussed by Musah (2018), Abubakari (2018, 2019) and Eddyshaw (2025). All three agree on the in-situ marking of focus with *nɛ*. However only Musah and Abubakari describe *ka* as the ex-situ focus marker in the language. Eddyshaw glossed it as ‘and’ CONJ in his grammar. He did not describe it as an ex-situ focus marker. The discussion so far on the focus system in Kusaal has largely been descriptive. As far as I know, only Abubakari (2018) provided an analysis for focus constructions in Kusaal, and this is based on the Lexical-Functional Grammar (LFG) framework. In what follows, unless otherwise noted, I will provide original fieldwork data on the focus realization patterns in Kusaal. Afterwards, I will propose a theoretical analysis similar to what I proposed for Kasem; all within the Generative Grammar framework.

²¹ See Assmann et al. (2023) for a proposal on how to capture morphosyntactic and prosodic focus marking languages. There is a review of their proposal in chapter 2, § 2.1.2.

3.3.1 In-situ focus

As stated above, in-situ focus is marked with *nɛ* in Kusaal. It immediately precedes the constituent that is in focus. The direct object in (71-b) is focused in the in-situ position. Interestingly, the in-situ focus marker *nɛ* does not follow the focused constituent in this case; it immediately precedes it instead (cf. (72) and (73)). This does not become obvious if we consider only a transitive construction like (71-b).²²

- (71) a. Adam kɔdig bɔ?
 Adam kill what
 ‘What did Adam slaughter?’
 b. Adam kɔdig **nɛ** [nua]_F.
 Adam kill FOC fowl
 ‘Adam slaughtered A FOWL.’

However, if we consider ditransitive constructions like (72) and (73), the pre-focal position of the focus marker *nɛ* becomes obvious. If the *wh*-question requires a direct object, the structure in (72-b), and not (72-c), is felicitous. On the other hand, if it is the indirect object that is in focus, the structure in (73-b), and not (73-c) is appropriate because of the question-answer congruence.

- (72) a. Azuure tis Adam bɔ?
 Azuure give Adam what
 ‘What did Azuure give to Adam?’
 b. O tis Adam **nɛ** [nua]_F.
 3SG give Adam FOC fowl
 ‘He gave Adam A FOWL.’
 c. #O tis **nɛ** Adam [nua]_F.
 3SG give FOC Adam fowl
 ‘He gave Adam A FOWL.’
- (73) a. Azuure tis nua la anɔ’ɔn?
 Azuure give fowl DEF who
 ‘Who did Azuure give the fowl?’
 b. O tis **nɛ** [Adam]_F nua la.
 3SG give FOC Adam fowl DEF
 ‘He gave ADAM the fowl.’
 c. #O tis [Adam]_F **nɛ** nua la.
 3SG give Adam FOC fowl DEF
 ‘He gave ADAM the fowl.’

Adjuncts are also focused in an in-situ position. In the example in (74-b), *nɛ* precedes the adverb *su’os* ‘yesterday’. If it precedes the direct object instead, as in (74-c), the sentence becomes infelicitous to the current question in (74-a).

²² The in-situ *wh*-questions are mostly unmarked, morphologically. See (71-a), for example.

- (74) a. Saŋkanε **ka** Adam sa da' nua la?
 when FOC Adam PST buy fowl DEF
 'When did Adam buy the fowl?'
 b. Adam sa da' nua la **nε** [su'os]_F.
 Adam PST buy fowl DEF FOC yesterday
 'Adam bought the fowl YESTERDAY.'
 c. #Adam sa da' **nε** nua la [su'os]_F.
 Adam PST buy FOC fowl DEF yesterday
 'Adam bought the fowl YESTERDAY.'

Consider the following example from Abubakari (2018, p.149) too. *Nε* follows the direct object but precedes the PP adjunct focus.

- (75) Pu'a la sa da' nua la **nε** [PP da'a-n la.]_F
 woman DEF PST buy fowl DEF FOC market-LOC DEF
 'The woman bought the fowl IN THE MARKET'

From the examples above, we can conclude that the in-situ focus marker *nε* precedes the focused constituent. For now, we may assume that the two form a constituent. We will return to this in the next section.

3.3.2 Ex-situ focus

The different categories discussed above can undergo focus fronting to the left periphery. This is the ex-situ focus strategy. In Kusaal, ex-situ non-subject and long distance subject focus employ a different morphological focus marker *ka* which immediately follows the fronted focus constituent.²³ As we will see in this section, *ka* can as well be seen as a non-local subject focus marker. This is because similar to what we saw in Kasem, there is a difference between ex-situ local subject focus and non-subject and long distance subject focus. Kusaal actually shows a clear morphological difference between these ex-situ foci.

First, consider the direct object focus in (76-b). The direct object is fronted to the left periphery and is followed by *ka*. The in-situ focus marker *nε* can neither occur alone in the left periphery, nor can it co-occur with the ex-situ focus marker *ka*, as in (76-c).

- (76) a. Bɔ **ka** Adam kɔdig-ε?
 what FOC Adam kill-FV
 'What did Adam slaughter?'
 b. [Nua]_F **ka** Adam kɔdig.
 fowl FOC Adam kill
 'Adam slaughtered A FOWL.'
 c. *Nε [nua]_F (**ka**) Adam kɔdig.
 FOC fowl FOC Adam kill
 'Adam slaughtered A FOWL.'

A narrow adjunct focus can also be fronted to the left periphery of the clause (77-b). The adverb *teyin* is immediately followed by the ex-situ focus marker *ka*.

²³ Both the ex-situ question and answer make use of *ka*.

- (77) a. Yaanε **ka** Adam mε yir-ε?
 where FOC Adam build house-Q
 ‘Where did Adam build a house?’
 b. [Tejin]_F **ka** Adam daa mε yir.
 community FOC Adam PST build house
 ‘Adam built a house IN THE VILLAGE.’

Long-distance object focus shows the same marking (78). When in focus, the embedded object can be fronted to either the matrix left periphery (78-b) or the embedded clause left periphery (78-c).

- (78) a. Bə **ka** Peter yel ye John kədig-ya?
 What FOC Peter say COMP John kill-DISJ
 ‘What did Peter say that John slaughtered?’
 b. [Nua]_F **ka** Peter yel ye John kədig-ya.
 fowl FOC Peter say COMP John kill-DISJ
 ‘Fowl, Peter said that John slaughtered.’
 c. Peter yel ye [nua]_F **ka** John kədig-ya.
 Peter say COMP fowl FOC John kill-DISJ
 ‘Peter said that fowl, John slaughtered.’

So, whether local or long distance, non-subject ex-situ focus are always morphologically marked with *ka*. The case is different with subject focus, however.

Subject focus:

In subject focus, the morphological marking differs between local subject and long-distance subject focus. While the local subject focus is marked with *n*, non-local subject focus patterns along the line of non-subject focus with *ka*-marking.²⁴

In (79-b) *n* follows the local subject focus.²⁵ This is different from what was just shown for non-subject ex-situ focus marking above.

- (79) a. Anɔɔn-(ε) kədig nua la?
 who-FOC slaughter fowl DET
 ‘Who slaughtered the fowl?’
 b. [Azuure]_F **n**(*ε) kədig nua la.
 Azuure FOC slaughter fowl DET
 ‘AZUURE slaughtered the fowl.’

²⁴ Musah (2018, p.235) believes that both *n* and *nε* are one and the same marker. The former is only a reduced form of the latter. But the position with respect to the focus is reversed.

²⁵ It is worth pointing out here that a second subject focus marker *i* was found during the data collection (i-b). It immediately follows the subject focus. What conditions the use of either *n* or *i* for subject focus is unknown at this stage. One reason could be dialect variation because either of the markers is used by speakers of different dialects of Kusaal. There were also cases where the subject is morphologically unmarked.

- (i) a. Anɔ'ɔn kədig nua?
 who kill fowl
 ‘Who slaughtered a fowl?’
 b. [Adam]_F **i** kədig nua.
 Adam FOC kill fowl
 ‘ADAM slaughtered a fowl.’

Unlike local subject focus, long-distance subject focus involves not only resumption, but also *ka*-marking. In (80-b), the fronted embedded subject is immediately followed by *ka* and resumed in its base position. The embedded subject can be focused locally as well, as in (80-c). In this case, the subject the focus marker *n* is used; the same as in the matrix subject focus above.

- (80) a. Anɔ'ɔn **ka** Peter yɛl ye *(**o**) kɔdig nua?
 who FOC Peter say COMP 3SG kill fowl
 ‘Who did Peter say slaughtered a fowl?’
 b. [Azuure]_F **ka** Peter yɛl ye *(**o**) kɔdig nua.
 Azuure FOC Peter say COMP 3SG kill fowl
 ‘Peter said that AZUURE slaughtered a fowl.’
 c. Peter yɛl ye [Azuure]_F **n** kɔdig nua.
 Peter say COMP Azuure FOC kill fowl.
 ‘Peter said that AZUURE slaughtered a fowl.’

Based on the discussion so far, we can say that the distinction with respect to focus marking is between the local subject focus and the others (non-subject and LD subject foci). In other words, while local subject focus is marked with *n*, other non-subject focus fronting and long-distance subject focus fronting are marked with *ka*.

Clefts: Interestingly, Kusaal also uses *clefts* for focus constructions. It has a cleft structure that looks similar to that of English (see (81)). The focused constituent is preceded by a pleonastic pronoun *li* and a copula *anɛ*. The only difference here is the presence of an overt focus particle immediately after the clefted (focus) constituent. A cleft construction can be used for both subject focus (81-b) and non-subject focus (82-b). This suggests that the usual focus constructions are different from clefts. The latter produce exhaustive interpretations. See Abubakari (2019) for a discussion on the different interpretations of focus in Kusaal.

- (81) a. Anɔ'ɔn di sa'ab la
 who.FOC eat cornflour DET
 ‘Who ate the corn flour?’
 b. Li anɛ [Azuure]_F **n** di sa'ab la.
 3SG COP Azuure FOC eat corn-flour DET
 ‘It is AZUURE who ate the corn flour.’
 (82) a. Bɔ **ka** Azuure di?
 what FOC Azuure eat
 ‘What did Azuure eat?’
 b. Li anɛ [sa'ab]_F *(**ka**) Azuure di.
 3SG COP corn-flour FOC Azuure eat
 ‘It is CORN-FLOUR that Azuure ate.’

3.3.3 Anti-pied-piping in Kusaal

Similar to Kasem, there is VP and TP anti-pied-piping in Kusaal. The example in (83) represents a case of TP anti-pied-piping.²⁶ The subject is followed by the local subject

²⁶ Interestingly, Abubakari (2018, p.129) gave an example of clausal focus, which is a correction to a previous sentence (He is hungry), with a sentence-final morphological focus marking (i). However, I did not encounter such data in my elicitation.

focus marker *n* even though the logical focus is the TP following the current question under discussion.

- (83) a. Bɔ naam?
 what do
 ‘What happened?’
 b. Biig la n kɔdig nua la.
 child DEF FOC kill fowl DEF
 ‘The child slaughtered the fowl.’

If the subject is an indefinite, the same TP anti-pied-piping structure is realized. In other words, the indefinite subject is followed by the marker *n* (84-b). Recall that this is different from prosodic focus marking languages, where indefinites or weak elements are often exempted from anti-pied-piping focus marking.

- (84) a. Bɔ naam?
 what do
 ‘What happened?’
 b. Sɔ’ɔ n kɔdig nua la.
 someone FOC kill fowl DEF
 ‘Someone slaughtered the fowl.’

For VP anti-pied-piping, the direct object can be morphologically marked in the in-situ position or it can be fronted to the left periphery, as in (85-b) and (85-c). The in-situ focus marker *nɛ* cannot precede the verb, as shown in (85-b).

- (85) a. Bɔ ka Adam sa tum su’osɛ?
 what FOC Adam PST do yesterday
 ‘What did Adam do yesterday?’
 b. Adam sa (*nɛ) kɔdig nɛ nua su’os.
 Adam PST FOC kill FOC fowl yesterday
 ‘Adam slaughtered a fowl yesterday.’
 c. Nua ka Adam sa kɔdig su’os.
 fowl FOC Adam PST kill yesterday
 ‘Adam slaughtered a fowl yesterday.’

There is a further evidence for anti-pied-piping from the ditransitive constructions below. The *wh*-question in (86-a) requires an answer with the focus on VP. Interestingly, for VP focus, the in-situ focus marker *nɛ* precedes the indirect object in (86-b), or the direct object in (86-c). These examples suggest VP anti-pied-piping in Kusaal. This is slightly different from Kasem, where the focus marker *mo* only follows the direct object in VP anti-pied-piping (see example (22) above).

- (86) a. Bɔ ka Awini sa tum.
 what FOC Awini PST do
 ‘What did Adam do?’

-
- (i) Ayei, kɔm pɔ mɔr o, o ligidi bɔrig nɛ.
 no, hunger NEG have 3SG, 3SG.POSS money lose FOC
 ‘No, he is not hungry; IT IS HIS MONEY THAT IS MISSING/LOST.’

- b. Awini [tis **nε** Sam ligidi la]_F.
 Awini give FOC Sam money DEF
 ‘Awini gave Sam the money.’
- c. Awini [tis Sam **nε** ligidi la]_F.
 Awini give Sam FOC money DEF
 ‘Awini gave Sam the money.’

Furthermore, the fronted direct object clause in (87-a) felicitously answers the current question in (86-a). This example presents us with a case where a sub-part (the direct object) of the logical focus (VP) is morphosyntactically marked, i.e., VP focus anti-pied-piping. Of course, the construction in (87-a) can also answer a direct object *wh*-question such as ‘What did Awini give to Sam?’ On the other hand, if the indirect object is fronted, as in (87-b), it does not answer the VP question. It is only felicitous as an answer to an indirect object *wh*-question (e.g., To whom did Awini give the money?). Importantly, I want to point out here that more research is needed to elaborate on what is exactly going on in Kusaal. First, we saw that both direct and indirect object can be morphologically marked for VP focus anti-pied-piping (cf. (85)), i.e., showing no sign of the leftmost requirement we saw in Kasem. Secondly, it is the fronting of the direct object, and not the indirect object, that can felicitously answer the VP focus question (cf. (86)). This remains unclear and calls for further investigation.

- (87) a. Ligidi la **ka** Awini [tis Sam ____]_F.
 money DEF FOC Awini give Sam
 ‘Awini gave Sam the money.’
- b. #Sam **ka** Awini [tis ____ ligidi la]_F.
 Sam FOC Awini give money DEF
 ‘Awini gave Sam the money.’

In summary, focus marking in Kusaal may involve ex-situ versus in-situ focus. Ex-situ focus uses the morphological focus markers *ka* for non-subject and long-distance subject focus, but *n* for local subject focus. Both focus markers immediately follow the fronted focus constituents in the left periphery. In-situ focus, on the other hand, is morphologically marked with *nε* which precedes the in-situ focus constituent. Additionally, long-distance subject focus involves resumption in its base position, while local subject focus and non-subject foci do not involve resumption whatsoever. These morphological marking patterns are summarized in table 3.1 below. Lastly, similar to Kasem, we saw cases of VP and TP anti-pied-piping in Kusaal.

	Ex-situ	In-situ
Local subject	n	—
Non-subject	ka	nε
Non-local subject	ka	—

Table 3.1: Overview of focus marking in Kusaal

In the next section, I provide an analysis for in-situ focus marking in Kusaal which is similar to the one given for Kasem above. Thereafter, I propose an analysis for ex-situ focus in the language. I show that the ex-situ focus constructions in Kusaal can

broadly be grouped into two based on the position they occupy in the left periphery: local subject versus non-subject and non-local subject foci. I argue, following Bošković (2024), that although local subject focus is *ex-situ* (i.e. occupies a position higher than TP), it occupies a different (lower) position from non-subject and long-distance subject foci. I will not repeat the analysis for the anti-pied-piping cases here since the same analysis has been provided for Kasem above.

3.3.4 A proposed analysis

On the surface, the structural position of the in-situ focus particle *nɛ* is not immediately apparent. On the surface, there may be potentially two options. The first option, which I will argue against, may be to assume that *nɛ* is a low focus head in the clause-internal periphery- the vP/VP edge. Several studies have argued for a clause-internal FocP in the vP periphery (see Tuller, 1992; Belletti, 2004; Collins and Essizewa, 2007; Kandybowicz, 2008; Aboh, 2009; Belletti and Rizzi, 2017, a.o.). The second option is to assume that *nɛ* forms a constituent with the focused constituent, just like we saw in Kasem. I argue that the second option is the right analysis for Kusaal by comparing it with Kabiye (Kɛwɛ dialect), another Mabilia (Gur) language spoken in Togo (Collins and Essizewa, 2007), and Məɖamba, a Grassfields Bantu language spoken in Cameroon (Zimmermann and Kouankem, 2024).

Just like what we have seen in Kusaal, in-situ focus in Kabiye and Məɖamba involves a morphological in-situ focus marking which follows the (object) focus in Kabiye (88-b) (Collins and Essizewa, 2007, p.193), and precedes the focus in Məɖamba (89-b) (Zimmermann and Kouankem, 2024, p.2).

- (88) a. What language do you understand?
b. Ma-ní-ʊ [kabiye]_F **na**. *Kabiye: in-situ focus*
1SG-understand-IMPV Kabiye FOC
‘I understand KABIYE.’
- (89) a. ‘To whom did Nana give food?’
b. Nana na fa chaŋ **á** [num Nami]_F. *Məɖamba: in-situ focus*
nana PST.REM give food FOC to nami
‘Nana give food TO NAMI.’

While Collins and Essizewa (2007) propose a low focus position at the vP edge for Kabiye, Zimmermann and Kouankem (2024) propose that the focus marker forms a constituent with the focus. The first empirical argument for Collins and Essizewa (C&E) is with respect to the surface position of the focus marker *na* which typically follows non-subject focus constituents (cf. (88-b) above). In *ex-situ* focus however, the final vowel of the the focused constituent is lengthened, and cannot co-occur with *na*, as in (90) (Collins and Essizewa, 2007, p.193). According to C&E, if *na* forms a constituent with focus, then both should be able to co-occur (i.e., move as a single constituent) in the left periphery. However, this is not the case. Thus, C&E propose that while the head of the left peripheral FocP is overtly realized by lengthening, the head of the clause internal FocP is overtly realized as *na*.

- (90) [Kabiye]_F-**ɛ** (*na) má-ní-ʊ.
Kabiye-FOC FOC 1SG-understand-IMPF
‘I understand KABIYE.’

Another empirical evidence is the post-verbal occurrence of *na* in local subject focus (91-b). With the local subject example, the position of the focus marker *na* seems to be fixed in a lower position within the clause. If *na* forms a constituent with whatever is in focus, then there is no way to explain why subject focus behave differently. C&E propose the structures in (92). In both structures, the focus phrase is projected above the VP, and object movement to Spec,FocP realizes object focus (92-a), subject movement through Spec,FocP to Spec,IP realizes subject focus (92-b) respectively. In each cases, the verb also undergoes V-I movement.

- (91) a. Who understands Kabiye?
 b. [Esɔ]_F ní-ʊ **na** kabiye.
 Esso understand-IMPV FOC Kabiye
 ‘ESSO understands KABIYE.’
- (92) a. [_{IP} [_{DP} Ma] [_{I'} [_I ni-u] [_{FocP} [_{DP} Kabiye] [_{Foc'} [_{Foc} na] [_{VP} ⟨_V ni⟩ ⟨_{DP} Kabiye⟩]]]]]]
 b. [_{IP} [_{DP} Esɔ] [_{I'} [_I ni-u] [_{FocP} ⟨_{DP} Esɔ⟩ [_{Foc'} [_{Foc} na] [_{VP} [⟨_{DP} Esɔ⟩] [_{V'} ⟨_V ni⟩ [_{DP} Kabiye]]]]]]]]

As for Məɖamba, Zimmermann and Kouankem (2024) (Z&K) argue that it is easy to differentiate it from Kabiye. First, while Kabiye marks in-situ focus and ex-situ focus differently, Məɖamba uses the same *a-marking* (93) (Z&K adapted the examples from Kepudjio (2020, p.18)). Secondly, unlike Kabiye, in Məɖamba, the subject focus is preceded by *a* just as in non-subject foci (94). The focus marker *a* cannot be stranded post-verbally (as in the Kabiye subject focus case). All these point to a constituent analysis of the focus marker according to Z&K.

- (93) a. Wàtɛt náʔ swɛɛn á [Nùŋgɛ]_F
 Watat PST.REM sell FOC Nuga
 ‘Watat betrayed NUGA.’
 b. á [Nùŋgɛ]_F Wàtɛt náʔ swɛɛn lá
 FOC Nuga Watat PST.REM sell C.DEF
 ‘NUGA, Watat betrayed.’
- (94) á [Wàtɛt]_F náʔ swɛɛn (*á) Nùŋgɛ.
 FOC Watat PST.REM sell FOC Nuga
 ‘WATAT betrayed Nuga.’

The Kusaal data discussed in the previous section shows some properties similar to Məɖamba. The focus marker *nɛ* precedes in-situ focus. However, both languages are different in that Kusaal marks ex-situ focus differently with *ka*. In other words, *nɛ* and *ka* do not co-occur in the left periphery of the clause (a pattern similar to Kabiye) (95).

- (95) ***Nɛ** [Nua]_F **ka** Adam kɔdig.
 FOC fowl FOC Adam kill
 ‘Adam slaughtered A FOWL.’

However, there is empirical evidence to believe that both *nɛ* and the focused constituent form a single constituent in in-situ focus in Kusaal. First, if we compare the examples (72-b) and (73-b) above, repeated below as (96-a) and (96-b), *nɛ* obligatorily precedes whatever is in focus. If *nɛ* is in a fixed post-verbal (after V-Asp movement) position, then it is unclear why the post-verbal occurrence of *nɛ* in (96-a) for instance, is

infelicitous with the current question (What did he give to Adam?).

- (96) a. O tis (**#nɛ**) Adam **nɛ** [nua]_F.
 3SG give FOC Adam FOC fowl
 ‘He gave Adam A FOWL.’
 b. O tis **nɛ** [Adam]_F (**#nɛ**) nua la.
 3SG give FOC Adam fowl DEF
 ‘He gave ADAM the fowl.’

Alternatively, we might assume that *nɛ* is actually in a fixed position, only that the indirect object moves to a higher position. However, such an assumption would raise the question of the position of the raised indirect object. Moreover, the fact that in adjunct focus, *nɛ* must precede the adjunct (cf. (97)) compounds the challenges that would face a fixed position assumption or analysis. In this case, we would have to assume the raising of the direct object in order to realize the linear order.

- (97) a. Adam sa da' nua la **nɛ** [su'os]_F.
 Adam PST buy fowl DEF FOC yesterday
 ‘Adam bought the fowl YESTERDAY.’
 b. #Adam sa da' **nɛ** nua la [su'os]_F.
 Adam PST buy FOC fowl DEF yesterday
 ‘Adam bought the fowl YESTERDAY.’

If the sentence is ditransitive, as in (98), we would have to assume multiple movements of both the indirect object and the direct object. The question is: to what positions? Different non-canonical positions would have to be assumed in order to realize the linear order in an in-situ adjunct focus context.

- (98) Adam tis Adam nua la **nɛ** [su'os]_F.
 3SG give Adam fowl DEF FOC yesterday
 ‘He gave Adam the fowl YESTERDAY.’

Even more compelling evidence for the constituent analysis of *nɛ* comes from manner adverbials. According to Enoch Aboh (p.c.), having *nɛ* precede a manner adverb is stronger evidence for constituency than the ones with time adverbials discussed above. This is because in some West African languages, time adverbials have been analyzed as nominals, and not as proper adverbs. Manner adverbs, on the other hand, are known to be stable with respect to their adverbial status. So, if *nɛ* truly forms a constituent with whatever is in focus, then it should be able to precede a manner adverb that is focused as well. This prediction is borne out. With the question in (99-a), which requires the manner in which Adam ate the food, the in-situ focus answer has *nɛ* preceding the manner adverb *tɔ'ɔtɔ* ‘quickly’, as in (99-b). If *nɛ* precedes the object, the answer becomes infelicitous with the current question under discussion.²⁷ Also, if we assume that adverbs adjoin to

²⁷ Like other non-subject focus, the manner adverb can also be realized ex-situ in the left periphery.

- (i) a. Wala **ka** Adam di diib la?
 how/why FOC Adam eat food DEF
 ‘How did Adam eat the food?’
 b. [Tɔ'ɔtɔ]_F **ka** Adam di diib la.
 quickly FOC Adam eat food DEF
 ‘Adam ate the food QUICKLY.’

vP in the language, then it only makes sense to say that *nε* forms a constituent with the adverb in a separate workspace before adjoining to the clause.

- (99) a. Adam di diib la wala?
 Adam eat food DEF how
 ‘How did Adam eat the food?’
 b. Adam di (#**nε**) diib la **nε** [tə’otə]_F.
 Adam eat FOC food DEF FOC quickly
 ‘Adam ate the food QUICKLY.’

Thus, I propose that *nε* actually forms a constituent with the in-situ focus constituent.

Evidence for the ex-situ realization of subject focus

Due to the form of the local subject focus marker *n*, we might say that the local subject focus is realized in an in-situ position. Recall that the in-situ focus marker in Kusaal is *nε*. We might hypothesize that the local subject focus marker *n* is only a shortened form of *nε*. If this is true, then we have to explain why *nε* does not precede the local subject focus in (100-a) in comparison with direct object in-situ focus (100-b).

- (100) a. ***Nε** Azuure kɔdig nua la.
 FOC Azuure slaughter fowl DET
 ‘AZUURE slaughtered the fowl.’
 b. Adam kɔdig **nε** [nua]_F.
 Adam kill FOC fowl
 ‘Adam slaughtered A FOWL.’

One possible explanation is to say that *nε* is a clitic that is ‘shy’ of occurring in the first position of the clause.²⁸ Therefore, in a local subject focus context, *nε* encliticizes to the local subject focus, and gets reduced to *n* (as in (79-b) above). If we follow this explanation, then the prediction is that *nε* should precede the subject focus when the latter is preceded by other elements (since it would no longer be in the first position of the clause). This prediction is however not borne out. Consider the subject of the embedded clause in (101). If the embedded subject is in focus, the grammatical construction is (138), and not (101-b). Therefore, it is not a case of encliticization. Instead, both *n* and *nε* only seems to be homophonous with respect to their consonant.

- (101) a. Peter yɛl ye [Azuure]_F **n** kɔdig nua.
 Peter say COMP Azuure FOC kill fowl.
 ‘Peter said that AZUURE slaughtered fowl.’
 b. *Peter yɛl ye **nε** [Azuure]_F kɔdig nua.
 Peter say COMP FOC Azuure kill fowl.
 ‘Peter said that AZUURE slaughtered fowl.’

Moreover, there are other Mabilia languages which mark local subject focus with the same *n* that Kusaal uses, but their in-situ focus markers are entirely different (i.e. no homophony). Two of such languages are Dagbani and Gurene.²⁹ In Dagbani, while the

²⁸ In some Indo-European languages, some clitics are called *second occurrence clitics* because they occupy a position after the first (prosodic) phrase or word in a sentence. This is called the *Wackernagel’s Effect* (cf. Wackernagel, 1892; Klavans, 1985; Halpern, 1995; Anderson, 2005, a.o.).

²⁹ The Dagbani and Gurene data is gotten from the WikiMabilia database (<https://mabilia-vp.com/tiki-index.php?page=Database>).

local subject focus (including the subject *wh*-phrase) is marked with *n* (cf. (102)), in-situ non-subject focus is marked with the morpheme *la* which is pre-focal, as in (103).

(102) **Local subject focus in Dagbani**

- a. ɲuni **n** kɔrigi noo?
 who FOC slaughter.PFV fowl
 ‘Who slaughtered a fowl?’
- b. [Adam]_F **n** kɔrigi noo.
 Adam FOC slaughter.PFV fowl
 ‘ADAM slaughtered a fowl.’

(103) **In-situ non-subject focus in Dagbani**

- a. Adam kɔrigi **la** bo?
 Adam slaughter FOC what
 ‘Adam slaughtered what?’
- b. Adam kɔrigi **la** [noo]_F.
 Adam slaughter.PFV FOC fowl
 ‘Adam slaughtered a fowl.’

The same thing happens in Gurene where although the local subject is marked with *n* (104), the in-situ non-subject focus is marked with *la* (105).

(104) **Local subject focus in Gurene**

- a. Ani **n** ku nu’a?
 who FOC slaughter fowl
 ‘Who slaughtered fowl?’
- b. [Adam]_F **n** ku nu’a.
 Adam FOC slaughter fowl
 ‘ADAM slaughtered fowl.’

(105) **In-situ non-subject focus in Gurene**

- a. Ziema bo Napog **la** beni?
 Ziema give Napog FOC what
 ‘What did Ziema give to Napog?’
- b. Ziema bo Napog **la** [duko]_F.
 Ziema give Napog FOC pot
 ‘Ziema gave A POT to Napog.’

Returning to the local subject focus in Kusaal, I propose that the local subject focus in Kusaal is not realized in an in-situ position; it is realized outside of the TP (*pace* Musah, 2018; Abubakari, 2018). One argument for this claim comes from sub-extraction data. The reasoning behind this is that if we have a complex DP in the subject position (Spec,TP), the possibility of sub-extracting a sub-constituent from the complex DP into a higher position suggests distinct surface positions for the two phrases (i.e., the complex DP and the sub-extracted constituent). This data is presented in the example below. This goes for both subject *wh*-questions and subject focus answers. Notice the use of resumption, which is an argument for the fact that the possessor has been sub-extracted.

- (106) a. Anɔ’ɔn **ka**/(***n**) o baa wɛs?
 who FOC 3SG dog bark
 ‘Whose dog barked.’

- b. Ada **ka**/(***n**) o baa wɛs.
 Ada FOC 3SG dog bark
 ‘ADA’S dog barked.’

An interesting observation from the examples above, however, is that the sub-extracted focused constituent (the possessor) is followed by the focus marker *ka*, instead of *n*, which is used for local subject focus marking. This reason for this is discussed in the next section, which discusses the focus marking asymmetry between local subject focus and everything else (i.e., non-subject focus and non-local subject focus). However, the main emphasis here is that subject foci occupy a different (higher) position from non-focused subjects. This is discussed further in the next section.

Another evidence for the different positions between the focused subject and the non-focused subject is the noticeable difference in the scopal behaviour between them. A scope test shows that a non-focused universally quantified subject is ambiguous with respect to scope in the presence of negation. In other words, (107-a) is ambiguous between inverse scope (with *biis la wusa* ‘all the children’ scoping below negation) and surface scope (with *biis la wusa* ‘all the children’ scoping above negation). In contrast, when the local subject is in focus, the scope ambiguity disappears. The only available reading is when the universal quantified subject scopes above negation, as in (107-b). A structural explanation for this difference in scopal behaviour is that in the two instances, the focus subject occupy a structurally different (higher) position than the canonical subject position in Spec,TP.

- (107) a. Biis la wusa pɔ di’ busa. [¬ > ∀; ∀ > ¬]
 children DET all NEG eat yam
 ‘All the children did not eat yam.’
 (i) It is not the case that all the children ate yam (meaning some did not eat while some might have).
 (ii) It is the case that none of the children ate yam.
- b. [Biis la]_F wusa **n** pɔ di’ busa. [*¬ > ∀; ∀ > ¬]
 children DET all FOC NEG eat yam
 ‘ALL THE CHILDREN did not eat yam.’
 (i) *It is not the case that all the children ate yam (meaning some did not eat while some might have).
 (ii) It is the case that none of the children ate yam.

In the next section, I propose an analysis which captures the focus marking (morphological) asymmetry between local subject focus and non-subject/non-local subject foci.

Proposal for ex-situ focus and the focus marking asymmetry in Kusaal

Recall that in section 3.3.2, we presented an interesting focus marking asymmetry between the local subject focus and non-subject focus plus non-local subject focus in Kusaal. This asymmetry is repeated below. While ex-situ local subject focus is morphologically marked with *n*, ex-situ non-subject (objects, PPs, adverbs) foci are marked with *ka*. Compare examples (108-b) and (109-b) below.³⁰

³⁰ Notice that ex-situ non-subject *wh*-question is also marked with *ka*, as in (109-a). We also have reasons to believe that local subject *wh*-question is marked with *n*. The *wh*-phrase in (108-a) *anɔɔn* does not show this because it already ends with an *n*. So, it is difficult to perceive the local subject focus morpheme *n*. However, if we use a different *wh*-phrase like *what*, we get the *n* marking (i). Similarly, if we use a D-linked local subject question, we have the *n* overtly realized (ii-a). (ii-b) is

(108) **Local subject focus**

- a. Anɔɔn kɔdig nua la?
who slaughter fowl DET
'Who slaughtered the fowl?'
- b. [Azuure]_F n kɔdig nua la.
Azuure FOC slaughter fowl DET
'AZUURE slaughtered the fowl.'

(109) **Local object focus**

- a. Bɔ ka Adam kɔdig-ɛ ____?
what FOC Adam kill-FV
'What did Adam slaughter?'
- b. [Nua]_F ka Adam kɔdig ____.
fowl FOC Adam kill
'Adam slaughtered A FOWL.'

As expected, ex-situ non-local or embedded object focus is marked with *ka* (110). The embedded object can be extracted to the embedded left periphery (110-c), or all the way to the matrix left periphery (110-b). It does not matter; it always gets a *ka*-marking. Adjuncts pattern exactly like object focus, i.e. marked with *ka* (see example (77) above).

(110) **Non-local object focus**

- a. Bɔ ka Peter yɛl ye John kɔdig-ya ____?
What FOC Peter say COMP John kill-DISJ
'What did Peter say that John slaughtered?'
- b. [Nua]_F ka Peter yɛl ye John kɔdig-ya ____.
fowl FOC Peter say COMP John kill-DISJ
'Peter said that John slaughtered A FOWL.'
- c. Peter yɛl ye [nua]_F ka John kɔdig-ya ____.
Peter say COMP fowl FOC John kill-DISJ
'Peter said that John slaughtered A FOWL.'

Importantly, long-distance subject focus behaves differently from local subject focus. Embedded subject focus patterns like non-subject foci when it is non-local, i.e. it is followed by *ka* (and not *n*) (111-b). However, if the embedded subject is extracted locally

ungrammatical because local subject *wh*-question and focus can only be marked with *n*, and not *ka*. In this section, I will concentrate on the focused answers, but I assume that the pattern carries over to *wh*-questions as well.

- (i) Bɔ *(n) kɔ nua la?
what FOC kill fowl DEF
'What killed the fowl?'

(ii) **D-linked question**

- a. Sakurbil kanɛ *(n) da' gbaun?
student which FOC buy book
'Which student bought a book?'
- b. *Sakurbil kanɛ ka da' gbaun?
student which FOC buy book
'Which student bought a book?'

to the embedded left periphery, it is followed by *n* (111-c). Thus, the restriction in marking of the subject focus seems to be conditioned by locality. When it is local (i.e. the left periphery of a simple sentence or the embedded periphery of an embedded clause), it is followed by *n*. But if it is non-local (as in the long-distance extraction), it is followed by *ka*, patterning with non-subject focus.

(111) **Non-local subject focus**

- a. Anɔ'ɔn **ka** Peter yɛl ye *(ɔ) kɔdig nua?
 who FOC Peter say COMP 3SG kill fowl
 'Who did Peter say slaughtered the fowl?'
- b. [Azuure]_F **ka** Peter yɛl ye *(ɔ) kɔdig nua.
 Azuure FOC Peter say COMP 3SG kill fowl
 'Peter said that AZUURE slaughtered fowl.'
- c. Peter yɛl ye [Azuure]_F **n** kɔdig nua.
 Peter say COMP Azuure FOC kill fowl
 'Peter said that AZUURE slaughtered fowl.'

Overall, the pattern shows that the split is not between subject focus versus non-subject focus, but rather between local subject focus and everything else. Interestingly, this pattern is not peculiar to only Kusaal. A few other Mabia languages exhibit the same asymmetry between local subject focus and non-subject/non-local subject focus. Two such languages are Dagbani and Gurene. I will only present the Gurene data here because in Dagbani, the morphological form of the asymmetry is the same as in Kusaal.³¹

Similar to Kusaal, in Gurene, while the local subject focus (or *wh*-phrase) is morphologically marked with *n*, non-subject focus and non-subject focus are marked with *ti*. The examples in (112) and (113) show this marking asymmetry for local subject focus and non-subject (object) focus respectively.

(112) **Local subject *wh*/focus in Gurene**

- a. Ani **n** ku nu'a?
 who FOC slaughter fowl
 'Who slaughtered fowl?'
- b. [Adam]_F **n** ku nu'a.
 Adam FOC slaughter fowl
 'ADAM slaughtered fowl.'

(113) **Local object *wh*/focus in Gurene**

- a. Beni **ti** Adam ku ____?
 what FOC Adam slaughter
 'What did Adam slaughter?'
- b. [Nu'a]_F **ti** Adam ku ____.
 fowl FOC Adam slaughter
 'Adam slaughtered A FOWL.'

If a PP adjunct is also fronted, it gets the *ti*-marking since adjunct focus falls within the category of non-subject focus (114).

³¹ Issah and Smith (2020) have discussion and a potential analysis of this asymmetry in Dagbani, which I will return to later.

(114) **Adjunct *wh/focus* in Gurene**

- a. Besa **ti** Adam mε yire ____?
where FOC Adam build house
'Where is Adam building a house?'
- b. Atiṇa **ti** a mε yire la ____.
village FOC 3SG build house DET
'He build the house IN THE VILLAGE.'

In the embedded clause constructions in (115) and (116), non-subject focus (represented by object focus) is still marked with *ti* in both *wh*-question or a focus answer. The *wh* phrase or the focus answer can be fronted to the embedded periphery with *ti*-marking, as in (115), or they can move all the way to the left periphery of the matrix clause, as the examples in (116) show. In all the cases, the *wh*-question and focused answers are marked with *ti*.

(115) **Embedded local object *wh/focus* in Gurene**

- a. Ama yeli beni **ti** John korigε ____?
Ama say what FOC John slaughter
'What did Ama say that John slaughtered?'
- b. Ama yeli [nua]_F **ti** John korigε ____?
Ama say fowl FOC John slaughter
'Ama said that John slaughtered a FOWL.'

(116) **Non-local embedded object *wh/focus* in Gurene**

- a. Beni **ti** Ama yeli ti John korigε ____?
what FOC Ama say COMP John slaughter
'What did Ama say that John slaughter?'
- b. [Nua]_F **ti** Ama yeli ti a korigε ____?
fowl FOC Ama say COMP 3SG slaughter
'Ama said that he slaughtered a FOWL.'

However, the case of embedded subject focus is different with respect to focus marking. When the embedded subject is fronted to the left periphery of the embedded clause, it gets the *n*-marking (117). But when it is fronted to the left periphery of the matrix clause, it gets the *ti*-marking instead (118). Obviously, here the locality of movement of the fronted constituent is at play.

(117) **Embedded local subject *wh/focus* in Gurene**

- a. Peter yeli ti ani **n** korigε nua?
Peter say COMP who FOC slaughter fowl
'Who did Peter say slaughtered a fowl?'
- b. Peter yeli ti [John]_F **n** korigε nua.
Peter say COMP John FOC slaughter fowl
'Peter said that JOHN slaughtered a fowl.'

(118) **Non-local embedded subject *wh/focus* in Gurene**

- a. Ani **ti** Peter yeli ti a korigε nua?
who FOC Peter say COMP 3SG slaughter fowl
'Who did Peter say slaughtered a fowl?'

- b. [John]_F **ti** Peter yeli ti a korige nua.
 John FOC Peter say COMP 3SG slaughter fowl
 ‘Peter said that JOHN slaughtered a fowl.’

I assume that the observed focus marking asymmetry in Kusaal (and potentially the other Maba languages which exhibit the same asymmetry) is as a result of the feature specification on the probing focus head. The probe (Foc) is restrictive and is conditioned in such a way that it determines which constituent occupies its Spec, and this is also reflected in the morphological realization of Foc. The consequence of such a restricted probe is that when *n* is the morphological realization of Foc, then it is the closest DP (in this case the local subject) that moves to Spec,FocP. But when *ka* is realized the ‘closest DP movement condition’ is out, and any other constituent bearing a focus feature is extracted to Spec,FocP. Importantly, the probe that licenses the closest DP movement is also specified with a focus feature which is also satisfied by the closest DP. The feature specifications and their respective morphological reflexes are given in (119). Before elaborating and going into the details of the analysis, it is worth discussing the idea of restricted probing in the literature.

(119) **Feature specifications on Foc and morphological reflexes**

- a. $\text{FOC}_{[\text{Foc}+\text{uD}, \text{EPP}]} \leftrightarrow n$
 b. $\text{FOC}_{[\text{Foc}, \text{EPP}]} \leftrightarrow ka$

The idea that a probe can be restrictive, or specified to attract the closest goal, has been widely discussed in the literature, especially for cases which are called *syntactic ergativity* (see Aldridge, 2004, 2008, 2017; Deal, 2015; Coon et al., 2014; Deal, 2016, 2017; Polinsky, 2017; Levin, 2018; Coon et al., 2021, i.a.). Syntactic ergativity is an extraction restriction property that is peculiar to the ergative-absolutive languages where the extraction of a transitive subject is banned. In other words, syntactic ergativity only allows the extraction of an absolutive argument in ergative-absolutive languages. This restriction is exemplified in the Tagalog example in (120).³² In Tagalog, only the absolutive argument can be extracted in a relative clause (120-a) (cf. Aldridge, 2017). An ergative subject cannot be extracted (120-b).

(120) **Absolutive-only extraction restriction in Tagalog**

- a. isda=ng [binili ng babae]
 fish=LK bought ERG woman
 ‘fish that the woman bought’
 b. *babae=ng [binili ang isda]
 woman=LK bought ABS fish
 ‘woman who bought the fish’

[from Branan and Erlewine (2024, p.377, ex. (4)) citing Aldridge (2017)]

The analysis that Aldridge (2017) assumes is based on C probing for the closest DP. However, in order to get the grammatical structure in (120-a), the absolutive argument (i.e. object) has to undergo movement to the outer specifier of the little vP, with the external argument i.e. subject base-generated in the inner specifier. This movement

³² Branan and Erlewine (2024) made a few adjustments w.r.t. the glossing of the example in (120) compared to what is given in the original source (Aldridge, 2017, p.25, ex. (61b-c)). For consistency of the discussion, I follow Branan and Erlewine’s (2024) glossing.

makes the absolutive object the closest goal, and then accessible to the probe C. The object then moves to Spec,CP, as the structure in (121) shows. Note that there is v to T movement as well which gets us the final surface order.

- (121) $[_{CP} DP_{obj} [_{TP} T+v [_{VP} \langle DP_{obj} \rangle [_{VP} DP_{subj} \langle v \rangle [_{VP} \langle DP_{obj} \rangle]]]]]]$

Branan and Erlewine (2024), on the other hand, argue that such an extraction restriction is not limited to ergative languages alone. It is a more general phenomenon which is attested even in non-ergative languages as well. They formulated the restriction that A'-probing targets the closest DP as (122).

- (122) **A'-probing for the closest DP** (Branan and Erlewine, 2024, p.376)
An A'-probe *can be specified* to target the closest DP.

(122) restricts a probe to only attract the DP that is closest to it, where the notion of closeness is defined based on Chomsky's (2000) notion of closeness for agreement phenomena (123).

- (123) **Closest** (Branan and Erlewine, 2024, p.376)
A potential goal G for probe P is closest if no other potential goal for P c-commands G.

Probing for the closest DP predicts cases where not only objects, but also subjects are attracted by the probe. In fact, this would be the normal outcome if the subject is the higher argument. Branan and Erlewine (2024) provided an example from Turkish (124) where the subject relative clause is morphologically marked differently from the non-subject relative clause. Branan and Erlewine (2024) (following Çağrı (2005, 2009)) gloss these forms as SR = subject relative and NSR = non-subject relative. Note that the relative morphology is linearly to the right but high in the clause.

- (124) **Two relative clause forms in Turkish**

- a. $[_{\text{__}} \text{ k} \ddot{\text{I}}\text{Z-1} \quad \text{sok-}\mathbf{an}] \text{ ar} \ddot{\text{I}}$
girl-ACC sting-SR bee
'the bee that stung the girl'
b. $[\text{ar} \ddot{\text{I-n}} \ddot{\text{I}} \quad \text{__} \text{sok-}\mathbf{tu}\check{\text{ğ-u}}] \text{ k} \ddot{\text{I}}\text{Z}$
bee-GEN sting-NSR-3SG girl
'the girl that the bee stung'

[Branan and Erlewine (2024, p.381, ex.12) citing Çağrı (2005)]

Branan and Erlewine (2024) assume that the SR form is tied to probing for the closest DP, and not necessarily the subject per se. They argue for this with the example in (125-a), with (125-b) as the source. They show that when the subject is indefinite, it occurs lower in the clause (see the source sentence (125-b)), which allows the object to occur in a higher position where it becomes the closest DP, and then accessible for relativization with SR morphology. In such a situation, the SR form is still realized showing that the SR form is not only for subjects but specifically anything that is the closest DP.

- (125) **Low indefinite subject makes object highest**

- a. $[_{\text{__}} \text{ ar} \ddot{\text{I}} \text{ sok-}\mathbf{an}] \text{ adam}$ *Object relative with SR form*
bee sting-SR man
'the man that a bee stung'

- b. Adam-ı arı sok-tu. *Source structure*
man-ACC bee sting-PST
‘A bee stung the man.’

[Branan and Erlewine (2024, p.382, ex.13) citing Temürcü (2001)]

Branan and Erlewine (2024) propose that the SR form is a reflection of a restricted probe [PROBE:REL+D] which requires the extraction of the closest nominal in Turkish.³³

With the brief discussion on the possibility of a restrictive or specified probe, we are now in a position to elaborate our analysis for the focus marking asymmetry in Kusaal. I will begin with the local subject focus, and then discuss the non-subject and non-local subject focus. I will also give some of the implications of the analysis. At the end, I will compare my analysis with two other analysis: one alternative analysis by Bošković (2024), and one similar analysis by Keine and Zeijlstra (2025). I will show that while the first alternative analysis faces a few challenges, the second one is very similar to the analysis here, and may even account for the Kusaal data. But it is not adopted due to a few complexities.

Following the idea that a probe can be specified and restricted to probe for the closest DP, I assume that the focus head that probes for the local subject is restricted in this regard. But the probe for the other constituents (including non-local subjects) does not have such a restriction. The assumption is that the different feature combinations on Foc is reflected morphologically as in (119) above, repeated as (126) below for convenience. (126) is a Vocabulary Insertion (VI) (as assumed in Distributed Morphology) rule which is sensitive to the interface between syntax and morphology (cf. Halle and Marantz, 1993; Harley and Noyer, 1999; Embick and Noyer, 2007; Bobaljik, 2012; Embick, 2015, a.o.). While (126-a) is a complex probe which probes for the closest focus DP, (126-b) is a single probe which only probes for any constituent that has a focus feature. Basically, the complex probe is specified for both an A'-feature [FOC], and an A-feature [uD].³⁴

(126) **Feature specifications on Foc and the morphological reflexes**

- a. $\text{Foc}_{[\text{Foc}+\text{uD}, \text{EPP}]} \leftrightarrow n$
b. $\text{Foc}_{[\text{Foc}, \text{EPP}]} \leftrightarrow ka$

For the complex probe, I assume that both the Foc feature and the uD feature probe together for a single goal, which satisfies them. The requirement of [uD] on Foc is also stated in (127), where what is regarded as closest is already defined in (123) above.

(127) **Probing for the closest DP statement**

The [uD] feature on Foc requires that the closest DP be attracted to Spec,FocP.

The assumption here anchors on the idea that agreement of a more specified (richer) probe outranks that of a less specified probe (cf. Béjar and Rezac, 2009). I assume that if the complex probe in (126-a) finds a partial goal (i.e. a goal that meets only one of the features), the probe will crash. Such a condition excludes the possibility of agreeing with non-focused subjects and non-subjects, even if they are specified with Foc, but are not the closest DP. So, once [uD] is present on Foc, it attracts the closest DP in its c-command

³³ Branan and Erlewine (2024) provided further evidence with temporal relatives and possessor relatives for a probe which is restricted to the closest argument.

³⁴ That a probe can be specified for both an A'-feature and an A-feature has been argued for in the literature. See Erlewine (2018b); Coon et al. (2021); Branan and Erlewine (2024); Keine and Zeijlstra (2025) and the references they cite.

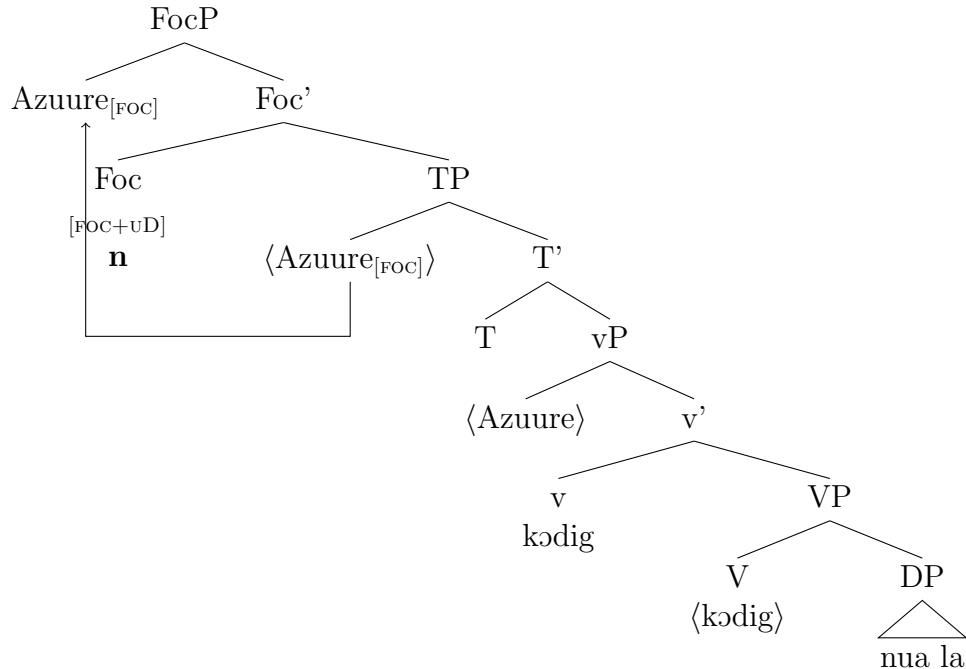
domain, if the DP is focused (cf. (127)). If not, the operation crashes with no further probing. This is why only the local subject focus is marked with *n*.

On the other hand, when the head is specified with the features in (126-b), the [uD] is absent, and the probe searches for any constituent that is focused in the sentence, and not the closest DP. In principle, this also predicts that if the local subject is focused, it can be attracted. However, this possibility must be excluded because we know that local subjects never occupy the Spec of a focus head that is morphologically realized as *ka*. Moreover, if the local subject is focused, it is the complex probe in (126-a) that is activated. Put differently, both *n* and *ka* are competing for the same head, and where both are possible, the complex *n* head blocks the *ka* head. When the *n* head cannot attract the local subject due to lack of focus, another search is initiated by the *ka* head. The *ka* head cannot attract the local subject because the *n* head would have done that, if possible, in the first place. With this, the *ka* head can only attract non-subjects and non-local subjects.

We are now in a better position to derive the local subject focus structure. Taking the local subject focus sentence in (128) as a reference, I propose the structure in (129) for local subject focus. First, the head of Foc is realized as *n*, which signifies the presence of the complex probe, as in (126-a). Thus, the probe searches for the closest DP which happens to be the subject in (129). The subject also has a focus feature, which means that the complex probe checks all its features with the subject and attracts it to Spec,FocP.

- (128) [Azuure]_F **n** kɔdig nua la.
 Azuure FOC slaughter fowl DET
 ‘AZUURE slaughtered the fowl.’

- (129) **Local subject focus structure**



The restricted probe with *n* morphological marking will always attract the local subject in Kusaal because the language has a rigid word order, with any variation in word order usually due to information structuring. So, for instance, unlike what we saw in the Turkish example in (125-a), where the subject can occupy a lower position if it is indefinite, allowing the object to be the closest goal, this is never the case in Kusaal. The language

has a strict SVO order, regardless of the nature of the subject. The restriction which requires probing for the closest DP predicts that the extraction of an object (or any other constituent for that matter) can reflect the *n*-marking if the object occupies a higher position than the subject, thus becoming the closest goal. However, I do not know of or find any of such cases. So, this prediction cannot be tested.

The analysis also accounts for why unaccusative subjects are marked with *n* since the unaccusative DP is the closest DP in this case. This is shown with the examples in (130) and (131) below. The subjects *laa* and *Azuure* start out as thematic objects of the verb, before moving to the structural subject position. Since, they happen to be the closest DP, they are attracted to Spec,FocP and the Foc head is reflected as *n*.

- (130) a. Bə n kɔ'ɔg-ɛ?
 what FOC break-FV
 ‘What broke?’
 b. Laa n/(*ka) kɔ'ɔg.
 plate FOC break
 ‘A plate broke.’
- (131) a. Anɔɔn lu?
 who fall
 ‘Who fell?’
 b. Azuure n/(*ka) lu.
 Azuure FOC fall
 ‘Azuure fell.’

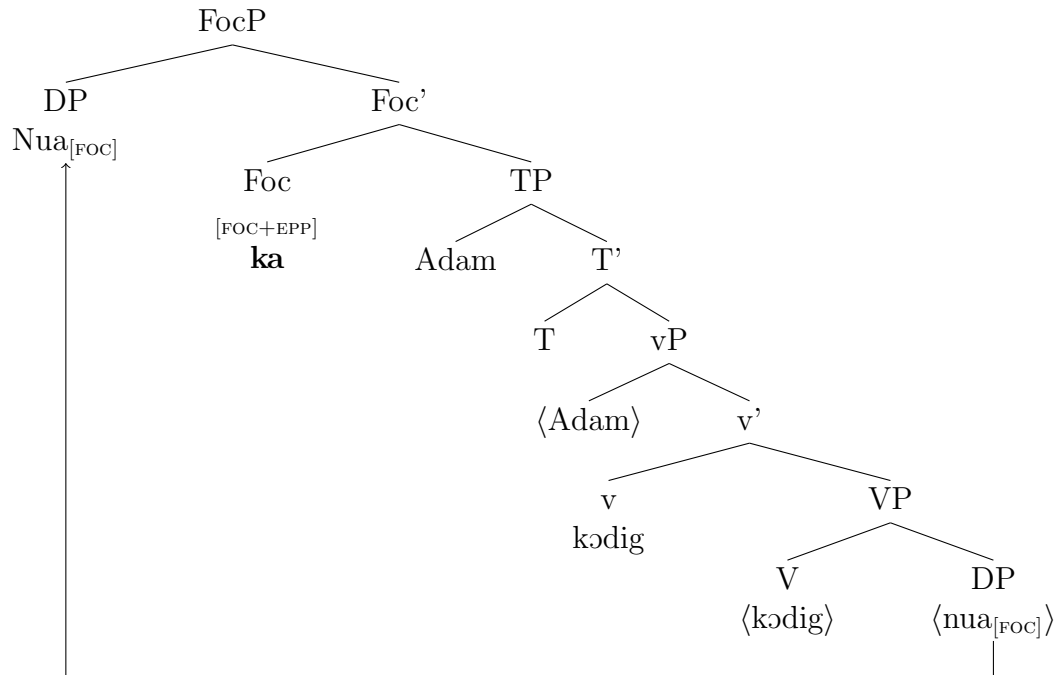
Moving on to the ex-situ local object focus, and by extension the other non-subject focus like PP and adverbs, according to the feature specification in (126-b), Foc is specified with a [Foc] feature with an EPP requirement.³⁵ The structural derivation of the object focus sentence in (132) is given in (133). The probe searches for a goal with a focus feature and finds the object. The probe Agrees with the object and attracts it to Spec,FocP.³⁶

- (132) [Nua]_F ka Adam kɔdig ____.
 fowl FOC Adam kill
 ‘Adam slaughtered A FOWL.’

³⁵ As would be discussed later, this is different from the in-situ focus cases where Foc lacks a movement requirement. In such cases, I assume that the in-situ focus only involves agreement without overt movement. However, I do not rule out covert or LF movement which has been argued for some languages with in-situ focus (see Chomsky, 1976; Huang, 1982; Aoun and Li, 1993; Richards, 1997; Nissenbaum, 2000; Bruening and Tran, 2006; Sulemana, 2019, a.o.).

³⁶ For convenience, I ignore the successive movement step through vP here, where the object movement would have to go through an outer Spec of vP. Nothing hinges on this here.

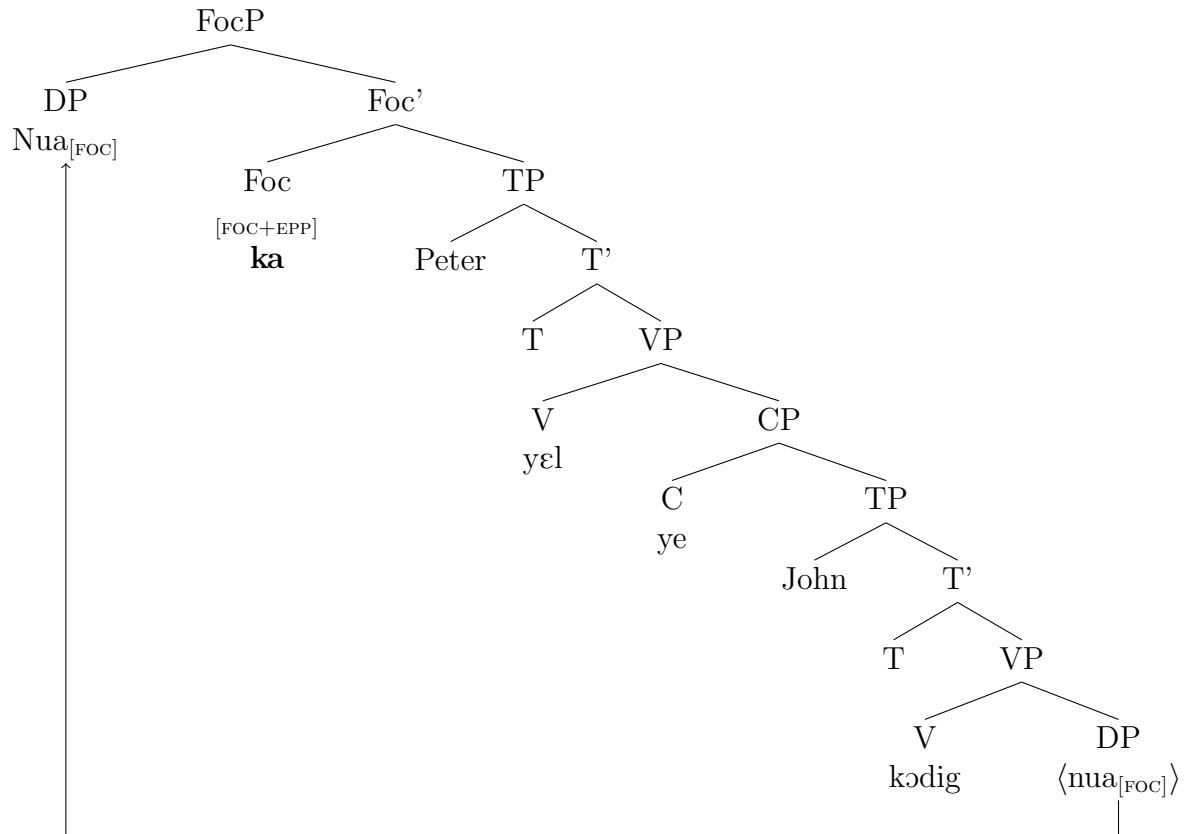
(133) **Local object focus structure**



This derivation in (133) also goes for the other non-subject focus like adjunct since they are also marked with *ka*. Long-distance non-subject focus will also follow from this derivation. The only difference would be the complex structure that the latter has. Consider again the non-local object focus in (110-b) above, repeated in (134). I give a simplified structure in (135) excluding the unnecessary traces and the intermediate movement steps. Foc probes for a goal with focus feature and attracts it to its Spec. Foc is morphologically realized as *ka*, in line with (126-b) since there is no [uD] requirement.

- (134) [Nua]_F **ka** Peter yel ye John kɔdig-ya ____.
 fowl FOC Peter say COMP John kill-DISJ
 ‘Peter said that John slaughtered A FOWL.’

(135) **Non-local object focus structure**

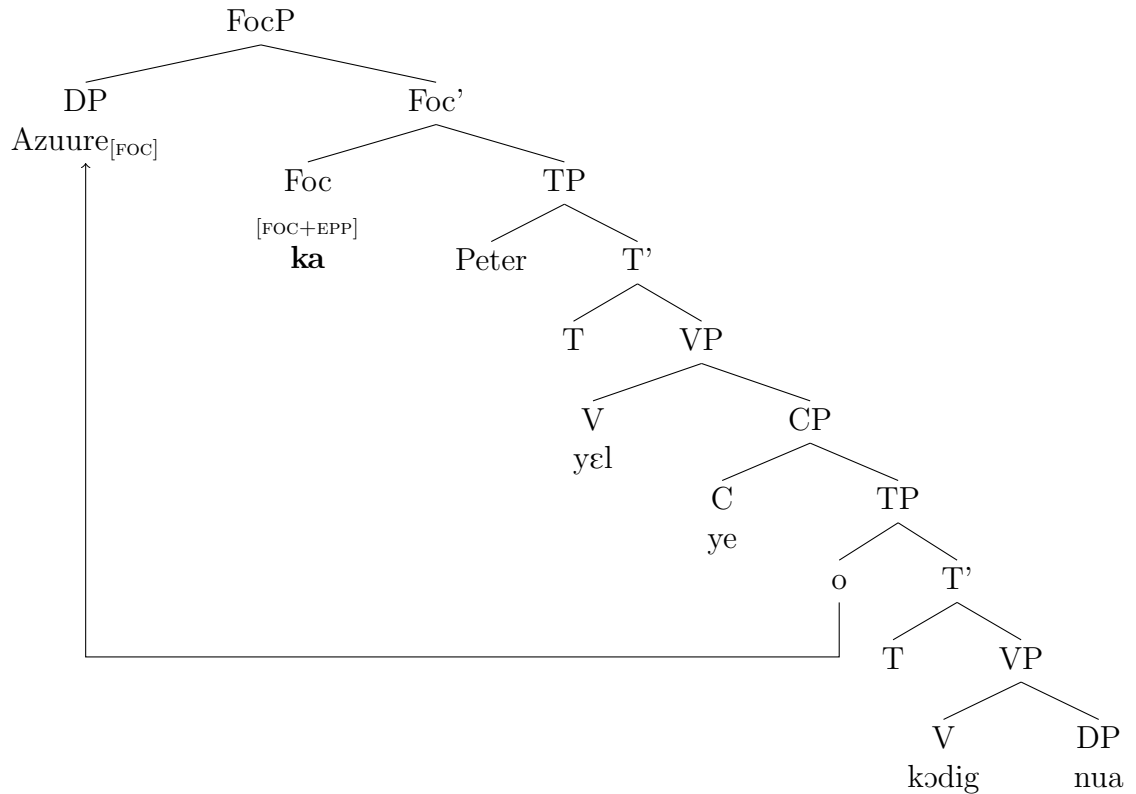


The next part of the analysis has to do with accounting for the non-local subject focus which also involve *ka*-marking like the non-subject foci. This simply follows from the analysis given for the non-subject foci. The matrix Foc probes for a goal with a focus feature, and attracts the latter to Spec,FocP. The structure for (136) is given in (137). Again, the intermediate movement steps are excluded in the structure.³⁷

- (136) [Azuuɛ]_F **ka** Peter yɛl ye *(o) kɔdig nua.
 Azuuɛ FOC Peter say COMP 3SG kill fowl.
 ‘Peter said that AZUURE slaughtered fowl.’

³⁷ See Section 3.5 for a discussion on the resumptive pronoun in the embedded subject position in Kusaal.

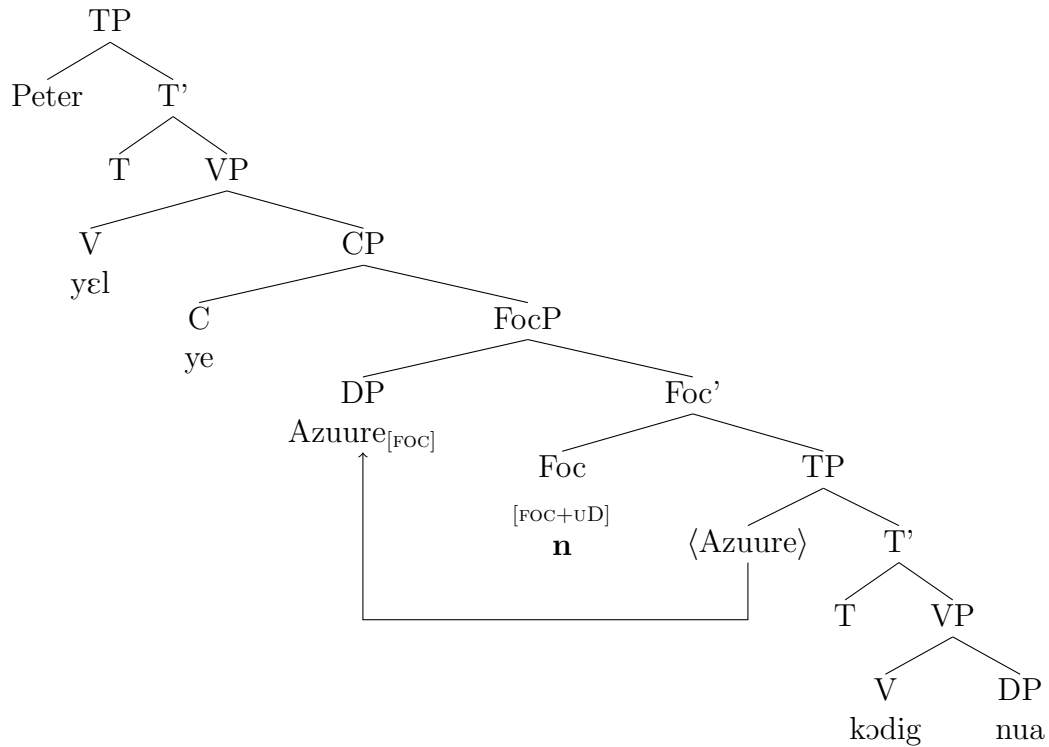
(137) Non-local subject focus structure



However, recall that if the embedded subject is fronted locally, i.e., to the embedded left periphery, it gets the *n*-marking, and not the *ka*-marking (138). Therefore, going by the feature specification in (126-a), Foc possesses a complex probe with a [uD] feature which requires probing for the closest DP (cf. (127)). And since the closest DP to an embedded Foc is the embedded subject, as shown in (139), the latter is attracted to the embedded Spec,FocP. The head of the embedded FocP is then morphologically realized as *n*.

- (138) Peter yel ye [Azuure]_F **n** kɔdig nua.
 Peter say COMP Azuure FOC kill fowl.
 ‘Peter said that AZUURE slaughtered fowl.’

(139) **Embedded Local subject focus structure**



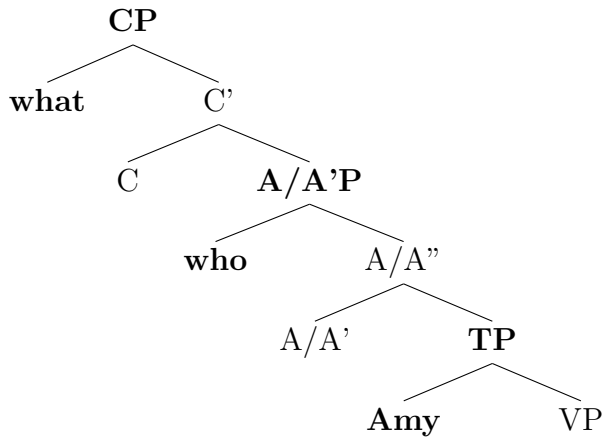
In summary, the implication of the proposal so far is that if Foc does not bear [uD], then (i) no agreement with the closest DP is required, and thus, (ii) the constituents other than the local subject can be extracted to Spec,FocP in as much as they bear a focus feature, and (iii) the morphological reflex of the focus feature on Foc would be *ka* (and not *n*). Conversely, if Foc bears [uD], then (i) agreement with the closest DP is required, and thus (ii) the local subject is extracted to Spec,FocP, and (iii) the morphological reflex of Foc would be *n*.³⁸

Two alternative analyses

As mentioned above, there are at least two alternative analyses that might be considered. The first is based on Bošković's (2024) proposal that treats local subject *wh*-phrases or focus as located in a position higher than the canonical TP position, but lower than the position occupied by non-subject and non-local subject *wh*-phrases or foci. The structure in (140) shows these different positions with the constituents that may occupy the positions. Bošković (2024) argues that while the local subject *wh*-phrase/focus occupies the Spec of a projection, which he assumes to have both A and A' properties (thus A/A'P), every other A'-moved phrase (apart from the local subject) occupies Spec,CP.

³⁸ Considering the fact that the local subject in Kusaal seems to always bear a nominative case, one may alternatively say that the probe is required to Agree with a goal that bears a specific case. This idea would be similar to the case discrimination analysis that Deal (2017) proposes for ergative-absolutive languages that show syntactic ergativity. The analysis requires that the probe that drives movement Agrees only with a goal bearing a specific case. However, we will not commit ourselves to such an analysis because this approach makes an incorrect prediction about long distance subject focus. First, it predicts that any DP with a nominative case should be attracted by the *n* head. While this would account for the local subject focus, it cannot account for long distance focus because (as we have seen) it is marked with *ka*, and not *n*. On the other hand, based on a case discrimination analysis, we may say that non-subject focus (with accusative case, for example) are marked with *ka*. However, this undergenerates the *ka*-marking on the non-local subject. So, this analysis cannot be adopted for Kusaal, and the other languages like it.

(140) **Bošković's different landing positions**



Generally, this analysis is not only interesting, but (at first glance) seems to suit the asymmetry observed in Kusaal. We can just assume that the different focus markers occupy the heads of the different projections above TP. So, this will mean that A/A' is realized as the local subject marker *n*, and C is realized as *ka* respectively (141).

(141) $[_{CP} \textit{wh} \textbf{ka} [_{A/A'P} \textit{wh} \textbf{n} [_{TP} \textit{Adam vP}]]]$

Although this may seem to be a straightforward analysis for the focus marking asymmetry in Kusaal (and the other Mabilia languages like Dagbani and Gurene), it predicts at least two things. First, it predicts the possibility of having multiple foci in the left periphery. If these two distinct positions exist, then they should be able to co-occur in a language that allows multiple foci. Second, if these two positions co-occur, then their morphologically realized heads should be overtly present as well. However, these predictions are not borne out. While multiple foci constructions are possible in Kusaal, only one of the foci can be fronted to the left periphery. Consider the *wh*-question and focus sentence in (142).³⁹ Both focus markers cannot be present at the same time. The absence of *n* suggests that the local subject focus is in-situ in multi-foci contexts.

(142) **Multiple focus with ex-situ local object *wh*/focus**

- a. Bɔ **ka** sakurbil kanɛ (***n**) kɔdig-ɛ ____?
 what FOC student which FOC kill-FV
 Lit. 'What did which student slaughter?'
- b. [Nua]_F **ka** sakurbil Kusaal (***n**) kɔdig ____.
 fowl FOC student Kusaal FOC kill
 'It was fowl that a Kusaal student slaughtered.'

Interestingly, the lack of co-occurrence of the focus markers is not peculiar to ex-situ focus alone. Recall that Kusaal also has the option to morphologically mark a focus in-situ (cf. section 3.3.1 above). Such an option presents the possibility of having both an ex-situ subject focus and an in-situ object focus co-occurring in a multi-foci context. Even in such a context, it is the high (fronted) focus that is morphologically marked; the in-situ focus cannot be marked, as in (143). (142-a) shows multiple *wh*-questions, and (143-b) is its felicitous answer.

³⁹ I use the D-linked subject question here because it shows clearly the presence of the local subject focus marker *n*. See fn. 30 above.

(143) **Multiple focus with ex-situ local subject *wh/focus***

- a. Sakurbil kanɛ **n** kɔdig (***nɛ**) bɔ.
 student which FOC kill FOC what
 ‘Which student slaughtered what?’
- b. [Azuure]_F **n** kɔdig (***nɛ**) [nua]_F.
 Azuure FOC kill FOC fowl
 ‘AZUURE slaughtered A FOWL.’

The multiple foci in (143-b) involves an ex-situ subject and an in-situ object. In a ditransitive construction, one of the objects can be realized ex-situ while the other is realized in-situ. (144) shows that the in-situ direct object focus cannot be morphologically marked.⁴⁰

- (144) [Adam]_F **ka** Azuure tis (***nɛ**) [nua]_F.
 Adam FOC Azuure give FOC fowl
 ‘Azuure gave ADAM A FOWL.’

All of these data suggest the presence of a single focus position in the left periphery of Kusaal. The different morphological realization of this focus head is then determined by the feature(s) on it, which in turn leads to a complex probe or a single probe. Therefore, it remains to be seen how Bošković’s analysis would account for the asymmetry in Kusaal without making special and/or complex assumptions in order to account for the absence of multiple fronting and the lack of co-occurrence of the focus markers.

The second approach, which is similar to the one provided here, is the one proposed by Keine and Zeijlstra (2025) for Defaka, a Niger-Congo language that also displays a focus marking asymmetry between local subject focus and everything else (see Bennett, 2009; Bennett et al., 2012, for an earlier proposal).

Keine and Zeijlstra (2025) argued for what they called a *DP intervention approach* to account for (i) (subject-only) extraction restriction in Standard Indonesian, (ii) successive cyclicity in Dinka, and (iii) successive cyclicity and focus marking asymmetry in Defaka. I will only briefly discuss their proposal for Defaka since it possesses a similar focus marking asymmetry that we saw in Kusaal. See Keine and Zeijlstra (2025) for the analysis of the other two cases. Keine and Zeijlstra (2025) argue in favour of a DP intervention analysis instead of a phase-based analysis that was initially provided by Bennett (2009); Bennett et al. (2012).

Consider the focus marking asymmetry in (145). The sentence in (145-a) is the baseline. When a local subject focus is fronted, it is followed by the focus marker *kò* (145-b).⁴¹ But if it is the local object that is fronted, it is followed by *ndò* (145-c). Additionally, non-local subject focus comes with a *kè*-marking on the verbal complex (cf. (145-c)).

⁴⁰ If the two foci stay in-situ, it is the structurally higher object that is morphologically marked, (i) (answers *Who did Azuure give what?*).

- (i) Azuure tis **nɛ** [Adam]_F (***nɛ**) [nua]_F.
 Azuure give FOC Adam FOC fowl
 ‘Azuure gave ADAM A FOWL.’

⁴¹ I maintain the glossing provided by the authors here.

(145) **Focus marking asymmetry in Defaka**

- a. Ì Bòmá ésé-kà-rè. *No focus-fronting*
 I Boma see-FUT-NEG
 ‘I will not see Boma.’
- b. [Ì]_F kò Bòmá ésé-kà-rè. *Local-subject focus*
 I FOC.SBJ Boma see-FUT-NEG
 ‘I will not see Boma.’
- c. [Bòmá]_F ndò ì ésé-kà-rè-kè. *Object focus*
 Boma FOC I see-FUT-NEG-KE
 ‘I will not see BOMA.’

[Keine and Zeijlstra (2025, p.42 ex.80) citing Bennett et al. (2012)]

Ndò focus marking plus the *kè*-marking are also present in PP-adjunct focus (146), non-local object focus (147-a) and non-local subject focus (147-b) respectively. In all these cases *ke* is present whenever the fronted constituent crosses the verbal complex. So, in example (147-a), both the embedded verbal complex and the matrix verbal complex bear *ke*. In example (147-b), on the other hand, only the matrix verb bears *ke* and the focus marker is *ndo*, and not *ko*. So, non-local subject focus patterns with non-subject focus in both *ndo* focus marking, and the presence of *ke* on the higher verb.

- (146) [Ándù kíkìà]_F ndò à èbèrè rì bòi-mà-kè. *PP-adjunct focus*
 canoe under FOC the dog RE hide-NFUT-KE
 ‘The dog is hiding under the canoe.’

(147) **Non-local object and subject focus**

- a. [Ándù_i]_F ndò Bòmá fàà-kè [CP ìní _____i été-kè].
 canoe FOC Boma say-KE they have-KE
 ‘It’s a CANOE that Boma said that they have.’
- b. [Bruce_i]_F ndò/*kò Bòmá jírí-*(kè) [CP _____i á ésé-mà].
 Bruce FOC/*FOC.SBJ Boma know-*(KE) her see-NFUT
 ‘Boma knows (that) BRUCE saw her.’

[Keine and Zeijlstra (2025, p.43 ex.(81a), (82) & (83)) citing Bennett et al. (2012)]

Bennett (2009); Bennett et al. (2012) adopt a phase-based approach to argue that *ke* is a reflex of intermediate movement steps, i.e. as evidence for successive cyclic movement. Keine and Zeijlstra (2025), however, argue that a phase-based account is unnecessary and does not even account for a few challenges that emerge. For instance, if *ke* reflects intermediate movement, it only seems to suggest vP as a phase, but not CP. In other words, why is *ke* present on only the verb and not C, since both CP and vP are traditional phases?

Keine and Zeijlstra’s (2025) own solution is to assume that both the change in focus marking (between local subject and everything else) and *ke*-marking can be reduced to a DP intervention effect. The asymmetry that exists in the morphological focus marking and the *ke*-marking reflects whether there is a DP intervening between probe C and a potential goal. They adopt the restriction in (122) (repeated below as (148)), where a probe searches for the closest DP. The specifications of the different focus markers and *ke*-marking are given in (149).

- (148) **A'-probing for the closest DP**
 An A'-probe *can be specified* to target the closest DP.

- (149) **Defaka focus & *ke* marking specifications**

- a. /ndò/ $\leftrightarrow$ C_[uFoc] / [CP XP ____]
 b. /kò/ $\leftrightarrow$ {C-Subj}_[uFoc] / [{C-Subj]P XP ____]
 c. /kè/ $\leftrightarrow$ Subj_[uFoc]

Beginning with (149-a), Keine and Zeijlstra (2025) assume that the object can be the closest DP if it moves to an outer specifier of what they called a SubjP (following Bennett et al. (2012)). Consider the structure in (150). The object began from the complement of V, and gets attracted to an outer Spec of Subj. This happens after the subject has moved to the inner Spec. Subj is then realized as *ke* because of the intermediate movement of the object DP to the outer Spec,SubjP (cf. (149-c)). At this position, the object becomes the closest goal for the probe on C and gets attracted to Spec,CP. This led to C being realized as *ndo* as well.⁴²

- (150) **Defaka's local object fronting**

[CP DP_{obj[Foc]} [C' C_[uFoc+uX]=**ndo** [SubjP t_{obj[Foc]} [Subj' DP_{subj} [Subj' Subj_[uFoc]=**ke**
 [TP T [vP t_{subj} [v' v ... t_{obj[Foc]}]]]]]]]]]

For the local subject focus, Keine and Zeijlstra (2025) adopt Bennett et al.'s (2012) idea of a conflated C+Subj head that has the VI rule in (149-b). They assume that both C and Subj are conflated to realize the local subject focus marking *ko* (151). In this case, the local subject is the closest goal, and it gets attracted directly from Spec,vP to Spec,C-SubjP. It is worth mentioning here that Keine and Zeijlstra's (2025) analysis is similar to Issah and Smith's (2020) proposal for Dagbani, especially with respect to a conflated or a fused head. They, however, assume two CPs and the conflation of the two Cs in a distributed morphology (post-syntactic) manner (see Issah and Smith, 2020, and the references they cited). They assumed that the two Cs are an extension of a single focus head with a single focus semantics.⁴³

- (151) **Defaka's local subject fronting**

[{C-Subj]P DP_{subj[Foc]} [{C-Subj'} {C-Subj}]_[uFoc+uX] **ko** [TP T [vP t_{subj} [v' v ...]]]]]

Furthermore, since there is no DP intervention in local subject fronting, there is no need for an intermediate movement to any outer Spec,SubjP, and thus no *ke*-realization.

While the analysis provided for Defaka is similar to the analysis given for Kusaal, there are a few differences. First, I do not assume a conflated head for Kusaal. The different morphological markers are inserted depending on the feature specifications on Foc. Secondly, while the DP intervention analysis seems to be able to account for the Kusaal data, we do not need to assume an outer Spec to make the object accessible to the probe on Foc. Unlike Defaka's *ke*, there is no morphological reflex in Kusaal for what would have been a high intermediate movement step. Moreover, the [uD] feature which restricts probing for the closest DP is only present with local subject focus, and not just DP fronting.

⁴² See Keine and Zeijlstra (2025) for more details, especially the evidence for a high subject position (Spec,SubjP) and the realization of Subj as *ke*.

⁴³ I will not go into the details of their proposal here.

Proposal for the in-situ focus

Before concluding this section, we need to propose an analysis for the in-situ focus strategy. I repeat the example in (71) above as (152). As described in section 3.3.1 above, the in-situ answer focus in (152-b) is preceded by an in-situ focus marker *nɛ*.

- (152) a. Adam kɔdig bɔ?
 Adam kill what
 ‘What did Adam slaughter?’
 b. Adam kɔdig **nɛ** [nua]_F.
 Adam kill FOC fowl
 ‘Adam slaughtered A FOWL.’

Similarly to the proposal for the in-situ focus in Kasem above, I assume the presence of a high Foc in the left periphery, which licenses the in-situ focus in Kusaal. Foc projects a FocP whose specifier is not occupied. I assume that this is because there is no movement (EPP) requirement on Foc. So, the in-situ focus constituent only Agrees with Foc without overtly moving to its Spec. As a result of this, we need to update our specifications in (126) (repeated below as (153)) to include (153-c). So, in in-situ focus context, Foc lacks EPP, and as such, we do not get *ka*-marking that is present in ex-situ object focus, for instance (see section 3.3.1 above).⁴⁴

(153) **Feature specifications on Foc and morphological reflexes**

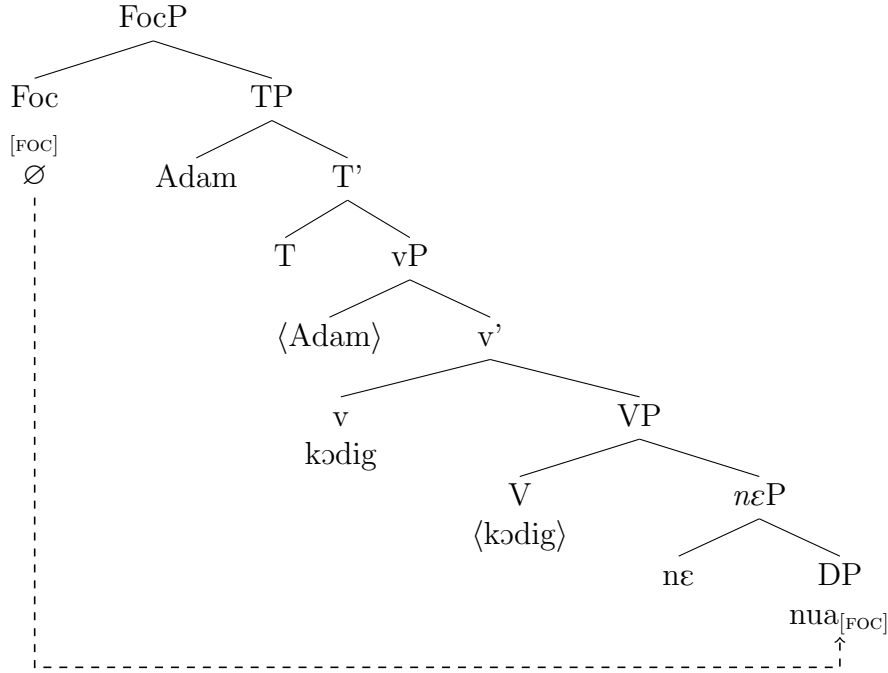
- a. $\text{Foc}_{[\text{Foc}+\text{uD}]} \leftrightarrow n$
 b. $\text{Foc}_{[\text{Foc}+\text{EPP}]} \leftrightarrow ka$
 c. $\text{Foc}_{[\text{Foc}]} \leftrightarrow \emptyset$

The structure of an in-situ focus like (152-b) is given in (154).⁴⁵ The assumption here is that *nɛ* is the exponent of the agreement of the focus feature on the focus constituent in (154).

⁴⁴ Recall that we already argued that a focus-marked subject is always ex-situ. So, our comparison is between (153-b) and (153-c) which are specifications for foci that can be realized ex-situ as well as in-situ.

⁴⁵ Recall that in section 3.3.4, I already argued that *nɛ* forms a constituent with the in-situ focus.

(154) **In-situ object focus structure**



Assuming Relativized Minimality (cf. Rizzi, 2001, a.o.), the agreement process in (154) predicts that when another focus phrase intervenes between the high Foc and the in-situ object focus, then agreement should be interrupted, and this would result in the absence of *nɛ* on the object focus.⁴⁶ This prediction is borne out when we consider multiple foci construction in Kusaal. Consider the examples in (143-b) and (144) above, repeated below as (155). The two examples involve multiple foci of both subject and direct object (155-a), and of both indirect object and direct object (155-b).⁴⁷

- (155) a. [Azuure]_F **n** kɔdig (***nɛ**) [nua]_F.
 Azuure kill FOC fowl
 ‘AZUURE slaughtered A FOWL.’
 b. [Adam]_F **ka** Azuure tis (***nɛ**) [nua]_F.
 Adam FOC Azuure give FOC fowl
 ‘Azuure gave ADAM A FOWL.’

In both examples, the in-situ focus lacks *nɛ* in the presence of a higher focus in the left periphery. This can be explained if we consider that in (155-a) for example, the focus head attracts the subject focus that is closer to it. So, the latter constitute an intervener

⁴⁶ In RM, “Y cannot be related to X if Z intervenes and Z has certain characteristics in common with X.” (i) (Rizzi 2001:89). Assuming X as an agreement probe in (i-a), Z intervenes in the agreement between X and Y because the latter has a common feature with X. However, agreement between X and Y is possible in (i-b) because Z does not possess a common feature with X, and thus, does not constitute intervention.

- (i) a. [X_{+f} ... Z_[+f] ... Y_[+f]]
 └───┬───┘
 X
 b. [X_{+f} ... Z_[-f] ... Y_[+f]]
 └───┬───┘
 ✓

⁴⁷ We know that we are dealing with a multiple foci case here because the created contexts are a multiple *wh*-questions context: *who ate what?* and *who did Azuure give what?*

which restricts Foc from agreeing with the object. This results in the absence of *nɛ* in the example. The same can be said for (155-b). This result supports our proposal of a high FocP licensing in-situ foci in Kusaal.

In principle, if there is a low FocP (and not the high one) or if the in-situ focus is not licensed by Foc in the left periphery, then the in-situ focus marker *nɛ* should be available in multiple focus context where a higher DP (the subject) intervenes between the high Foc and the in-situ object. In other words, if the in-situ focus marking is independent of the high Foc, then it should not matter if there is an intervention. But since that is not case, I assume that in-situ focus is licensed by a high Foc in the left periphery, and that the in-situ focus marker *nɛ* is a reflex of the focus feature on the in-situ focus in the absence of movement.

The same reasoning may be extended to the unavailability of multiple morphologically marked in-situ foci, as in (156). Only the hierarchically high focus is morphologically marked.

- (156) Azuure tis **nɛ** [Adam]_F (***nɛ**) [nua]_F.
 Azuure give FOC Adam FOC fowl
 ‘Azuure gave ADAM A FOWL.’

On the surface, this seems to support the idea that there is a (clause-internal) low FocP (à la Belletti (2004), a.o.). However, I already ruled out such an assumption above, in favour of the assumption that *nɛ* actually forms a constituent with the focus phrase. How then do we account for the focus marking asymmetry that is observed in multiple in-situ foci cases such as (156)? I assume that the explanation here is the same as the one provided for the multiple foci (involving an ex-situ focus and an in-situ focus) cases above. The high Foc head has a single probe which Agrees with the closest goal, with the realization of the in-situ focus marker the result of an Agree relation on the goal. Under Relativized Minimality, the higher focus DP, which in (156) is *Adam*, will Agree with the high Foc head, and at the same time intervene Foc from agreeing with the lower DP *nua* (‘fowl’). Since *nɛ* realizes the agreement process on the goal, the higher, but not lower, DP has *nɛ* marking. The implication of this analysis is that the distribution of the morphological marking of focus is syntactically conditioned and/or controlled by agreement and locality constraints.⁴⁸

To summarize this section, I have discussed the focus construction patterns in Kusaal, and I have proposed an analysis for the focus marking asymmetry that exist between local subject focus on the one hand, and non-subject plus non-local subject focus on the other. The vocabulary insertion rules, which realize different morphological markers based on different feature specifications, form the basis for our analysis. I included a specification for in-situ focus as well which realizes a null focus head in the left periphery. Therefore, it is the specification on Foc that determines whether it is restricted to attract the closest DP or others, or even cases where there is no attraction at all, as in the in-situ focus.

We now turn our attention to the focus construction in Yorùbá in the next section.

⁴⁸ I want to note here that the analysis provided is not in any way exhaustive. Further investigation is definitely required to attain a concrete proposal for the multiple-foci cases in Kusaal (as well as Kasem). This will be the focus of a future research.

3.4 Focus realization in Yorùbá

While focus construction in Yorùbá has been widely studied both descriptively and theoretically, there remains a debate on the syntactic or structural status of Yorùbá ex-situ focus (Owolabi, 1981, 1983; Awoyale, 1985; Awobuluyi, 1987; Yusuf, 1989; Bisang and Sonaiya, 2000; Adesola, 2005; Ilori, 2010, among others). For instance, Adesola (2005) analyzes the Yorùbá ex-situ focus construction as involving operator movement with the focus base-generated in the Spec of a PredP. According to Adesola (2005), it is the null Op that moves from the focus constituent’s canonical position to Spec,CP. One of the arguments for such an analysis is the absence of weak crossover in Yorùbá. Basically, his analysis (and Bisang and Sonaiya’s (2000) description) points towards a bi-clausal structure for the Yorùbá focus construction. However, in this section, I will argue for a mono-clausal structure for the focus construction in the language in line with some of the studies listed above. But before that, a description of the focalization strategies in the languages is necessary.

3.4.1 Subject and non-subject focus

Focus can be realized both in-situ and ex-situ in Yorùbá. However, subject focus is always realized in an ex-situ position. I will begin with the non-subject focus realization.

Non-subject focus

As in Kasem and Kusaal, local object focus can be realized both in an in-situ position (157-b), and in an ex-situ position (157-c). However, unlike Kasem and Kusaal, only ex-situ focus is morphologically marked, with *ni*, in Yorùbá. In-situ foci are not morphologically marked. In such cases, the focus can only be determined from context. This is unsurprising since many West African languages like Akan, Ga, Gungbe, and so on mark only ex-situ focus morphologically (Grubic et al., 2019; Aboh, 1998, 2004b, 2007). In-situ foci are usually unmarked. With the *wh*-question context in (157-a), the answer may be realized in-situ (157-b) or ex-situ (157-c).

- (157) a. Kí **ni** Adé rà?
 what FOC Adé buy
 ‘What did Adé buy?’
 b. Adé ra [bàtà]_F (***ni**).
 Adé buy shoe FOC
 ‘Adé bought A SHOE.’
 c. [Bàtà] **ni** Adé rà.
 shoe FOC Adé buy
 ‘Adé bought A SHOE.’

Although both the in-situ and the ex-situ focus strategies may be used to answer a question, a closer look at them shows that they are different with respect to the contexts in which they are used. For example, while the ex-situ focus is exhaustive, the in-situ focus is not. In the following, I repeat the example above, but now with an exhaustive test using the additive focus sensitive particle *pèlu*. The ex-situ focus in (158-c) is infelicitous with a continuation that contains an additive particle. This is because ex-situ focus is exhaustive. However, a sentence with an additive particle can felicitously follow the

sentence with the in-situ focus because it does not involve exhaustivity.

- (158) a. Kí **ni** Adé rà?
 what FOC Adé buy
 ‘What did Adé buy?’
 b. Adé ra [bàtà]_F (***ni**). Ó ra așo **pèlu**
 Adé buy shoe FOC 3SG buy cloth also
 ‘Adé bought A SHOE. He also bought a cloth.’
 c. #[Bàtà] **ni** Adé rà. Ó ra așo **pèlu**
 shoe FOC Adé buy 3SG buy cloth also
 ‘Adé bought A SHOE. He also bought a cloth.’

Adjuncts also follow the same pattern of zero in-situ morphological marking, and an overt morphological ex-situ focus realization (159).

- (159) a. Igba wo **ni** Adé rà bàtà?
 time which FOC Adé buy shoe
 ‘When did Adé buy a shoe?’
 b. Adé rà bàtà ní [àná]_F.
 Adé buy shoe LOC yesterday
 ‘Adé bought a shoe YESTERDAY.’
 c. [Àná]_F **ni** Adé rà bàtà.
 yesterday yesterday FOC Adé buy shoe
 ‘Adé bought a shoe YESTERDAY.’

The in-situ verb focus in (160-b) is realized if the question in (160-a) is considered. This is the most natural way of answering a question that requires a verb focus answer in the language. There is no prosodic prominence or difference here whatsoever.

- (160) a. Ki **ni** Adé ɕe sí fèrèsé?
 what FOC Adé do to window
 ‘What did Adé do to the window?’
 b. Ó [fó]_F o.
 3SG.NOM break 3SG.ACC
 ‘He BROKE it.’

Verb focus can also be realized ex-situ. However, this has to undergo a nominalization process and verb copying. In other words, a copy of the verb remains in its canonical position within the clause.

- (161) [Fífó]_F **ni** Adé fọ fèrèsé.
 NMLZ.break FOC Adé break window
 ‘Adé BROKE the window.’

The verb phrase (VP) is usually focused in an in-situ position, as in (162-b). Just like the in-situ focus cases above, there is no overt focus marker and no prosodic difference here as well. The current question must be used to determine which constituent is in focus and which one is not. Ex-situ VP focus is also possible, as in (162-c).

- (162) a. Ki **ni** Adé ɕe?
 what FOC Adé do
 ‘What did Adé do?’

- b. Adé fọ fèrèsé.
Adé break window
'Adé BROKE A WINDOW.'
- c. [Fífọ fèrèsé]_F **ni** Adé fọ fèrèsé.
NMLZ.break window FOC Adé break window
'Adé BROKE THE WINDOW.'

Subject focus

Lastly, as mentioned above, subject focus in Yorùbá are always ex-situ (163-b). This is evident due to the presence of a resumptive pronoun in the argument position of the subject.⁴⁹ Just as we saw for the ex-situ non-subject foci, the focused subject precedes the focus marker *ni* which is the only way to answer a subject *wh*-question as in (163-a). The example in (163-c) with the subject in an in-situ position is not a felicitous answer to the current question.

- (163) a. Ta **ni** ó rà bàtà?
who FOC 3SG buy shoe
'Who bought a shoe?'
- b. [Adé]_F **ni** ó rà bàtà.
Adé FOC 3SG buy shoe
'ADÉ bought a shoe.'
- c. #[Adé]_F rà bàtà.
Adé buy shoe
'ADÉ bought a shoe.'

The subject of an embedded clause is also resumed regardless of whether the subject focus is realized in the left periphery of the matrix clause (164-a) or the embedded clause (164-b).

- (164) a. [Adé]_F **ni** Bọlá rò wípé ó rà bàtà.
Adé FOC Bọlá think COMP 3SG buy shoe
'Bọlá thinks that ADÉ bought a shoe.'
- b. Bọlá rò wípé [Adé]_F **ni** ó rà bàtà.
Bọlá think COMP Adé FOC 3SG buy shoe
'Bọlá thinks that ADÉ bought a shoe.'

Similar to what was discussed for Kasem and Kusaal above, the structural position of local subject focus in Yorùbá is higher than the canonical Spec,TP non-focused subject position. The simple fact that local subject focus is morphologically marked with *ni* and also involves resumption proves that it is not realized in Spec,TP.

⁴⁹ There have been some controversies around the status of what I called resumptive pronoun here. While people like Yusuf (1989); Adesola (2005) believe that it is indeed a resumptive pronoun, others like Awobuluyi (1992); Ilori (2010) argue that it is more of a non-future tense marker. There are also those who see it as just pure subject marker (Oyelaran, 1970; Fresco, 1970). Another group sees it as a perfective aspect marker (Adewole, 1988; Bisang and Sonaiya, 1999). Others see it as a subject concord prefix (Stalhke, 1974), and so on. For some discussions and an overview of the different claims, see Ilori (2010). However, I will label it as a resumptive pronoun in this dissertation. But see Aremu (2024) for arguments in support of the resumptive status of the pronoun.

3.4.2 Anti-pied-piping in Yorùbá

Unlike Kasem and Kusaal which have both TP and VP anti-pied-piping (the (morpho)syntactic marking of a sub-part of a logical focus), Yorùbá only has TP focus anti-pied-piping, and not VP focus anti-pied-piping. In other words, while a sub-part of a TP focus can be morphosyntactically marked, a sub-part of a VP logical focus is not morphosyntactically marked. In example (165), the logical focus is the TP, but it is the subject that is morphosyntactically marked (165-b).⁵⁰ Notice that the sentence without the focus marker can also be used to answer the question (165-c).

- (165) a. Kí ni ó sẹ̀lẹ̀?
 what FOC 3SG happen
 ‘What happened?’
 b. Tolú **ni** [ó jẹ̀ iṣu]_F.
 Tolú FOC 3SG eat yam
 ‘TOLÚ ATE YAM.’
 c. [Tolú jẹ̀ iṣu]_F.
 Tolú eat yam
 ‘TOLÚ ATE YAM.’

On the other hand, if the VP is the logical focus, the direct object cannot be morphosyntactically marked (compare examples (166-b) with (166-c)). The direct object remains in its in-situ position, post-verbally.

- (166) a. Kí ni Tolú se?
 what FOC Tolú do
 ‘What did Tolú do?’
 b. Tolú [pa eku]_F.
 Tolú kill rat
 ‘Tolú KILLED A RAT.’
 c. #Eku_i **ni** Tolú [pa t_i]_F.
 rat FOC Tolú kill
 ‘Tolú killed a rat.’

Furthermore, when the clause is unaccusative or unergative, the subject is still morphosyntactically marked for TP focus. Both sentences in (167-b) and (167-c) can be TP focus answers to the question in (167-a).

⁵⁰ Bisang and Sonaiya (2000) provided an example that we can classify as TP anti-pied-piping in Yorùbá. The question-answer pair in (i) (which has been modified) from Bisang and Sonaiya (2000, p.185) involves a TP focus which can be deduced from the question. As an answer, the subject is followed by the focus marker *ni* (i-b).

- (i) a. Kí ni ó ẹ̀ se é?
 what FOC 3SG make/do 2SH
 ‘What’s wrong with you?’
 b. Màmá mi ni ó kú lánàá.
 mother POSS FOC 3SG die yesterday
 ‘My mother died yesterday.’

- (167) a. Kí **ni** ó sẹ̀lẹ̀?
 what FOC 3SG happen
 ‘What happened?’
 b. Tolú **ni** [ó subú]_F.
 Tolú FOC 3SG fall
 ‘TOLÚ FELL.’
 c. Tolú **ni** [ó sùn]_F.
 Tolú FOC 3SG sleep
 ‘TOLÚ SLEPT.’

Unlike the prosodic focus marking languages discussed in section 3.2 above, where indefinites are not targeted by anti-pied-piping, this is not the case in Yorùbá. Similar to Kasem and Kusaal, an indefinite subject can be morphosyntactically marked in TP focus (168).

- (168) Enikan **ni** [ó jí ọ̀kọ̀]_F.
 someone FOC 3SG steal car
 ‘SOMEONE STOLE A CAR.’

The absence of VP anti-pied-piping in Yorùbá may be due to the possibility of VP fronting. In (169), the VP is fronted with the verb nominalized. Such a possibility may make VP anti-pied-piping unnecessary.⁵¹

- (169) [Pípa eku]_F **ni** Tolú pa eku.
 kill.NMNL rat FOC Tolú kill rat
 ‘Tolú SLAUGHTERED A RAT.’

Before providing a structural analysis for focus realization in Yorùbá, in the next section, I discuss the view in the literature with the claim that the focus marker *ni* can also function as a copula marker.

3.4.3 The focus marker *ni* as a copula

The focus particle *ni* can be used in copula constructions in Yorùbá, as in the examples in (170).

- (170) a. Ọ̀lẹ́ **ni** Tolú.
 lazy COP Tolú
 ‘Tolú is lazy.’

⁵¹ Bisang and Sonaiya (2000) gave examples (like the one below) that involve *ni* in a post-VP position. They however claim that such constructions are only possible if the subject is a pronoun and not an overt noun, with the pronoun functioning like an agreement marker (Bisang and Sonaiya, 2000, p.182).

- (i) a. Kí ni Bọ́lá ní ẹ̀?
 what FOC Bọ́lá PROG do
 ‘What is Bola doing?’
 b. [Ó ní sùn]_F **ni**.
 he PROG sleep FOC
 ‘HE IS SLEEPING.’

- b. Obìnrin **ni** Tolú.
 woman COP Tolú
 ‘Tolú is a woman.’

There are two copulas in Yorùbá, *ni* and *jé*. One difference between them is quoted below by Bisang and Sonaiya (2000, p.171). While the predicate of *ni* is seen to be part of a preconstructed context, *jé*’s predicate does not. By preconstruction, Bisang and Sonaiya (2000) mean “a domain of concepts presupposed by the speech-act participants”. This has led to the claim that *ni*, as a focus marker, is used for identificational focus and not new information focus since identificational focus is also perceived to involve preconstruction/

“The difference between *ni* and *jé* will be seen to be a question of PRECONSTRUCTION, that is, the existence of a domain of concepts presupposed by the speech-act participants. Where a certain domain is preconstructed the copula *ni* must be used, and where such is not the case, then *jé* is employed.”

The copula *je* does not only occur in a non-preconstructed context, but also maintains the canonical word order (SVO) in the language (171). The sentences below do not involve exhaustive identification. They are not contrasted with any set of alternatives.

- (171) a. Tolú **jé** onígberága (èniyàn).
 Tolú COP proud person
 ‘Tolú is a proud person.’
 b. Tolú **jé** olùkó iléewé girama.
 Tolú COP teacher house.book secondary
 ‘Tolú is a secondary school teacher.’

In fact, *jé* can optionally occur in constructions involving *ni*, as in the repeated copula constructions in (172) below.

- (172) a. Òlẹ́ **ni** Tolú (**jé**).
 lazy COP Tolú COP
 ‘Tolú is lazy.’
 b. Obìnrin **ni** Tolú (**jé**).
 woman COP Tolú COP
 ‘Tolú is a woman.’

3.4.4 Focus construction in Yorùbá as mono-clausal structures

The view that *ni* is a copula opens up the possibility of a cleft-like focus structure. Since clefts are often bi-clausal structures (consisting of a matrix clause with a copula and a subordinate clause that is introduced by a relativizer), this would mean that focus constructions in Yorùbá are bi-clausal. In fact, this is what Bisang and Sonaiya’s (2000) account suggests.⁵² However, several studies have analysed ex-situ focus constructions in Yorùbá as mono-clausal structures. Among them are Yusuf (1989), Awobuluyi (1987, 2008), Ilori (2010), Aremu (2021, 2024), Aremu and Weisser (2024). In this study, I follow the mono-clausal view of Yorùbá focus construction.

⁵² It is worth noting here that Bisang and Sonaiya (2000) themselves do not in any way propose a bi-clausal analysis for Yorùbá focus construction. They only discuss both the distribution of *ni* and its interpretive effects. However, in the literature (and among Yorùbá scholars) their work is often cited when the option of Yorùbá bi-clausal focus structure is discussed.

While a cleft-like focus structure is not impossible (considering the copula clauses with *ni*), there are cross-linguistic studies that show the diachronic development of focus constructions from bi-clausal (cleft-like) to mono-clausal structures. This transition is often associated with the grammaticalization of cleft constructions. This usually happens when a focus marker develops from either the copula of a matrix clause or the relativizer of an embedded relative clause (cf. Heine and Reh, 1984; Harris and Campbell, 1995; Heine and Kuteva, 2002; Schwarz, 2007; Cruschina, 2012; Khan, 2019, a.o.). Khan (2019, p.3) says that in such cases “the boundary between the matrix copula clause and the subordinate clause is lost and the construction becomes a single matrix clause.” In Kikuyu (a Bantu language spoken in Kenya) for instance, Schwarz (2007) argues that a mono-clausal focus phrase analysis is preferred to a cleft analysis. Kikuyu uses ex-situ focalization like the one in (173) below. The focus marker, *ne* precedes both the ex-situ *wh*-phrase and the ex-situ focus answer, respectively.

- (173) a. **Ne- kee** Abdul a- ra nyu- ir- ε.
 FOC- what Abdul SM- T- drink- ASP- FV
 ‘What did Abdul drink?’
 b. **Ne- mae** Abdul a- ra nyu- ir- ε.
 FOC- 6.water Abdul SM- T- drink- ASP- FV
 ‘Abdul drank WATER.’

(adapted & modified from Schwarz, 2007, p.140)

But *ne* obligatorily occurs in a copula construction as well, as in (174).⁵³ Although, this led to a cleft analysis by Bergvall (1987), however, using empirical arguments, Schwarz maintains that focus construction in Kikuyu involves a mono-clausal focus phrase analysis (cf. Clements, 1984; Schwarz, 2003).

- (174) Abdul *(**ne**) ∅ mo- rutani.
 Abdul FOC COP 1- teacher
 ‘Abdul is a teacher.’

There are other languages where both a bi-clausal (thus a cleft) and a monoclausal focus construction co-exist. This is the case of Tilapa Otomi, an endangered language of the Otomi group (an Amerindian language family from Mexico). Palancar (2021) proposes that Tilapa Otomi has, at least, two specialized focus constructions which are bi-clausal and mono-clausal in nature. In fact, the latter “... shares structure with clefts, to the extent that superficially could be taken to be an instance of a cleft with a zero copula.”

Coming back to Yorùbá, there are reasons to believe that the ex-situ focus construction in Yorùbá is mono-clausal. The first evidence has to do with long distance movement out of an embedded clause which contains a fronted focus with *ni*. Since clefts involve relative clauses, we predict a lack of movement out of such relative clauses, since the latter is an island in Yorùbá (175). However, considering the examples in (176), movement (of the

⁵³ Schwarz (2007, p.141), however, shows that *ne* becomes optional once we consider examples that are not in the third person present tense form (i).

- (i) Abdul (**ne**) a- a -re mo- rutani.
 Abdul FOC SM T -be 1- teacher
 ‘Abdul is a teacher.’

embedded object) out of an embedded clause consisting of *ni* is possible. This should be impossible if the *ni* construction is a cleft.

- (175) a. Tolú rí obìnrin tí ó ra bàtà.
 Tolú see woman REL 3SG buy shoe
 ‘Tolú saw the woman who bought a shoe.’
 b. *Bàtà_i ni Tolú rí obìnrin [tí ó rà t_i].
 shoe FOC Tolú see woman REL 3SG buy
 ‘Tolú saw the woman who bought A SHOE.’
- (176) a. Adé mò wípé Tolú ni ó ra bàtà.
 Adé know COMP Tolú FOC 3SG buy shoe
 ‘Adé knows that TOLÚ bought a shoe.’
 b. Bàtà_i ni Adé mò wípé Tolú ni ó ra t_i.
 shoe FOC Adé know COMP Tolú FOC 3SG buy
 ‘Adé knows that TOLÚ bought a SHOE.’

Similarly, it is possible to have a topic preceding an ex-situ focus in the language (177). If it was a cleft construction, the prediction is that this should be impossible because the topic movement would be from a relative clause. This is what the topicalized PP adjunct shows in (178-b). The sentence is ungrammatical because the PP is extracted from a relative clause.⁵⁴

- (177) Ní iwájú ilé, Tolú **ni** ó n ta iṣu.
 LOC front house Tolú FOC 3SG PROG sell yam.
 ‘At the front of the house, TOLÚ is selling yam.’
- (178) a. Ọkùnrin tí ó n ta iṣu ní iwájú ilé wá.
 man REL 3SG PROG sell yam LOC front house come
 ‘The man who sells yam at the front of the house came.’
 b. *Ní iwájú ilé, ọkùnrin tí ó n ta iṣu wá.
 LOC front house man REL 3SG PROG sell yam come
 ‘At the front of the house, the man who sells yam came.’

Another evidence that supports a mono-clausal analysis of Yorùbá focus construction comes from Aremu and Weisser (2024). They show that the presence of (future) tense and (perfective) aspect markers in constructions with *ni* is not permitted (179). If *ni* is a real copula, then this might be surprising because copula and cleft constructions are usually expected to co-occur with tense and aspect (as in Adam is a teacher/Adam *will* be a teacher). Importantly, the examples in (180) show that these temporal and aspect morphemes cannot occur in copular sentence as well. In contrast, the other copula morpheme *jé* co-occurs with tense and aspect (181). This justifies our claim that the focus construction is not a cleft (see Aremu and Weisser, 2024, for further arguments in support of a mono-clausal analysis).

⁵⁴ Although Aremu (2024) argues that topics are base-generated in Yorùbá, however, this seems to only hold for DPs (only topicalized DPs were discussed in the paper). PPs and adverbials seem to involve topic movement instead. That PPs/adverbials and DPs may differ with respect to A'-movement is not new. For instance, in some languages, the realization of the C head differ depending on whether it is a DP that is fronted or it is a PP/adverb. See Keine and Zeijlstra (2025), and the references there, for a discussion on the PP/adverb versus DP extraction asymmetries in Indonesian/Malay (cf. Hsieh, 2020).

- (179) a. *Aṣọ **yóò ni** mo rà
 clothes FUT FOC 1SG buy
 Intended: ‘I will buy CLOTHES.’
 b. *Aṣọ **ti ni** mo rà
 clothes PERF foc 1SG buy
 Intended: ‘I have bought CLOTHES.’
- (180) a. *Olùkó **yóò ni** Adé.
 teacher FUT COP Adé
 ‘Adé will be a teacher.’
 b. *Olùkó **ti ni** Adé.
 teacher PERF COP Adé
 ‘Adé has been a teacher (for some time).’
- (181) a. Adé **yóò jé** olùkó.
 Adé FUT COP teacher
 ‘Adé will be a teacher.’
 b. Adé **ti jé** olùkó.
 Adé PFV COP teacher
 ‘Adé has been a teacher.’

Lastly, we already saw above that copula construction with *ni* does not follow the canonical word order of the language, but *je* does. We saw that the latter can even optionally occur in constructions with *ni* (cf. ex. (172) above).

3.4.5 A proposed analysis

The proposed analysis is that in Yorùbá, the local subject focus, the non-subject focus and long-distance subject focus all occupy the same position in the left periphery: Spec,FocP. Ex-situ focus, whether local subject or non-subject/long-distance subject, are consistently and obligatorily marked with *ni*. Thus, *ni* occupies a single head which is that of FocP. I will begin with a structural analysis for subject focus and then non-subject focus.

Ex-situ subject focus is base-generated

That the local subject focus is realized in an ex-situ position is supported by the presence of a resumptive pronoun. Again, consider the example in (182). It is obvious that the local subject focus *Adé* is not in its canonical A-position, SpecTP. It is dislocated. The question is whether this dislocation is by movement or base-generation.

- (182) [Adé]_F **ni** ó ra bàtà.
 Adé FOC 3SG buy shoe
 ‘ADÉ bought a shoe.’

Both strategies for dislocation have their implications. For instance, if the dislocation is by movement, then this would be a violation of the *Spec-to-Spec Anti-Locality Constraint*: a constraint that prohibits movement that is *too short*, as defined in (183) below (see Erlewine, 2016, 2020)

- (183) **Spec-to-Spec Anti-Locality:**
 A'-movement of a phrase from the Specifier of XP must cross a maximal projection other than XP.

(Erlewine, 2016, p.431)

However, I argue for a base-generation approach for ex-situ local subject focus by applying the Coordinate Structure Constraint (CSC) test.⁵⁵ If one of the conjuncts of a conjoined subject (cf. (184-a)) is extracted, the result is an ungrammatical sentence (184-b). However, when a resumptive pronoun occupies the dislocated conjunct's position, as in (184-c), the sentence becomes grammatical. If indeed the local subject focus involves movement, then the prediction is that resumption should not repair CSC.

(184) **Coordinate Structure Constraint:**

- a. [Adé àti Bólá] jẹ ịṣu.
Adé CONJ Bólá eat yam
'Adé and Bólá ate yam.'
- b. *[Adé]_F **ni** [___ àti Bólá] jẹ ịṣu.
Adé FOC CONJ Bólá eat yam
'ADÉ and Bólá ate yam.'
- c. [Adé]_F **ni** [òun àti Bólá] jẹ ịṣu.
Adé FOC 3SG CONJ Bólá eat yam
'ADÉ and Bólá ate yam.'

A similar result is found with adjunct island, where the resumption of the dislocated subject of the adjunct clause in (185) makes the sentence grammatical (compare (185-c) with (185-b)). The behaviour of dislocated subject in the complex noun phrase construction in (186) also supports the base-generation analysis.

(185) **Adjunct island**

- a. Tolú parí ịṣu [kí Adé tó ra omi].
Tolú finish yam before Adé able buy water
'Tolú finished (eating) yam before Adé bought water.'
- b. *[Adé]_F **ni** Tolú parí ịṣu [kí ___ tó rà omi].
Adé FOC Tolú finish yam before able buy water
Lit. 'It was Adé that Tolú finished (eating) yam before he bought water.'
- c. ?[Adé]_F **ni** Tolú parí ịṣu [kí ó tó rà omi].
Adé FOC Tolú finish yam before 3SG able buy water
Lit. 'It was Adé that Tolú finished (eating) yam before he bought water.'

(186) **Complex Noun Phrase Constraint**

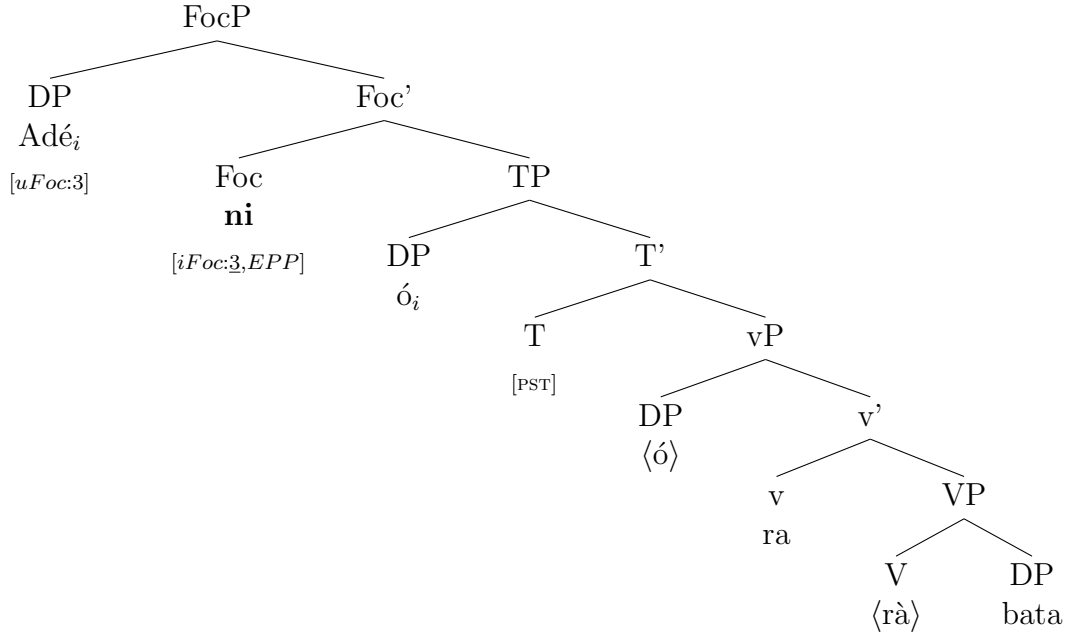
- a. Tolú jẹ oúnjẹ tí Adé rà.
Tolú eat food REL Adé buy
'Tolú ate the food that Adé bought.'
- b. [Adé]_F **ni** Tolú jẹ [oúnjẹ tí *(ó) rà].
Adé FOC Tolú eat food REL 3SG buy
Lit. 'It was Adé that Tolú ate the food that he bought.'

I propose a simple structure for local subject focus in (187). The focused subject is base-generated in Spec,FocP, followed by the focus marker *ni* realizing the focus head. This results in a Spec-head agreement between the focused constituent and the focus head. At the same time, the presence of the focused subject in Spec,FocP satisfies the EPP requirement on the focus head.⁵⁶

⁵⁵ For further arguments which shows that subject focus is a base-generated in the language, see Aremu (2024).

⁵⁶ The EPP feature in the structure in (187) only requires that an element fill the Spec of the focus

(187) **Subject focus structure in Yorùbá**



Looking at the structure in (187), the assumption is that the resumptive pronoun is merged at Spec,vP for theta assignment, and then moves to Spec,TP to satisfy the EPP on T. At this position, it c-commands its trace (in Spec,vP) and also forms a binding relation with the base-generated focused subject in Spec,FocP. This relation is supported by the possibility of phi-feature agreement between the focused subject and its resumptive pronoun. In example (188-a) and (188-b), the resumptive pronouns agree in number with the focused subject that they resume.⁵⁷

- (188) a. [Àwọn ọmọ]_F **ni wọn** ra bàtà.
 PL child FOC 3PL buy shoe
 ‘THE CHILDREN bought a shoe.’
 b. [Adé]_F **ni ó** ra bàtà.
 Adé FOC 3SG buy shoe
 ‘ADÉ bought a shoe.’

The same analysis can be extended to the long-distance subject focus. I assume that it also involves the base-generation of the embedded subject in the left periphery of the matrix clause, and a resumptive pronoun is present in its embedded clause position.⁵⁸

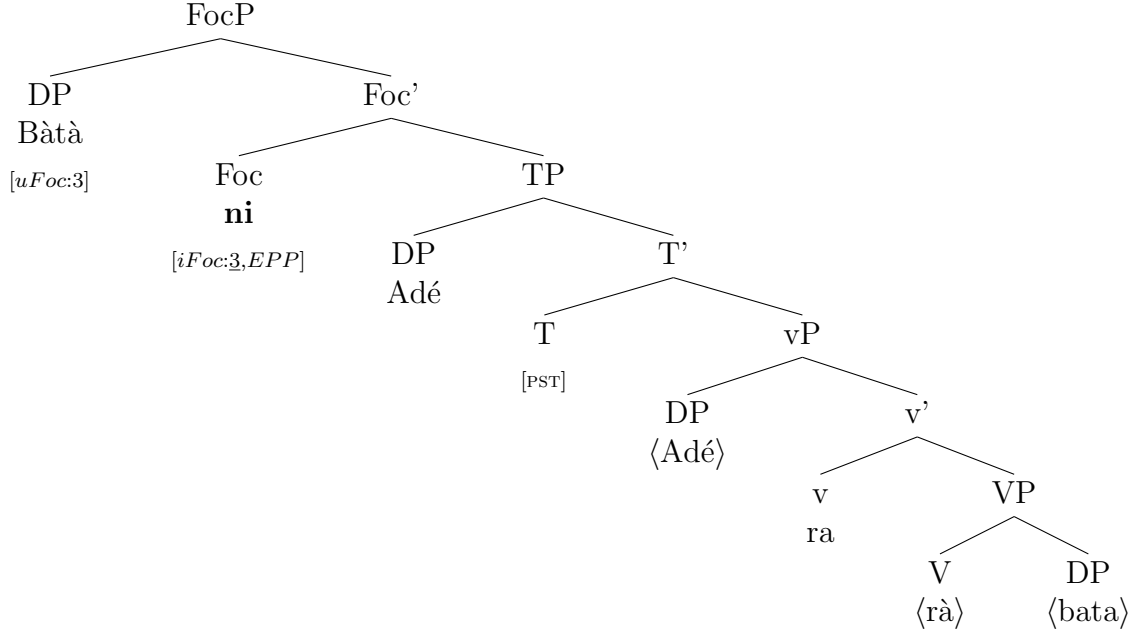
Ex-situ object focus involves movement

For non-subject focus (specifically object focus) on the other hand (see ex. (189)), I assume that the object moves from its base position (the complement of V position) to Spec,FocP after agreeing with the focus head, as in (190).⁵⁹ Again, the movement is triggered by the need to satisfy the EPP feature on Foc.

head. This can be by movement or by base-generation (cf. Landau, 2007).
⁵⁷ Yorùbá also has instances non-agreeing resumptive pronouns. See Adesola’s (2005) thesis on a comprehensive study on agreeing and non-agreeing resumptive pronouns in Yorùbá.
⁵⁸ See Section 3.5 for a further discussion.
⁵⁹ The movement passes through a higher Spec,vP (assuming two Specs for vP) assuming the little vP as a phase.

- (189) [Bàtà]_F **ni** Adé ra.
 shoe FOC Adé buy
 ‘Adé bought A SHOE.’

- (190) **Ex-situ object Focus**



That non-subject focus involves movement (and not base-generation like the subject focus) is evident in its sensitivity to islands. Consider the relative clause island (Complex Noun Phrase Constraint) and the adjunct clause island in (191) and (192). The ungrammaticality of (191-b) and (192-b) shows that extraction from these islands are sensitive to focus movement. Notice that unlike what we find with subject focus (see (184-c) above), resumption does not repair these islands. Sensitivity to island supports our proposal for ex-situ object focus movement.⁶⁰

- (191) **Complex Noun Phrase Constraint**

- a. Tolú rí [ọmọ tí ó jẹ ịṣu].
 Tolú see child REL 3SG eat yam
 ‘Tolú saw the child who ate yam.’
- b. *[Iṣu]_F **ni** Tolú rí [ọmọ tí ó jẹ (é)].
 yam FOC Tolú see child REL 3SG eat 3SG
 Lit. ‘It was yam that Tolú saw the child who ate.’

- (192) **Adjunct island**

- a. Tolú jẹ ịṣu [kí Adé tó ra omi].
 Tolú eat yam before Adé TO buy water
 ‘Tolú ate yam before Adé bought water.’
- b. *[Omi]_F **ni** Tolú jẹ ịṣu [kí Adé tó rà (á)].
 water FOC Tolú eat yam before Adé TO buy 3SG
 Lit. ‘It was Water that Tolú ate yam before Adé bought.’

⁶⁰ The lack of object focus resumption in Yorùbá may be associate with the lack of object criterion going by the Criterial Approach (Rizzi, 1997, 2006; Rizzi and Shlonsky, 2007). Subject criterion, on the other hand, is available, and thus Yorùbá satisfies this criterion with the subject focus resumption (cf. Shlonsky, 2014). See Aremu (2024) for a more detailed discussion on this.

We now turn to the proposal for the in-situ focus construction. Recall that Yorùbá only marks ex-situ focus morphologically (with *ní*). In-situ focus is morphologically unmarked (cf. (193)). In the discussion that follows, I will treat in-situ focus in a similar manner with in-situ *wh*-questions. The only difference here is that unlike in-situ focus answers, in-situ (argument) *wh*-questions obligatorily appear with *ní*.⁶¹

a. Adé ra kí **ni**?
Adé buy what FOC
'What did Adé buy?'

b. Adé ra [bàtà]_F (***ni**).
Adé buy show FOC
'Adé bought A SHOE.'

Some of these studies argue for covert movement of in-situ *wh*-phrases by applying popular movement tests which are used to argue for overt *wh*-movement constructions. Some of these tests include island sensitivity, intervention (blocking) effects, scope, and binding theory. Below, I present evidence from island insensitivity of in-situ *wh*/focus construction in Yorùbá, scope, and binding theory to argue for the absence of LF movement in the language.

(194) **Complex Noun Phrase Constraint for in-situ focus**

- a. Tolú rí [ɔmɔ tí ó jẹ kí **ni** láàná]?
Tolú see child REL 3SG eat what FOC yesterday
'Tolú saw the child who ate what yesterday?'
- b. Tolú rí [ɔmɔ tí ó jẹ [iɕu]_F láàná].
Tolú see child REL 3SG eat yam yesterday
'Tolú saw the child who ate yam yesterday.'

⁶² The result is that at the end of the derivation, both instances of overt *wh*-movement and covert *wh*-movement would produce similar structures at LF (i).

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We get the same result for adjunct island in (195). The presence of the in-situ *wh*-phrase and its focus answer in (195-a) and (195-b) respectively does not violate the adjunct clause island. If both the *wh*-phrase and the focus involve LF movement, then we expect them to be sensitive to adjunct island. But this is not the case.

(195) **Adjunct island for in-situ focus**

- a. Tolú jẹ iṣu [kí Adé tó ra kí **ni** láàná]?
Tolú eat yam before Adé TO buy what FOC yesterday
'Tolú ate yam before Adé bought what yesterday?'
- b. Tolú jẹ iṣu [kí Adé tó ra [omi]_F láàná].
Tolú eat yam before Adé TO buy water yesterday
'Tolú ate yam before Adé bought water yesterday.'

Lastly, the in-situ *wh*-phrase and focus do not violate the coordinate structure constraint (196). It does not matter whether the *wh*-phrase is the first conjunct or the second conjunct (196-a)-(196-b), the structure is preserved. The same result is true of the in-situ focus counterpart (196-c).

(196) **Coordinate Structure Constraint for in-situ focus**

- a. Tolú ra [kí ni àti iṣu]?
Tolú buy what FOC CONJ yam
'Tolú bought what and yam?'
- b. Tolú ra [iṣu àti kí ni]?
Tolú buy yam CONJ what FOC
'Tolú bought yam and what?'
- c. Tolú ra [iṣu_(F) àti èwà_(F)].
Tolú buy yam CONJ beans
'Tolú bought yam and beans.'

The fact that in-situ *wh*-question and focus are insensitive to these islands is evidence against LF movement in Yorùbá.

Similarly, assuming that scope involves (covert) movement, the prediction would be that an in-situ *wh*-phrase or focus should take wide scope above a universal quantifier (cf. Aoun and Li, 1993; Sulemana, 2019). This prediction is not borne out in Yorùbá. The examples in (197) show that both in-situ *wh*-phrase and in-situ focus do not take wide-scope above the universal quantifier *gbogbo èyàn* (everyone). This shows that they do not undergo covert movement.

- (197) a. Gbogbo èyán rí ta ni?
all person see who FOC
'Everyone saw who?'
(i) 'For each person *y*, which person *x* did *y* see?'
(ii) *'Who is *x* such that everyone saw *x*.'
- b. Gbogbo èyán rí [ènikàn]_F.
all person see someone
'Everyone saw someone.'
(i) 'For each person *y*, there is *x* such that *y* saw *x*.'
(ii) *'There exist an individual *x* such that everyone saw *x*.'

The argument from Binding theory was inspired by Nissenbaum (2000). Nissenbaum

links the different grammatical status of the examples in (198) to the possibility of a covert *wh*-phrase movement in (198-a), and not in (198-b). Binding Principle A requires that a reflexive must be bound locally. This is why example (198-b) is ungrammatical because the anaphor and its antecedent are not in the same clause. However, Nissenbaum argues that (198-a) is grammatical because the reflexive is contained inside an in-situ *wh*-phrase which undergoes covert movement. This movement ends up licensing the anaphor. Since (198-b) does not involve such movement, it is ungrammatical. The import of this test is that if the in-situ *wh*-phrase undergoes covert movement, then such movement should feed Principle A of the binding theory (see also Sulemana, 2019). If, on the other hand, the in-situ *wh*-phrase does not undergo covert movement, then Principle A is not fed. This is the case with in-situ *wh*-phrase in Yorùbá.

- (198) a. Who_i thinks Mary was looking at which picture of himself_i?
 b. *Who_i thinks Mary was looking at a picture of himself_i.

(Nissenbaum, 2000, p.126)

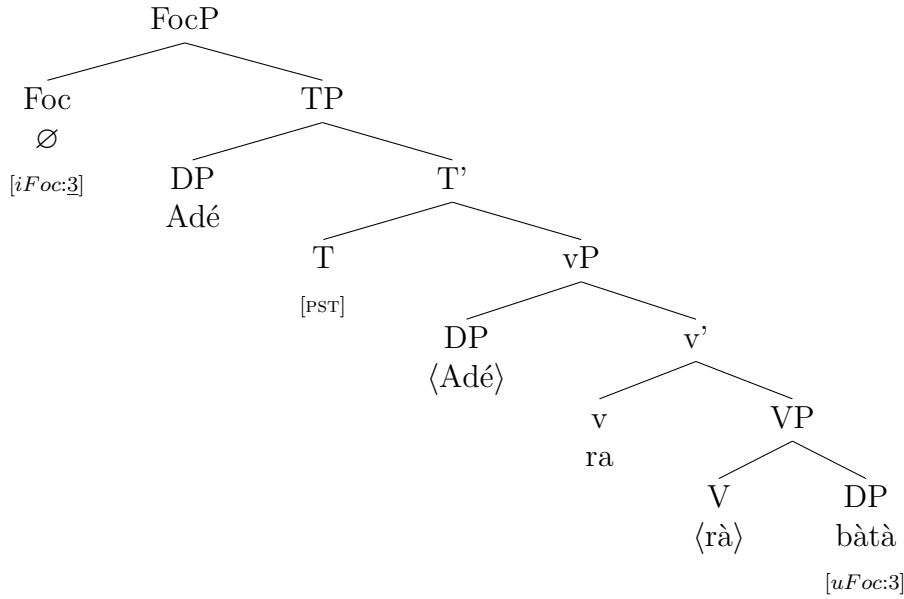
Going by the co-indexation in the examples in (199), the sentence with an in-situ *wh*-phrase in (199-a), as well as the one without a *wh*-phrase (199-b), would be ungrammatical if the antecedent of the reflexive anaphor is the matrix subject. In other words, in both cases, the antecedents of the reflexive anaphor can only be the two potential candidates within the embedded clause: the agent *Bólá*, and the goal *Şégún*. The subject of the matrix clause *Tolú* cannot serve as the antecedent. Thus, the prediction for covert *wh*-movement is not borne out. If the in-situ *wh*-phrase which embeds the reflexive undergoes covert movement, then the anaphor should be able to co-index with the matrix subject *Tolú* in (199-a). This is what Nissenbaum's 'feeding of Principle A by covert *wh*-movement' would predict.

- (199) a. Tolú_i sọ wípé Bólá_j fún Şégún_k ní [àwòrán ara a rẹ_{*i/j/k} wo]?
 Tolú say COMP Bólá give Şégún SEC picture body HTS 3SG.REFL
 which
 'Tolú said that Bólá gave Şégún which picture of himself?'
 b. Tolú_i sọ wípé Bólá_j fún Şégún_k ní [àwòrán ara a rẹ_{*i/j/k}].
 Tolú say COMP Bólá give Şégún SEC picture body HTS 3SG.REFL
 'Tolú said that Bólá gave Şégún a picture of himself?'

Based on the evidence above, I argue that in-situ *wh*-question and focus constructions in Yorùbá do not involve covert movement. Therefore, my proposal for in-situ focus constructions in Yorùbá is that there is a null focus head in the left periphery that only Agrees with the focus constituent without involving any form of LF movement. I assume that the lack of overt movement is due to the absence of an EPP feature on the focus head. I will return to the formalization of a focus head with EPP (as in the ex-situ focus cases) and the one without EPP below. The structure for the in-situ object focus in (200) is given in (201) below. The interpretable null focus head probes for a valued focus phrase. Since there is no EPP, the derivation stops at probing for valuation, with the direct object interpreted as focus in the in-situ position.

- (200) Adé ra [bàtà]_F.
 Adé buy show
 'Adé bought A SHOE.'

(201) **In-situ object focus structure**



The assumption here is that in ex-situ cases, the overt realization of the focus head *ni* is licensed by the presence of an EPP. In other words, while *ni* is specified for interpretable focus feature and an EPP, a null focus head is specified for only an interpretable focus feature. Thus, the setting of this parameter is based on the presence or absence of EPP on Foc (cf. Bonan, 2021). See the formalization in (202). The summary is that when the focus head is morphologically realized, then it must contain a movement-inducing feature, and when it is covertly realized, it does not contain a movement-inducing feature. With this, we can account for the distinction between the focus head and the focus realization strategies (i.e. in-situ and ex-situ).

(202) **Different realization of the Foc head**

- a. Foc:[iFoc:__, EPP] $\leftrightarrow$ *ni*
- b. Foc:[iFoc:__] $\leftrightarrow$ $\emptyset$

Before concluding this chapter, in the next section, I briefly discuss the use of resumptive pronouns in subject focus contexts in the three languages under investigation.

3.5 Resumption in the three languages

As discussed above, while Yorùbá uses resumptive pronouns for both local and non-local subject focus, Kasem and Kusaal only use resumptive pronouns for long-distance (non-local) subject focus. A local subject focus is not resumed in Kasem and Kusaal. The phenomenon of resumption has received various analysis in different languages. See, among many others, McCloskey (1990, 2002); Korsah and Murphy (2019); Amaechi and Georgi (2019, 2024); Georgi and Amaechi (2020); Hein and Georgi (2021); Aremu (2021); Scott (2021); Jarvis (2025). Since this will not be an integral part of the overall discussion in this dissertation, I only provide a sketchy explanation of the use of resumption in these languages.

3.5.1 Resumption as a repair strategy in Yorùbá

Recall that, in Yorùbá, resumption does not only occur in the canonical argument position of the dislocated local focused subject. Non-local focused subject that is realized in the left periphery of the matrix clause is also resumed in the embedded canonical subject position, as in (203). I also showed that the two elements can agree in number feature. The singular embedded focused subject in (203-a) agrees in number feature with the resumptive pronoun that occupies the subject position (Spec,TP) in the embedded clause. The other example in (203-b) shows a plural agreement between the embedded focused subject and its resumptive pronoun (cf. Adesola, 2005, 2010).

(203) **Long distance focused subject in Yorùbá**

QUD: Who does Bólá think bought a shoe?

- a. [Adé]_F **ni** Bólá rò wípé *(ó) rà bàtà.
Adé FOC Bólá think COMP 3SG buy shoe
'Bólá thinks that ADÉ bought a shoe.'
- b. [Adé àti Tolú]_F **ni** Bólá rò wípé *(wón) rà bàtà.
Adé CONJ Tolú FOC Bólá think COMP 3SG buy shoe
'Bólá thinks that ADÉ AND TOLÚ bought a shoe.'

The assumption is that the use of the resumptive pronoun in the embedded clause is to repair what would otherwise have been a *that-trace effect* (see Perlmutter, 1968, 1971; Bresnan, 1977; Chomsky and Lasnik, 1977; Pesetsky, 1982, 2017; Rizzi, 1990, 2001; Pesetsky and Torrego, 2001; Rizzi and Shlonsky, 2007; Bayer and Salzmann, 2013, a.o.). In English, the *that-trace effect* is observed in examples such as (204-a), where the trace of an extracted *wh*-question follows the complementizer *that*. This effect is not available for long-distance extraction of non-subjects, as in (204-b).

- (204) a. *Who_i did Adé say that t_i bought a shoe?
b. What_i did Adé say that Sam bought t_i?

The effect is obviated if the complementizer is covertly realized, as in (205).⁶³

- (205) Who_i did Adé say t_i saw Daniel?

In Yorùbá, on the other hand, the complementizer *wípé* cannot be covert. It is obligatorily overt. So, the resumptive pronoun in the embedded clause is a last resort strategy in a context where a gap is prohibited (cf. McCloskey, 2006; Sichel, 2014).⁶⁴

⁶³ It has also been argued that the *that-trace effect* is suspended at the introduction of intervening materials between the complementizer and the trace. This is exemplified in (i).

- (i) Who_i did Adé say that [after many years] t_i saw Daniel?

⁶⁴ An alternative analysis is the Criterial freezing approach (Rizzi, 2006; Belletti and Rizzi, 2017). Here, the assumption is that the subject position is a criterial position, and once a subject moves to this position, it is 'frozen in place', and can no longer move further. However, languages have ways of 'escaping' this criterial position. One of the ways, according to Rizzi and Shlonsky (2007), is via resumption. Thus, we may assume that the resumptive pronoun in Yorùbá is a repair strategy for the subject criterion, and since there is no object criterion, there is no need for an object resumption. See Aremu (2024) for an analysis in this direction. I leave a thorough investigation of resumption in Yorùbá for the future (see Adesola, 2005, 2010, for some discussions).

There are two properties that are typical of a base-generation derivation of long distance A'-dependency (Georgi and Amaechi, 2023; Amaechi and Georgi, 2024). These are (a) the use of an agreeing resumptive pronoun, and (b) the use of epithet in the place of the resumptive. Thus, the prediction is that a non-agreeing resumption is most likely a result of an A'-movement (and not base-generation), and secondly, the ungrammaticality of an epithet precludes base-generation (and not movement). In Yorùbá, the prediction in support of a base-generation approach is borne out, i.e., both the use of agreeing resumptive pronouns and epithet are available.

For the first property, the examples in (203) above show phi-feature agreement between the embedded subject and the resumptive pronoun. In (206), example (206-a) only serves as the baseline. Example (206-b) involves the grammatical use of epithet.

(206) **Long distance focused subject with epithet in Yorùbá**

- a. Bólá rò wípé Adé rà bàtà.
Bólá think COMP Adé buy shoe
'Bólá thinks that Adé bought a shoe.'
- b. [Adé]_F **ni** Bólá rò wípé [ò̀mùgò ara rẹ̀] rà bàtà.
Adé FOC Bólá think COMP fool body self buy shoe
'Bólá thinks that ADÉ bought a shoe.'

This is further supported by topicalization in Yorùbá which has been argued to involve base-generation in the language (see Aremu, 2024). The embedded subject contrastive topic (CT) in (207-b) can occur with an agreeing resumptive pronoun or an epithet phrase.

(207) **Long distance subject topic with epithet in Yorùbá**

- a. Bólá rò wípé Adé àti Tolú rà bàtà.
Bólá think COMP Adé CONJ Tolú buy shoe
'Bólá thinks that Adé and Tolú bought a shoe.'
- b. [Ní ti Adé àti Tolú]_{CT}, Bólá rò wípé [wọn/ò̀mùgò ara wọn]
as for Adé CONJ Tolú Bólá think COMP 3PL/fool body 3PL
rà bàtà.
buy shoe
'As for Adé and Tolú, Bólá thinks that they/the fools bought a shoe.'

For the local focused subject resumption in Yorùbá, see Section 3.4.5, especially the structure in (187). The use of epithet is also grammatical in local focused subject context, as in (208).

(208) **Local subject focus with epithet in Yorùbá**

- a. Adé rà bàtà.
Adé buy shoe
'Adé bought a shoe.'
- b. [Adé]_F **ni** [ò̀mùgò ara rẹ̀] rà bàtà.
Adé FOC fool body self buy shoe
Lit. 'ADÉ, the fool, bought a shoe.'

3.5.2 Resumption as a repair strategy in Kasem and Kusaal

Recall that in Kasem and Kusaal, the resumptive pronoun is only used in long-distance cases. In the local subject focus examples below, the extracted local subjects are not resumed in Kasem (209-b) and Kusaal (210-b), respectively.

(209) **Local subject focus in Kasem**

- a. Wɔ **mo** go chworo kom?
 who FOC kill fowl DEF
 ‘Who slaughtered the fowl?’
- b. [Adam]_F **mo** ___ go chworo kom.
 Adam FOC kill fowl DEF
 ‘ADAM slaughtered the fowl.’

(210) **Local subject focus in Kusaal**

- a. Anɔɔn-(ɛ) kɔdig nua la?
 who-FOC slaughter fowl DEF
 ‘Who slaughtered the fowl?’
- b. [Azuure]_F **n** ___ kɔdig nua la.
 Azuure FOC slaughter fowl DEF
 ‘AZUURE slaughtered the fowl.’

However, in long distance dependencies, an embedded local subject focus is resumed in the embedded subject position. This is shown for Kasem (211-b) and Kusaal (212-b) respectively. Similarly to Yorùbá, the use of resumption in long distance subject focus extraction in Kasem and Kusaal is also a strategy to circumvent the *that*-trace effect. The subject position after the complementizer cannot be a gap. The resumptive pronoun satisfies both the theta criterion of the embedded verb, as well as the EPP requirement on the embedded T. Following what Issah and Smith (2020) propose for Dagbani, I assume that long distance subject extraction in Kasem and Kusaal involves base-generation of the focused subject in the left periphery. It then binds the resumptive pronoun that is now occupying the embedded subject position (embedded Spec,TP). The binding relation is derived by Principle B of the binding principle, which allows binding between an anaphoric pronoun and an antecedent that are not clause-mate.

(211) **Long distance focused subject in Kasem**

- a. Wɔ **mo** Peter boje we *(o) go chworo kom?
 who FOC Peter think COMP 3SG kill fowl DEF
 ‘Who did Peter think slaughtered the fowl?’
- b. [Adam]_F **mo** Peter boje we *(o) go chworo kom.
 Adam FOC Peter think COMP 3SG kill fowl DEF
 ‘Peter said that ADAM slaughtered the fowl.’

(212) **Long distance focused subject in Kusaal**

- a. Anɔ’ɔn **ka** Peter tɛn’ɛs ye *(o) kɔdig nua la?
 who FOC Peter think COMP 3SG kill fowl DEF
 ‘Who did Peter think slaughtered a fowl?’
- b. [Adam]_F **ka** Peter tɛn’ɛs ye *(o) kɔdig nua la.
 Adam FOC Peter think COMP 3SG kill fowl DEF
 ‘Peter thought that ADAM slaughtered a fowl.’

Similarly to Yorùbá, the resumptive pronoun also agrees in phi-feature with the base-generated embedded subject focus in Kasem and Kusaal. The example in (213-b) is from Kasem, and it shows agreement in number feature between the embedded plural subject focus (in the left periphery of the matrix clause) and its resumptive pronoun. The use of a singular resumptive pronoun is disallowed. The same thing is replicated in Kusaal, see (213-b).

(213) **Long distance focused subject in Kasem and phi-agreement**

- a. Who did Peter think slaughtered the fowl?
- b. [Adam de Sam]_F **mo** Peter boje we **ba/*o** go chworo kom.
Adam CONJ Sam FOC Perter think COMP 3PL/3SG kill fowl DEF
'Peter said that ADAM AND SAM slaughtered the fowl.'

(214) **Long distance focused subject in Kusaal and phi-agreement**

- a. Who did Peter think slaughtered a fowl?
- b. [Adam nɛ Sam]_F **ka** Peter tɛn'ɛs ye **ba/*o** kɔdig nua la.
Adam CONJ Sam FOC Peter think COMP 3PL/3SG kill fowl DEF
'Peter thought that ADAM AND SAM slaughtered a fowl.'
- c. [Tinam]_F **ka** Peter tɛn'ɛs ye **ti** kɔdig nua la.
1PL.EMPH FOC Peter think COMP 1PL kill fowl DEF
'Peter thought that WE slaughtered a fowl.'

The extraction of long distance object foci does not make use of resumption because it does not involve a *that*-trace effect.⁶⁵ It is fine to leave the embedded object focus position as a gap. This is shown in (215-b) for Kasem, and (216-b) for Kusaal.

(215) **Long distance focused object in Kasem**

- a. What did Peter think that Adam slaughtered?
- b. [Chworo]_F **mo** Peter boje we Adam go ____.
fowl FOC Perter think COMP Adam kill
'Peter thought that Adam slaughtered A FOWL.'

(216) **Long distance focused object in Kusaal**

- a. What did Peter think that Adam slaughtered?
- b. [Nua]_F **ka** Peter tɛn'ɛs ye Adam kɔdig ____.
fowl FOC Peter think COMP Adam kill
'Peter thought that Adam slaughtered A FOWL.'

Apart from the use of resumption, there are other repair strategies (for the *that*-trace effect) that have been discussed in the literature. Some of them include: complementizer drop in Levantine Arabic (also in English, as in (205)) (Kenstowicz, 1989) and Igbo (Amaechi and Georgi, 2024), the use of different forms of the complementizer (*que* vs. *qui*) in French (Perlmutter, 1968, 1971) and (*sí*, instead of the regular *nà*) in Igbo (Amaechi and Georgi, 2024). See Rizzi and Shlonsky (2007) for some other strategies.

In conclusion, different underlying derivations have been proposed for resumption as a repair strategy for long distance subject focus. For example, the proposals have involved, a base-generation approach (Issah and Smith, 2020), an A'-movement approach

⁶⁵ Based on the Empty Category Principle (ECP) (Chomsky, 1982), it is argued that the trace position of the extracted object focus is properly governed. Thus, there is no need for a repair strategy. See Rizzi (1990, 2006) for an explanation based on Relativized Minimality.

(McCloskey, 2002; Kandybowicz, 2009; Erlewine and Lim, 2023), and a prolepsis approach (Amaechi and Georgi, 2024). Whether this is a repair strategy of a PF-constraint or a syntactic constraint is a discussion for a future occasion (cf. Perlmutter, 1968; Kandybowicz, 2006, 2008; Amaechi and Georgi, 2024; Rizzi and Shlonsky, 2007; Pesetsky, 2017, a.o.).⁶⁶ Overall, this section only introduces the phenomenon of resumption in Yorùbá, Kasem and Kusaal. I do not, in any way, claim to have analyzed the phenomenon in these languages. In fact, the aim is to inspire further research into this area of these languages. For example, in Yorùbá, apart from the agreeing resumptive pronouns, there are also cases of non-agreeing resumptive pronouns discussed by Adesola (2005, 2010). Similarly, a deeper investigation of Kasem and Kusaal might present us with more data that might guide our analysis in the future.

3.6 A typological summary

This chapter set out to achieve two goals. The first was to describe the focus realization strategies employed by the languages under the current study, and the second was to provide analyses for the observed strategies: Kasem (§3.2), Kusaal (§3.3) and Yorùbá (§3.4). The table 3.2 below summarizes the major findings and proposals in this chapter, where XP = focus constituent.

	Ex-situ	In-situ
Kasem	$[_{\text{FocP}}[_{\text{XP}_F} mo] \text{ FOC } [_{\text{TP}} \dots]]$	$[_{\text{FocP}} \text{ FOC } [_{\text{TP}} \dots [_{\text{XP}_F} mo]]]$
Kusaal	$[_{\text{FocP}}[_{\text{XP}_F}] \text{ FOC}=ka [_{\text{TP}} \dots]]$	$[_{\text{FocP}} \text{ FOC } [_{\text{TP}} \dots [n\epsilon \text{ XP}_F]]]$
Yorùbá	$[_{\text{FocP}}[_{\text{XP}_F}] \text{ FOC}=ni [_{\text{TP}} \dots]]$	—

Table 3.2: Overview of the focus markers and their positions in Kasem, Kusaal, & Yorùbá

I have shown that the three languages make use of the ex-situ focus marking strategy with dedicated focus markers. However, the relation between the displaced focus constituent and the focus markers varies among the languages. While Kusaal and Yorùbá pattern similarly in that their ex-situ focus markers, *ka/n* and *ni* respectively, realize the head of FocP in the left periphery, the focus marker *mo* in Kasem is argued to form a constituent with the focus constituent not only in the in-situ position, but also in the ex-situ position. This explains the ex-situ column in table 3.2. The interesting fact here is that the two Mabia languages (Kasem and Kusaal) show a clear difference with respect to the positions of their focus markers in the ex-situ focus contexts.

For the in-situ strategy on the other hand, the two Mabia languages share a lot in common. Unlike Yorùbá, Kasem and Kusaal mark in-situ focus morphologically. In Kasem and Kusaal, the focus marker forms a constituent with the focus element (a common property shared with Mə̀dumba, see Zimmermann and Kouankem, 2024). There were only two noticeable differences between them. First, while the focus marker *nε* precedes the focus constituent in Kusaal, *mo* follows the focus constituent in Kasem (see the in-situ column of table 3.2). Secondly, the form of the in-situ focus marker in Kusaal is different from the one used in the ex-situ focus, *ka/n*. In Kasem, however, the form does not change. *Mo* is used in both in-situ and ex-situ focus.

⁶⁶ For Yorùbá, Adesola (2010) argues that resumption is used because of the presence of an EPP on T, which is phonological in nature.

Furthermore, I showed that the phenomena of *anti-pied-piping* and *pied-piping* are present in all the three languages. This is when (a) the morphologically focus marked constituent is contained within the logical focus (217-a), or (b) the logical focus is contained within the morphologically focus marked constituent (217-b) (cf. Fanselow and Lenertová, 2011; Branen and Erlewine, 2023). While Kasem and Kusaal show cases of TP and VP anti-pied-piping, Yorùbá only has TP anti-pied-piping.

- (217) a. $[_{YP} Y XP_{PRT} \dots]_F$
 b. $[_{YP} Y XP_F \dots]_{PRT}$

I propose that such (anti-)pied-piping cases can be captured by leaning on the syntax-pragmatic interface. This requires that we assume, based on the QUD approach, that at least one of the focus alternatives must be the logical focus or answer to a current question under discussion. So, in Kasem and Kusaal, where in-situ focus is morphologically marked, the focus markers (*mo* in Kasem, and *nɛ* in Kusaal) are supposed to be exponents of the focus feature on the focus constituents. However, due to language-specific properties such markings can either be contained within the logical focus or on a constituent that contains the logical focus. In Yorùbá, on the other hand, we saw that the absence of an in-situ morphological focus marking makes it difficult to know whether there is (for example) a VP anti-pied-piping or not. But we saw that even the option of fronting the sub-constituent (e.g. the direct object) of the VP is not felicitous with VP focus (see ex. (166-c) above); an option that is available in Kasem and Kusaal.

In the next chapter (and the ones that follow), we will now discuss association with focus in the three languages. We will concentrate on the focus-sensitive operators that Beaver and Clark (2008) classified as the conventional focus-sensitive operators: *only*, *even* and *also* (see §2.2.1 for a brief introduction).

Chapter 4

Association with exclusive focus operators

In this chapter, I discuss the different properties of the exclusive focus sensitive operators in Kasem, Kusaal, and Yorùbá. I show that the three languages distinguish between adfocal exclusive particles and adverbial exclusive operators. In the three languages, I show that in addition to individual occurrences, exclusive operators can be doubled (i.e. adfocal and adverbial exclusive doubling) in the same sentence, resulting in a single exclusive semantic interpretation. These are discussed in Sections 4.1: Kasem, 4.2: Kusaal, and 4.3: Yorùbá, respectively. In Section 4.4, I discuss the scalar property of exclusives in the three languages, and provide data which supports the scalar analysis of exclusives. After that, in Section 4.5, I discuss the behaviour of the exclusive operators with respect to locality and clause-boundedness in the languages, and argue that while the exclusives can scope across embedded clauses, they cannot scope out of an island, and expectedly so, because of the restriction that islands place on syntactic or semantic operations.

I wish to note here that I will not cover exclusives whose scope is within the noun phrase (NP-*only*) (1). Here, my focus is on the exclusive operator *only* whose scope is propositional (2). However, for a discussion on the different exclusive elements and the semantic types of their focus associates, see Coppock and Beaver (2014).

- (1) a. The students met the *only* professor in the department.
b. The *only* child of his friend is graduating.
- (2) Adam *only* ate RICE.

4.1 Exclusive association with focus in Kasem

This section discusses the properties of the exclusive focus sensitive particles *yerane* and *weeni* in Kasem. I show that while *yerane* is adfocal, i.e., it adjoins to the focus associate, *weeni*, on the other hand, functions as an adverbial exclusive operator. The equivalents of both operators have the same form in English: *only*. Thus, Kasem provides clear cross-linguistic evidence for the use of different forms for adfocal-*only* and adverbial-*only* respectively.¹

¹ There are already studies that show morphological difference between adfocal-*only* and sentential-*only*. Among such languages are Ngamo (West Chadic) (Grubic, 2015); Mandarin Chinese (Sun, 2021); Vietnamese (Erlewine, 2017); Cantonese (Yip, 2025), and others.

4.1.1 Adfocal-*yerane*

The position of the adfocal exclusive particle *yerane* within a sentence is dependent on the focused constituent of the sentence. For example, if a direct object is in focus, then *yerane* follows (adjoins to) it whether in an in-situ position (3-a), or an ex-situ position (3-b). Notice the position of *yerane* which is between its focus associate and the focus particle *mo*. This supports *yerane*'s adfocal nature.² Additional evidence comes from the ex-situ example in (3-b) which shows that *yerane* is fronted with its focus associate to the left periphery of the clause. This is only possible if *yerane* directly adjoins to the focus associate, otherwise, the construction in (4), where *yerane* and/or the focus marker *mo* are stranded, should be grammatical. However, these constructions are ungrammatical. For the interpretation, the sentences in (3) are true iff Adam slaughtered only *x*, and *x* is a fowl.

- (3) a. Adam go [chworō]_F **yerane** *(**mo**).
 Adam kill.PFV fowl only FOC
 'Adam slaughtered only A FOWL.'
 b. [Chworō]_F **yerane** *(**mo**) Adam go-a.
 fowl only FOC Adam kill-DISJ
 'Adam slaughtered only A FOWL.'
- (4) a. *[Chworō]_F **mo** Adam go **yerane**.
 fowl FOC Adam kill.PFV only
 'Adam slaughtered only A FOWL.'
 b. *[Chworō]_F Adam go **yerane mo**.
 fowl Adam kill.PFV only FOC
 'Adam slaughtered only A FOWL.'

In a ditransitive construction, *yerane* can adjoin to either the direct object or the indirect object, depending on which one is focused. In each of the examples in (5) and (6), the sentences in (5-a) and (6-a) serve as the given context, while (5-b) and (6-b) show adfocal-*yerane* adjoining to the DO focus and the IO focus respectively.

- (5) a. I think that Adam gave the money and the bag to John yesterday.
 b. Aawo, Adam pe [sabu kom]_F **yerane mo** John diim.
 no Adam give money DEF only FOC John yesterday
 'No, Adam gave only THE MONEY to John yesterday.' (He didn't give him the bag.)
- (6) a. I think that Adam gave the money to John and Sam yesterday.
 b. Aawo, Adam pe sabu kom [John]_F **yerane mo** diim.
 no Adam give money DEF John only FOC yesterday
 'No, Adam gave the money only to JOHN yesterday.' (He didn't give the money to Sam.)

The following examples show that DO/IO focus with adfocal-*yerane* can also be realized in an ex-situ position (see (7-a) and (7-b)).

² Recall that in the last chapter, I already argued that *mo* forms a constituent with the focus. So, having the focus sensitive particle *yerane* occurring between the focus and *mo* is only possible if the former also forms a complex constituent with the focus.

- (7) a. [Sabu kom]_F **yerane mo** Adam pɛ ____ John diim.
 money DEF only FOC Adam give John yesterday
 ‘Adam gave only THE MONEY to John yesterday.’
 b. [John]_F **yerane mo** Adam pɛ sabu kom ____ diim.
 John only FOC Adam give money DEF yesterday
 ‘Adam gave the money only to JOHN yesterday.’

The sentence in (8) exemplifies association with subject focus in Kasem. The focus sensitive particle *yerane* and the focus marker *mo* follow the focused subject associate. The sentence is true iff only *y* slaughtered a fowl, and *y* is Adam. So, a continuation such as ‘Sam also slaughtered a fowl’ would be infelicitous.

- (8) [Adam]_F **yerane** *(mo) go chworo.
 Adam only FOC kill.PFV fowl
 ‘Only ADAM slaughtered a fowl.’

The adfocal property of *yerane* is also seen in its occurrence with focused prepositional phrases as modifiers (9-a) and adverbial modifiers (9-b). In both cases, *yerane* forms a constituent with the focus associate. This is even more evident in (9-c) where *yerane* is scrambled together with the adverbial modifier to the pre-verbal domain. PP-modifiers do not have such a possibility in Kasem. They are rigid with respect to their post-verbal or clause-final position.³ The sentence in (9-a) presupposes that, in a given context, Adam did not eat rice at any other place (e.g., in the market, or in the house), but only in the farm.

- (9) a. Adam di mumuna [gaa kom ne]_F **yerane mo**.
 Adam eat rice farm DET in only FOC
 ‘Adam ate rice only IN THE FARM.’
 b. Adam di mumuna [diim]_F **yerane mo**.
 Adam eat rice yesterday only FOC
 ‘Adam ate rice only YESTERDAY.’
 c. Adam [diim]_F **yerane mo** di mumuna.
 Adam yesterday only FOC eat rice
 ‘Adam ate rice only YESTERDAY.’

It is also important to point out here that the focus marker *mo* cannot precede *yerane* as shown with association with V(P) focus in (10) below. (10) is an intransitive construction. However, if we have a transitive construction, then the position of *yerane* presents us with an ambiguous sentence where the focus associate could be the entire VP or the direct object (11).

³ While adverbial modifiers can scramble to the pre-verbal position, PP-modifiers cannot occupy a pre-verbal position in Kasem. Compare (i) and (ii) below.

- | | | | |
|-----|---|------|--|
| (i) | a. Adam di mumuna diim .
Adam eat rice yesterday
‘Adam ate rice yesterday.’
b. Adam diim di mumuna.
Adam yesterday eat rice
‘Adam ate rice yesterday.’ | (ii) | a. Adam di mumuna de foone .
Adam eat rice with fear
‘Adam ate rice with fear.’
b. *Adam de foone di mumuna.
Adam with fear eat rice
‘Adam ate rice with fear.’ |
|-----|---|------|--|

- (10) Adam [toŋe]_F (*mo) **yerane** *(mo).
 Adam work.PFV FOC only_{adf} FOC
 ‘Adam only WORKED.’
- (11) Adam [go [chworɔ]_F]_F **yerane** *(mo).
 Adam kill.PFV fowl only_{adf} FOC
 ‘Adam (only) [SLAUGHTERED (only) [A FOWL]].’

4.1.2 Adverbial/sentential-*weeni*

Similar to English where *only* can occur in the pre-verbal position (12), Kasem also has a pre-verbal *only*, or what has been called the sentential-*only*. The form, however, is different from that of the adfocal particle. The sentential-*only* in Kasem is *weeni* (cf. (13)). The constructions with *weeni* are unique in that they seem to be more complex than normal mono-clausal structures. If we consider the examples in (13), apart from the logical subject that precedes *weeni*, there is a pronoun that follows *weeni*.⁴ Interestingly, this pronoun agrees in number and person feature with the subject. Compare (13-a) and (13-b).

- (12) Adam **only** ate YAM.
- (13) a. Adam **weeni** *(o) go [chworɔ]_F *(mo).
 Adam only 3SG kill.PFV fowl FOC
 ‘Adam slaughtered only A FOWL.’
- b. Bia bam **weeni** *(ba) go [chworɔ]_F *(mo).
 children DEF.PL only 3PL kill.PFV fowl FOC
 ‘The children slaughtered only A FOWL.’

The structural position of *weeni* is always preverbal. This is regardless of which constituent is the focus associate. For instance, in the ditransitive construction below (14), the focus can be the direct object (14-a), the indirect object (14-b) or even the adverb (14-c), respectively. The choice of which constituent is the focus associate is easily accessible because, as we have seen from the last chapter (cf.3.2), the language employs an in-situ morphological focus marker *mo* which forms a constituent with the focus associate. So, *weeni* can interact with its focus associate at a distance.

- (14) a. Adam **weeni** *(o) pɛ [tɔnɔ]_F **mo** John diim.
 Adam only 3SG give cup FOC John yesterday
 ‘Adam gave only A CUP to John yesterday (and nothing else).’
- b. Adam **weeni** *(o) pɛ tɔnɔ [John]_F **mo** diim.
 Adam only 3SG give cup John FOC yesterday
 ‘Adam gave a cup to only JOHN yesterday (and no one else).’
- c. Adam **weeni** *(o) pɛ tɔnɔ John [diim]_F **mo**.
 Adam only 3SG give cup John yesterday FOC
 ‘Adam gave a cup to John only YESTERDAY (and not any other day).’

With respect to subject focus, Kasem presents interesting data that appear to involve what has been called *Backward Association with Focus* (BAwF) in the literature (see Jackendoff, 1972; Beaver and Clark, 2008; Erlewine, 2014b; Hirsch and Wagner, 2019, among others). This is when the adverbial focus-sensitive operator associates with a focus

⁴ I will briefly discuss the structure of *weeni*-clauses in section 4.1.4 below.

constituent that is not c-commanded by the operator in the surface structure.⁵ Consider the sentence in (15). Based on this example, adverbial-*weeni* seems to associate with the subject focus. I will return to the phenomenon of BAwF in Chapter 5 (see Section 5.5 specifically).

- (15) [Adam]_F **mo weeni** *(o) go chworo diim.
 Adam FOC only 3SG kill.PFV fowl yesterday
 ‘Only ADAM slaughtered a fowl yesterday (no one else did).’

In each of these focus sensitive contexts, exclusive association requires that the pre-jacent containing the focus associates is the only true alternative among the set of alternatives.⁶ Thus, we can tentatively say that the exclusive particles (adfocal-*yerani* and adverbial-*weeni*) in Kasem (just like the English *only*) are truth-conditional operators that demand truthfulness of their propositions, while all other relevant alternative propositions are false. I will refine this position in the next Chapter 5.

4.1.3 Exclusive doubling cases

Based on the discussion so far, we might want to ask whether (a) both the adverbial exclusive particle *weeni* and the adfocal exclusive particle *yerane* can co-occur (i.e. be doubled), and (b) we then can still get a single exclusive interpretation (similar with negative concord cases). This is in fact possible. Adverbial-*weeni* can co-occur with adfocal-*yerane*, and interestingly, the structure gets a single exclusive interpretation. Such focus concord phenomena have been observed in quite a number of languages. Some of these languages include German (Hole, 2015; Bayer, 2020), Dutch (Barbiers, 2014), Korean (Lee, 2005), Cantonese (Lee, 2019; Yip, 2025), Mandarin (Hole, 2017; Sun, 2021), and Vietnamese (Erlewine, 2017; Hole, 2017). For instance, Erlewine (2017, p.133) provided the example of exclusive doubling in Vietnamese in (16). Both (16-a) and (16-b) are instances of the individual occurrence of both adverbial-*chi* and adfocal-*moi* respectively. Importantly, in both cases, the particles associate with the object focus, and the structure is said to produce a single exclusive reading.

- (16) **Vietnamese single and double occurrence of adv. & adf. *onlys***
- a. Nam **chi** mua [cuon sách]_F.
 Nam only buy CL book
 ‘Nam only bought THE BOOK.’
 - b. Nam mua **moi** [cuon sách]_F.
 Nam buy only_{adf} CL book
 ‘Nam only bought THE BOOK.’
 - c. Nam **chi** mua **moi** [cuon sách]_F.
 Nam only buy only_{adf} CL book
 ‘Nam only bought THE BOOK.’

In the discussion that follows, I present more data from Kasem showing the single occurrence of the adverbial particle and the adfocal particle respectively, and importantly examples which shows the co-occurrence of both of them. In (17), assuming speaker A

⁵ There is a separate section 5.5 for the discussion of BAwF in the languages under the current study.

⁶ By pre-jacent, I mean the sentence without the presence of the focus-sensitive operator *only*. I return to this in the next Chapter 5.

utters the sentence in (17-a), speaker B could reply with any of the sentences from (17-b)-(17-d). (17-d) presents us with a sentence where both adverbial-*weeni* and adfocal-*yerane* co-occur. All three options (adfocal, adverbial, and doubling) produce a single exclusive reading where it is not the case that Adam married a Gonja woman or any other woman for that matter. He married only a Kasem woman.

- (17) a. A boje we Adam swo Gonja bukɔ de Kasena bukɔ
 1.SG think COMP Adam marry Gonja woman CONJ Kasem woman
 maama.
 both
 ‘I think that Adam married both a Gonja woman and a Kasem woman.’
 b. Aawo, Adam swo [Kasena]_F bukɔ **yerane mo**.
 no Adam marry Kasem woman only_{adf} FOC
 ‘No, Adam only married a KASEM woman.’
 c. Aawo, Adam **weeni** *(o) swo [Kasena]_F bukɔ **mo**.
 no Adam only 3SG marry Kasem woman FOC
 ‘No, Adam only married a KASEM woman.’
 d. Aawo, Adam **weeni** *(o) swo [Kasena]_F bukɔ **yerane mo**.
 no Adam only 3SG marry Kasem woman only_{adf} FOC
 ‘No, Adam only married a KASEM woman.’

The examples also represent an instance of a DP-internal focus. The focus associate is functioning as an adjective in this case, and yet both the adfocal-*yerane* and the focus marker *mo* occur at the edge of the DP. Having them follow Kasem (i.e. inside the DP) in the repeated example in (18) would lead to ungrammaticality.

- (18) Aawo, Adam swo [[Kasena]_F (***yerane mo**) bukɔ] **yerane mo**.
 no Adam marry Kasem only FOC woman only_{adf} FOC
 ‘No, Adam only married a KASEM woman.’

The data in (19) provide further instances of both single and double occurrences of *yerane* and *weeni* respectively.

- (19) a. Bia bam go [chworɔ]_F **yerane** *(mo).
 children DEF.PL kill.PFV fowl only_{adf} FOC
 ‘The children slaughtered only A FOWL.’
 b. Bia bam **weeni** *(ba) go [chworɔ]_F *(mo).
 children DEF.PL only 3PL kill.PFV fowl FOC
 ‘The children slaughtered only A FOWL.’
 c. Bia bam **weeni** *(ba) go [chworɔ]_F **yerane** *(mo).
 children DEF.PL only 3PL kill.PFV fowl only_{adf} FOC
 ‘The children slaughtered only A FOWL.’

4.1.4 The structure of *weeni*-clauses

The sentences with *weeni* have unique structures because they involve an overt copy of the subject below *weeni*. The sentences in (20) are repeated from (13) above. The pronoun is a lower copy of the subject and can agree in number and person feature with the latter. Compare (20-a) and (20-b) (from examples (13-a) and (13-b) above).

- (20) a. Adam **weeni** *(o) go [chworō]_F *(mo).
 Adam only 3SG kill.PFV fowl FOC
 ‘Adam only slaughtered A FOWL.’
 b. Bia bam **weeni** *(ba) go [chworō]_F *(mo).
 children DEF.PL only 3PL kill.PFV fowl FOC
 ‘The children only slaughtered A FOWL.’

That the presence of *weeni* triggers the use of a resumptive pronoun is evident when we consider the example in (21-a), where the resumptive pronoun is obligatorily absent in a declarative clause without *weeni*, or in (21-b) when only the adfocal exclusive particle *yerane* occurs.

- (21) a. Adam (*o) go chworō.
 Adam 3SG kill.PFV fowl
 ‘Adam slaughtered a fowl.’
 b. Adam (*o) go [chworō]_F **yerane mo**.
 Adam 3SG kill.PFV fowl only FOC
 ‘Adam slaughtered only A FOWL.’

This kind of structure is not peculiar to only adverbial exclusive cases in Kasem. As highlighted in Chapter 1, progressive clauses also have the same structure, where the subject has a lower copy below the morpheme that has always been classified as progressive. In fact, there are more constructions with this structure. Based on my elicitation, there are at least five constructions with this structure. They include: (a) adverbial exclusive cases (22-a), (b) progressive constructions (22-b), (c) modal constructions (modal auxiliaries and necessity modal verbs) (22-c), and (d) scalar-additive constructions (22-d).

(22) **Structures licensing subject lower copy in Kasem**

- a. Adam **weeni** *(o) go [chworō]_F *(mo).
 Adam only 3SG kill.PFV fowl FOC
 ‘Adam only slaughtered A FOWL.’
 b. Adam **wora** *(o) go chworō.
 Adam PROG 3SG kill fowl
 ‘Adam is slaughtering a fowl.’
 c. Adam **wae** *(o) go chworō.
 Adam may 3SG kill fowl
 ‘Adam may slaughter a fowl.’
 d. Adam **ta-maje** *(o) go chworō.
 Adam even 3SG kill.PFV fowl
 ‘Adam even slaughtered a fowl.’

The first impression that these constructions give is that they are bi-clausal structures. This is because other declarative sentences do not have such a structure in Kasem, as show above in (21). Similarly, if we consider focus constructions, where subject focus are known to involve resumption, only the long distance subject focus are resumed in Kasem. This is what I already showed in the last chapter. Compare the examples in (23) with those in (24). The local subject *wh* and focus do not involve resumption. However, long distance subject *wh*-questions and subject focus answer involve resumption of the dislocated subject in the embedded clause. What this suggests is that bi-clausal structures may involve pronoun copies in Kasem.

(23) **Local subject focus in Kasem**

- a. Wɔ **mo** go chworɔ kom?
 who FOC kill fowl DEF
 ‘Who slaughtered the fowl?’
- b. [Adam]_F **mo** ___ go chworɔ kom.
 Adam FOC kill fowl DEF
 ‘ADAM slaughtered the fowl.’

(24) **Long distance focused subject in Kasem**

- a. Wɔ **mo** Peter boje we *(o) go chworɔ kom?
 who FOC Peter think COMP 3SG kill fowl DEF
 ‘Who did Peter think slaughtered the fowl?’
- b. [Adam]_F **mo** Peter boje we *(o) go chworɔ kom.
 Adam FOC Peter think COMP 3SG kill fowl DEF
 ‘Peter said that ADAM slaughtered the fowl.’

However, this is just a stipulation, and does not provide an absolute answer to the nature of the constructions in (22). In fact, we may be dealing with cases whereby the constructions that were once bi-clausal structures are now mono-clausal.

There are reasons to believe that the constructions with the adverbial exclusive particles, as in (22-a), are mono-clausals. One argument comes from the placement of PP-adjunct in such constructions. I use PP-adjunct here, instead of the usual adverbials, because time adverbials in Kasem, as in (25-a), have a freer distribution, i.e., they can occur clause-finally, or even clause-medially. However, PP-adjuncts do not have such a freedom, see (25-b). They can only occur clause-finally in Kasem. With this, the use and position of PP-adjuncts would be a good test for bi-clausality. The prediction is that if the construction with the adverbial exclusive particle is bi-clausal, then, both clauses should be able to host PP-adjuncts. This prediction is not borne out. I will compare the discussion on the adverbial exclusive structure with that of the progressive clauses, which has been argued to involve bi-clausal structures (cf. Aremu and Apebuga, 2025).

- (25) a. Adam (**diim**) toje (**diim**).
 Adam yesterday work.PFV yesterday
 ‘Adam worked yesterday.’
- b. Adam (*[**gaao kom ne**]) toje [**gaao kom ne**].
 Adam farm DEF in work.PFV farm DEF in
 ‘Adam worked in the farm.’

I already showed that the progressive clause in Kasem also have the construction that looks like a bi-clausal structure, see (22-b) above. If we apply the PP-adjunct test on that sentence, we would have the following constructions in (26). In (26-a), the PP-adjunct (in square bracket) occupies the expected clause-final position. Interestingly, this PP-adjunct can also occur high in the clause, after what has, hitherto, been called the progressive *wora*, see (26-b). The possibility of such a structure supports a bi-clausal analysis for the progressive clauses.⁷

⁷ Of course, the nature of the bi-clausal structure is still up for discussion, which I leave for a future occasion, but see Aremu and Apebuga (2025) for some ideas.

- (26) a. Adam **wora** o goe chworokom [gaao kom ne].
 Adam PROG 3SG kill.IPFV fowl DEF farm DEF in
 Adam is slaughtering the fowl in the farm.
- b. Adam **wora** [gaao kom ne] o goe chworokom.
 Adam PROG farm DEF in 3SG kill.IPFV fowl DEF
 Lit. ‘Adam is there in the farm, slaughtering the fowl.’

However, if we apply the same test to the adverbial exclusive sentence, we get a different result, see (27). The PP-adjunct can only be realized clause-finally, (27-a). When it is realized after the exclusive particle *weeni*, then the sentence becomes ungrammatical, (27-b). This suggests that the *weeni*-clause might be a mono-clausal structure. It might be the case that the structure was once a bi-clausal structure, but is now a mono-clausal structure, with a residue of the bi-clausality.

- (27) a. Adam **weeni** o go chworomo [gaao kom ne].
 A. only 3SG kill.PFV fowl FOC farm DEF in
 ‘Adam only slaughtered A FOWL in the farm.’
- b. *Adam **weeni** [gaao kom ne] o go chworokom.
 A. only farm DEF in 3SG kill.PFV fowl
 ‘Adam only slaughtered A FOWL in the farm.’

I must say that this discussion is not in any way conclusive. There is indeed more to be investigated; the same with the other sentences in (22). I leave a comprehensive investigation of these constructions for a future study. For our purposes here, I will assume that the *weeni*-clauses are mono-clausal.

4.2 Exclusive focus-sensitive operators in Kusaal

We now shift our focus to Kusaal where both the adfocal and adverbial exclusive particles are also realized with different forms. In Kusaal, while the adfocal exclusive particle is realized as *ma’aa*, the adverbial exclusive particle is realized as *kudim*.⁸ Similar to what I did for Kasem, I will describe the distribution of the two exclusive focus-sensitive operators one after the other, and that Kusaal also shows instances of exclusive doubling.

4.2.1 *Ma’aa* as an adfocal exclusive particle in Kusaal

Recall from chapter 3 (specifically, §3.3) that Kusaal makes use of an in-situ focus marker *ne* which precedes the focus constituent. In the event of a context which requires the exclusion of other alternatives, Kusaal makes use of the focus-sensitive particle *ma’aa*, which must immediately follow the focus associate. This is exemplified with the following sentences below. In (28), adfocal-*ma’aa* immediately follows the direct object both in-situ (28-a) and ex-situ (28-b). The contiguous nature of *ma’aa* is confirmed by the fact that it cannot occur at a distance (separate) from the focus constituent (28-a), nor can it be stranded after the focus fronting of the focus associate (28-b).

⁸ It is worthy of note here that Abubakari (2018, p.130-131) briefly described the focus-sensitive particles *ma’aa* ‘only’ and *men* ‘also’.

- (28) a. Adam kɔdig (*ma'aa) **nɛ** [nua]_F **ma'aa**.
 Adam kill.PFV only_{adf} FOC fowl only_{adf}
 'Adam only slaughtered A FOWL.'
 b. [Nua]_F **ma'aa** *(ka) Adam kɔdig (*ma'aa).
 fowl only_{adf} FOC Adam kill only_{adf}
 'Adam only slaughtered A FOWL.'

The same property can be observed in (29) and (30) respectively. Both examples involve in-situ and ex-situ focus. Example (30) is a case of adjunct focus with a null preposition.

- (29) a. Ti mɛ **nɛ** [yir]_F **ma'aa** ti tɛɲin.
 1.PL build.PFV FOC house only_{adf} our village
 'We only built A HOUSE in our village.'
 b. [Yir]_F **ma'aa** *(ka) ti mɛ ti tɛɲin.
 house only_{adf} FOC 1.PL build.PFV our village
 'We only built A HOUSE in our village.'
- (30) a. Ti mɛ yir **nɛ** [tɛɲin la]_F **ma'aa**.
 1.PL build.PFV house FOC village DEF only
 'We only built a house IN THE VILLAGE.'
 b. [Tɛɲin la]_F **ma'aa** *(ka) ti mɛ yir.
 village DEF only_{adf} FOC 1.PL build.PFV house
 'We only built a house IN THE VILLAGE.'

Further examples showing the adfocal property of *ma'aa* are found in the Kusaal Bible. See (31) and (32) below. The in-situ focus marker *nɛ* was not used.

- (31) ... amaa ba [yinne]_F **ma'aa** di piini la. (1 Corinthians 9:24)
 but 3PL one only_{adf} eat.IPFV gift DEF
 '... but only ONE of them wins the prize.'
- (32) Faangid kae zin'isi'a aɛɛɛ [mam]_F **ma'aa**. (Isaiah 43:11)
 saviour NEG.LOC anywhere unless me only_{adf}
 'There is no saviour anywhere apart from only ME.'

Association with subject focus is also possible with *ma'aa* (33). In this case, *ma'aa* follows the focus associate but precedes the local subject focus marker *n*.⁹

- (33) a. Anɔ'ɔn tɔm: Adam bɛɛ John?
 who work Adam DISJ John
 'Who worked: Adam or John?'
 b. Adam **ma'aa n** tɔm
 Adam only_{adf} FOC work.PFV
 'Only ADAM worked.'

All the examples presented above entail that the focus associates are the only true alternatives in their respective contexts. In other words, any other focus alternative is excluded.

⁹ Abubakari (2018) claims that in association with subject focus context, the focus marker *n* is obligatorily absent while in non-subject in-situ focus context, *nɛ* is optional. However, based on my data elicitation session with other Kusaal speakers, *n* is only optional in association with focus context, and not obligatorily absent.

4.2.2 Adverbial-*kudim* and exclusive doubling in Kusaal

Apart from the adfocal-*ma'aa*, Kusaal also has an adverbial form of ‘only’ *kudim*. *Kudim* can associate with the focus constituent at a distance. In this context (just like Kasem), the morphological marking of the in-situ focus makes it easy to know which constituent in the clause is the focus associate. Consider the examples below. The adverbial particle *kudim* seems to occupy a fixed position, while its focus associates can be the object of a transitive verb (34-a), object of an ditransitive verb (34-b), or even a manner adverb (34-c) respectively. In all of these cases, the adverbial focus-sensitive *kudim*, unlike the adfocal-*ma'aa*, is distant from its focus associate.

- (34) a. Adam **kudim** de **ne** [mui]_F.
 Adam only eat FOC food
 ‘Adam **only** ate RICE.’
 b. Adam **kudim** tis John **ne** [duk]_F.
 Adam only give John FOC pot
 ‘Adam only gave John A POT.’
 c. Adam **kudim** di diib la **ne** [tɔ'ɔtɔ]_F.
 Adam only eat food DEF FOC quick-quick
 ‘Adam only ate the food QUICKLY.’

Importantly, both adfocal-*ma'aa* and adverbial-*kudim* can co-occur associating with the same focus constituent. This is exemplified below with the repeated examples from (34). But now the adfocal particles are also present (35).

- (35) a. Adam **kudim** di **ne** [mui]_F **ma'a**.
 Adam only eat FOC rice only_{adf}
 ‘Adam only ate RICE.’
 b. Adam **kudim** tis John **ne** [duk]_F **ma'a**.
 Adam only give John FOC pot only_{adf}
 ‘Adam only gave John A POT.’
 c. Adam **kudim** di diib la **ne** [tɔ'ɔtɔ]_F **ma'a**.
 Adam only eat food DEF FOC quick-quick only_{adf}
 ‘Adam only ate the food QUICKLY.’

Just like we saw for Kasem above, in all these cases we seem to have a form-meaning mismatch between the morphology (exclusive doubling) and the interpretation, i.e. the focus is the only true alternative that fits into the current context. Other alternatives are excluded. This same meaning can also be expressed by *ma'aa* or *kudim* alone.

4.3 Exclusive focus-sensitive operators in Yorùbá

We now turn to the third language, Yorùbá, which shows properties similar to the exclusive focus-sensitive particle of the other two languages discussed above.

There is minimal discussion in the literature about exclusive focus-sensitive operators (and association with focus in general) in Yorùbá. The few previous discussions were purely descriptive which involved brief mentions when discussing adverbs in the language (Awobuluyi, 1987) and focus interpretation (Bisang and Sonaiya, 2000). However, there is a recent paper by Yip and Adedeji (2024) on Yorùbá exclusive particles which I return to in Section 5.6 of Chapter 5.

In this section, I describe the structural properties of the exclusive focus-sensitive operators in the language. Similar to the earlier discussions on Kasem and Kusaal, Yorùbá also uses different forms for both the adfocal and the adverbial exclusive particles: *nìkan* and *kàn* respectively.

4.3.1 Yorùbá singleton cases

Both the adfocal-*nìkan* and the adverbial-*kàn* can occur individually in a sentence involving focus. Consider the examples below. Assuming a question context like (36-a), the focus associate can be followed by the adfocal-*nìkan* thereby excluding the other alternative *iṣu* ‘yam’ present in the common ground. The structure can involve the focus associate plus *nìkan* in an in-situ position (10-c), or an ex-situ position (36-c). Recall that unlike the Mabia languages (Kasem and Kusaal), Yorùbá does not make use of an in-situ focus marker.

- (36) a. Šé Tolú jẹ iṣu àti ìrẹ̀sì?
 Q Tolú eat yam CONJ rice
 ‘Did Tolú eat yam and rice?’
 b. Rárá, ó jẹ [ìrẹ̀sì]_F **nìkan**.
 no 3SG eat rice only_{adf}
 ‘No, he ate only RICE.’
 c. Rárá, [ìrẹ̀sì]_F **nìkan** ni ó jẹ ____.
 no rice only_{adf} FOC 3SG eat
 ‘No, he ate only RICE.’

Further data in (37) shows that *nìkan* is indeed adfocal and always follows the focus associate. In (37-a), the in-situ direct object is the focus associate. It forms a constituent with *nìkan*. This means that Tolú bought *x* for Adé, and *x* is a shoe. When *nìkan* follows the PP instead (37-b), the interpretation changes to mean that Tolú bought a shoe for *x*, and *x* is Adé. Tolú did not buy a shoe for any other person. So, depending on the constituent that *nìkan* follows, the exclusive quantification applies.

- (37) a. Tolú ra [bàtà]_F **nìkan** fún Adé.
 Tolú buy shoe only_{adf} for Adé
 ‘Tolú bought a SHOE for Adé.’
 b. Tolú ra bàtà fún [Adé]_F **nìkan**.
 Tolú buy shoe for Adé only_{adf}
 ‘Tolú bought a shoe for ADÉ.’

Association with subject focus is also possible with *nìkan*. Assuming that the example in (38-a) is an answer to the subject *wh*-question ‘Who gave a shoe to Adé?’, example (38-b) can be used to correct the assertion that two persons (Bòla and Tolú) gave Adé a book. (38-b) means that only Tolú (excluding Bòla) gave Adé a book. Expectedly, the adfocal-*nìkan* follows the subject focus associate.

- (38) a. [Bòla àti Tolú]_F **ni** wòn ra bàtà fún Adé.
 Bòla CONJ Tolú FOC 3PL buy shoe for Adé
 ‘BÒLA AND TOLÚ bought a shoe for Adé.’
 b. Rárá, [Tolú]_F **nìkan ni** ò ra bàtà fún Adé.
 no Tolú only_{adf} FOC 3SG buy shoe for Adé
 ‘No, only TOLÚ bought a shoe for Adé.’

If *nìkan* happens to occur lower in the clause, as in (39-b), then the result would be a multiple focus context. Here, both Tolú and Adé are in focus, but only Adé receives an exclusive quantification. It means that only Adé, and no other person, got a shoe from Tolú. It does not mean that only Tolú gave Adé a shoe. Of course, there is the option of having *nìkan* follow each of the focus constituents, which results in a multiple association with focus context. We will return to this in chapter 5 below.

- (39) a. Who bought a shoe only for Adé?
 b. [Tolú]_F **ni** ò ra bàtà fún [Adé]_F **nìkan**.
 Tolú FOC 3SG buy shoe for Adé only_{adf}
 ‘TOLÚ bought a shoe only for ADÉ.’

Yorùbá also uses an adverbial exclusive focus sensitive particle *kàn*. Adverbial-*kàn* can associate with focus at a distance. In comparison to Kasem and Kusaal, this may result in ambiguous association with focus. This is due to the fact that Yorùbá does not have an in-situ focus marker that distinguishes the focus from the other constituents within the background. So, a construction such as (40) is ambiguous between exclusive association with the verb *ra*, the direct object *bàtà*, or the object of the PP *Adé*, but not the subject *Tolú*. The structural position of *kàn* and the lack of in-situ focus marking in the language result in the observed ambiguity. Unlike English, Yorùbá does not make use of prosody for focus marking. So, prosody does not disambiguate the focused constituent from the rest of the constituents.

- (40) Tolú **kàn** ra bàtà fún Adé.
 Tolú only buy shoe for Adé
 ‘Tolú only bought a shoe for Adé.’

However, with a proper context like the question in (41-a), the focus associate of *kàn* is easily determined. The answer to the question involves *kàn* associating with the direct object in this case. In this context, the focus associate can only be the direct object *bàtà*, and not the recipient *Adé* for example. The other disambiguating question contexts include: Did Tolú buy or rent a shoe for Adé? = association with the verb; Did Tolú buy a shoe for Adé or Sam? = association with the indirect object.

- (41) a. Šé Tolú ra bàtà àti ìwé fún Adé?
 Q Tolú buy shoe CONJ book for Adé
 ‘Did Tolú buy a shoe and a book for Adé?’
 b. Rárá, ó **kàn** ra [bàtà]_F fún Adé.
 no 3SG only buy shoe for Adé
 ‘No, he only bought A SHOE for Adé (not a shoe and a book).’

4.3.2 Exclusive doubling in Yorùbá

The exclusive focus sensitive particles *kàn* and *nìkan* can be doubled, as in (42). The construction does not only produce a single exclusive reading, but the focus associate is also easily recognizable (i.e. no ambiguity) since the adfocal-*nìkan* would have to immediately follow whatever is the focus associate.

- (42) a. Tolú **kàn** ra [bàtà]_F **nìkan** fún Adé.
 Tolú only buy shoe only_{adf} for Adé
 Lit. ‘Tolú only bought only A SHOE for Adé (and not a book, for example).’
 b. Tolú **kàn** ra bàtà fún [Adé]_F **nìkan**.
 Tolú only buy shoe for Adé only_{adf}
 Lit. ‘Tolú only bought a shoe for only ADÉ (and not Sam, for example).’

As we will see below, such doubling cases are interesting because they seem to result in a form-meaning mismatch where, although the forms are doubled, the interpretation is single. Whether this is truly the case or to what extent this is true will be part of the discussion in chapter 5.

The table in 4.1 summarizes the exclusive focus-sensitive particles (adfocal and adverbial) that we have discussed so far in Kasem, Kusaal and Yorùbá. The three languages allow exclusive doubling which results in a single exclusive association and interpretation.

	Adfocal- <i>only</i>	Adverbial- <i>only</i>	Doubling
Kasem	<i>yerane</i>	<i>weeni</i>	✓
Kusaal	<i>ma’aa</i>	<i>kɔdim</i>	✓
Yorùbá	<i>nìkan</i>	<i>kàn</i>	✓

Table 4.1: Exclusive focus sensitive particles in Kasem, Kusaal, & Yorùbá

So far, we have looked at the syntactic distribution of *only* elements in Kasem, Kusaal and Yorùbá. In the next section, we will see that *only* in these three languages do not only have the primary exclusive quantificational property, but also other scalar and evaluative properties (Jacobs, 1983; König, 1991; Beaver and Clark, 2008; Coppock and Beaver, 2014; Hole, 2015, a.o.).

4.4 The scalarity of *only*

That *only* has an evaluative scalar property (in addition to its exclusive quantificational property) has been a subject of discussion from as far back as the works of Lerner and Zimmermann (1981); Jacobs (1983); König (1991); Bayer (1996), and others. The interpretation of *only* that is derived in a scalar context is that which presupposes that the focus associate is ranked lower, perceived as little or insufficient compared to other contextually relevant alternatives. This would also mean that among the contextually relevant propositions, the prejacent proposition has a lower value on a relevant scale (cf. Beaver and Clark, 2008; Hole, 2015; Greenberg, 2022, a.o.).

There are two groups of scholars based on whether *only* is ambiguous between scalar and non-scalar meanings, or whether it always has a scalar property as part of its meaning. The first group believes that the meaning component of *only* is primarily non-scalar, but

it may introduce an additional scalar meaning in appropriate contexts (Groenendijk and Stokhof, 1984; Rooth, 1985, 1992; Krifka, 1992, a.o.). The second group believes that *only* is always scalar, i.e., not only its evaluative property, but also its quantificational (merely exclusive) property is underlyingly scalar (Krifka, 1993; van Rooij, 2002; Klinedinst, 2004; Beaver, 2004; Beaver and Clark, 2008; Coppock and Beaver, 2014; Hole, 2015, a.o.). I adopt the view that *only* is uniformly scalar.

There have been different views on the types of scalarity of *only*. I adopt the view of Coppock and Beaver (2014) and Hole (2015) to assume that there are at least three types of scalarity of *only* (cf. Horn, 1996, 2000; Krifka, 1993; Gast, 2004; Hole, 2004). They are given in (43).

- (43) **Type of scalarity of *only***
- a. Complement exclusion reading
 - b. Rank-order reading
 - c. Evaluation of a Proposition

There is, at least, one important point of divergence between Coppock and Beaver’s (2014) view of exclusive semantic composition, and that of Hole (2015). This has to do with the ‘presuppositional component’ of *only*. Coppock and Beaver (2014) assume that *only* has both a presuppositional component and an at-issue entailment component. The presuppositional component of *only* asserts that the answer to a current question is at least as strong as the prejacent. Coppock and Beaver (2014) call this $\text{MIN}\pi$, where π represents the prejacent. However, this does not exclude stronger alternatives than the prejacent. It is the at-issue entailment component of *only* that excludes stronger alternatives than the prejacent. In other words, the prejacent is the strongest relevant answer to the current question. Coppock and Beaver (2014) call this $\text{MAX}\pi$ (see Coppock and Beaver, 2014, p.376). Hole’s (2015) exclusive operator, on the other hand, has no presupposition, but just an at-issue entailment component, which entails that only the prejacent is true, and every other alternative is false; thus, excluded.¹⁰

I will now discuss these scalar types of *only* one after the other.

4.4.1 Complement exclusion

The complement exclusion reading of *only* relies on the informational strength of the prejacent in relation to the excluded alternatives. By ‘informational strength’ here, we mean ‘logical entailment’. Consider the example in (44). *Only* entails that the prejacent is the informationally strongest true proposition (i.e., that answers a current question such as: Who did Adam invite to the party?). This means that any stronger alternative proposition is excluded, as in the set of alternatives below (cf. Beaver and Clark, 2008; Coppock and Beaver, 2014). The excluded alternative propositions entail the prejacent.

- (44) Adam only invited JOHN to the party.
Presupposes: (That) Adam invited John to the party (is true).
Entails: Adam invited no other person than John.
Alternative propositions: {Adam invited John, ~~Adam invited John and Sam~~,
~~Adam invited John, Sam and Paul~~, ...}

The exclusive examples discussed in sections 4.1 (Kasem), 4.2 (Kusaal), and 4.3

¹⁰ I am grateful to Daniel Hole (p.c.) for pointing this distinction out for me more clearly.

(Yorùbá) are instances of complement exclusion usage of *only*. If we take for instance, the Yorùbá example in (37-b) (repeated below as (45)), we may assume the following sets of alternative propositions (including the prejacent) in (46). The scalarity of the complement exclusion reading of *only* is obtained in (45) by presupposing that the alternative that is as strong as the prejacent [Tolú bought a shoe for Adé] is true. This is the *at least* component (MIN) of the complement exclusion reading according to Coppock and Beaver (2014). So, we may say that Tolú bought a shoe for *at least* Adé.¹¹ The at-issue content, on the other hand, is the exclusion of every alternative that is logically stronger than the prejacent. This is the *at most* component (MAX) of *only* in a complement exclusion reading (cf. Beaver and Clark, 2008; Coppock and Beaver, 2014).¹² In other words, if we rank the alternatives on a scale, the prejacent is ‘at least’ and ‘at most’ the true answer to the current question. Thus, (46-b) and (46-c) are excluded.

- (45) Tolú ra bàtà fún [Adé]_F **nìkan**.
 Tolú buy shoe for Adé only_{adf}
 ‘Tolú bought a shoe for ADÉ.’
- (46) a. Tolú bought a shoe for Adé.
 b. Tolú bought a shoe for Adé and Sam.
 c. Tolú bought a shoe for Adé, Sam and Paul.

4.4.2 Rank-order reading

For the rank-order reading, the alternatives are ordered on a scale of contextual ranking, and the prejacent (containing the focus associate) is ranked lower on this scale. The difference between the complement exclusion reading and the rank-order reading is the nature of the scale involved. While the complement exclusion reading involves a scale that is based on logical entailment, where the excluded alternatives are those that asymmetrically entail the prejacent (thus stronger than it), the rank-order reading involves a (non-logical) contextual scale, where the excluded alternatives are ranked higher on the given scale. Consider the sentence in (47), for example. The prejacent proposition (Adam is a cleaner) is ranked lower on the scale of prestige, while the excluded alternatives are ranked higher than the prejacent.

- (47) Adam is only a CLEANER.
Presupposes: Adam is a cleaner (is true).
Entails: He is no more than a cleaner.
Alternative Propositions: {Adam is a cleaner, ~~Adam is a manager~~, ~~Adam is a CEO~~, ...}

Another example is given in (48), from Alxatib (2020, p.30, ex.20). The sentence means that Jess did not interview officers with higher rank than a first lieutenant. It does not, however, imply that he did not interview officers of lower rank than a first lieutenant. In other words, the sentence can be uttered if Jess interviewed a second lieutenant. However, the sentence in (48) would be infelicitous if Jess interviewed any

¹¹ This also leaves room for alternatives that are stronger than the prejacent. But the next component eliminates those.

¹² I do not commit to whether the prejacent proposition that answers a CQ is the lowest-ranked, since there may be other lower-ranked alternatives which are of course false if the prejacent is true (cf. Beaver and Clark, 2008).

officer that has a higher rank (e.g., a captain).

(48) Jess only managed to interview a FIRST LIEUTENANT.

Kasem, Yorùbá, and Kusaal offer a few notable patterns to rank-order reading of *only*. Unlike the complement exclusion reading where the adfocal particles can independently occur without the adverbial exclusive operators in Kasem and Kusaal, in rank-order readings, the adfocal particles cannot occur independently of the adverbial operators. This suggests that the exclusive interpretation comes from a position other (higher) than the adfocal particles. This is the central discussion for the next chapter 5. I will leave this for now. Yorùbá, on the other hand, provides us with a different adfocal particle that occurs in rank-order reading contexts. However, similar to Kasem and Kusaal, the exclusive adverbial particle in Yorùbá can occur independently of the adfocal particle, and still licenses a rank-order reading.

Rank-order reading in Yorùbá

Yorùbá has an adfocal particle that is specifically used in rank-order contexts. This is *lásán*. The sentence in (49-a) produces a rank-order reading, where the prejacent is ranked lower on a relevant scale of alternative propositions. *Lásán* can also be doubled with the adverbial operator *kàn*, as show in (49-b). The two sentences mean Tolú has no more prestigious occupation than being a farmer. Thus, other relevant alternatives (e.g. Tolú is a professor, Tolú is an engineer, ...) are ranked higher than the prejacent.

(49) [There are different candidates with different pedigrees running for the election. Tolú is one of them. What is Tolú's profession compared to the others?]

- a. Tolú jẹ́ [àgbè]_F **lásán**.
Tolú COP farmer only_{adf}
'Tolú is only A FARMER (and nothing more).'
≈ The property of Tolú being a farmer is lowly-ranked.
- b. Tolú **kàn** jẹ́ [àgbè]_F **lásán**.
Tolú only COP farmer only_{adf}
'Tolú is only A FARMER.'
≈ The property of Tolú being a farmer is lowly-ranked.
- c. Tolú **kàn** jẹ́ [àgbè]_F (**#nìkan**).
Tolú only COP farmer only_{adf}
'Tolú is only a farmer.'
≠ The property of Tolú being a farmer is lowly-ranked.

However, the occurrence of adfocal-*nìkan*, does not produce a rank-order reading, and is thus infelicitous in the given context, as in (49-c). Rather, it may mean that Tolú has only one occupation, and that is farming. This is the complement exclusion reading. This would be felicitous in a context where maybe someone thought Tolú was a farmer and a plumber, for instance.

The two adfocal particles (*lásán* and *nìkan*) can further be distinguished based on the following context and examples. In (50-a), it is not possible to use *nìkan* in order to get a ranking where coming fourth in an exam is lower on the scale of possible positions. *Lásán* is the appropriate adfocal particle for the rank-order context. On the other hand, *lásán* is not felicitous in example (50-b), because coming first cannot be ranked low on a relevant scale. By conventional implicature, first position is considered the highest grade. So, it cannot be lower on a scale of possible positions. If *nìkan* has a rank-order property, then

it should be felicitous in (50-a). I will return to this distinction in Section 5.4 of Chapter 5.

- (50) [The examination result is out, and (of course) whoever comes first tops the class.]
- a. Tolú **kàn** gbé ipò [kẹ̀rìn]_F **lásán**/(#**nìkan**).
 Tolú only carry position four only
 ‘Tolú only came FOURTH. (and it is a lower-ranked position)’
 - b. Tolú **kàn** gbé ipò [kíní]_F (**#lásán**).
 Tolú only carry position first only
 ‘Tolú only came FIRST. (*and it is a lower-ranked position)’

Consider another example below. Based on the context, only the example with adfocal-*lásán* in (51-a) is felicitous because the deputy governor is ranked lower than the invited governor. The example with adfoal-*nìkan* is infelicitous, given the current context. It can only mean that out of those invited for the project commissioning, only the deputy governor came. Others did not come. This is a complement exclusion reading.

- (51) [The local government wanted to commission a new completed project. So, they invited the governor of the state. Unfortunately, the governor could not make it. Only the deputy governor came.]
- a. [Igbákejì gómìnà]_F **lásán** ni ó wá.
 assistant governor only FOC 3SG come
 ‘Only the deputy governor came (for the commissioning).’
 - b. #[Igbákejì gómìnà]_F **nìkan** ni ó wá.
 assistant governor only FOC 3SG come
 ‘Only the deputy governor came (for the commissioning).’

Rank-order reading in Kasem

A similar pattern is found in Kasem. Consider the context in (52). For a rank-order reading, the use of adfocal-*yerane* is infelicitous, see (52-a). To get the rank-order reading, where the property of John as a farmer is ranked lower on the scale of possible alternative propositions, the use of the adverbial particle *weeni* is necessary (52-b). However, unlike Yorùbá, Kasem (and Kusaal, as would be shown soon) does not have a distinct adfocal particle for rank-order reading contexts, as adfocal *yerane* appears to be present only in complement exclusion readings.

- (52) [It’s been long since John and his friends saw one another. Fortunately, they are now having a get-together after so many years apart. Among them are bankers, engineers, and pilots, but ...]
- a. #John ye [valo]_F **yerane** mo.
 John COP farmer only_{adf} FOC
 ‘John is only A FARMER (and nothing more).’
 ≈ John being a farmer is ranked low.
 - b. John **weeni** o ye [valo]_F (**#yerane**) mo.
 John only 3SG COP farmer only_{adf} FOC
 ‘John is only A FARMER (and nothing more).’
 ≈ John being a farmer is ranked low.

The examples in (53) provide a further support for a rank-order reading for exclusives. The sentence in (53-a) has a context where the speaker is not Adam, who in this case

is assumed to have a higher rank or position than the speaker, who happens to just be Adam's secretary. To get such a rank-order reading, the adverbial exclusive particle *weeni* must be used, see (53-a). (53-b) shows that the single occurrence of the adfocal exclusive particle *yerane* is infelicitous with the rank-order context.

- (53) a. Amo dage Adam, a **weeni** a ye o [tɔn-poponnu]_F mo.
 1SG NEG.COP Adam 1SG only 1SG COP 3SG.POSS book-writer FOC
 'I am not Adam, I am just his secretary.'
 b. #Amo dage Adam, a ye o [tɔn-poponnu]_F **yerane** mo.
 1SG NEG.COP 1SG only 1SG COP 3.SG.POSS book-writer FOC
 Int. 'I am not Adam, I am just his secretary.'

Rank-order reading in Kusaal

Kusaal shows the same pattern as Kasem. The rank-order reading is not derived in the presence of only adfocal-*ma'aa*. So, (54-a) is infelicitous in the context where the property of Adam being a farmer is ranked lower on the scale of relevant alternative propositions. This is based on a scale of likely professions of those present at the get-together. The reading that (54-a) gives, is that of complement exclusion where Adam is only a farmer, and nothing other than that. However, the example in (54-b), with the introduction of the adverbial operator *kudim* is felicitous with a rank-order reading.

- (54) [It's been long since Adam and his friends saw one another. Fortunately, they are now having a get-together after so many years apart. Among them are bankers, engineers, and pilots, but ...]
 a. #Adam anɛ [kpaad]_F **ma'aa**.
 Adam COP farmer only_{adf}
 'Adam is just A FARMER (≈ and being a farmer is low on the scale of relevant professions).'
- b. Adam **kudim** anɛ [kpaad]_F (#**ma'aa**).
 Adam only COP farmer only_{adf}
 'Adam is just A FARMER (≈ and being a farmer is low on the scale of the relevant professions based on the context).'

Furthermore, considering the context in (55), the sentence with *ma'aa* is infelicitous because the context is not about the number of medals won by Adam. Rather, it is about the quality of the medal won by Adam in comparison to that of Sam, who won a higher medal. The infelicity of *ma'aa* in this context supports its complement exclusion status. Example (55-b), with *kudim*, is felicitous as it expresses the rank-order meaning that Adam's bronze medal is lower than Sam's gold medal on a scale of medals.

- (55) [Two participants (Adam and Sam) from Ghana went to the Olympics. While Sam won a gold medal, Adam only won a bronze medal.]
 a. #Adam di-nɛ [kɔnt-zɛn'ug]_F **ma'aa**.
 Adam eat-PERF medal-red only
 'Adam only won A BRONZE MEDAL. (≈ which is lower than Sam's gold medal).'
- b. Adam **kudim** di-nɛ [kɔnt-zɛn'ug]_F (#**ma'aa**).
 Adam only eat-PERF medal-red
 'Adam only won A BRONZE MEDAL. (≈ which is lower than Sam's gold medal).'

Another rank-order context is given in (56). The sentence in (56-a) is infelicitous with the adfocal particle *ma'aa*. It does not give the rank-order reading. Instead it means that the person being spoken about has or plays no other role in the office apart from being Adam's assistant. This gives the complement exclusion interpretation. However, the example with *kudim* in (56-b) produces the desired rank-order reading. The sentence means that the person involved has a lower rank than Adam's rank (who may be the manager).

- (56) a. #O kae Adam. o anε [o nya'andɔl]_F **ma'aa**.
 3SG NEG.COP Adam, 3SG COP 3SG.POSS assistant only
 'He is not Adam. He is just his assistant.'
- b. O kae Adam. o **kudim** anε [o nya'andɔl]_F.
 3SG NEG.COP Adam, 3SG only COP 3SG.POSS assistant
 'He is not Adam. He is just his assistant.'
- c. O kae Adam. o **kudim** anε [o nya'andɔl]_F (#**ma'aa**).
 3SG NEG.COP Adam, 3SG only COP 3SG.POSS assistant only
 'He is not Adam. He is just his assistant.'

4.4.3 Evaluation of a proposition

The third notion of scalarity, according to Hole (2015, p.50), "relates to a low, or the lowest, value on some *speaker-dependent scale*". The component of exclusive meaning here is similar to that of a rank-order reading, in that the alternatives are ranked on a scale, and the prejacent is low on the relevant scale. However, one major difference is that the nature of the exclusive in the evaluative context presupposes some kind of standard which the prejacent is lower than. The prejacent is below the standard or a contextual norm on a relevant scale (cf. Alxatib, 2013, 2020; Greenberg, 2022). In the rank-order component of *only*, the alternatives' scale has an inherent ordering. Thus, *only* is simply interacting with alternatives that are already hierarchical (see ex. (48)). Consider the example in (57) below. There is already a standard (4 papers). Adam only having 2 papers is below this standard, or it is insufficient. Outside the context and the expectation, the sentence could mean that Adam has written no more than 2 papers, and that's not little, at least for his level.

- (57) [*Context*: The committee is interviewing candidates for a postdoctoral position. One major requirement is that the successful candidate must have at least 4 papers. Unfortunately, Adam only has 2 papers]
 Adam only has [2]_F papers. (So he cannot be employed).

Consider another example in (58) with the context that Peter had fasted for three days. With this context, the sentence conveys that what Peter ate is considered little and/or bad in comparison to, for instance, what is expected of him, or the norm for someone who just ended a three days fasting.¹³

- (58) Peter **only** ate PEANUTS (after fasting for three days).
 (where to eat peanuts is considered little, bad, or both after such an exercise)

¹³ However, literally, the sentence (Peter only ate peanuts) conveys the fact that Peter ate only peanuts, and nothing else (i.e., the simple quantificational or complement exclusion reading).

It is important to note that not all cases of ‘evaluation of a proposition as little’ entail ‘evaluation of a proposition as bad’, and vice versa. The example in (59) is given by Hole (2015, p.51) to show that the evaluation of *only*’s prejacent as little or bad can be independent of each other. The evaluation in (59) is only based on how little the particular boy is, and not how bad, since being four years old is not a bad thing.

(59) You can’t blame him for this. He’s *only* four years old.

EVAL in Yorùbá

Yorùbá interestingly distinguishes between ‘evaluation as little’ and ‘evaluation as bad’ by employing distinct adfocal particles. While the particle *lásán* (to be glossed as ‘only_{lit/bad}’ here) may be felicitous in an evaluation as ‘bad’ and/or ‘little’ reading, the particle *péré* (to be glossed as ‘only_{lit}’) is felicitous only in an ‘evaluation as little’ reading. The regular complement exclusion’s adfocal particle *nìkan* is infelicitous in these contexts as well. In example (60-a) with *péré*, the evaluation can only be as little, and not bad. However, in (60-b) (with *lásán*), the evaluation is both little and bad. Example (60-c), with *nìkan*, is infelicitous with both ‘evaluation as little’ and ‘evaluation as bad’ readings.

- (60) [Tolú promised his wife to buy five fish on his way home. So, his wife was expecting five fish. However, he only got two fish.]
- a. Tolú (kàn) rà ẹja [méjì]_F **péré**.
 Tolú only buy fish two only_{lit}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is little/insufficient (not bad), contrary to expectation.
 - b. Tolú (kàn) rà ẹja [méjì]_F **lásán**.
 Tolú only buy fish two only_{lit/bad}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is both little and bad, contrary to expectation.
 - c. #Tolú rà ẹja [méjì]_F **nìkan**.
 Tolú buy fish two only_{lit}
 ‘Tolú bought only TWO fish.’
Cannot mean: That Tolú bought two fish is little and/or bad, contrary to expectation.

In a different context where Tolú promised his wife to buy two fish, and he got exactly two fish, only the sentence with adfocal-*nìkan* is felicitous, see (61-c). This is because Tolú got exactly the number of fish he promised to buy; not more and not less.¹⁴ However, the sentences with *péré* and *lásán* are not felicitous in this context (see (61-a) and (61-b) respectively). This is expected because the prejacent of *péré* and/or *lásán* is expected to be lower on the scale of relevant alternatives or the standard.

¹⁴ The sentence with *nìkan* (61-c) can also have a subtle pragmatic implicature, where although Tolú fulfilled his promised by buying exactly two fish, however, buying more would probably have been better or seen as generous.

- (61) [Tolú promised his wife to buy two fish on his way home.]
- a. #Tolú rà ẹja [méjì]_F **péré**.
 Tolú buy fish two only_{lit}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is little/insufficient (not bad), contrary to expectation.
- b. #Tolú rà ẹja [méjì]_F **lásán**.
 Tolú buy fish two only_{lit/bad}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is both little and bad, contrary to expectation.
- c. Tolú rà ẹja [méjì]_F **nìkan**.
 Tolú buy fish two only_{lit}
 ‘Tolú bought only TWO fish.’
Cannot mean: That Tolú bought two fish is little and/or bad, contrary to expectation..

On the other hand, if Tolú got more than two fish (e.g., six), the sentences with the three adfocal particles will be infelicitous, see (62). The sentence with adfocal-*nìkan* is infelicitous because the set of alternatives is already explicit, i.e., given by the context. The alternatives can only be ‘Tolú bought one fish’ and ‘Tolú bought two fish’. So, if Tolú buys more fish than two, this will not be a member of the set of contextually-determined alternatives. The infelicity of the evaluative particles *péré* and *lásán* is as a result of the contradiction between their evaluative contributions and the given context.

- (62) a. Tolú sọ wípé òun máa ra ẹja méjì.
 Tolú say COMP 3SG FUT buy fish two
 ‘Tolú said that he will buy two fish.’
- b. #Sùgbón, ó ra ẹja [méfà]_F **péré**.
 but 3SG buy fish six only
 ‘But he bought only SIX fish.’
- c. #Sùgbón, ó ra ẹja [méfà]_F **lásán**.
 but 3SG buy fish six only
 ‘But he bought only SIX fish.’
- d. #Sùgbón, ó ra ẹja [méfà]_F **nìkan**.
 but 3SG buy fish six only
 ‘But he bought only SIX fish.’

The ‘evaluation as little’ property of *péré* and the ‘evaluation as little and bad’ property of *lásán* can further be argued for if we consider the context in (63) below. Based on this context, the use of *péré* and *lásán* makes the sentences in (63-a) and (63-b) infelicitous because buying three fish cannot be considered little and/or bad when what is usually the norm (buying of two fish) is even less.¹⁵

¹⁵ Note that the use of *nìkan* is felicitous here, since it only shows a complement exclusion reading. See example (64), for instance.

- (63) [Tolú always buys two fish on his way home. But today, he wanted to surprise his wife. So, he bought three fish.]
- a. #Tolú rà ẹja [mẹta]_F **péré**.
 Tolú buy fish three only_{lit}
 ‘Tolú bought only THREE fish.’
 (#That Tolú bought three fish is little/insufficient (not bad), contrary to expectation.)
- b. #Tolú rà ẹja [mẹta]_F **lásán**.
 Tolú buy fish three only_{lit/bad}
 ‘Tolú bought only THREE fish.’
 (#That Tolú bought three fish is both little and bad, contrary to expectation.)

Another good distinguishing context is given in (64), where the reading of three books is not considered little, even though Tolú set out to read four. This relies on the fact that reading two books is already a lot or sufficient. While the sentence with the complement exclusion *nìkan* is felicitous in this context (64-a), the sentences with *péré* and *lásán* (see (64-b) and (64-c) respectively) are not.

- (64) [Reading two books is already a lot, but Tolú said he wanted to read four. But in the end, he only read three.]
- a. Tolú ka ìwé [mẹta]_F **nìkan**.
 Tolú read book three only
 ‘Tolú only read THREE books (#but this is not little).’
- b. #Tolú ka ìwé [mẹta]_F **péré**.
 Tolú read book three only
 ‘Tolú only read THREE books (and this is little).’
- c. #Tolú ka ìwé [mẹta]_F **lásán**.
 Tolú read book three only
 ‘Tolú only read THREE books (and this is little/bad).’

The example in (65) only conveys an ‘evaluation as bad’ reading. Only *lásán*, and not *péré*, is felicitous in this context.

- (65) [Olójúkòkòrò]_F **lásán**/(#**péré**) ni Tolú jé.
 covetous only_{bad}/only_{lit} FOC Tolú COP
 ‘Tolú is only A COVETOUS PERSON (and it’s bad).’
Meaning: That Tolú is covetous is bad.

To my knowledge, there is a little difference between *péré* and *lásán* with respect to the ‘evaluation as little’ reading. While *péré* seems to be more natural with cases where the focus associate is a numeral quantifying a DP (66-a), *lásán* seems to be unrestricted (66-b). Meanwhile, if *lásán* replaces *péré* in (66-a), then the evaluation of eating only two bananas can be considered bad (see ex. (60-b) above).

- (66) a. Tolú jẹ ògèdè [mėjì]_F **péré**.
 Tolú eat banana two only_{lit}
 ‘Tolú ate only TWO bananas.’
Cannot mean: That Tolú ate two bananas is considered bad.

- b. Tolú jẹ̀ [ògèdè]_F **lásán**/ʔ**péré**.
 Tolú eat banana only_{lit/bad}/only_{lit}
 ‘Tolú ate only BANANAS.’

The use of the evaluative particles *péré* and *lásán* can be dissociated. Consider the following context and the examples that follow. (67-a), and not (67-b), is felicitous because, considering the given context, having to carry just two bags from the post office is a desirable outcome. The example with *lásán*, on the other hand, is infelicitous because it produces a reading that having to carry two bags from the post office is bad or insufficient, i.e., more is expected. This is contrary to the desire of the participant who wishes to have a few bags to carry because he is using a bicycle.

- (67) [Sam asked you to help him pick up some bags from the post office on your way back. However, you were worried because you only have a bicycle, and you didn’t know how many bags there are. Luckily for you, you only got two bags from the post office.]
- a. O gbé àpò [méjì]_F **péré** láti ilé ìfiwé ránsé.
 2SG carry bag two only_{lit} from house use.book send
 ‘You carried only/just TWO bags from the post office.’
 (i.e., This is (positively) little, but not bad)
- b. #O gbé àpò [méjì]_F **lásán** láti ilé ìfiwé ránsé.
 2SG carry bag two only_{bad} from house use.book send
 ‘You carried only/just TWO bags from the post office.’
 (i.e., This is (positively) little, but not bad)

EVAL in Kasem

In Kasem, evaluation as little is only felicitous in the presence of the adverbial-*weeni*. If the sentence contains only adfocal-*yerane*, the sentence becomes infelicitous in the context of an ‘evaluation as little’ reading. Consider the following context and the examples that follow. (68-a) is not felicitous because the use of *yerane* does not necessarily produce an ‘evaluation as little’ reading based on the given context. On the other hand, example (68-b), with *weeni*, produces a an ‘evaluation as little’ reading.

- (68) [Tolú promised his wife to buy five fish on his way home. So, his wife was expecting five fish. However, he only got two fish.]
- a. #Adam nɛ ywoana [yalei]_F **yerane** mo.
 Adam buy fish two only FOC
 ‘Adam only bought two fish.’
- b. Adam **weeni** o na ywoana [yalei]_F (**#yerane**) mo.
 Adam only 3SG buy fish two only FOC
 ‘Adam only bought two fish.’

The sentence in (69-a) is judged as infelicitous for the current context. It does not mean that what Adam slaughtered for Christmas is little in comparison to what Sam and John slaughtered. It only means that Adam slaughtered nothing other than a fowl. This is simply a complement exclusion reading without any speaker-dependent evaluation. However, at the introduction of the adverbial operator *weeni* (69-b), the sentence becomes felicitous for the current context.

- (69) [It is the Christmas period, and it is normal for everyone to plan properly, in order to slaughter animals that are considered big enough for the celebration. Adam, Sam, and John bought different animals for their families. While Sam and John slaughtered a cow and a goat respectively, Adam only slaughtered a fowl.]

- a. #Adam go [chworɔ]_F **yerane** mo biriyɔm dem ŋwaane.
 Adam slaughter fowl only FOC Christmas DEF because.of
 ‘Adam slaughtered only A FOWL for Christmas.’
- b. Adam **weeni** o go [chworɔ]_F (**#yerane**) mo biriyɔm dem
 Adam only 3SG slaughter fowl only FOC Christmas DEF
 ŋwaane.
 because.of
 ‘Adam only slaughtered A FOWL for Christmas.’

Unlike Yorùbá where the adfocal exclusive particles *lásán* and *péré* are used in evaluation as little or bad contexts, there are no such adfocal particles in Kasem. My assumption is that they are covertly realized in Kasem and Kusaal. This would be argued for in chapter 5 (section 5.4, precisely).

The example in (69-b) can also have the ‘evaluation as bad’ reading. This would mean that Adam slaughtering only a rat for a Christmas celebration is bad.

The context in (70), and the examples that follow it, present a case where the evaluation of the proposition is only little, and not bad. That Bill wrote only 2 papers in comparison to John’s 6 and Sandra’s 4 is not bad, but little.

- (70) [John, Sandra, and Bill are all recent PhD graduates applying for the same job position. A major requirement for such a position is the number of papers written. John has written 6 papers, Sandra 4 papers, and Bill only 2 papers.]

- a. #Bill popone tɔnnɔ telei **yerane** mo.
 Bill write book two only FOC
 ‘Bill wrote **only** [2]_F papers.’
- b. Bill **weeni** o popone tɔnnɔ telei (**#yerane**) mo.
 Bill only 3SG write book two only FOC
 ‘Bill wrote **only** [2]_F papers.’

The same can be said for example (71). For example, in a traditional African community, having two kids is considered little, but not bad.

- (71) a. #John jege bia balei **yerane** mo.
 John have children two only FOC
 ‘John has only [2]_F kids.’
- b. John *(**weeni**) o jege bia balei (**yerane**) mo.
 John only 3SG have children two only FOC
 ‘John has only [2]_F kids.’

In both (70) and (71), the (a) examples without the adverbial operator are infelicitous. Only the (b) examples with the adverbial exclusive operators are felicitous.

Another context is provided below. This context, like the one above, is to show that in Kasem, evaluation of an exclusive proposition as little does not necessarily contain an ‘evaluation as bad’ interpretation. In fact, it is actually a good thing that the box is little. So, both evaluation of a proposition as little and evaluation of a proposition as bad are

distinguishable. For such a context, it is the sentence with the adverbial exclusive particle *weeni* in (72-a) that is felicitous. The sentence with the adfocal exclusive particle *yerane* in (72-b) is not felicitous.

- (72) [I need to pick up some packages at the post office. I worry it'll be too big, because I'm biking home. But thankfully, it was only one small box.]
- a. Ko **weeni** ko ye [bɔxi balaŋa]_F mo.
 3SG only 3SG COP box small FOC
 'It is only a small box.'
- b. #Ko ye [bɔxi balaŋa]_F **yerane** mo.
 3SG 3SG COP box small only FOC
 'It is only a small box.'

EVAL in Kusaal

The evaluation of a proposition by *only* in Kusaal displays the same structural pattern as in Kasem. Only in the presence of the adverbial-*kudim* can the 'evaluation as little' or 'evaluation as bad' reading be realized. The example in (73-a) with only the adfocal-*ma'aa* is infelicitous with an evaluation as little or bad reading. The sentence cannot imply that Adam's buying of a fish for the Christmas celebration is little or bad. It can only have a complement exclusion reading, where Adam only bought a fish and nothing else. However, the sentence in (73-b), which has adverbial-*kudim*, is felicitous with both evaluation as little and/or bad readings.

- (73) [Adam has a family of six, five children and his wife. Yet, he only got a fish for Christmas.]
- a. #Adam da' ziiŋ **ma'aa** di burinya.
 Adam buy fish only_{adf} eat Christmas
 'Adam only bought A FISH to celebrate Christmas.'
Cannot mean: Adam getting a fish for Christmas is little (or bad).
- b. Adam **kudim** da' ziiŋ (ma'aa) di burinya.
 Adam only buy fish only_{adf} eat Christmas
 'Adam only bought A FISH to celebrate Christmas.'
Meaning: Adam getting a fish for Christmas is little (or bad).

Similarly, (74) can have both 'evaluation as little' and 'evaluation as bad' interpretations, as long as the adverbial operator is present. The infelicitous example in (74-a) only has the complement exclusion interpretation whereby Adam ate only peanuts and nothing other than that. However, it does not imply that that is little or bad.

- (74) [Adam decided to take a fast for three days. He then broke his fast with a peanut.]
- a. #Adam di suma **ma'aa**.
 Adam eat peanut only_{adf}
 'Adam **only** ate PEANUTS.'
Cannot mean: To eat peanuts is considered little, bad, or both after fasting for three days.
- b. Adam **kudim** di suma (ma'aa).
 Adam only eat peanut only_{adf}
 'Adam **only** ate PEANUTS.'
Meaning: To eat peanuts is considered little, bad, or both after fasting for three days.

In a context where having many children is considered the norm, the sentence in (75) can only mean that Adam having only two children, is considered little. It does not mean that it is bad to have two children.

- (75) a. #Adam mɔri biisa [yi']_F **ma'aa**.
Adam have child.PL two only_{adf}
‘Adam only has TWO children’
b. Adam **kudim** mɔri biisa [yi']_F (ma'aa).
Adam only have child.PL two only_{adf}
‘Adam only has TWO children.’

In summary, I have presented data showing the different types of scalarity with association with *only*. The data that I described in Sections 4.1, 4.2, and 4.3 are instances of complement exclusion reading. Considering Table 4.2, the adfocal particles *yerane* (=Kasem), *ma'aa* (=Kusaal), and *nikan* (=Yorùbá) are all felicitous in complement exclusion reading. However, only Yorùbá has an overt morphological adfocal particles for the other two scalar types: *lásán* = rank-order reading or evaluation of a proposition reading, and *péré* = evaluation of a proposition reading. Although Kasem and Kusaal also realize these scalar interpretations, they do not have distinct adfocal morphology for them. One important thing that is common to the three languages is the fact that the adverbial operators, *weeni* (=Kasem), *kudim* (=Kusaal), and *kàn* (=Yorùbá), are felicitous for all the scalar contexts. I will present a proposal for these scalar ambiguities of *only* in the next Chapter 5 (particularly in Section 5.4).

	Complement excl.	Rank-order	EVAL
Yorùbá	<i>nikan</i>	<i>lásán</i>	<i>lásán/péré</i>
Kasem	<i>yerane</i>	—	—
Kusaal	<i>ma'aa</i>	—	—

Table 4.2: Adfocal particles and scalar types

In the next Section, I show further properties of these exclusive particles (particularly the adverbial and the adfocal complement exclusion particles), in relation to their scopal behaviours in embedded clauses and syntactic locality domains such as islands.

4.5 Exclusive scope, locality and clause-boundedness

In this Section, we want to see whether the interpretation of *only* is restricted by locality conditions in these languages. We will consider clausal embedding and locality constraints, specifically islands, to see how the exclusive operators in these languages behave when they are inside embedded clauses and islands.

That *only* can involve scope ambiguity has been known, at least, since Taglicht (1984). Subsequent and cross-linguistic studies have confirmed this property for *only* words (cf. Rooth, 1985, 1992; Lee, 2005; Quek and Hirsch, 2017; Sun, 2021; Branan and Erlewine, 2023, a.o.). The idea is that *only* can scope below or above certain predicates, modals, and other quantificational operators like negation. Look at the examples in (76) and (77), for instance. The adfocal-*only* is ambiguous between a narrow scope and a wide scope

with the bridge verb in (76-a) & (76-b). It can also take a narrow or a wide scope with the modal in (77-a) & (77-b).¹⁶

- (76) Adam knew that Sam ate **only** RICE. [know > only, only > know]
- a. Adam knew that Sam only ate rice (*meaning*: Sam did not eat anything else).
 - b. Adam only knew that Sam ate rice (*meaning*: Adam did not know whether Sam ate any other food).
- (77) Adam may eat **only** RICE. [may > only, only > may]
- a. Adam may only eat rice (*meaning*: Adam has permission to eat nothing but rice, but might also be permitted to eat other food).
 - b. Adam only may eat rice (*meaning*: Rice is the only thing that Adam is allowed to eat. No other permission).

So, this Section discusses whether there is scope ambiguity in embedded clauses and islands. Specifically, the question is whether the adfocal exclusives in Kasem, Kusaal, and Yorùbá can (a) scope widely above the matrix predicate of an embedded clause, and (b) scope across island boundaries. The prediction is that while we may find scope ambiguity in embedded clause contexts, the same should be banned in island contexts (i.e. only narrow scope should be possible when *only* is inside an island). This is simply because islands are known to be opaque to outside operations

4.5.1 Locality and exclusive scope in Kusaal

The prediction that embedded clauses should allow scope ambiguity in the presence of an embedded adfocal-*only*, and that islands should not, is fully borne out in Kusaal.

Starting with the embedded clause in (78), the adfocal *only* in (78-a) may take a narrow scope below the matrix predicate *mi'i* 'know', as in (78-a)-(i). It may also take a wide scope above the matrix predicate (78-a)-(ii). Recall that this cannot be because of what might seem to be a clause-final position for *ma'aa* in (78-a). We have already established the fact that adfocal-*ma'aa* (including the adfocal particles in the other languages) forms a constituent with its focus associate, and it is not a clause-final particle that may be argued to be high in the clause and, therefore, able to take wide scope. The scope can be restricted to a narrow scope when the adverbial-*kɔdim* is present inside the embedded clause (i.e., precedes the embedded verb) (78-b) (cf. (78-b)-i). Here, the wide scope reading is absent (78-b)-(ii). However, if the adverbial-*kɔdim* precedes the matrix verb, the only scope present is the wide scope (78-c). We will return to the matter of scope with exclusive particles in more detail in the next chapter.

¹⁶ This property of *only* will be important to our proposal in Chapter 5. So, we will discuss it further there.

(78) **Exclusive amb. scope in Kusaal's embedded clause**

- a. Adam *mi'i* ye Sam *kɔdig* nɛ [nua]_F **ma'aa**.
 Adam know COMP Sam slaughter FOC fowl only
 'Adam knows that Sam killed only A FOWL.' [know>only, only>know]
 (i) Narrow scope: Adam knows that Sam slaughtered only a fowl (and no other animal).
 (ii) Wide scope: Adam only knows that Sam slaughtered a fowl (but does not know about other animals).
- b. Adam *mi'i* ye Sam **kɔdim** *kɔdig* nɛ [nua]_F **ma'aa**.
 Adam know COMP Sam only slaughter FOC fowl only
 'Adam knows that Sam only killed A FOWL.' [know>only, *only>know]
 (i) Narrow scope: Adam knows that Sam slaughtered only a fowl (and no other animal).
 (ii) *Wide scope: Adam only knows that Sam slaughtered a fowl (but does not know about other animals) .
- c. Adam **kɔdim** *mi'i* ye Sam *kɔdig* nɛ [nua]_F **ma'aa**.
 Adam only know COMP Sam slaughter FOC fowl only
 'Adam **only** knows that Sam killed A FOWL.' [*know>only, only>know]
 (i) *Narrow scope: Adam knows that Sam slaughtered only a fowl (and no other animal).
 (ii) Wide scope: Adam only knows that Sam slaughtered a fowl (but does not know about other animals).

As predicted, exclusive scope ambiguity is unavailable in island contexts. I make use of the relative clause island here. That relative clauses are islands in Kusaal is shown with the restriction on movement extraction, in both the *wh*-question and focus answer examples in (79). The construction in (79-a) is ungrammatical because the *wh*-phrase is extracted from a relative clause (in square bracket). The same restriction applies to the focus answer construction in (79-b). The examples clearly show that relative clauses are island in Kusaal. Now, we can see examples with adfocal-*ma'aa* and its scopal property in a sentence involving a relative clause.

- (79) a. *Bɔ ka Adam nyɛ dau [kanɛ kɔdigɛ ____]?
 what FOC Adam see man REL slaughter
 'What did Adam see the man who slaughtered?'
 b. *Nua ka Adam nyɛ dau [kanɛ kɔdigɛ ____].
 fowl FOC Adam see man REL slaughter
 'A fowl, Adam saw the man who slaughtered.'

The examples with a relative clause in (80) show different behaviours with respect to scope. Unlike the embedded clause example in (78-a) above, where the adfocal-*ma'aa* can have ambiguous scope with the matrix predicate, the same cannot be said of *ma'aa* when it occurs inside a relative clause, as in (80-a). Consistent with what is expected in an island context, wide scope reading above the matrix verb *mi'i* is unavailable. Only the narrow scope reading is available. In other words, the construction is unambiguous with respect to scope. Expectedly, when the adverbial-*kɔdim* occurs inside the relative clause, only the narrow scope reading is available (80-b).

- (80) a. Adam mi'i dau [kanε kɔdig nε [nua]_F **ma'aa**]. *unambiguous*
 Adam know man REL slaughter FOC fowl only
 'Adam knows the man who slaughtered only A FOWL.'
 b. Adam mi'i dau [kanε **kɔdim** kɔdig nε [nua]_F].
 Adam know man REL only slaughter FOC fowl
 'Adam knows the man who only slaughtered A FOWL.'

4.5.2 Locality and exclusive scope in Kasem

As in Kusaal, the same scopal behaviour of exclusives between embedded clause and island is found in Kasem. Recall that the prediction above is that, while we may find cases of scope ambiguity in embedded clauses that are not islands, scope ambiguity should be unavailable in island constructions. This prediction is borne out in Kasem.

In example (81-a), the narrow scope reading between the adfocal particle *yerane*, and the bridge verb *yei* ('know') is available, as in (81-a)-(i). The wide (matrix) scope reading over the bridge verb is also available, as in (81-a)-(ii). These availability of the two readings are further confirmed by the continuations that follow. If the exclusive particle scopes below the bridge verb, as in a narrow scope, then the continuation in (81-b) is felicitous. If it takes a wide scope above the bridge verb, then (81-c) is a felicitous continuation.

(81) **Ambiguous scope in embedded clause**

- a. Adam yei we Sam go [chworɔ]_F **yerane mo**.
 Adam know COMP Sam slaughter fowl FOC only
 'Adam knows that Sam killed **only** A FOWL.' [know > only; only > know]
 (i) Narrow scope: Adam knows that Sam killed nothing but a fowl.
 (ii) Wide scope: The only thing Adam knows is that Sam killed a fowl.
 b. ... Adam ta yei we Sam wo go kolokolo daga. (narrow)
 Adam also know COMP Sam NEG kill anything else
 '... Adam also knows that Sam did not slaughter anything else.'
 c. ... amaa Adam yeiri we Sam ta go boŋo. (wide)
 but Adam know.NEG COMP Sam also kill goat
 '... but Adam does not know that Sam also slaughtered a goat (for instance).'

However, if the adverbial exclusive operator *weeni* is introduced, depending on the position, either of the two readings is derived. If *weeni* occurs within the embedded clause, only the narrow scope reading is available (82-b). But if *weeni* precedes the bridge verb in the matrix clause, only the wide scope reading is available (82-a).

(82) **Fixed scope depending on *weeni*'s placement**

- a. Adam yei we Sam **weeni** o go [chworɔ]_F **yerena mo**.
 Adam know COMP Sam only 3SG kill fowl only FOC
 'Adam knows that Sam killed only A FOWL.' [know > only; only > know]
 (i) Narrow scope: Adam knows that Sam killed nothing but the fowl (i.e. he knows that Sam didn't kill anything else).
 (ii) *Wide scope: The only thing Adam knows is that Sam killed the fowl (i.e. he doesn't know about other things).
 b. Adam **weeni** o yei we Sam go [chworɔ]_F **yerena mo**.
 Adam only 3SG know COMP Sam kill fowl only FOC
 'Adam only knows that Sam killed A FOWL.' [*know > only; only > know]

- (i) *Narrow scope: Adam knows that Sam killed nothing but the fowl (i.e. he knows that Sam didn't kill anything else).
- (ii) Wide scope: The only thing Adam knows is that Sam killed the fowl (i.e. he doesn't know about other things).

The scope ambiguity observed above is constrained by syntactic locality. If the adfocal particle *yerane* associates with a DP_F within an island, the scope flexibility is unavailable. Particularly, wide scope is unavailable. This is because the exclusive particle cannot scope out of an island. I show this using the Complex Noun Phrase Constraint (with a relative clause).

That relative clauses constitute an island in Kasem is supported by the examples in (83). Both *wh*-movement and focus answer movement are disallowed from the relative clause in (83-a) and (83-b), respectively.

(83) **Relative clause as island in Kasem**

- a. *Bε mo Adam nε [baaro wolo (wom) na go ___ to].
 what FOC Adam see.PFV man REL the NA kill REL.DET
 ‘What did Adam see the man who slaughtered?’
- b. *[Chworo]_F mo Adam nε [baaro wolo (wom) na go ___ to].
 fowl FOC Adam see.PFV man REL the NA kill REL.DET
 ‘A FOWL, Adam saw the man who slaughtered.’

Now, if the exclusive adfocal particle *yerane* associates with the object of the relative clause, as in (84-a), the sentence can only produce a narrow scope reading where *yerane* only scopes inside the relative clause. It cannot scope above *nε* (‘see’). So, the continuation in (84-b) is the only felicitous one. (84-c) is not a felicitous continuation because it is not the case that Adam only saw *x*, and *x* is the man who slaughtered a fowl.

(84) **No flexible scope with island** [**only > see; see > only*]

- a. Adam nε [baaro wolo (wom) na go [chworo]_F **yerane** to
 Adam see.PFV man REL the NA kill fowl only REL.DET
 mo].
 FOC
 ‘Adam saw the man who slaughtered only A FOWL.’
- b. ...and the man didn't slaughter anything else.
- c. *...but he didn't see the man who slaughtered a goat.

As can be seen in (85), if *weeni* occur inside the relative clause, only the narrow scope reading is available. See the continuations in (84).¹⁷

- (85) Adam nε [[baaro wolo (wom) na **weeni** o go [chworo]_F (**yerane**)
 Adam see.PFV man REL the NA only 3SG kill fowl only
 to] **mo**].
 REL.DET FOC
 ‘Adam saw the man who only slaughtered A FOWL.’

The difference in the scope behaviour of the exclusive particle between the island data

¹⁷ The focus marker *mo* appears to be in the periphery of the relative clause here, following the relative determiner. This was part of the discussion in Chapter 3, where I assume that the adjunction of *mo* is sensitive to locality and phases (cf. 3.2.4).

and the embedded clause is that, while the latter allows wide scope, the former does not. However, in both contexts, the narrow scope reading is available.

4.5.3 Locality and exclusive scope in Yorùbá

In Yorùbá, adfocal-*only* cannot scope out of an island. Adfocal-*only* can however scope out of an embedded clause. Take, for example, the embedded clause in (86-a). Adfocal-*nìkan* can take a narrow scope below the bridge verb *mò* ('know') (86-a)-(i), or above it (86-a)-(ii). For the narrow scope reading, the continuation in (86-b) is felicitous, while the continuation in (86-c) is felicitous for the wide scope reading respectively.

(86) Flexible scope in embedded clause

- a. Tolú mò wípé Adé jẹ [iṣu]_F **nìkan**. [know > only, only > know]
 Tolú know COMP Adé eat yam only
 'Tolú knows that Adé ate only YAM.'
 (i) Narrow scope: It is the case that Tolú knows that Adé only ate *x*, and *x* is yam.
 (ii) Wide scope: Within a relevant context, Tolú only knows one thing, and that is the fact that Adé ate yam (he does not know about other things).
- b. ... Adé ò jẹ ẹwà.
 Adé NEG eat beans
 'Adé did not eat beans.'
- c. ... Tolú ò mò wípé Adé jẹ ẹja pẹlú.
 Tolú NEG know COMP Adé eat fish also
 'Tolú does not know that Adé also ate fish.'

The ambiguous scope of adfocal-*nìkan* can be disambiguated in the presence of adverbial-*kàn*. If *kàn* occurs inside the embedded clause, the exclusive scope can only take a narrow scope, as in (87-a). However, if *kàn* occurs in the matrix clause (i.e. before the bridge verb), then we have a wide scope reading (87-b).

(87) Rigid scope in the presence of adverbial-*kàn*

- a. Tolú mò wípé Adé **kàn** jẹ [iṣu]_F (**nìkan**).
 Tolú know COMP Adé only eat yam only
 'Tolú knows that Adé only ate YAM.' [know > only, *only > know]
- b. Tolú **kàn** mò wípé Adé jẹ [iṣu]_F (**nìkan**).
 Tolú only know COMP Adé eat yam only
 'Tolú only knows that Adé ate YAM.' [*know > only, only > know]

The scope of adfocal-*nìkan* is sensitive to island. When it is embedded inside a relative clause island such as (88), it can only have a narrow scope reading. While (88)-(i) is a felicitous interpretation, (88)-(ii) is not. This is the case because of the absence of a wide scope reading in the context of an island.

- (88) Tolú mọ [obìnrin tí o jẹ [iṣu]_F] **nìkan** [know > only, *only > know].
 Tolú know woman REL 3SG eat yam only
 'Tolú knows the woman who ate only YAM.'
 (i) Tolú knows the woman who ate nothing but yam.
 (ii) *Tolú only knows the woman who ate yam (but does not know the woman who ate beans).

4.6 Summary

In this chapter, I started out by describing the exclusive focus sensitive operators in Kasem, Kusaal and Yorùbá, and highlighting their distinct structural positions in the clause, and the possibility of exclusive doubling. Furthermore, I presented data showing the scalar nature of *only* by distinguishing between complement exclusion reading, rank-order reading, and evaluation of a proposition reading. Among the three languages, I showed that Yorùbá offers distinct adfocal morphology for the three types of scalarity discussed. Although the other two languages, Kasem and Kusaal, do not show distinct adfocal morphology, they, however, show distinction between the scalar types through the felicity of the adfocal particles based on the scalar reading that is at-issue. In other words, I showed that while the singleton occurrence of the adfocal particles are felicitous in the complement exclusion reading, they are not felicitous in the rank-order reading and the evaluation of a proposition (as little and/or bad) reading, without the adverbial exclusive operators. This suggests that the adverbial exclusive operators are responsible for the scalar interpretations of *only*. Furthermore, I presented arguments showing the scopal relations of the exclusive operators. In the presence of an embedded clause, the adfocal particles can scope across an embedding verb. However, in an island context, the adfocal particles cannot scope across an island. These results are found across the three languages.

In the next chapter, I will propose a uniform analysis for the different behaviours and properties of the exclusive focus sensitive operators in Kasem, Kusaal and Yorùbá. I will argue that we can uniformly account for the different properties that we have seen in this chapter, by assuming that the exclusive interpretation comes from a high exclusive operator, and not the adfocal particles themselves. This same idea would also have bearing on my analysis for the other two focus sensitive operators (*even* and *also*) in these languages.

Chapter 5

Proposal for exclusive focus sensitive particles

The last chapter mainly describes the syntax-semantic properties of the exclusive focus sensitive particles in Kasem, Kusaal and Yorùbá. In this chapter, I propose a unified analysis for these properties. In order to make sense out of the discussions that follow, I will first elaborate on the meaning of *only* that I adopt here, which is generally propositional in nature (i.e. operates over a proposition, which is its prejacent that contains the focus associate). Then, I will discuss a challenge that would come from assuming that the exclusive meaning comes from the adfocal exclusive particles in the three languages. This challenge has to do with a mismatch between the syntactic nature of adfocal exclusive particles and the general propositional semantic property of exclusive operators. All these are discussed in Section 5.1. Afterwards, Section 5.2 summarizes two potential approaches (i.e., the Quantifier Approach and the Propositional Approach) to addressing the challenge. In Section 5.3, I provide arguments in support of the propositional approach, and propose an analysis which can uniformly account for not only the adfocal exclusive particles, but also the adverbial exclusive particles and the exclusive doubling cases. This approach affords us with the tools to analyze the different exclusive scalar types that we discussed in the last chapter. This is the goal of Section 5.4. In Section 5.5, I discuss the phenomenon of Backward Association with Focus (BAwF), and show how the chosen approach (the propositional approach) accounts for the presence of exclusive BAwF in Kasem, Kusaal and Yorùbá respectively.

5.1 Introducing *only* and the problem of association

5.1.1 Introducing *only*

I established in Chapter 2 that *only* is a propositional operator because it composes with a proposition (Horn, 1969; Rooth, 1985, 1992; Beaver and Clark, 2008; Coppock and Beaver, 2014, among many others) (1). *Only* presupposes that the prejacent proposition is true, and asserts that all other alternative propositions p' that are true must be entailed by the prejacent proposition p . This condition rules out any alternative proposition that is not entailed by p as false. Since the relevant alternatives are usually not entailed by p , then they are automatically excluded.

$$(1) \quad \llbracket \text{only} \rrbracket_{ALT} = \lambda p_{\langle s, t \rangle} \cdot \lambda w : p(w) \cdot \forall p' \in ALT[p'(w) \rightarrow p \subseteq p']$$

It is important to note here that based on Rooth’s alternative semantics approach, it is the proposition that *only* composes with, and not the individual focus associate itself. However, it is the latter’s focus semantic value that produces the alternative set of propositions, which *only* quantifies over. Therefore, we can say that *only* takes a proposition as its syntactic argument. The focus associate reflects the point of variation between the alternative propositions in a contextually-relevant set (cf. Horn, 1969, 1996; Rooth, 1985, 1992; Lee, 2005, a.o.). In a propositional set of alternatives, the focus associate is substituted with elements of the same semantic type.¹

The example in (2) is used to explain the discussion further. The prejacent is true, but all the non-prejacent alternative propositions are false. The set of alternatives consist of the prejacent and the propositions that are different only in the substituted position of the focus.

- (2) Adam only ate RICE.
Prejacent: Adam ate rice.
Alternatives: {Adam ate rice, Adam ate beans, Adam ate yam, Adam ate noddles ...}

Technically, *only* (cf. (2)) has both a positive and a negative component (Horn, 1969; Rooth, 1992; Alxatib, 2020). Based on (2), the positive component is the prejacent, and the negative component is the exclusion of the set of alternatives.²

- (3) Adam only ate RICE.
 $\rightsquigarrow$ Adam ate rice
 $\rightsquigarrow$ Adam ate nothing else

The contextually-relevant set of alternatives C that we have been referring to is a subset of the focus-semantic value. The context that determines the domain of restriction of the alternative set can be explicit, as in (4), or implicit, as in (5). In the explicit context examples in (4), the context is restricted to only the set of people in the church (4-a), and only the set of people that are already given in (4-b). On the other hand, in the implicit alternative context in (5-a), the restriction is based on the animals that are in the sea. So, based on our knowledge of the world, Adam could not have caught an elephant in the sea, for instance. Similarly, in (5-b), Sam could not have spoken to a shark (for example) at the gym.³

- (4) a. In my church, *only* THE PASTOR preaches.
b. Among Sam, John, and Peter, *only* SAM bought a car.

¹ There is an ongoing debate on whether it is always the case that the focus associate (and by extension the prejacent) and the evoked alternatives are of the same semantic type (see Rooth, 1985, 1992; Beaver and Clark, 2008; Fox and Katzir, 2011; Hirsch and Schwarz, 2023; Hirsch, 2025, a.o.). However, for our purpose here, it suffices to say that the focus associate and its alternatives are of the same semantic type.

² This means that *only* presupposes its prejacent. Although there are discussions in the literature on whether *only* indeed presupposes its prejacent or not, I will not get into this discussion here. See Horn (1996); van Rooij (2002); van Rooij and Schulz (2007); Beaver and Clark (2008); Ippolito (2008) for some discussions.

³ There is also the option of constraining the domain of the set of alternative by the use of questions. This is the Question Under Discussion (QUD) view, where the set of alternatives consists only of alternatives that answer a current question (cf. Roberts, 1996; Beaver and Clark, 2008; Coppock and Beaver, 2014). See Chapter 2.

- (5) a. Adam *only* caught A SHARK in the sea.
 b. Adam *only* spoke to SAM at the gym.

Constraining the set of alternatives to relevant ones is important so as to exclude alternatives that are not relevant in a given context. Such a constraint on the focus-semantic value is licensed by a squiggle operator ($\sim$) which Rooth (1992) included in his compositional mechanism for focus semantics. Importantly, $\sim$ also requires that the prejacent (the ordinary semantic value) be part of the contextually-determined set. Thus, the primary function of the $\sim$ operator is to mediate between focus semantic values and the relevant context.⁴ The requirement of the $\sim$ operator is summarized in (6) with α representing a proposition. So, if we consider (6-a) and (6-b), then the prejacent is an alternative itself (as in: $\llbracket \alpha \rrbracket^o \in \llbracket \alpha \rrbracket^f$). I will not go deeper into the details of the compositional analysis here.

- (6) a. $C \subseteq \llbracket \alpha \rrbracket^f$
 b. $\llbracket \alpha \rrbracket^o \in C$

In simple terms, what we have seen so far is that *only*'s relation with its prejacent entails that every other contextually-relevant alternative proposition, apart from its prejacent, is false, and it presupposes the truth condition of its prejacent (Horn, 1969; Rooth, 1992).

5.1.2 The problem of association

If *only* is a propositional operator, then how do we account for cases where *only* is syntactically adfocal, as in (7) in English and (8) in Kasem, Kusaal and Yorùbá respectively? We already had a brief introduction to this challenge for the syntax-semantics interface in Section 2.3 of Chapter 2. The semantics of *only* requires a proposition in the scope of *only*. But the adfocal forms of *only* in the examples below are not in a propositional scope position. They are all adjoined to direct object focus DPs which are of course not propositions.

- (7) Adam slaughtered $[_{DP} \text{only A FOWL}]$.
- (8) **Adfocal-particles associating with focus**
- | | | | |
|----|-----------------------------------|--|---------------|
| a. | Adam go | $[_{DP} [\text{chworò}]_F \text{yerane}^*(\text{mo})]$. | <i>Kasem</i> |
| | Adam kill.PFV | fowl only _{adf} FOC | |
| | 'Adam slaughtered only A FOWL.' | | |
| b. | Adam kɔdig | $[_{DP}^*(\text{nɛ}) [\text{nua}]_F \text{ma'aa}]$. | <i>Kusaal</i> |
| | Adam kill.PFV | FOC fowl only _{adf} | |
| | 'Adam slaughtered only A FOWL.' | | |
| c. | Ádààmù pa | $[_{DP} [\text{adìe}]_F \text{nìkan}]$. | <i>Yoruba</i> |
| | Ádààmù kill | fowl only _{adf} | |
| | 'Ádààmù slaughtered only A FOWL.' | | |

Adverbial-*only*, on the other hand, does not constitute a problem for the syntax-semantic properties of *only* because it occupies a propositional position by adjoining to

⁴ It does not only ensure that the focus-sensitive operator gets the relevant alternatives, but also see to the question-answer congruence.

vP in the examples in (9) and (10).⁵

(9) Adam [_{vP} only [_{vP} slaughtered A FOWL]].

(10) **Adverbial-particles associating with focus**

- | | | |
|----|---|---------------|
| a. | Adam [_{vP} weeni [_{vP} *(o) go [chworɔ] _F mo]]. | <i>Kasem</i> |
| | Adam only 3SG kill.PFV fowl FOC | |
| | ‘Adam only slaughtered A FOWL.’ | |
| b. | Adam [_{vP} kudim [_{vP} de nɛ [mui] _F]]. | <i>Kusaal</i> |
| | Adam only eat FOC food | |
| | ‘Adam only ate RICE.’ | |
| c. | Ádààmù [_{vP} kàn [_{vP} pa [adìɛ] _F]]. | <i>Yoruba</i> |
| | Ádààmù only kill fowl | |
| | ‘Ádààmù only slaughtered A FOWL.’ | |

5.2 Two approaches

At least two approaches have been proposed to account for the challenge posed by the adfocal-*only* cases: (a) the Quantifier Approach (henceforth QA), and (b) the Propositional (bipartite) Approach (henceforth PA). In this section, I will summarize these approaches (see also Ch. 2), and in the next section, I will provide arguments in support of a propositional approach to exclusives in Kasem, Kusaal and Yorùbá.

5.2.1 The Quantifier Approach

For QA, the adfocal particle remains the exclusive operator which composes with its DP associate to form a single quantifier (see Rooth, 1985; Wagner, 2006, a.o.). In more technical terms, QA assumes a flexibility in *only*’s semantics which allows it to type-shift from being a propositional operator to a quantifier (11). This allows it to be able to compose with the focused DP, forming a single quantifier in the process.⁶ The derived meaning of *only* in such a context is given in (12).

(11) **Type-shifting**

$$\llbracket \text{only} \rrbracket = \llbracket \text{only}_Q \rrbracket$$

(12) **Derived meaning**

$$\llbracket \text{only}_Q \rrbracket^{ALT} = \lambda Q_{\langle \langle e, st \rangle, st \rangle} \cdot \lambda f_{\langle e, st \rangle} \cdot \llbracket \text{only} \rrbracket^{ALT}(Q(f))$$

(Hirsch, 2017, 2022; Hirsch and Wagner, 2019)

(12) may be read as the type-shifted-*only* having two arguments, one that involves a quantifier (DP), and the other that involves a function *f*. The latter forms an input to the former. The result is that the meaning of *only* excludes every other alternative of the quantifier DP with a given function. Therefore, what we see in this version of *only* is that instead of operating directly on propositions, it operates as a quantifier, and still maintains its alternative-sensitive semantics. However, recent studies have cast doubt

⁵ The smallest propositional unit is assumed to be vP, bearing in mind that the subject originates from vP. This is where the argument structure is determined according to Chomsky (2000, 2001).

⁶ This predicts that depending on the semantic type of *only*’s complement, it can type-shift.

on the possibility of type-shifting in semantics (see Heim, 2017; Hirsch, 2017; Sauerland, 2018, a.o.).

Thus, in QA, the adfocal-*only* example in (13) means that only rice x and nothing other than x is affected by the function of eating.

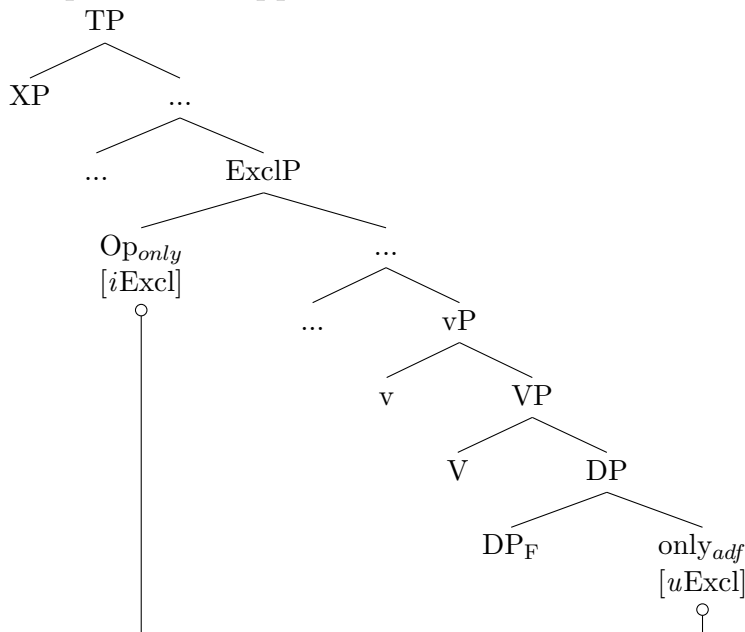
- (13) Adam ate [DP only RICE].
 = [Adam ate nothing other than rice]

5.2.2 The Propositional Approach

The second approach offers a uniform analysis for *only*'s syntax-semantics properties, by discarding the type-shifting requirement that is peculiar to an adfocal syntactic adjunction. The approach has been called differently in the literature: Operator-Particle Approach, Bipartite Approach, and Concord Approach (see Bayer, 1996, 2020; Lee, 2005; Quek and Hirsch, 2017; Sun, 2020, 2021; Hirsch, 2019, 2022; Bassi et al., 2022; Branan and Erlewine, 2023; Yip, 2025, a.o.). Regardless of the name, what all these analyses have in common is that (a) there is a high (c)overt exclusive operator (Op_{only}) which takes a proposition as its complement, and is responsible for the exclusive interpretation, and (b) the adfocal particle is semantically inert. The exclusive Op_{only} and the adfocal particle are then connected via Agree. Whether the adfocal particles in the three languages we are considering here are indeed semantically inert will be the subject of discussion in Section 5.4 below. For now, and importantly for the next section 5.3, I do not assume anything for them.

The derived structure that I assume for the analysis here was given in the example (62) of Chapter 2, which is repeated and modified as (14) below. The adfocal particle adjoins to the focus DP projecting an extended DP, and Agrees with the exclusive operator which adjoins to a propositional position projecting an Exclusive Phrase (ExclP).⁷ The adfocal particle adjoins to the right because, unlike English, the adfocal particles in Kasem, Kusaal and Yorùbá are post-focal.

- (14) **Propositional Approach structure**



⁷ Such a position has been called an Extended Verbal Projection (EVP) by Buring and Hartmann (2001).

The advantage of such an analysis is that it provides a uniform propositional treatment for both adverbial-*only* cases and adfocal-*only* cases, especially in the languages we are considering. It also predicts more flexibility for the position of Op_{only} . This is captured by the use of [...] which means that Op_{only} can occupy a position that is higher than the projections above vP such as NegP, AspP, MoodP, and so on. Op_{only} may also occur in between these projections. In addition, in principle, nothing stops it from occurring above TP, etc. The only condition is that Op_{only} must scope over a proposition, of which the vP is the minimal constituent that makes a proposition, and the scope will contain the focus. It is assumed that if an adverbial-*only* is present, it may overtly realize Op_{only} . This nature of Op_{only} will become evident when I discuss the scope relations of *only* in the next section.

If the adverbial-*only* can realize the exclusive operator in ExclP, then this predicts that we can have cases of concord or doubling of the adverbial-*only* and the adfocal-*only* which produce a single exclusive reading. As we have already seen in the discussion in Chapter 4, this prediction is borne out in the three languages (15).⁸

- (15) a. Adam **weeni** o go [chworɔ]_F **yerane mo**. *Kasem*
 A. only 3SG kill.PFV fowl only_{adf} FOC
 ‘Adam only slaughtered A FOWL.’
 b. Adam **kudim** di **nɛ** mui **ma’a**. *Kusaal*
 Adam only eat FOC rice only_{adf}
 ‘Adam ate only RICE.’
 c. Tolú **kàn** jẹ [lɛ̀sì]_F **nìkan**. *Yoruba*
 Tolú only eat rice only_{adf}
 ‘Tolú only ate RICE.’

The kind of analysis we are discussing is similar to the negative-concord analysis that has been proposed for negative concord languages (cf. Zeijlstra, 2004a, 2006, 2007, 2008, 2022; Penka, 2011; Biberauer and Zeijlstra, 2012, a.o.).⁹ The negative-concord analysis assumes a high interpretable negative operator (with a [*i*NEG] feature) which is responsible for the negative interpretable. It may Agree with multiple occurrences of n-words (each with [*u*NEG]), which produces a structure with a single negative reading. For instance, in Italian, a negative concord language, Zeijlstra (2007) argues that the multiple occurrences of n-words in an example like (16) results in a single negative reading. Such an example is analyzed through a negative concord agreement between the negative operator *non* and n-words *niente* and *nessuno*.

- (16) Gianni non_[iNEG] ha detto niente_[uNEG] a nessuno_[uNEG]. *Italian*
 Gianni NEG has said n-thing to n-body
 ‘Gianni did not call anybody’

Zeijlstra (2007, p.513)

Apart from the morphological doubling cases in (15), in the next section, I provide

⁸ This is, however, not possible in English where both the adverbial-*only* and adfocal-*only* cannot associate with one and the same focus associate.

(i) *Adam **only** killed **only** A FOWL.

⁹ Biberauer and Zeijlstra (2012) proposed a typology for the negative concord phenomena and languages.

further evidence for PA showing that the exclusive interpretation comes from the high exclusive Op_{only} . Thus, the next section does not only propose an analysis for the syntax-semantic properties of exclusive particles in Kasem, Kusaal and Yorùbá, but also provides cross-linguistic support for PA as a better approach, which provides a uniform treatment of exclusives in comparison to QA.

5.3 Evidence for a high exclusive operator

In this section, I present three arguments for PA in Kasem, Kusaal and Yorùbá, viz:

- (17)
- a. Split-scope phenomenon
 - b. Ellipsis and exclusive association
 - c. Multiple association and concord reading

Each of these arguments shows that PA offers a better analysis of *only* in these languages, in terms of both its afocal and adverbial syntactic properties.

5.3.1 Split-scope phenomenon

Recall that in section 4.5 of chapter 4, I introduced the scopal behaviour of *only* that is well-known and cross-linguistically confirmed (cf. Taglicht, 1984; Rooth, 1985, 1992; Lee, 2005; Quek and Hirsch, 2017; Sun, 2021; Branan and Erlewine, 2023, a.o.). We saw that while adfocal-*only* can involve scope ambiguity in the presence of certain embedding predicates (like *know*, *require*, etc.), modals, and other quantificational operators like negation, adverbial-*only* is scope-rigid. For instance, in example (77) of Chapter 4, repeated below as (18), the adfocal particle may scope below (narrow scope) or above (wide scope) the modal.

- (18) Adam may eat **only** RICE. [may > only, only > may]
- a. Narrow scope: Adam may only eat rice (*meaning*: for this particular permission, Adam is allowed to eat nothing but rice, but there could also be other permissions).
 - b. Wide scope: Adam only may eat rice (*meaning*: Rice is the only thing that Adam is allowed to eat. No other permission).

Similarly, in the presence of the auxiliary *have*, the same ambiguity is derived with adfocal-*only*. Consider the sentence in (19). The sentence can either have the exclusive scoping below *have* (19-a), or above it (19-b) respectively.

- (19) Adam has to eat **only** RICE. [have > only, only > have]
- a. Narrow scope: Adam has to only eat rice (i.e., he cannot eat any other thing).
 - b. Wide scope: Adam only has to eat rice (i.e., he may eat other things if he wants).

However, if *only* occupies an adverbial position, the ambiguity disappears as exemplified in (20). In (20-a), the wide scope is unavailable, and in (20-b), the narrow scope is unavailable. So, in each cases, *only* takes scope in its surface position; a behaviour that was not seen with the adfocal-*only* in (19).

- (20) a. Adam has to **only** eat only RICE. [have > only, *only > have]
 (i) Narrow scope: Adam has to only eat rice (i.e., he cannot eat any other thing).
 (ii) *Wide scope: Adam only has to eat rice (i.e., he may eat other things if he wants).
 b. Adam **only** has to eat only RICE. [*have > only, only > have]
 (i) *Narrow scope: Adam has to only eat rice (i.e., he cannot eat any other thing).
 (ii) Wide scope: Adam only has to eat rice (i.e., he may eat other things if he wants).

The same facts are replicated in the three languages that we are considering. In the presence of a modal, the sentence with an adfocal-*only* is ambiguous. Adfocal-*only* in Kasem, Kusaal and Yorùbá takes either narrow scope below the modal ‘can’ or wide scope above it. See (21), (22), and (23) respectively.¹⁰

(21) **Kasem exclusive ambiguous scope with a modal**

Adam **wae** *(o) go [chworɔ]_F **yerane mo**. [may > only; only > may]
 Adam may 3SG kill.PFV fowl only_{adf} FOC
 ‘Adam may slaughter only A FOWL.’

- (i) Narrow scope: It is okay for Adam to only slaughter a fowl (i.e., other animals are allowed).
 (ii) Wide scope: Adam is only allowed to slaughter a fowl (i.e., no other animal is allowed).

(22) **Kusaal exclusive ambiguous scope with a modal**

Adam **tun’e** di-d **nɛ** [mui]_F **ma’a**. [may > only; only > may]
 Adam may eat-IPFV FOC rice only_{adf}
 ‘Adam may eat only RICE.’

- (i) Narrow scope: It is okay for Adam to only eat rice (i.e., other food allowed).
 (ii) Wide scope: Only rice is such that Adam is allowed to eat (i.e., no other food is allowed)..

(23) **Yorùbá exclusive ambiguous scope with a modal**

Tolú **lè** jẹ [iṣu]_F **nìkan**. [may > only; only > may]
 Tolú may eat yam only_{adf}
 ‘Tolú may eat only YAM.’

- (i) Narrow scope: Tolú is allowed to eat only rice (i.e., other food may be allowed).
 (ii) Wide scope: Only rice is such that Tolú is allowed to eat (i.e., no other food is allowed).

The same ambiguity is observed with negation in the three languages. However, I only present a representative example from Yorùbá in (24).

¹⁰ Recall the discussion in Chapter 4, where the different constructions involving a pronoun copy of the subject were mentioned. The construction with modal auxiliaries is one of such.

(24) **Yorùbá exclusive ambiguous scope with negation**

Tolú ò jẹ [iṣu]_F nìkan. [neg > only; only > neg]

Tolú NEG eat yam only_{adf}

‘Tolú did not eat only YAM.’

(i) Narrow scope: he didn’t only eat yam (i.e., he ate something else in addition to yam)

(ii) Wide scope: he only didn’t eat yam (i.e., he ate something else apart from yam)

For each of the possible meanings of *only*, the proposal is that Op_{only} occupies either of the scope positions. For the narrow scope reading, it occupies a position below the modal, as in (25-a). While for the wide scope reading, it occupies a position above the modal (25-b).

- (25) a. LF = Narrow scope: [TP DP_{subj} may [ExclP Op_{only} [vP V [DP only [DP_F]]]]]
 b. LF = Wide scope: [TP DP_{subj} [ExclP Op_{only} [may [vP V [DP only [DP_F]]]]]]]

Unsurprisingly, adverbial-*only*s in these languages take only surface scope. In the sentences in (26), the adverbial exclusives *weeni*, *kùdim*, and *kàn* take surface scope (i.e., wide scope) based on their pre-modal positions. Note that the adfocal-*only* particles may optionally co-occur, with no change in meaning.

(26) **Adverbial-*only* takes surface scope = [only > may; *may > only]**

- a. Adam **weeni** *(o) **wae** *(o) go [chworó]_F (yerane) **mo**. **Kasem**
 Adam only 3SG may 3SG kill.PFV fowl only_{adf} FOC
 ‘Adam only may slaughter A FOWL.’
- b. Adam **kùdim tun’e** di-d **nè** [mui]_F (ma’a). **Kusaal**
 Adam only may eat-IPFV FOC rice only_{adf}
 ‘Adam only may eat RICE.’
- c. Tolú **kàn lè** jẹ [iṣu]_F (nìkan). **Yorùbá**
 Tolú only may eat yam only_{adf}
 ‘Tolú may eat only YAM.’

Similarly, the adverbial exclusive particles can also occur below the modals in the three languages, which derive a narrow scope exclusive reading, see (27).

(27) **Adverbial-*only* takes surface scope = [*only > can; can > only]**

- a. Adam **wae** *(o) **weeni** *(o) go [chworó]_F (yerane) **mo**. **Kasem**
 Adam can 3SG only 3SG kill.PFV fowl only_{adf} FOC
 ‘Adam can only slaughter A FOWL.’
- b. Adam **tun’e kùdim** di-d **nè** [mui]_F (ma’a). **Kusaal**
 Adam can only eat-IPFV FOC rice only_{adf}
 ‘Adam can only eat RICE.’
- c. Tolú **lè kàn** jẹ [iṣu]_F (nìkan). **Yorùbá**
 Tolú can only eat yam only_{adf}
 ‘Tolú can only eat YAM.’

As promising as this evidence from scope ambiguity is to the propositional approach, the data may be accounted for with the quantifier approach as well (cf. Hirsch, 2019, 2022). With QA, the cases we just discussed may involve the derived single quantifier (consisting

of the adfocal-*only* and the focus associate) undergoing quantifier raising below the modal (narrow scope) or above it (wide scope). (28) and (29) show how the derivations look like with quantifier raising.¹¹

- (28) **Adfocal-*yerane*+DP_F QR below the modal *wae* at LF**
 $[_{TP} \text{Adam } [_{Mod} \text{wae } [_{XP} \text{o } [_{QP} [\text{chwor}o]_F \text{yerane mo}]_i [_{vP} \text{go } t_i]]]]]. \quad [can > only]$
- (29) **Adfocal-*yerane*+DP_F QR above the modal *wae* at LF**
 $[_{TP} \text{Adam } [[_{QP} [\text{chwor}o]_F \text{yerane mo}]_i [_{Mod} \text{wae } [_{XP} \text{o } [_{vP} \text{go } t_i]]]]]]. \quad [only > can]$

However, we can argue for PA with a split-scope possibility (à la Hirsch, 2017; Quek and Hirsch, 2017; von Fintel and Iatridou, 2007). The assumption here is that (according to PA) if the exclusive semantics is dissociated from adfocal-*only*, then we might be able to find a scenario where the focus associate would take scope independently of the exclusive particle. One way to do this is to have a focus associate which can scope itself. The prediction of PA in such a situation is that the latter should be able to take scope independently of the semantics of *only*. This is in fact borne out in (30).¹²

- (30) **Split-scope in Kasem, Kusaal and Yorùbá = [only > require > two]**
- a. Adam jege se o go cheeni [delei]_F **yerane mo.** **Kasem**
Adam require to 3SG kill fowl.PL two only_{adf} FOC
- b. Adam nar ye o kɔdig nɛ nuas [ayi]_F **ma’a.** **Kusaal**
Adam require to 3SG kill FOC fowl two only_{adf}
- c. Adam ní lati pa adìḡ [méjì]_F **nìkan.** **Yorùbá**
Adam has to kill fowl two only_{adf}
‘Adam is required to slaughter only TWO fowls.’
Meaning: ‘Adam only has to slaughter (any) TWO fowls (but could slaughter more if he wishes).’

With the derived meaning in (30), where the syntactic position of the adfocal exclusive particles is lower than its scope, QA cannot account for such a split. While the exclusive particles in the languages scope above *require*, the associate of the adfocal-particles scope below *require*. Since QA assumes that the adfocal-*only* forms a single quantifier with its focus associate, it predicts that this derived quantifier will take scope together either below or above another operator. However, it does not predict the possibility of split-scope. Therefore, QA *under-generates* split-scope (Hirsch, 2017; Quek and Hirsch, 2017; Hirsch, 2019). The predicted QA’s LF derivations are given in (31) and (32), respectively. While the former has a non-specific reading for which two fowls are involved, the latter has a specific reading for the exact fowls involved.

- (31) **Narrow scope QA at LF: [require > only > two fowls]**
 $[_{TP} \text{Adam}_i \text{ is } [_{vP} \text{required } [_{TP} t_i \text{ to } [_{vP} [_{QP} \text{only } \textbf{two fowls}]_k [t_i \text{ }_{vP} \text{slaughter } t_k]]]]]]$
Meaning: Adam is required to only slaughter (any) two fowls (and no more).
- (32) **Wide scope QA at LF: [only > two fowls > require]**
 $[_{TP} \text{Adam}_i \text{ is } [_{vP} [_{QP} \text{only } \textbf{two fowls}]_k [_{vP} \text{required } [_{TP} t_i \text{ to } [t_i \text{ }_{vP} \text{slaughter } t_k]]]]]]$
Meaning: There are only two particular fowls that Adam is required to slaughter.

¹¹ For QA, adfocal-*only* and the DP associate form a Quantifier Phrase. Hence the QP in the derivation.

¹² Note that the narrow scope exclusive reading [require > only] is also possible in these examples. However, we are highlighting the wide scope exclusive reading [only > require] here because that is what is important for our current argument for PA. See Section 5.6.5 for a further discussion.

However, PA can account for the split-scope data because the exclusive meaning is dissociated from the adfocal-particles in the languages we are investigating. In (33), the exclusive interpretation is introduced by the exclusive *Op_{only}* in a position above *require*, but the quantified DP (two fowls) scopes below it at the same time. The quantified DP only needs to raise to a position above the embedded verb for the sake of scope.

- (33) **Split-scope PA LF: [only > require > two fowls]**
 [TP Adam_i is [ExclP **Op_{only}** [vP *required* [TP t_i to [vP [DP *only two fowls*]_k [t_i vP slaughter t_k]]]]]]
Meaning: The only requirement for Adam is to slaughter (any) two fowls (may be more).

The point here is that the possibility of split-scope in these languages supports PA. QA does not predict split scope because, in such an approach, both the adfocal particle and the quantified DP (two fowls) would have to scope at the same time either below the modal or above the modal (cf. (31) and (32)). QA therefore undergenerates the split scope possibility.¹³ In the next section, I examine the behaviour of another independent phenomenon (ellipsis) that favours the propositional approach over the quantifier approach.

5.3.2 Ellipsis and exclusive AwF

Based on Tancredi’s (1990, p.30) Principle of Lexical Association (PLA) which requires that a focus-sensitive operator like *only* must have its focus associate in its scope, Beaver and Clark (2008) argue that a conventional focus-sensitive operator and its focus associate can therefore not be separated by ellipsis. They gave the following example in (34). The reason clause (*because Sandy only does*) has the focus-sensitive operator *only* dissociated from its focus associate (*salute*) which is part of the VP ellipsis site. Due to the eliding of the focus associate, the interpretation where “Kim salutes (and does nothing else) because Sandy salutes (and does nothing else)” is not available.

- (34) *Kim only SALUTES because Sandy only does ~~salute~~.
 [cannot mean: ‘Kim salutes (and does nothing else) because Sandy salutes (and does nothing else).’]

(from Beaver and Clark, 2008, p.117)

With the PLA in mind, we may simply state this constraint as (35) below.

- (35) **Beaver & Clark ellipsis constraint on conventional FSPs**
 A focus-sensitive operator like *only* cannot be separated from its focus associate by ellipsis.

Extending such an idea to PA, Bassi et al. (2022) formulated the constraint in (36) based on the ellipsis facts. It requires that the focus associate of the exclusive *Op_{only}* should always be in its scope.¹⁴

¹³ See Hirsch (2017), pp.239-245 for a more elaborate discussion on split-scope in exclusive association with focus.

¹⁴ Notice that the overt position of the adfocal *only* here does not matter. What matters is the position of the exclusive *Op_{only}*.

(36) **PA’s version of Beaver & Clark’s Constraint:**

Op_{only} and its associated Foc cannot be separated by a node targeted for ellipsis.

(adapted & modified from Bassi et al., 2022, p.819)

If we take the ambiguous example in (37) (emphasis on the second conjunct) with the different scope structures in (38) for instance, the constraint in (36) predicts the absence of the wide scope reading because while the exclusive Op_{only} is higher than the modal, the associated Foc is part of the ellipsis site, as in (38-b). This violates (36). This prediction is borne out, according to Bassi et al. (2022), because only the narrow scope reading is available in sentences like (37). The narrow scope reading is available because both the exclusive Op_{only} and its associated Foc are part of the ellipsis site, as in (38-a); they are not separated by a node targeted for ellipsis.

(37) Adam may eat **only** RICE and Sam may [~~eat-only-riee~~] too. [may > only, *only > may]

(38) a. LF = Narrow scope: [TP DP_{subj} may [_{ExcIP} Op_{only} [_{VP} V [~~DP-only~~ [DP_F]]]]]
 b. LF = *Wide scope: [TP DP_{subj} [_{ExcIP} Op_{only} [may [_{VP} V [~~DP-only~~ [DP_F]]]]]]

The languages under investigation provide the same result. The first two examples in (39) and (40) are from Kasem and Kusaal respectively. The (b) examples (see (39-b) and (40-b)) involving VP ellipsis are our concern here. They show that unlike their (a) counterparts (see (39-a) and (40-a)), they only allow for a narrow scope reading, but not a wide scope reading.

(39) **Ellipsis and AwF in Kasem**

- a. Adam **wae** o yeigi [paano]_F **yerane mo**. [can > only, only > can]
 Adam can 3SG buy bread only_{adf} FOC
 ‘Adam can buy only bread.’
- b. [John de wae [_{VP} o yeigi paano ~~yerane mo~~]]. [can > only, *only > can]
 John too can
 ‘John may too.’
Cannot mean: It is only bread x , such that John is allowed to buy x (and no other permission).
But: John is allowed to only buy bread (and nothing else).

(40) **Ellipsis and AwF in Kusaal**

- a. Adam **tun’e** da’ad **ne** [bodobodo]_F **ma’a**.
 Adam may buy.IPFV FOC bread only_{adf}
 ‘Adam may buy only bread.’ [may > only, only > may]
- b. [John me tun’e [_{VP} da’ad ~~ne bodobodo~~_F ~~ma’a~~]].
 John also may
 ‘John may too.’ [may > only, *only > may]
Cannot mean: It is only bread x , such that John is allowed to buy x (and no other permission).
But: John is allowed to only buy bread (and nothing else).

The idea here is that with QA, the wide scope reading is predicted to be available, since the adfocal particles and the DP are assumed to form a single quantifier which can quantifier-raise together above the modals in the languages. So, the unavailability of the

wide scope reading is unexpected under QA. In other words, QA predicts the availability of not only the narrow scope reading, but also the wide scope reading. The structural representations are given in (41-a) and (41-b) (based on the Kusaal example in (40-b)). Based on (41-b), the wide scope reading is predicted to be available by QA, contrary to fact.

- (41) a. Narrow scope: $[_{TP} \text{ John m}\epsilon [_{tun'e} [_{VP} [_{QP} [_{DP} \text{ n}\epsilon \text{ bodobodo}_F] \text{ ma'aa}]_i [_{VP} \text{ da'ad } t_i]]]]]$
 b. Wide scope: $*[_{TP} \text{ John m}\epsilon [_{QP} [_{DP} \text{ n}\epsilon \text{ bodobodo}_F] \text{ ma'aa}]_i [[tun'e [_{VP} \text{ da'ad } t_i]]]]]$

The PA, on the other hand, is able to handle the scopal behaviour of *only* under ellipsis by assuming a dissociation of the exclusive interpretation from the adfocal-particles. Consider the structural representations of the two readings in (42) (using the Kusaal data in (40-b)). The wide scope reading, with the structure in (42-b), is predicted to be unavailable because the exclusive Op_{only} is separated from its focus associate by a node targeted for ellipsis, as in the constraint in (36) above. Conversely, the narrow scope reading is available because the exclusive Op_{only} is not separated from its focus associate by a node targeted for ellipsis; both of them are part of the ellipsis site.

- (42) a. Narrow scope: $[_{TP} \text{ John m}\epsilon [_{tun'e} [_{ExclP} Op_{only} [_{VP} \text{ da'ad n}\epsilon [_{bodobodo}_F] \text{ ma'a}]]]]]]]$
 b. Wide scope: $*[_{TP} \text{ John m}\epsilon [_{ExclP} Op_{only} [_{tun'e} [_{VP} \text{ da'ad n}\epsilon [_{bodobodo}_F] \text{ ma'a}]]]]]]]$

In Yorùbá, the ellipsis condition on the scope of *only* is also realized. Comparing the non-elliptical example in (43-a) with the elliptical example in (43-b), while the former has an ambiguous scope with *nìkan*, the latter only allows a narrow scope, as shown in (43-b)-i-ii.

(43) **Ellipsis and AwF in Yorùbá**

- a. Tolú şèlérí lati wá [ní òla]_F **nìkan**.
 Tolú promise to come LOC tomorrow only_{adf}
 ‘Tolú promised to come only TOMORROW.’ [prom. > only, only > prom.]
 b. Dàpò nàà şèlérí pèlú [TP-lati-wá [ní òla]_F **nìkan**].
 Dàpò too promise also
 ‘Dàpò promised also.’
 (i) [promise > only]: Dàpò made a promise that he would come only tomorrow (the content of the promise is about coming only tomorrow).
 (ii) [*only > promise]: Only tomorrow is such that Dàpò promised to come then (tomorrow is the only day for which Dàpò made a promise to come, i.e., he made no other promise about another day).

This is predicted by the constraint in (36) above. The narrow scope reading is available because both the exclusive Op_{only} and the focus associate are inside the ellipsis site, and are not separated by ellipsis, as in (44-a). However, the wide scope reading is unavailable because Op_{only} is separated from its focus associate by ellipsis, as shown in (44-b).¹⁵

¹⁵ I wish to mention here that there are disparities among the Yorùbá speakers concerning the (un)availability of the wide scope exclusive reading under ellipsis. So, this makes this test a bit uncertain, at least for Yorùbá. The judgment here is by me and two other Yorùbá speakers.

- (44) a. Narrow scope: [TP Dàpò náà [vP $\text{\textcircled{r}}\text{\textcircled{e}}\text{\textcircled{r}}\text{\textcircled{i}} \text{\textcircled{p}}\text{\textcircled{e}}\text{\textcircled{l}}\text{\textcircled{u}} [\text{\textcircled{TP}}\text{\textcircled{-}}\text{\textcircled{lati}} [\text{\textcircled{ExclP}}\text{\textcircled{-}}\text{\textcircled{Op}}_{\text{only}} \text{\textcircled{wá}} [\text{\textcircled{ní}}\text{\textcircled{-}}\text{\textcircled{ò}}\text{\textcircled{la}}]_F \text{\textcircled{nìkan}}]]]]]
 b. Wide scope: *[TP Dàpò náà [ExclP Op_{only} [vP $\text{\textcircled{r}}\text{\textcircled{e}}\text{\textcircled{r}}\text{\textcircled{i}} \text{\textcircled{p}}\text{\textcircled{e}}\text{\textcircled{l}}\text{\textcircled{u}} [\text{\textcircled{TP}}\text{\textcircled{-}}\text{\textcircled{lati}} \text{\textcircled{wá}} [\text{\textcircled{ní}}\text{\textcircled{-}}\text{\textcircled{ò}}\text{\textcircled{la}}]_F \text{\textcircled{nìkan}}]]]]]$$

In summary, based on the ellipsis test, QA over-generates the possible scope reading because it predicts the availability of a wide scope reading under ellipsis. However, PA offers an analysis for the observed scope interactions.

5.3.3 Multiple association and concord reading

In section 5.2.2, I briefly discussed how the propositional approach is similar to the concord analysis proposed for negative concord languages. In the negative concord example from Italian above, repeated below (45), the high interpretable negative Op Agrees with multiple occurrences of uninterpretable n-words according to Zeijlstra (2007). The sentence produce a single negative (concord) reading.

- (45) Gianni non_[iNEG] ha detto niente_[uNEG] a nessuno_[uNEG]. Italian
 Gianni neg has said n-thing to n-body
 ‘Gianni did not call anybody’

(Zeijlstra, 2007, p.513)

Similar to the Neg-concord analysis (Zeijlstra, 2004b, 2006, 2007; Biberauer and Zeijlstra, 2012), PA also predicts the possibility of multiple occurrences of adfocal-*only* with a concord or a single exclusive interpretation. Lee (2004, 2005) was the first to use this argument for Korean adfocal exclusive particle *man*. This prediction is borne out in Kasem, Kusaal and Yorùbá. Consider the context below, with a current question such as *Who ate what?* The sentences in (46) involve multiple foci, where both the subject and the direct object are focused and are associated with adfocal-*only* in the languages.¹⁶

Context: *There are three people in a room [Adam, John and Sam], and there are three dishes in the room as well [rice, bean, yam].*

- (46) [Who ate what?]
 a. Adam **yerane mo** di mumuna **yerane**. Kasem
 Adam only_{adf} FOC eat rice only_{adf}
 b. Adam **ma’a** pa’a di (nɛ) mui **ma’a**. Kusaal
 Adam only_{adf} PST eat FOC rice only_{adf}
 c. Ádààmù **nìkan ni** *(ó) jẹ ìrẹ̀sì **nìkan**. Yorùbá
 Adam only_{adf} FOC 3SG eat rice only_{adf}
 Meaning: ‘The pair <Adam, rice> is the only alternative that satisfies the eating relation.’

The concord reading is derived if we assume an alternative set like (47), where each pair may satisfy the property of eating. However, only the pair <Adam, rice> is true, others are false. In short, this reading may be paraphrased as *nobody ate anything except Adam who ate rice*.

¹⁶ See Chapter 3 on a potential reason for the presence of only one *mo* (and usually on the structurally higher focus) in multiple focus constructions. This is anchored on focus agreement and relativized minimality.

(47) ALT: {Adam ate rice, Adam ate beans, John ate rice, John ate beans, ... }

As in the negative concord languages, the exclusive concord reading requires a single interpreted exclusive operator that quantifies over the alternative pair ($\langle \text{Adam, rice} \rangle$). Structurally, this is represented in (48) (using the Kusaal example), where a single Op_{only} Agrees with the multiple adfocal-*only* cases in order to realize the *only*-concord reading.

(48)
$$[\text{ExclP } \text{Op}_{\text{only}} [\text{TP } [\text{DP } \text{Adam}_F \text{ma}'a] [\text{T}' \text{pa}'a [\text{vP } \text{di} [\text{DP } (\text{n}\epsilon) \text{mui}_F \text{ma}'a]]]]]$$



In addition, there is a second reading, which is in fact the obvious one, resulting in the English sentence “Only Adam ate only rice”. This is the reading where, although only Adam chose to eat only rice, there are others who ate other things (maybe including rice). In other words, Adam is the only one who has the property of eating only rice. Others (Sam and John) have the property of eating other things (including rice). This reading would involve the presence of double exclusive Op_{only} , as in (49).

(49)
$$[\text{Op}_{\text{only}} [\text{TP } [\text{DP } \text{Adam}_F \text{ma}'a] [\text{T}' \text{pa}'a [\text{Op}_{\text{only}} [\text{vP } \text{di} [\text{DP } (\text{n}\epsilon) \text{mui}_F \text{ma}'a]]]]]]]$$



So far, we have seen how multiple instances of adfocal-*only* are associated with a single exclusive Op_{only} . This is typical of a concord analysis which PA predicts. QA only predicts the double exclusive reading, and not the concord exclusive reading. This is because in QA, each of the adfocal particles has an exclusive semantics which composes with its respective focus associate. This under-generates the concord reading.

So far, I have argued that the exclusive semantics is on a high exclusive Op_{only} , which may be overtly realized as the adverbial operators in the languages under investigation. In other words, the primary exclusive semantics does not come from the adfocal exclusive particles. Therefore, I assume the semantic entry in (50) for the covert exclusive Op_{only} and the adverbial exclusive operators in Kasem, Kusaal and Yorùbá respectively (Horn, 1969; Rooth, 1992; Hirsch, 2022, a.o.).

(50) **Lexical denotation of the propositional exclusive operators**

$$\llbracket \text{Op}_{\text{only}} / \text{weeni} / \text{kudim} / \text{kàn} \rrbracket^{\text{ALT}} = \lambda p_{\langle s, t \rangle} . \lambda w : p(w) . \forall p' \in \text{ALT}[p'(w) \rightarrow p \subseteq p']$$

The semantic entry for the adverbial exclusive operators in (50) shows that they take a proposition p of type $\langle s, t \rangle$ in a possible world w , and presupposes that p must be true in w . They then assert that any alternative proposition p' that is true must be entailed by p . In other words, alternatives p' that are not entailed by the prejacent p are false in w . Thus, much like the behaviour of *only* in English, the adverbial exclusive operators in Kasem (= *weeni*), Kusaal (= *kudim*), and Yorùbá (= *kàn*) presuppose that their prejacent is true, while the alternative propositions are false.

In the next section, I will discuss the nature of the adfocal exclusive particles and their contribution to the overall exclusive interpretation.

5.4 The nature of the adfocal particles and a distributed syntax for scalar *only* sentences

Previous accounts have analyzed adfocal particles as semantically vacuous elements. The question is whether this is true, especially for the languages under investigation here. In this section, I show that this is not the case. The adfocal exclusive particles do more than just concord marking. They are involved in the exclusive interpretation of their sentences. I summarize their function as involving the ordering or ranking of the alternative propositions to align with the type of scalarity of *only* in the next Section 5.4.1. After the summary, I elaborate on their function by describing the patterns and differences in the alternative scale involved in each case in Section 5.4.2. Finally, in Section 5.4.3, I propose a distributed syntactic analysis for the scalarity of *only* in Kasem, Kusaal and Yorùbá.

5.4.1 A synopsis of the assumptions

Recall that in Section 4.4 of Chapter 4, I adopt the scalar view of *only* in the languages under investigation. I also discussed the different types of scalarity of *only* in the three languages. In this section, we will elaborate on this view of *only*, and try to embed these types into the analysis that we have proposed so far. In this way, the analysis will also capture the nature or function of the adfocal particles that have, hitherto, been classified as vacuous (agreement) concord elements in the bipartite or propositional approach to *only* (Lee, 2005; Quek and Hirsch, 2017; Hirsch, 2019, 2022; Sun, 2021; Branan and Erlewine, 2023; Yip, 2025, a.o.).¹⁷

I propose a distributed syntax for exclusives in Kasem, Kusaal and Yorùbá by showing that the adfocal particles are not vacuous semantic elements. They encode the scalar type that is under discussion: complement exclusion, rank-order reading, or evaluation of a proposition (i.e., presupposes that the prejacent is little/insufficient). Of the three languages, I show that Yorùbá clearly shows these different exclusive scalarity by making use of different adfocal particles (*nìkan*, *lásán*, and *péré*). On the other hand, I assume that Kasem and Kusaal lack adfocal morphology for the rank-order and the evaluative readings. Their adfocal particles, *yerane* and *ma'aa* respectively, only mark the complement exclusion reading of *only*. However, I assume that in both rank-order and evaluative cases, there are covert adfocal particles in Kasem and Kusaal. In the syntax, the propositional operator (the covert Op_{only} or the overt adverbial exclusive operators) establishes an agreement relationship with the adfocal exclusive particles.

Semantically, I assume that the adverbial exclusive operators exclude alternatives that are higher ranked as false, and the adfocal particles determine the relevant ranking. This function of the adfocal particles can be connected to the strength ranking parameter of Coppock and Beaver (2014), where different exclusives place different restrictions on the strength ranking of alternative propositions that answer the current question under discussion.

¹⁷ However, see Yip (2024) for a distributed meaning approach to exclusive doubling, where he argues that the sentence final particle *zaa3* in Cantonese is not a vacuous concord element, but encodes the not-at-issue scalar semantics in exclusive doubling cases with *zinghai*. My analysis is similar to Yip's (2024) analysis for Cantonese. A major difference, however, is that the Cantonese doubling cases involve sentential particles, unlike the cases here which involve both adverbial/sentential and adfocal particles.

The strength ranking parameter is one of the three parameters proposed by Coppock and Beaver (2014) for all exclusive particles (51).

- (51) a. Type parameter
 b. Question parameter
 c. Strength ranking parameter

The type parameter has to do with the flexibility in the semantic type of the element modified by the exclusive particle. For example, if the exclusive particle as a modifier takes a property of type $\langle e, p \rangle$, the result is a property-modifier $\langle \langle e, p \rangle, \langle e, p \rangle \rangle$ (e.g. VP-*only*). On the other hand, if the exclusive particle modifies a quantifier of type $\langle \langle e, p \rangle, p \rangle$, then it results in a quantifier-modifier $\langle \langle \langle e, p \rangle, p \rangle, \langle \langle e, p \rangle, p \rangle \rangle$ (e.g. NP-*only*). We are dealing with property-modifier cases in this dissertation. Exclusives may also place a restriction on the possible current questions that the prejacent can answer. For example, the sentence: ‘Adam is only a farmer’ answers the question What is so special about Adam?, and not necessarily What does Adam do? This is the question parameter. The third parameter is our concern here. We will see how the answers to the current question (CQ) are ranked in order to derive different exclusive types.

5.4.2 Strength parameterizing with adfocal particles

In this section, I propose that the adfocal particles constrain the scale of the alternative propositions that answer the CQ (cf. Yip, 2024). The assumption is that, for the complement exclusion reading, the ranking of the relevant alternative propositions is that of *logical entailment*.¹⁸ On the other hand, the ranking of the relevant alternatives in the rank-order reading cases is based on contextual or inherent scale (cf. Beaver and Clark, 2008; Coppock and Beaver, 2014). I assume that, although the evaluation of a proposition reading is speaker-oriented, its alternatives’ ranking is that of a non-logical scale as well. In other words, in order to judge whether a prejacent counts as little and/or bad, the prejacent must primarily involve some form of ranking, where the prejacent is ranked lower and evaluated as little or insufficient.

As we have seen in Chapter 4, Section 4.4, the adverbial particles may occur in different exclusive scalar contexts. However, in the presence of the adfocal particles, the distinction becomes clear. For instance, consider the Yorùbá data in (52). The predicative sentence in (52-a), which overtly has only the adverbial-*kàn*, can be ambiguous between a complement exclusion reading and a rank-order reading. For the complement exclusion reading, this means that Tolú is only a student and *nothing other than* that. For the rank-order reading, this means that Tolú is *nothing more than* a student. For evaluation of a proposition reading, based on a speaker’s judgment, the sentence may mean that Tolú is only a student, and that is little or insufficient (maybe for a particular role).¹⁹ However, with the introduction of adfocal-*nikan* in (52-b), only the complement exclusion reading is available. That is why we cannot have the continuation that follows the sentence. Tolú cannot be any other thing, based on the context. Similarly, in (52-c), the presence of the

¹⁸ What I mean by ‘relevance’ here is tied to how close each of the alternative propositions answers the CQ.

¹⁹ Depending on the context, the prejacent (Tolú is a student) may also be considered as bad. For instance, this may be the case in Afghanistan, where the Taliban government discourages female education. So, in that context, Tolú (if female) being a student may be considered as bad. Although that is an extremist view. NB: *Tolú* is both a male-given name and a female-given name among the Yorùbás.

adfocal-*lásán* constrains the exclusive interpretation to be that of a rank-order reading. This means that Tolú does not have a higher status more than being a student. He is not a professor, for instance. This is supported by the continuation that follows the sentence. While the continuation that Tolú is also a cleaner is felicitous (because being a cleaner is ranked lower than being a student), the continuation that Tolú is also a professor is bad, because this is ranked higher than being a student. Thus, in the rank-order reading, as I will show below, it is the higher-ranked alternatives that are excluded.

- (52) [Several people in the meeting are discussing what they do.]
- a. Tolú **kàn** jẹ́ [akẹ̀kòṣ] _F.
Tolú only COP student
'Tolú is only/just A STUDENT.'
 - b. Tolú **kàn** jẹ́ [akẹ̀kòṣ] _F **nìkan**. #Ó tún jẹ́ agbálẹ̀/òjògbón
Tolú only COP student only 3SG also COP cleaner/professor
'Tolú is only A STUDENT.'
(i) *Continuation*: #He is also a cleaner.
(ii) *Continuation*: #He is also a professor.
 - c. Tolú **kàn** jẹ́ [akẹ̀kòṣ] _F **lásán**. Ó tún jẹ́ agbálẹ̀/#òjògbón
Tolú only COP student only 3SG also COP cleaner/professor
'Tolú is just A STUDENT.'
(i) *Continuation* He is also a cleaner.
(ii) *Continuation*: #He is also a professor.

The same observation is present with the examples in (53). While the example in (53-a) is ambiguous between the complement exclusion reading and the rank-order reading, the examples in (53-b) and (53-c) can only derive a complement exclusion reading and a rank-order reading, respectively. Again, see the continuation for the felicitous use of the different adfocal particles.

- (53) [The competition is over, and the medals have been awarded to the participants.
Q: What did Tolú win?]
- a. Tolú **kàn** gba m̀édàlì [fàdákà] _F.
Tolú only collect medal silver
'Tolú only won a SILVER medal.'
 - b. Tolú **kàn** gba m̀édàlì [fàdákà] _F **nìkan**. #Ó tún gba m̀édàlì
Tolú only collect medal silver only 3SG also collect medal
ide/wùrà.
bronze/gold
'Tolú only won a SILVER medal. (He did not win anything *other than* that).'
 - (i) *Continuation*: #He also won a bronze medal.
(ii) *Continuation*: #He also won a gold medal.
 - c. Tolú **kàn** gba m̀édàlì [fàdákà] _F **lásán**. Ó tún gba m̀édàlì
Tolú only collect medal silver only 3SG also collect medal
ide/#wùrà.
bronze/gold
'Tolú only won a SILVER medal. (He did not win anything *more than* a silver medal).'
 - (i) *Continuation*: He also won a bronze medal.
(ii) *Continuation*: #He also won a gold medal.

Complement exclusion and adfocal-*nìkan*: Under the complement exclusion reading, the logical entailment ranking of the relevant alternative propositions is illustrated with the alternatives of (54) in (55).

(54) [Many people were present at the party. So, who did Tolú invite?]

- a. Tolú pe [Adé]_F **nìkan**.
Tolú invite Adé only_{CER}
'Tolú invited only Adé (and no other person).'
- b. Tolú **kàn** pe [Adé]_F (**nìkan**).
Tolú only invite Adé only_{CER}
'Tolú only invited Adé (and no other person).'

(55) **Complement exclusion ALT ranking**

- a. Tolú invited Adé, Sam and John.
- b. Tolú invited Adé and Sam.
- c. **Tolú invited Adé.**

The excluded alternative propositions entail the prejacent that is in (55-c), and may be seen as stronger alternatives based on the strength ranking parameter. However, the at-issue component of *only* rules out any alternative that is stronger than the prejacent. So, this means that Tolú invited Adé has to be the strongest possible answer to the current question (cf. Beaver and Clark, 2008; Coppock and Beaver, 2014). The exclusion of the stronger alternatives in turn entails that Tolu did not invite Sam, and also that Tolu didn't invite John.

Therefore, the scalar nature of the complement exclusion reading is associated with the information strength of the evoked alternative propositions. I assume that the adfocal particle *nìkan* helps in disambiguating this alternative scale from the other scales. In a way, it has access to the set of alternatives that is already computed by the adverbial operator, and is able to reflect the scale of these alternatives in order to derive a complement exclusion reading.

Rank-order and adfocal-*lásán*:

In rank-order readings with exclusive operator *kàn* (cf. (53-c)), the evoked alternatives, including the focus associate itself, are ranked on a scale, as in (56), where the arrow head is the top of the scale.²⁰ The focus associate *silver* is ranked lower than gold on the scale. The claim is that the person in question did not win a medal that is ranked higher than a silver medal. So, the alternative (a gold medal) that is ranked higher than silver is excluded (cf. (56)). Notice that the lower ranked alternative, the bronze medal, is not excluded. I will return to this in a moment below.

(56) **Rank-order ALT ranking**
 $\overbrace{\{\text{bronze} \quad \text{silver} \quad \text{gold}\}}$

I assume that the adfocal particle *lásán* disambiguates the strength ranking of the alternatives in the rank-order reading from that of the complement exclusion reading. The scalar analysis for the complement exclusion reading is associated with the informational strength of the alternatives based on logical entailment. However, the scalar analysis

²⁰ Although I only gave the focus associate and the possible alternatives in the set, these alternatives are whole propositions, and not just individual words.

for the rank-order reading is associated with the ranking of the alternatives based on a contextual or inherent scale. I present two pieces of evidence for the difference between the complement exclusion meaning of *only*, and the rank-order reading below.

Differentiating Rank-order meaning from complement exclusion meaning:

Unlike the complement exclusion property of *nìkan* that excludes every other alternative minus the prejacent, in rank-order meaning (with *lásán*), lower alternatives (than the prejacent) may not be excluded (Alxatib, 2020). With the provided context below, the sentence with *lásán* in (57-a) is felicitous, even though there is a weaker alternative (a bronze medal) to the focus associate (a silver medal). However, a higher medal like gold is excluded. Meaning, Tolú did not win a gold medal. But this does not exclude the fact that he won a bronze medal as well. The highest medal he won at the competition was a silver medal, so he needs to be humble. On the other hand, the sentence with *nìkan* in (57-b) is infelicitous because this would exclude every other alternative, including winning a bronze medal, contrary to the facts of the provided context.

- (57) [Tolú won 1 bronze medal and 1 silver medal, while Ámúsàn won 2 gold medals. Yet, Ámúsàn remains humble, but Tolú displays pride. So, you say the following.]
- a. Tolú **kàn** gba mэдàlì [fàdákà]_F **lásán**.
 Tolú only collect medal silver only
 ‘Tolú just won a SILVER medal. (He did not win anything *more than* a silver medal).’
- b. #Tolú **kàn** gba mэдàlì [fàdákà]_F **nìkan**.
 Tolú only collect medal silver only
 ‘Tolú only won a SILVER medal. (He did not win anything *other than* a silver medal).’

Furthermore, the different alternative scales between the complement exclusion reading and the rank-order reading can be distinguished by their inference under negation. Consider the Yorùbá example in (58). In the complement exclusion reading, the prejacent proposition is entailed under negation, such that ‘Tolú did not only eat yam’ entails that Tolú ate yam and something else.

- (58) Tolú ò **kàn** jẹ [iṣu]_F **nìkan**.
 Tolú NEG only eat yam only
 ‘Tolú did not only eat YAM. (He ate something else (maybe beans) as well)’

On the other hand, in the rank-order reading, see (59), the prejacent is not entailed under negation, such that ‘Tolú is not just a farmer’ does not entail that Tolú is a farmer and something else. It means that Tolú has a more prestigious (higher-ranked) profession.

- (59) Tolú ò **kàn** jẹ [àgbè]_F **lásán**.
 Tolú NEG only COP farmer only
 ‘Tolú is not just a FARMER (he is a more than a farmer).’

Evaluation of a proposition and adfocal-*lásán*/péré:

We saw in Chapter 4 that both *lásán* and *péré* can be used as adfocal particles in the context where a proposition is evaluated as either little or bad, or even both. The example is repeated below, where *péré* only conveys that the prejacent is little, as in (60-a), and *lásán* can convey that the prejacent is both little and bad (60-b).

- (60) [Tolú promised his wife to buy five fish on his way home. But he did not keep his promise. He only got two fish.]
- a. Tolú rà ẹja [méjì]_F **péré**.
 Tolú buy fish two only_{lit}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is little (not bad) compared to the expected number.
- b. Tolú rà ẹja [méjì]_F **lásán**.
 Tolú buy fish two only_{lit/bad}
 ‘Tolú bought only TWO fish.’
Meaning: That Tolú bought two fish is both little and bad compared to the expected number.

Repeated below is an example from the last chapter, which provides a context for the dissociation of *péré* from *lásán*. Based on the context, little is a good thing, and not bad. Therefore, it is *péré*, and not *lásán*, that is felicitous.

- (61) [Sam asked you to help him pick up some bags from the post office on your way back. However, you were worried because you only have a bicycle, and you didn’t know how many bags there are. Luckily for you, you only got two bags from the post office.]
- a. O gbé àpò [méjì]_F **péré** láti ilé ifiwé ránsé.
 2SG carry bag two only_{lit} from house use.book send
 ‘You carried only/just TWO bags from the post office.’
 (i.e., This is (positively) little, but not bad)
- b. #O gbé àpò [méjì]_F **lásán** láti ilé ifiwé ránsé.
 2SG carry bag two only_{bad} from house use.book send
 ‘You carried only/just TWO bags from the post office.’
 (i.e., This is (positively) little, but not bad)

I assume that the evaluation of a proposition reading makes use of the alternative ranking that is similar to that of the rank-order reading. The only difference is that the scale in the evaluation of a proposition reading could be speaker-oriented or context-dependent. In other words, what is considered little or bad is speaker-dependent, as well as it is context-dependent (cf. Hole, 2015). Therefore, the evaluative meaning is that the prejacent is little or insufficient (and maybe bad), and the assertion is that the relevant alternatives are excluded. Similar to the rank-order reading, it is the higher-ranked alternative that are excluded, as in (62).

- (62) **EVAL of a proposition ALT ranking**
 {one two three four five}

Alternative ranking in Kasem and Kusaal:

Unlike Yorùbá, recall that the Mabilia languages, Kasem and Kusaal, have only one adfocal exclusive particle each, and they are only felicitous for the complement exclusion meaning (cf. Chapter 4, Section 4.4). The adfocal particles can be doubled with the adverbial particles. However, in the rank-order readings (and the evaluation of a proposition readings of course), only the adverbial exclusive particles are usually used since Kasem and Kusaal do not have adfocal particles for rank-order readings and the evaluative read-

ings. These facts are supported with the following examples from Kasem (63) and Kusaal (64) respectively. Given the context, the (b) sentences have the rank-order reading in the presence of the adverbial operators. The (a) sentences, on the other hand, do not.²¹

- (63) **Kasem:** [Some old friends had a get-together after many years apart, and were mentioning their professions. Among them were bankers, engineers, and medical doctors, but...]
- a. #John ye [valo]_F **yerane** mo.
 John COP farmer only_{adf} FOC
 ‘John is only A FARMER (and nothing more).’
Cannot mean: John being a farmer is lowly ranked.
- b. John **weeni** o ye [valo]_F (#**yerane**) mo.
 John only 3SG COP farmer only_{adf} FOC
 ‘John is only A FARMER (and nothing more).’
Means: that John is a farmer is rank lower on the scale of relevant alternatives (such as being a banker, an engineer, etc).
- (64) **Kusaal:** [Some old friends are having a get-together after years apart, and are mentioning their professions. Among them were bankers, engineers, and medical doctors, but...]
- a. #Adam anε [kpaad]_F **ma’aa**.
 Adam COP farmer only_{adf}
Cannot mean: ‘Adam is just A FARMER (and being a farmer is low on the scale of relevant professions).’
- b. Adam **kudim** anε [kpaad]_F (#**ma’aa**).
 Adam only COP farmer only_{adf}
 ‘Adam is just A FARMER (and being a farmer is low on the scale of the relevant professions).’

I assume that the discussion on the ranking of alternatives in Yorùbá can be extended to Kasem and Kusaal because the three languages share at least two important properties in common: (a) the adverbial exclusive operators can occur across the different scalar meanings, and (b) the complement exclusion reading is ad focally marked in the three languages, and these particles are infelicitous in rank order meaning and evaluative meaning. The only difference is that while Yorùbá has more robust ad focal morphology (i.e., has distinct ad focal particles for the three scalar meanings), Kasem and Kusaal do not. Nonetheless, I assume that the ad focal particles for the rank-order reading and the ‘evaluation of a proposition (as little/bad)’ reading contexts are covertly realized in Kasem and Kusaal. This is indirectly supported by the fact that the only recorded ad focal exclusive particle in each of the languages (Kasem and Kusaal) is only felicitous with a complement exclusion interpretation, and not in the other scalar interpretations.

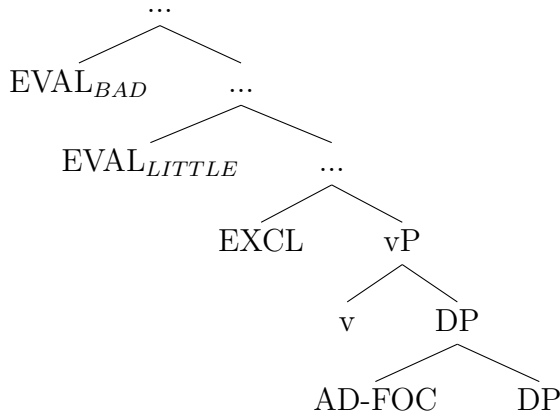
In the next section, I propose a syntactic analysis for these scalar types, and argue, following Hole (2015), that the exclusive contribution can be distributed across different syntactic projections.

²¹ These sentences can only realize complement exclusion readings, where John’s only occupation is farming, and Adam’s only occupation is farming. They are not both a farmer and a trader, for instance. But this is not relevant here.

5.4.3 A distributed syntax for exclusive scalarity

Structurally, building on the proposed Propositional Approach’s structure given in (14) above, I adopt an extended scalar-*only* structure from Hole (2015, p.48), as in (65). Hole assumes a C/T (C/I in his words) level evaluative projections, which he called $\text{EVAL}_{\text{LITTLE}}$ and EVAL_{BAD} , above what I have hitherto labeled as Op_{only} . Hole uses EXCL. Nothing hinges on this, as both are the same propositional exclusive operator which may be covert. According to Hole, both $\text{EVAL}_{\text{LITTLE}}$ and EVAL_{BAD} may constitute a single head or may be separated, as in (65). Both are responsible for the ‘evaluation of a proposition’ reading, while EXCL is responsible for both the complement exclusion reading (primary scalarity) and the rank-order reading (secondary scalarity).²² Hole further argues for the presence of a covert adfocal particle which adjoins to the focus associate. The C/T level evaluative projections Agree with the covert AD-FOC in what he assumes to be a purely syntactic process. In Hole’s analysis, the AD-FOC head functions as a focus marker, and it is semantically vacuous. It simply marks the constituent which is to be evaluated in the sentence.

(65) **Hole’s Scalar-*only* projections**



In the current analysis, I will use Yorùba as a case study because it provides us with distinct morphology for these different exclusive scalar meanings, but I assume that the same analysis can be extended to Kasem and Kusaal. I assume that AD-FOC is overtly realized by the different adfocal particles, and (as we have seen) they are *not* semantically vacuous, contrary to earlier claims (*pace* Hole, 2015; Quek and Hirsch, 2017; Branan and Erlewine, 2023; Bassi et al., 2022; Yip, 2025, a.o.). Which adfocal particle realizes AD-FOC is dependent on the scalar context and the relevant alternative scale that they parameterize. I summarize the function of the different AD-FOCs at the semantic level in (66), where the adfocal particles are assumed to be exponents of scalar agreement in the syntax.²³

(66) **Adfocal particles as exponents of scalar agreement in the syntax**

- a. $\text{AD-FOC}_{\text{CER}}$ = the prejacent proposition is the strongest possible alternative to the CQ on an entailment scale.
- b. $\text{AD-FOC}_{\text{ROR}}$ = the prejacent proposition is low on the relevant scale of alternatives.

²² Regardless of the scalarity type, the EXCL is responsible for the exclusion of the relevant alternatives.

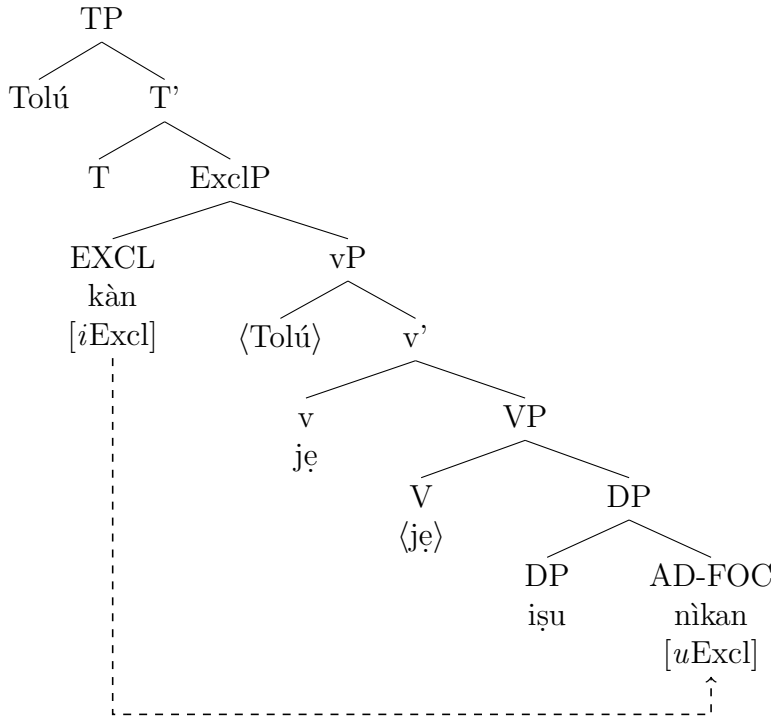
²³ The following are the subscripted abbreviations: CER = Complement Exclusion Reading; ROR = Rank-Order Reading; EPR = Evaluation of a Proposition Reading

- c. $\text{AD-FOC}_{\text{EP}R}$ = the prejacent proposition is low on the relevant scale, and is considered little/insufficient depending on the speaker's judgment, some kind of standard, or a contextual norm .

The structural representation for the complement exclusion scalar sentence is given in (68), based the repeated example in (67). The adverbial operator *kàn* Agrees with the AD-FOC head, which is realized as *nìkan*. At the syntactic level, the agreement simply connects the adverbial operator with the adfocal particle.²⁴

- (67) Tolú **kàn** pe [Adé]_F **nìkan**.
 Tolú only invite Adé only_{CER}
 'Tolú only invited Adé (and no other person).'

- (68) **Structural representation for CER**

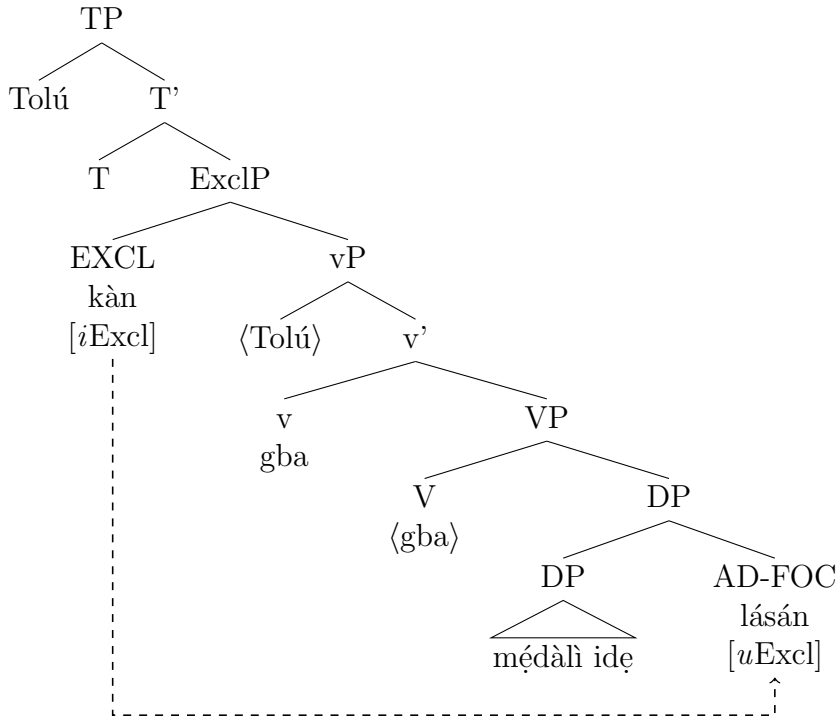


The structural representation for the rank-order scalar sentence is given in (70), based on the repeated example in (69). The adverbial exclusive particle Agrees with the adfocal particle.

- (69) Tolú **kàn** gba m̀ẹ̀dàlì [idẹ]_F **lásán**.
 Tolú only collect medal bronze only
 'Tolú only won a BRONZE medal. (He did not win anything *more than* a bronze medal).'

²⁴ I do not develop a compositional semantic analysis for this. For a compositional semantic analysis of ideas that are similar to the ones here, see Yip (2024).

(70) **Structural representation for ROR**

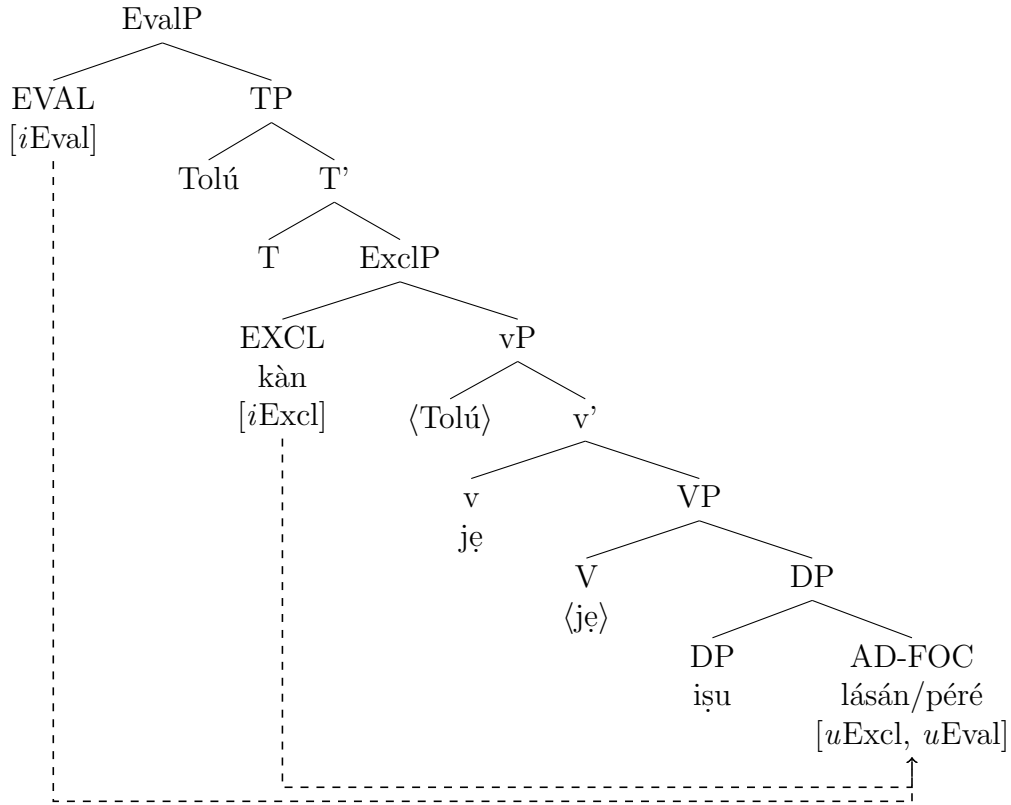


For the evaluation of a proposition as little/bad reading, I assume that since such an interpretation is speaker-oriented, then it also involves a C/T level projection which I call an Evaluative Phrase (EvalP) (Cinque, 1999, 2004; Rizzi, 2004; Hole, 2015).²⁵ I propose that EVAL probes and Agrees with the AD-FOC head, as in (71). Notice that the AD-FOC head is now realized as either *lásán* or *péré*, with the latter only felicitous with an ‘evaluation of a proposition as little’ reading. The EXCL head is still responsible for the alternatives exclusion. However, it is the EVAL head that is responsible for the evaluative presupposition, and it Agrees with the AD-FOC head.²⁶ The assumption here is that EXCL and EVAL Agree with the same goal. While the agreement between AD-FOC and EXCL creates the link for the exclusion of the alternatives, which is based on the scale introduced by the adfocal particle, the second agreement process involves an evaluative feature, which is responsible for evaluating the prejacent. The link that this agreement creates helps in constraining the specific part of the prejacent that is focused, and is evaluated.

²⁵ I provide some arguments for this projection in the next two sections below.

²⁶ Conceptually, having such a projection above the TP would allow the speaker to be able to evaluate or judge the proposition. There are already syntactic frameworks which postulate sentence-peripheral projections for speaker-oriented meanings (cf. Rizzi, 1997, 2004; Cinque, 1999, 2004; Speas and Tenny, 2003, a.o.). An alternative idea would be to assume that the evaluative presupposition is just a pragmatic inference, and has nothing to do with a dedicated syntactic structure.

(71) **Structural representation for EPR**



In the following sections (specifically, Sections 5.4.4 and 5.4.5), I provide arguments for (a) a C/T level evaluative projection (EvalP) in the languages under investigation, and (b) a separation between the projections of the evaluative heads in the form of $\text{EVAL}_{bad} > \text{EVAL}_{little}$.

5.4.4 EVAL_{little} & EVAL_{bad} are distinct heads that are separate from EXCL

Using arguments from reduced C domains, Hole (2015) argues that $\text{EVAL}_{little/bad}$ can further be divided as separate heads in the clausal periphery, where EVAL_{bad} is higher than EVAL_{little} . Haegeman (2010) argues that certain adverbial clauses are not compatible with some evaluative projections (in line with Cinque’s (1999; 2004) adverbial hierarchy). More specifically, she said that “evaluative adverbs are not easily compatible with temporal and conditional adverbial clauses”, and this is exemplified with the conditional adverbial clause in (72) (from Haegeman (2010), p.630).

(72) *If they *luckily* arrived on time, we will be saved.

However, Hole (2015, p.62) shows that this is not the case with reason clauses which allows the evaluative ‘good’ adverb *luckily*, as in (73) below. This means that while conditional clauses have impoverished C domains, reason clauses do not.

(73) Since they *luckily* arrived on time, we will be saved.

Based on our analysis which assumes that the evaluative inference of the exclusive particles comes from a C/T level EVAL projection, which then Agrees with the adfocal particles (*lásán* or *péré*) (cf. (71)), the prediction would be that the evaluative readings

should be unavailable in adverbial clauses which are said to be incompatible with evaluative adverbs: i.e., temporal and conditional adverbial clauses. This prediction is borne out. The reason clause in (74-a) has an interpretation, where just giving me money is a bad reason for getting married to you. So, the use of *lásán* is compatible with the evaluation here. However, the conditional clause in (74-b) lacks that same presupposition.²⁷ It cannot be interpreted as conveying that giving me money is a bad condition for getting married to you. This is why the use of *lásán* is bad here. This follows from Hole's (2015) analysis for German.

- (74) a. O fẹ n fẹ ọ [nítorípé o **kàn** fún mi [lówó]_F **lásán**].
 2SG want 1SG marry 2SG because 2SG only give 1SG money only
 'You want me to marry you just because you gave me MONEY.'
 [EVAL bad present: That you just gave me money is not a good reason for me to marry you.]
- b. Tolú fẹ n fẹ ọ [tí o bá **kàn** fún mi [lówó]_F #**lásán**].
 Tolú want 1SG marry 2SG if 2SG if only give 1SG money only
 'Tolú want me to marry you if you just give me MONEY.'
 [EVAL bad absent: *That you just gave me money is not a good condition for me to marry you.]

Interestingly, although the evaluation as bad reading is absent, the 'evaluation as little' reading can still be derived in example (74-b) above. This would mean that giving me money is not enough condition to marry you. Predictably, (74-a) also has the meaning that giving me money is not a sufficient reason to marry you.²⁸

We can further buttress our point here if we consider the other adverbial clause (i.e., the temporal adverbial clause) which has been claimed to lack certain evaluative projections. If we compare the reason clause in (75-a) with the temporal clause in (75-b), while the former has both evaluative as little and bad reading, the latter only has an evaluation as little reading.²⁹ (75-b) does not have an evaluation as bad reading. Recall that I already showed that *lásán* can mark an evaluation as little reading and/or evaluation as bad reading. So, its infelicity in (75-b) (and (74-b) above) is only with respect to the EVAL bad reading (and not little).

- (75) a. Ìyàwò Tolú n bínú [nítorípé ó jẹ ọ̀ṣu [mẹ̀jì]_F **lásán**].
 wife Tolú PROG angry because 3SG eat yam two only
 'Tolú's wife is angry because he ate only TWO slices of yam.'
 [EVAL bad present: Tolú's eating of just two slices yam is considered not only little, but also bad, and this has angered his wife.]
- b. Bólá bẹ Tolú [kí ó to jẹ ọ̀ṣu [mẹ̀jì]_F #**lásán**].
 Bólá beg Tolú before 3SG before eat yam two only
 'Bólá begged Tolú before he ate only TWO slices of yam.'
 [EVAL bad absent: *Tolú's eating of just two slices of yam is considered not only little, but also bad, after he was begged by Bólá.]

²⁷ In conditional clauses, Yorùbá uses a periphrastic word (tí ... bá) for the English equivalent *if*. Usually, the subject occupies the position between *tí* ... *bá*.

²⁸ Recall that the adfocal particle *lásán* can have both 'evaluation as little and bad' reading. It is *péré* that only has an 'evaluation as little' property. So, the prediction is that *péré* is equally acceptable with reason and temporal clause.

²⁹ Just like the conditional clauses, the Yorùbá equivalent of the temporal adverb *before* is also periphrastic in nature, as in *kí* ... *tó*.

The discussion in this section supports (a) the dissociation and the hierarchy that exist between the EVAL heads [$\text{EVAL}_{\text{bad}} > \text{EVAL}_{\text{little}}$], and (b) the dissociation of the evaluative presupposition from the EXCL head (à la Hole, 2015) (see the simplified structure in (65) above). Based on the adverbial clauses that were considered, the assumption would be that while the reason clauses are big enough to accommodate EVAL_{bad} projection (and $\text{EVAL}_{\text{little}}$ of course), the conditional and temporal clauses are not, as they seem to only have $\text{EVAL}_{\text{little}}$ projection.³⁰

Although the exact position of the EvalP is unknown, but we have reasons to believe that its position is quite low in the clausal periphery. This is corroborated by the fact that the evaluative readings are present in different reduced complement clauses. Consider the following control complement clauses in (76). In both subject control clauses, the evaluative readings are present despite the reduced status of the clauses. The result is the same in the object control clause, as in (77).

- (76) [After Tolù's wife took her time to cook for Tolù for more than 10 hours, Tolù just wants to eat two slices of yam.]
- a. Tolù **kàn** fẹ jẹ iṣu méjì **lásán/péré**.
Tolù only want eat yam two only
'Tolù wants to eat only two slices of yam. (i.e., eating two slices of yam is considered little and/or bad)'
- b. Tolù **kàn** ṣèlérí lati jẹ iṣu méjì **lásán/péré**.
Tolù only promise to eat yam two only
'Tolù promised to eat only two slices of yam. (i.e., eating two slices of yam is considered little and/or bad)'
- (77) Tolù **kàn** pè mi lati jẹ iṣu méjì **lásán/péré**.
Tolù only invite me to eat yam two only
'Tolù invited me just to eat two slices of yam. (i.e., eating two slices of yam is considered little and/or bad)'

What we can deduce (or hypothesize) from this is that the evaluative phrase may be positioned lower in the clausal architecture, at least below the parts that are typically truncated in reduced clauses. I leave this as an open issue for now.³¹

5.4.5 EvalP is separate from ExclP

The prediction that having distinct heads for both EXCL and EVAL makes is that, their interaction should be blocked in the presence of an intervening element. This prediction is borne out in the presence of an intervening 'evaluative reversal negation' according Hole (2015). Consider the following Yorùbá examples in (78). The sentence in (78-a) has the evaluation of a proposition as little reading. This means that the fact that Tolù only bought two fish is considered little in comparison to buying more fish, for instance. If the sentential negative element is introduced, as in (78-b), the evaluation of a proposition as little reading is blocked. This is supported by the infelicity of *péré* in the example.

³⁰ Apparently, the question of how far apart are these EVAL heads remains. However, I do not intend to go into that discussion here.

³¹ A potential analysis is to assume that EvalP is above TP, but below FinP.

- (78) a. Tolú **kàn** ra ẹja [méjì]_F **péré**.
 Tolú only buy fish two only
 ‘Tolú only bought TWO fish.’
 [EVAL little present: that Tolú did not buy more than two fish is considered little or insufficient]
- b. Tolú ò **kàn** ra ẹja [méjì]_F (**#péré**).
 Tolú NEG only buy fish two only
 ‘Tolú did not only buy TWO fish. (i.e. he bought more)’
 [EVAL little blocked: (That) he bought more fish than two (#is considered little or insufficient)]

We cannot use *péré* in (78-b) because the sentence now means that Tolú bought more fish, and this cannot be considered little. I assume that the lack of the evaluative reading is due to the intervention by the negative element between EVAL and EXCL, as in (79) (cf. Hole, 2015).

- (79) *[_{CP} ... EVAL ... NEG ... EXCL ...]

Interestingly, the evaluative reversal under negation is restricted to the clause-mate context. In an embedded clause construction, if the negative morpheme occurs in the matrix clause, and the complement clause contains the evaluative particle *péré*, then the ‘evaluation as little’ reading is present (80). This follows from the fact that the EVAL little’s projection occupies a position in the periphery of the embedded clause. Since the negative element occur in the matrix clause, it does not intervene between EVAL and EXCL in the embedded clause, see (81). This result supports the separation between the EVAL head and the EXCL head.

- (80) Mi ò gbàgbọ́ wípé Tolú **kàn** ra ẹja [méjì]_F **péré**.
 1SG NEG believe COMP Tolú only buy fish two only
 ‘I do not believe that Tolú only bought TWO fish.’
 [EVAL little present: that Tolú did not buy more than two fish is considered little or insufficient]
- (81) [_{TP} ... NEG ... [_{CP} ... EVAL ... EXCL ...]]

The negation in (80) only negates the perception verb *gbàgbọ́* (‘believe’), and not the embedded proposition that Tolú bought only two fish. So, the proposition that Tolú only got two fish is still considered little, and *péré* is felicitous. In other words, more fish was not gotten, and this is considered little/insufficient.

In summary, this section has expanded the discussion on both the nature of the adfocal particles and the different scalar types of *only*. I argued that, contrary to earlier claims, the adfocal particles are not semantically vacuous. They are involved in the strength parameterization of the relevant alternatives depending on the type of scalarity of *only* that is under discussion. I further proposed a syntactic analysis for these different scalar exclusives by arguing for distinct heads within the clause with agreement connecting these heads with the adfocal particles. In a complement exclusion and a rank-order context, the EXCL head Agrees with the adfocal particles involved. However, in the speaker-oriented evaluative readings, I assume a clausal periphery EvalP whose head is responsible for the evaluative reading, but must also establish an agreement with the relevant adfocal particles (*lásán* or *péré*). The EXCL head is always responsible for the semantic exclusion of the relevant alternatives. Therefore, the discussion here supports a distributed syntax

analysis for the scalarity of exclusives (à la Hole, 2015).

Before moving to the next section, I want to briefly mention a prediction that the proposed analysis seem to make, which is the fact that in the absence of both particles, the exclusive interpretation should still be present due to the presence of a covert *Op_{only}*. However, it is important to note here that under the proposed approach, the covert operator licenses the presence of the adfocal particle, even though both of them play distinct, yet interdependent, roles as discussed in the last section. This is required to make the focus association relation visible. On the other hand, when the covert operator is overtly realized (lexicalized) with the adverbial particle, the adfocal particle does not necessarily need to be overt since there is already an exclusive element in the structure. However, when it is present, as we have seen, it not only condition scale of the alternatives, but can also highlight or mark the constituent that is in focus, for instance in Yorùbá. Apparently, this discussion is in need of further consideration, which will be attempted in a future study.

In the next section, I discuss another interesting phenomenon that has, so far, not been investigated in this dissertation. This is the backward association with focus phenomenon. It has to do with a situation whereby a focus sensitive operator (like *only*) associates with a focused constituent that is not in its surface scope.

5.5 Backward Association with Focus (BAwF)

The data that we have discussed and analyzed in this chapter so far have primarily been based on association with in-situ focus. However, in Chapter 4, we saw that the focus associates can also be realized in an ex-situ position. In other words, the languages under investigation also allow for focus fronting to the left periphery. The question is whether such data can be analyzed using the propositional approach for *only* that has been advocated for and developed in this chapter (see Sections 5.2 and 5.3, especially). The examples in (82), (83), and (84) are cases of ex-situ focus with association with the adfocal focus-sensitive particles in Kasem, Kusaal and Yorùbá, respectively.

- (82) [Chworo]_F **yerane** *(**mo**) Adam go-a _____. **Kasem**
 fowl only FOC Adam kill-DISJ
 ‘Adam slaughtered only A FOWL.’
- (83) [Nua]_F **ma’aa** *(**ka**) Adam kɔdig _____. **Kusaal**
 fowl only_{adf} FOC Adam kill
 ‘Adam slaughtered only A FOWL.’
- (84) [Ìrẹ̀sì]_F **nìkan** *(**ni**) Tolú jẹ _____. **Yorùbá**
 rice only_{adf} FOC Tolú eat
 ‘Tolú ate only RICE.’

Since we have argued that the main exclusive semantics is not on the adfocal particles, but on a Propositional Operator (*Op_{only}*) which occupies a propositional position, the position of *Op_{only}* in relation to the fronted foci in these languages can be represented as in (85).³² A simplified schematic representation is given in (86). In both illustrations, the ex-situ focus is outside the surface scope of *Op_{only}*.

³² There is also another possible analysis, which I will not pursue here. This is the possibility of having a high *Op_{only}* (preceding the foci) in all the ex-situ focus association cases in (82)-(84) above.

(85) [CP/FocP [DP [DP]_F **Ad-foc**]₂ Foc [TP Subj₁ [ExclP **Op_{only}** [vP t₁ [v' v [vP V t₂]]]]]]

(86) [[[DP]_F Ad-foc] ... Op_{only} ... t]

Recall that we have shown that Op_{only} may be overtly realized as the adverbial exclusive focus-sensitive operators in these languages. So, the sentences with a covert Op_{only} in (82), (83), and (84) may involve an overt realization of Op_{only} as in (87). The simplified schematic representation will look like (88).

- (87) a. [Chworo]_F (**yerane**) *(**mo**) Adam **weeni** o go-a _____. **Kasem**
 fowl only FOC Adam only 3SG kill-DISJ
 ‘Adam slaughtered only A FOWL.’
 b. [Nua]_F (**ma’aa**) *(**ka**) Adam **kɔdim** kɔdig _____. **Kusaal**
 fowl only_{adj} FOC Adam only kill
 ‘Adam slaughtered only A FOWL.’
 c. [Ìrɛ̀sɪ]_F (**nìkan**) *(**ni**) Tolú **kàn** jẹ _____. **Yorùbá**
 rice only_{adj} FOC Tolú only eat
 ‘Tolú ate only RICE.’

(88) [[[DP]_F (Ad-foc)] ... weeni/kɔdim/kàn ... t]

Importantly, even though the focus associates are ex-situ in (87) (cf. (88)), we still get the exclusive associative reading, where the preajacent proposition (containing the focus) is the only true proposition in a relevant set of alternatives. Other alternative propositions are excluded.

The interesting fact here is that, although the Principle of Lexical Association (introduced in Chapter 2) requires that a focus-sensitive operator like *only* c-commands its focus associate (Tancredi, 1990), we have a case where the adverbial focus-sensitive operators in these languages seem to associate *backwards* with their ex-situ focus associate. This happens regardless of the presence or absence of the adfocal particles. As shown in the examples in (87), they are optional. Such a *Backward Association with Focus* (BAwF) configuration is said to be logically impossible (see Jackendoff, 1972; Jacobs, 1983; Büring and Hartmann, 2001; Erlewine, 2014b, a.o.).

In order to have a good understanding of what is going on here, I present a variation that exists between English and German with respect to BAwF in the next section (Section 5.5.1). After that, in Section 5.5.2, I propose an analysis for the phenomenon of BAwF in the languages under investigation. In summary, in order to account for the presence of BAwF in these languages, I assume an interaction between the propositional approach and reconstruction of the fronted focused constituents.

5.5.1 A variation in English and German

BAwF in English

In English, it has been argued that the (adverbial) focus-sensitive operator *only* must c-command its focus associate, as in (89-a). If the focus associate is not c-commanded by *only*, as in (89-b), the association is not possible. In other words, *only* cannot associate backwards.³³ However, this is only the case with *only*, and not *even*, as the sentence in (90) allows *even* to associate with a focus that has moved out of its c-command domain.

³³ Notice that the ungrammaticality is regarding the exclusive association that is at-issue, and not the sentence per se.

- (89) a. Adam *only* ate RICE.
 b. *RICE, Adam *only* ate.
- (90) RICE, Adam *even* ate.

This observation between *only* and *even* was put forward by Jackendoff (1972), and the requirement that operators like *only* must c-command their focus associate was introduced by Tancredi (1990) (cf. Rooth, 1985, 1992). Subsequently, Erlewine (2014a,b) gave extensive discussion about the nature of *only* and *even* with respect to backward association. He argues that what looks like backwards association with *even* in (90), is actually illusory, because *even* associates with “a lower copy of movement”, and this is within the scope of *even* (Erlewine, 2014a).

The absence of backwards association with *only* in English can be supported based on the scopal behaviour of *only*. Consider the repeated example from (18) below. With the adfocal-*only*, the sentence can have an ambiguous scope where *only* takes narrow scope below the modal, as in (91-a), or wide scope above the modal, as in (91-b). We have already established above that the different scope possibilities are due to the position in which the covert exclusive Op_{only} occurs in the sentence. If we have a narrow scope reading, Op_{only} is placed below the modal, which produces the structure in (92-a). In contrast, if we get a wide scope reading, Op_{only} occur above the modal, as in (92-b).

- (91) Adam has to eat **only** RICE. [have > only, only > have]
 a. Narrow scope: Adam has to only eat rice (i.e., he cannot eat any other thing).
 b. Wide scope: Adam *only* has to eat rice (i.e., he may eat other things if he wants).
- (92) a. [TP Adam [_{Aux} has to [_{ExclP} **Op**_{only} [_{vP} eat [_{DP} **only** RICE]]]]].
 b. [TP Adam [_{ExclP} **Op**_{only} [_{Aux} has to [_{vP} eat [_{DP} **only** RICE]]]]].

However, the covert exclusive Op_{only} can be overtly realized with adverbial-*only*. When this happens, the scope ambiguity disappears. The specific scope under discussion is determined by the position of adverbial-*only*. So, while (93-a) produces a narrow scope reading (cf. (92-a)), (93-b) produces a wide scope reading (cf. (92-b)). With adverbial-*only*, both sentence have surface scope readings.³⁴

- (93) a. Adam has to **only** eat RICE.
 b. Adam **only** has to eat RICE.

Based on the discussion above, we can now talk about the constraint on BAwF in English and its effect. Consider an example involving the fronting of the adfocal particle and its focus associate in (94).

- (94) **Only** RICE, Adam has to eat ____.

The propositional approach, assumed in this dissertation, predicts the possibility of

³⁴ It is already a known fact that English does not have *only*-doubling, as in (i), which produces an exclusive concord reading. (i) can only be felicitous on the ground where both exclusive operators associate with distinct focused constituent, such as the verb and the direct object, for instance (cf. Lee, 2004; Quek and Hirsch, 2017, a.o.).

- (i) #Adam *only* ate *only* RICE.

both wide scope and narrow scope readings of the fronted constituent in relation to the modal. With the wide scope exclusive reading, the position of the exclusive Op_{only} can immediately be above the modal, or above the whole TP, as in (95). What matters is that the modal is in the scope of Op_{only} .

- (95) **Wide scope** [only > have]
 Surface structure: $[_{ExclP} \textbf{Op}_{only} [_{CP} [_{DP} \text{only RICE}]_i \text{C} [_{TP} \text{Adam has to } [_{vP} \text{eat } t_i]]]]]$

With the structure in (95), the scope of Op_{only} is independent of whether the adfocal-*only*+DP reconstruct or not, the wide scope will always be derived because everything is in the scope of Op_{only} .³⁵ However, the reconstruction of the fronted constituent (adfocal-*only*+DP) is not required. We know this from the fact that in English, only the wide scope reading of the sentences like (94) is available (cf. Taglicht, 1984; Hirsch and Wagner, 2025). The narrow scope exclusive reading is unavailable. See the structures in (96). If there was a reconstruction of the fronted constituent in English, then it would be surprising not to have a narrow scope interpretation. In other words, suppose reconstruction is possible, then we predict both readings to be possible in (94), just like the non-movement versions (cf. (91)). But since only the wide scope is possible, this indicates that adfocal-*only*+DP does reconstruct in English.

- (96) ***Narrow scope** [have > only]
 a. Surface structure: $[_{CP} [_{DP} \text{Only RICE}]_i \text{C} [_{TP} \text{Adam has to } [_{ExclP} \textbf{Op}_{only} [_{vP} \text{eat } t_i]]]]]$
 b. LF reconstruction: $[_{CP} t_i \text{C} [_{TP} \text{Adam has to } [_{ExclP} \textbf{Op}_{only} [_{vP} \text{eat } [_{DP} \text{only RICE}]_i]]]]]$

The evidence from scope and the results show the absence of BAwF in English. This has been associated with a constraint that was introduced in Chapter 2 as the Principle of Lexical Association (PLA) (see Jackendoff, 1972; Tancredi, 1990; Smeets and Wagner, 2018; Hirsch and Wagner, 2019). A revised version of PLA, which takes into consideration particle doubling, is given in (97) by Hirsch and Wagner (2025).³⁶

- (97) **No Backwards Association with Focus**
 Op_{only} must c-command Adfoc in the surface syntax.

(adapted from Hirsch and Wagner, 2025, p.14)

This is clear if we look at the surface structure given in (96-a) above. The covert exclusive Op_{only} cannot associate backwards with adfocal-*only*+DP. Therefore, the constraints on BAwF prevents Op_{only} from associating with a focus that is not in its c-command

³⁵ If Op_{only} occurs just above the modal, which now means that it is below the fronted constituent, then the prediction may be that reconstruction of the fronted constituent is possible, see the surface and LF structures in (i). But that does not seem to be the case.

- (i) **Wide scope** [only > have]
 a. Surface structure: $[_{CP} [_{DP} \text{only RICE}]_i \text{C} [_{TP} \text{Adam } [_{ExclP} \textbf{Op}_{only} [\text{has to } [_{vP} \text{eat } t_i]]]]]]]$
 b. LF reconstruction: $[_{CP} t_i \text{C} [_{TP} \text{Adam } [_{ExclP} \textbf{Op}_{only} [\text{has to } [_{vP} \text{eat } [_{DP} \text{only RICE}]_i]]]]]]]$

³⁶ Since the adfocal particle forms a constituent (or occurs locally) with the focus associate, this automatically makes the constraint applicable to the focus associate as well.

domain in the surface syntax.

This same idea applies to cases where Op_{only} is overtly realized as adverbial-*only*, as in the example in (89-b) above, repeated in (98). The only difference here is that adfocal-*only* is absent. Nothing hinges on this since the exclusive semantics always comes from the propositional exclusive operator, in this case, the adverbial-*only*.

(98) *RICE, Adam **only** ate.

BAwF in German

In arguing for a consistently adverbial (extended verbal projection) analysis for *nur* in German, Büring and Hartmann (2001) argued that *nur*, in an example like (99), unambiguously takes wide scope above the universal subject quantifier *jeder* ('everyone'). They claimed that the inverse scope (i.e. narrow scope in this case) is unavailable, as in (99)-(ii) (see also Jacobs, 1983).

- (99) Nur MARIA liebt jeder _____. [nur > jeder, *jeder > nur]
 only Mary love everyone.NOM
 (i) 'Only Mary is loved by everyone.' (nur > jeder)
 (ii) NOT: 'Everyone loves only Mary.' (*jeder > nur)

(Adapted from Büring and Hartmann, 2001, p.260, ex.57a)

At first, this appears to be the same as what we just saw in English above, which will mean that the constraint on BAwF is operational in German as well. However, later studies have argued that there is indeed inverse scope for the kind of German example in (99), provided that the fronted focused constituent carries a nuclear stress (Reis, 2005; Meyer and Sauerland, 2009; Bayer, 2018; Smeets and Wagner, 2018).³⁷ Contrary to Büring and Hartmann (2001), these studies argue for the availability of the narrow scope reading in (99)-(ii) when MARIA receives primary prominence. The example in (99) is repeated below in (100), but not the narrow exclusive reading is claimed to be available (cf. Hirsch and Wagner, 2025).

- (100) Nur MARIA liebt jeder _____. [nur > jeder, jeder > nur]
 only Mary love everyone.NOM
 (i) 'Only Mary is loved by everyone.' (nur > jeder)
 (ii) 'Everyone loves only Mary.' (jeder > nur)

Based on the propositional approach, both readings are accounted for. In the wide scope exclusive reading, if (similar to English) Op_{only} occurs higher than the fronted constituent, as in (101), then reconstruction is not necessary. However, in order to get the narrow scope exclusive reading, reconstruction is necessary. Both the surface structure and the structure showing the LF reconstruction are represented in (102).

- (101) **Wide scope** [nur > jeder]
 Surface structure: [_{ExclP} **Op_{only}** [_{CP} [_{DP} nur MARIA]_k liebt_i [_{TP} jeder [_{vP} t_i t_k]]]]

³⁷ See also Büring and Hartmann's (2001) fn.21, where they mentioned an anonymous reviewer's reported judgment of inverse scope. Overall, there seems to be some variation on speakers' judgment as to whether German has BAwF or not (Katharina Hartmann, Johannes Mursell, Anke Himmelreich, p.c.).

- (102) **Narrow scope** [jeder > nur]
- a. Surface structure: $[_{CP} [_{DP} \text{Nur MARIA}]_k \text{liebt}_i [_{TP} \text{jeder} [_{\text{ExclP}} \text{Op}_{only} [_{vP} t_i t_k]]]]]$
 - b. LF reconstruction: $[_{CP} t_1 \text{liebt}_i [_{TP} \text{jeder} [_{\text{ExclP}} \text{Op}_{only} [_{vP} t_i [_{DP} \text{nur MARIA}]_k]]]]]$

The implication of this idea is that *nur* may also be realized ad focally. However, it is important to point out here that there have been a lot of debates on whether German *nur* is purely adverbial or also has an adfocal property (see Buring and Hartmann, 2001; Reis, 2005; Hole, 2015; Smeets and Wagner, 2018; Hirsch and Wagner, 2019; Mursell, 2021, a.o.). One major argument of those who argue for an additional adfocal use of *nur* is that a sentence with *nur* in a sentence-initial position, as in (99) and (100) above, creates a V3 structure if *nur* is always treated as an adverbial particle. Instead, the V2 property of German is retained if *nur* in such instances forms a constituent with the fronted phrase. However, Mursell (2021) provided arguments in support of the adverbial treatment of sentence-initial *nur*. He argued that a V3 structure in German is not empirically unattested, as several studies have shown (cf. Müller, 2003, 2005b; Frey, 2004; Shaer and Frey, 2004). Since this section is not meant to discuss this debate in the German literature, I will not go into the details here, but see Mursell (2021) for a more detailed discussion. For our discussion here, we assume that sentence-initial *nur* forms a constituent with the left-dislocated element.

The availability of the narrow scope reading in German suggests that the constraint on BAwF is not universal, and that languages can be parameterized across two spectrum: those that allows BAwF, and those that do not. The former group will violate the constraint in (97).

In the next section, I show that the three languages under investigation, Kasem, Kusaal, and Yorùbá, allows for exclusive BAwF (i.e., languages which violate the exclusive backward association with focus constraint in (97)).

5.5.2 Analyzing BAwF in Kasem, Kusaal and Yorùbá

As mentioned above, BAwF in Kasem, Kusaal and Yorùbá is possible. Consider the repeated sentences (from ex. (87) above) for the three languages in (103). It does not matter whether the adfocal particles are present or not, the sentences still produce the appropriate exclusive interpretation in each language. Although the focus associate, which is the direct object in all the example, is fronted to the left periphery, there is still exclusive association with the fronted constituents. In Kasem and Kusaal (ex. (103-a) and (103-b)), the sentences still mean that Adam slaughtered nothing but a fowl. In Yorùbá, despite the fronting of the focus, the sentence in (103-c) still has the interpretation that Tolú ate rice and nothing else. I will take Yorùbá as a case study for the following discussion, but then I will show later that the analysis that is proposed, also applies to Kasem and Kusaal.

- (103) a. [Chworo]_F (yerane) *(mo) Adam weeni o go-a _____. **Kasem**
 fowl only FOC Adam only 3SG kill-DISJ
 ‘Adam slaughtered only A FOWL.’
- b. [Nua]_F (ma’aa) *(ka) Adam kudim kɔdig _____. **Kusaal**
 fowl only_{adf} FOC Adam only kill
 ‘Adam slaughtered only A FOWL.’

- c. [Ìrẹ̀ṣì]_F (**nìkan**) *(**ni**) Tolú **kàn** jẹ́ ____.
 rice only_{adf} FOC Tolú only eat
 ‘Tolú ate only RICE.’

Yorùbá

I simply assume that the ‘backward association’ cases are as a result of syntactic reconstruction. The fronted focused constituent reconstructs back to its canonical argument position at LF in order to be in the scope of adverbial-*kàn* in Yorùbá. The same follows for Kasem and Kusaal, with *weeni* and *kudim* believed to associate with a reconstructed focused constituent respectively. So, even though the languages seem to violate the BAwF constraint in the surface syntax, at LF, they still obey the BAwF constraint, since there is reconstruction of the focused constituent.

We already saw in Section 5.3 that exclusives in Kasem, Kusaal and Yorùbá can take both narrow and wide scope in the presence of another scope-taking operator such as a modal. But the data presented there were mainly in-situ focus. Depending on the position of the covert exclusive *Op_{only}*, the specific scope is derived. Here, if we also assume the propositional operator approach, we might predict the ambiguity that the sentence produces.

- (104) [Ìrẹ̀ṣì]_F **nìkan** *(**ni**) Tolú **le** jẹ́ ____ [only > can, can > only] **Yorùbá**
 rice only_{adf} FOC Tolú can eat
 Lit. ‘Only RICE Tolú can eat.’

- (i) Wide scope: Tolú is only allowed to eat rice (i.e., he is not allowed to eat any other food).
 (ii) Narrow scope: Tolú is allowed to only eat rice (i.e., he’s allowed to eat other things too, if he wants).

Based on the surface structure of (104), only the surface (wide) scope exclusive reading is expected since the focused constituent and the adfocal particle are in an ex-situ position. However, as we have seen above, this is contrary to the facts on ground, as the ex-sit focus cases are equally ambiguous. I am now going to show that the presence of not only the surface (wide) scope, but also the inverse (narrow) scope is predicted by the propositional approach. For the wide scope reading, the covert exclusive *Op_{only}* occupies a position that is higher than the modal in the surface structure (SS), and at LF, the fronted focus + *nìkan* reconstruct to their canonical position in (105).³⁸ In the narrow scope reading,

³⁸ For the wide scope reading, nothing hinges on the covert *Op_{only}* being realized very high in the clause, e.g., above the fronted focus, as in (i). If the exclusive *Op_{only}* occupies this position, then reconstruction is not necessary because all the other constituents in the clause are in the scope of *Op_{only}*, including the modal and the fronted constituent.

(i) **Wide scope**

- a. SS: [ExclP **Op_{only}** [FocP [DP Ìrẹ̀ṣì_F **nìkan**]_i ni [TP Tolú [Mod **le** [vP jẹ́ t_i]]]]]

The occurrence of *Op_{only}* above the modal in (105) is supported by the position of the overt adverbial operator *kàn*. If adverbial-*kàn* overtly realizes *Op_{only}*, it occupies a position just above the modal, as in the example in (ii). However, the narrow scope reading is unavailable in this case.

- (ii) [Ìrẹ̀ṣì]_F (**nìkan**) *(**ni**) Tolú **kàn** le jẹ́.
 rice only_{adf} FOC Tolú only can eat
 Lit: ‘Only RICE Tolú only can ate.’
 (i) Wide scope: The only thing Tolú can eat is rice.

on the other hand, the covert exclusive Op_{only} is realized below the modal in (106-a), and at LF, the fronted constituent is reconstructed in (106-b). Therefore, in both cases, there is reconstruction of the fronted focus constituent, and this is independent from the distribution of the exclusive Op_{only} , which is primarily associated with the position of the respective scope.

(105) **Wide scope**

- a. SS: $[_{FocP} [_{DP} \text{Ìrẹ̀sì}_F \text{ nìkan}]_i \text{ ni } [_{TP} \text{ Tolú } [_{ExclP} \text{ Op}_{only} [_{Mod} \text{ le } [_{vP} \text{ jẹ } t_i]]]]]$
b. LF: $[_{FocP} t_i \text{ ni } [_{TP} \text{ Tolú } [_{ExclP} \text{ Op}_{only} [_{Mod} \text{ le } [_{vP} \text{ jẹ } [_{DP} \text{ ìrẹ̀sì}_F \text{ nìkan}]_i]]]]]$

(106) **Narrow scope**

- a. SS: $[_{FocP} [_{DP} \text{Ìrẹ̀sì}_F \text{ nìkan}]_i \text{ ni } [_{TP} \text{ Tolú } [_{Mod} \text{ le } [_{Op_{only}} [_{vP} \text{ jẹ } t_i]]]]]$
b. LF: $[_{FocP} t_i \text{ ni } [_{TP} \text{ Tolú } [_{Mod} \text{ le } [_{Op_{only}} [_{vP} \text{ jẹ } [_{DP} \text{ ìrẹ̀sì}_F \text{ nìkan}]_i]]]]]$

The surface structure of the narrow scope reading in (106-a) involves BAwF because the fronted focused constituent is not c-commanded by the covert exclusive Op_{only} . So, even though at LF the fronted focused constituent plus adfocal-*nìkan* reconstructs, at the surface syntax, we have a case of BAwF.

The discussion and the analysis that we just provided for Yorùbá can be extended to Kasem and Kusaal. The sentences with fronted focused constituent can also show scope ambiguity in relation to modal verbs. The examples for the two languages are given in (107) and (108).

- (107) $[Chworo]_F$ **yerane** *(**mo**) Adam **wae** o go-a _____. **Kasem**
fowl only FOC Adam may 3SG kill-DISJ
Lit. ‘Only A FOWL, Adam may slaughter.’
(i) Wide scope: Adam only may slaughter a fowl (he has no other permission).
(ii) Narrow scope: Adam may only slaughter a fowl (he is allowed to slaughter a fowl and a fish).
- (108) $[Nua]_F$ **ma’aa** *(**ka**) Adam **tun’e** kɔdig _____. **Kusaal**
fowl only_{adv} FOC Adam may kill
‘Only A FOWL, Adam may slaughter.’
(i) Wide scope: Adam only may slaughter a fowl (he has no other permission).
(ii) Narrow scope: Adam may only slaughter a fowl (he is allowed to slaughter a fowl and a fish).

The data discussed above involves clausemate BAwF. That is, the fronted focus and the adverbial particle or Op_{only} are within the same clause. BAwF can also occur in long distance dependency. Consider the examples in (109). Even though the embedded objects are fronted to the left periphery of the matrix clause, BAwF is still available.

- (109) a. $[Chworo]_F$ *(**mo**) Sam yei we Adam **weeni** o go-a _____. **Kas.**
fowl FOC Sam know COMP Adam only 3SG kill-DISJ
‘Sam knows that Adam only slaughtered A FOWL (✓and not a goat).’
b. $[Nua]_F$ *(**ka**) Sam mi’i ye Adam **kudim** kɔdig _____. **Kus.**
fowl FOC Sam know COMP Adam only kill
‘Sam knows that Adam only slaughtered A FOWL (✓and not a goat).’

(ii) *Narrow scope: Tolú has permission to eat nothing but rice.

- c. [Ìrẹ̀sì]_F *(**ni**) Adé mò wípé Tolú **kàn** jẹ _____. **Yor.**
 rice FOC Adé know COMP Tolú only eat
 ‘Adé knows that Tolú only ate RICE (✓and not yam).’

Therefore, based on the possibility of BA_wF in Kasem, Kusaal, and Yorùbá, we can group these languages under those who allow BA_wF.

Another Germanic language that has been argued to allow BA_wF (i.e., violates the BA_wF constraint in (97)), according to Erlewine (2014b), is Dutch. The Dutch example in (110) is argued to involve BA_wF where the fronted focus is licensed by the V2 property of Dutch (similar to the German case above). Although the focus associate is not within the surface scope of the exclusive operator, the sentence still means that John bought only two books and no more.

- (110) [TWEË boeken]_i denk ik dat Jan **pas/maar** _____i heeft gekocht.
 two books think I that John just/only has bought
 ‘TWO books, I think that John has bought just/only (✓and nothing more).’

(Adapted from Erlewine (2014b, p.187), based on Barbiers (1995, p.68))

Just like German, it is worth noting here that there is a debate on whether Dutch indeed allows BA_wF or not. For instance, while Beaver and Clark (2008) show that an example such as (110) may seem to involve BA_wF, they argued that this is not the case with a variant of the exclusive focus sensitive operator *alleen maar* (‘only’), as in (111). They claimed that BA_wF fail in (111), and the sentence does not mean that I think that John bought only two books and nothing more.³⁹

- (111) [TWEË boeken]_i denk ik dat Jan **alleen maar** _____i heeft gekocht.
 two books think I that John only has bought
 *‘I think that John has bought only TWO books (and nothing more).’

(Adapted from Beaver and Clark, 2008, p.176)

They further argue that *pas* in (110) is not an exclusive operator at all, and generally combines with quantifiers. In order to establish the facts for Dutch, more investigation would have to be done. But until then, I will group it under the languages that allow BA_wF.

On the other hand, languages that do not allow BA_wF include: Vietnamese (Erlewine, 2017), Swedish, Spanish, and Italian (see Beaver and Clark, 2008, p.166-169). These languages do not allow BA_wF. See the summary in Table 5.1.

In conclusion, in addition to German, the BA_wF cases in Kasem, Kusaal, and Yorùbá show that the constraint on BA_wF is not a universal constraint. However, it remains to be seen what other interactions could be at play that may determine the absence or presence of BA_wF in a language.⁴⁰ Important for us is the fact that the propositional approach can account for BA_wF in the three languages under investigation.

³⁹ Although they did not say what this sentence may mean. But it may mean that I think that John has only bought two books, and not sold two books. This is an association with the verb that is in the scope of the exclusive particle.

⁴⁰ One line of thought might be to relate BA_wF to obligatory focus movement in Kasem, Kusaal and Yorùbá. In German (and maybe Dutch), this may be linked to the V2 requirement. In English, however, there are no such requirements (V2 or obligatory focus movement) (cf. Erlewine, 2014b; Aremu et al., 2025). This calls for a further research, which I leave for a future occasion.

Languages with BA _w F	Languages without BA _w F
Kasem	English
Kusaal	Swedish
Yorùbá	Vietnamese
German	Spanish
Dutch	Italian

Table 5.1: A typology of languages and exclusive BA_wF

In the next section, I discuss the analysis that Yip and Adedeji (2024) proposed for Yorùbá’s exclusive doubling, where they argued for a different approach, instead of the Propositional Approach that I argued for in Section 5.3. I show that on a closer examination, the propositional approach is still the better approach for Yorùbá.

5.6 A reply to Yip and Adedeji (2024) on Yorùbá exclusives

Recall that in Section 5.3, I argued for a Propositional Approach to exclusives (doubling) in, not only Kasem and Kusaal, but also Yorùbá. I also mentioned earlier that the Propositional Approach is also called the Operator-Particle Approach (Op-PRT approach).⁴¹ However, in a recent study, Yip and Adedeji (2024) argue that exclusive doubling in Yoruba, such as (112), cannot be analyzed with the Op-PRT approach.⁴² That is, they claim that the exclusive semantics does not come from a high exclusive Op_{only}, which may be overtly realized by the adverbial *kàn*.

- (112) Tolú **kàn** jẹ́ [lɪɾɛ̀sɪ]_F **nìkan**.
 Tolú only eat rice only_{adf}
 ‘Tolú only ate RICE.’

Instead, they propose an alternative and novel approach where the exclusive interpretation actually comes from adfocal-*nìkan*, while the adverbial-*kàn* is assumed to only be a scope marker. In other words, the latter only marks the scope where the interpretation of the former applies. They called their approach the **Quantifier-particle approach** (Q-approach). In the Q-approach, even though the exclusive semantics is on the adfocal *nìkan*, *nìkan* + DP_F (QP) still needs to undergo a covert or overt movement to a propositional scope position, just over the adverbial-*kàn* in (113), for example. Also, they assume that the adfocal particle can be covertly realized in cases where only the adverbial-*kàn* is present. In their approach, QP’s movement is an A’movement. So, in (113), QP moves to Spec,FP, and then to Spec,FocP only in ex-situ focus cases. In ex-situ focus, the two movement steps would have to be overt. In in-situ focus cases, on the other hand, only the first movement is required. They assume a copy theory of movement, which allows for the pronunciation of the lower copy (cf. Bošković and Nunes, 2007). The higher copy

⁴¹ So, in this section, I will use the term Operator-Particle Approach (Op-PRT approach) in order to be consistent with the term used in Yip and Adedeji (2024).

⁴² In this section, I use my own data, and not the ones directly from Yip and Adedeji (2024). Anywhere their data is used, it would be referenced. I believe that the approach should be applicable regardless of the data, .

is not pronounced because of an independent PF constraint which allows only particles between the subject and the verb in the language (and not a PRT+DP). Thus, when the movement is overt (as in ex-situ focus), the highest copy is pronounced. However, when the movement is covert (as in in-situ focus), the lowest copy is pronounced.

- (113) **Yip & Adedjei's *only*-doubling Structure (with modifications)**
- $$[\text{FocP} [\text{QP} \dots] [\text{Foc}' \text{ni} [\text{TP} \text{DP}_{\text{subj}} [\text{FP} [\text{QP} \dots] [\text{F}' \text{kàn} [\text{vP} \text{v}+\text{V} [\text{QP} \text{DP}_F \text{nìkan}]]]]]]]]$$
- $\xrightarrow{\text{A}'\text{-movement}}$
 $\xrightarrow{\text{A}'\text{-movement}}$

They provided four arguments for their claim based on the following cases in (114).

- (114)
- a. Backward association with focus
 - b. Multiple associations with focus
 - c. Scopal interaction with negation
 - d. Wide scope under ellipsis

In the remaining part of this section, I discuss these arguments one after the other, and show how the Op-PRT (Propositional) approach can account for these arguments. I also discuss two further cases which the Q-approach cannot account for, but are straightforwardly captured by the Op-PRT approach. The first case has to do with the possibility of an *only*-concord reading (see Section 5.6.2), and the second case is the possibility of a split-scope reading in Yorùbá (see Section 5.6.5).

5.6.1 Backward association with focus

Yip & Adedjei rightly argue that the Q-approach would account for BAwF cases such as (115). The focus is ex-situ, but still gets an exclusive interpretation. So, going by the Q-approach, the adfocal-*nìkan* (which can be covert or overt), is already adjoined to the focus associate. Both the focus associate and the covert/overt particle are in a propositional operator position. Thus, if *kàn* is not an exclusive Op (but only a scope marker), there is no backward association to begin with. They claim that BAwF is unexpected under the Op-PRT approach, and that it is cross-linguistically rare.

- (115) [Ìṣù]_F (nìkan) **nì** Tolú **kàn** jẹ ____.
yam only FOC Tolu only eat
Lit. 'YAM, Tolu only ate.'

However, that is not true. On a closer look, the Op-PRT approach can also account for BAwF, and I already showed this in Section 5.5 above. In example like (115), the exclusive interpretation is available because the focus associate plus the (c)overt adfocal particle reconstruct at LF to their canonical position, within the scope of *kàn* (which overtly realizes Op_{only}) (116). Therefore, BAwF is only a surface notion, at LF the focus associate is c-commanded by the adverbial exclusive operator, and the principle of lexical association is respected.⁴³

- (116) LF reconstruction: [_{FocP} t_i ni [_{TP} Tolu **kan** [_{vP} jẹ [_{DP} ṣù_F (nìkan)]_i]]]].

⁴³ Therefore, we may say that the principle of lexical association, which requires that the focus associates of operators such as *only* should be in their scope, is more of an LF requirement than a surface requirement.

As a matter of fact, as we saw in Section 5.5, the Op-PRT approach captures BAwF not only in Yorùbá, but also in the Mabia languages (Kasem and Kusaal), and German. In these languages, I have argued that the exclusive semantics is on the propositional exclusive operator that may be overtly realized by the adverbial particles. The rarity of BAwF is not an argument against its existence. In fact, its existence in the languages under the current study only calls for a cross-linguistic investigation. Interestingly, English, that is said to lack BAwF seems to lack it only with exclusives. We saw in Section 5.5 that the scalar-additive operator *even* allows BAwF. However, Erlewine (2014a,b) argues that such a backwards association with *even* is ‘illusory’ in English. I will not go into the analysis here.

5.6.2 Multiple associations with focus

Yip & Adedjei’s second argument for a Q-approach comes from what they claim to be the behaviour of *kàn* in multiple-focus cases in Yorùbá. In multiple-focus cases, multiple operators should in principle be available to establish more than one AwF, in order to derive cases like (117-a). They argue that if the adverbial-*kàn* is an exclusive operator, it should be able to establish association (independently) with any focus in its scope. However, according to them, it does not, based on the interpretations in (117-a)-i-ii. This means that only association with subject focus is available. Association with object focus is not available (i.e., no multi-*only* reading). The assumption is that if *kàn* is an exclusive operator, it should be able to associate with the object focus in its c-command domain. If it does not, then it cannot be an exclusive operator. For the multi-*only* reading, they argue that an example such as (117-b), where adfocal-*nìkan* is doubled, is required.

- (117) [Context: Who only ate yam?]
- a. [Tolú]_F **nìkan** ni ó **kàn** jẹ [ìṣu]_{*F}. (*multi-*only* reading)
 Tolu only_{adf} FOC 3SG only eat yam
 Lit. ‘Only TOLU only ate YAM’
 (i) Tolu is the only person who just ate yam. (#other people ate both yam and something else)
 (ii) *NOT*: Tolu is the only person who only ate yam. (other people ate both yam and something else)
- b. [Tolú]_F **nìkan** ni ó (**kàn**) jẹ [ìṣu]_F **nìkan**. (multi-*only* reading)
 Tolu only_{adf} FOC 3SG only eat yam only_{adf}
Meaning: ‘Tolu is the only person who only ate yam (other people ate both yam and something else).’

While the logic behind the argument is true, the claim is not true for the data. In the example in (117-a), *kan* actually associates with the direct object *isu*, and the multi-*only* reading is available. This is confirmed with the repeated example in (118-a). The continuation in (118-c), and not (118-b), is felicitous. This is unexpected if *kàn* is not an exclusive operator.

- (118) a. [Tolú]_F **nìkan** ni ó **kàn** jẹ [ìṣu]_F.
 Tolu only_{adf} FOC 3SG only eat yam
Intended. ‘Only TOLU only ate YAM (other people ate both yam and something else).’

- b. #... ó (sì) jẹ ọgèdè pèlú.
 3SG CONJ eat banana also
 ‘... and he also ate banana’
- c. ... kò jẹ ọgèdè.
 NEG eat banana
 ‘... did not eat banana.’

The Q-approach predicts that the continuation in (118-b) should be felicitous, since *kàn* is assumed not to associate independently. So, it would not quantify over the object. However, that is not the case as we have seen. It is the continuation in (118-c) that is felicitous with the previous sentence. This supports the idea that *kàn* independently associates with the object.⁴⁴ For the Op-PRT approach, the multiple-*only* reading in (118-a) is realized by assuming a high covert Op_{only} that associates with the subject focus, while adverbial-*kàn* associates with the object focus (119). So, the presence of the two exclusive operators (one covert, and the other overt) captures the multiple-*only* reading.

- (119) [ExclP Op_{only} [DP_{SF} ... [ExclP **kàn** [vP ... DP_{OF}]]]]

Apart from a multi-*only* reading discussed above, there is another available reading that I already discussed in Section 5.3.3. This is the *only*-concord reading. The Q-approach does not predict such a reading. Consider the repeated example with multiple association with adfocal-*nìkan* in (120) (Context: Who ate what?). The sentence can have a concord reading where in a given context, the property of eating is satisfied only by the pair <Tolu, yam>. Such a concord reading is structurally represented in (121), where both adfocally marked phrases Agree with a single exclusive Op_{only}.

- (120) [Tolú]_F **nìkan** ni ó jẹ [iṣu]_F **nìkan**.
 Tolu only_{adf} FOC 3SG eat yam only_{adf}
 ‘Only TOLU ate only YAM.’

Concord reading: ‘In a given context, ONLY <Tolu, yam> satisfies the property of *eating*.’ [in other words: the property of *eating* only holds for only the alternative pair <Tolu, yam>, and not, for instance <Tolu, rice>, <John, yam>, <John, rice>, ...]

- (121) [ExclP Op_{only} [FocP [DP Tolú_F **nìkan**] ni [TP ó [T' [vP jẹ [DP iṣu_F **nìkan**]]]]]]
-

The Q-approach does not predict an *only*-concord situation. This is because adfocal-*nìkan* plus its focus associate would have to always move together (covertly or overtly), and interpreted independently. The Q-approach only predicts the multiple-*only* reading if the exclusive semantics is on adfocal-*nìkan*. If it was adfocal-*nìkan* that carried the exclusive semantics, then the concord reading would have been unavailable.

⁴⁴ The continuation in (118-b) is only possible if another speaker is trying to correct the previous statement or update the previous speaker’s knowledge. If not, the sentence in (118-a) and the continuation in (118-b) cannot be uttered by the same speaker.

5.6.3 Scopal interaction with negation

For the scopal interaction with negation argument, Yip & Adedeji used the flexibility in the occurrence of the adfocal particle *nìkan*, and the rigidity in the occurrence of the adverbial particle *kàn* to argue for their scopal interactions with negation. According to Adebayo (2021), Standard Yorùbá has only four true sentential negative markers: *kò*, *kìí*, *kó*, and *má*.⁴⁵ He subcategorized them based on the mood in which they are used. While the first three *kò*, *kìí*, *kó* are used in realis mood, the last one *má* is used in irrealis mood. The realis mood negation markers are used in different contexts. Consider the sentences in (122), with (122-a) as the positive sentence. *Kò* is used to negate a perfective sentence (122-b), *kìí* to negate an imperfective (habitual) sentence (122-c), and *kó* is used in the left periphery in a sentence with focus (122-d).

- (122) a. Tolú jẹ ịsu.
Tolú eat yam
'Tolú ate yam.'
- b. Tolú **kò** jẹ ịsu.
Tolú NEG eat yam
'Tolú did not eat yam.'
- c. Tolú **kìí** jẹ ịsu.
Tolú NEG eat yam
'Tolú does not eat yam.'
- d. Ịsu **kó** ni Tolú jẹ ____.
yam NEG FOC Tolú eat
'It was not yam that Tolú ate.'

Like *kó*, the negative marker *kìí* can also occur in the left periphery of a fronted focus, as in (123).

- (123) **Kìí** ẹ ịsu ni Tolú jẹ.
NEG do yam FOC Tolú eat
'It was not yam that Tolú ate.'

The irrealis negative marker *má* often occurs with imperatives or subjunctive cases, see (124). The realis mood negative markers will be of importance to us here.

- (124) **Má** jẹ ịsu.
NEG eat yam
'Don't eat yam.'

Using the examples involving ex-situ focus in (125), Yip & Adedeji argued that the position of the adfocal particle *nìkan* in relation to the negative markers determines the resulted scope. Therefore, in example (125-a), *nìkan* takes a narrow scope under negation when it is preceded by the negative marker *kìí* in the left periphery. In (125-b), *nìkan* takes a wide scope above the perfective negative marker *kò*. They argued that if the adverbial particle *kàn* is the exclusive operator, then both examples should have a wide scope negation reading. According to them, the Q-approach accounts for the observed scopal differences because the position of the adfocal particle *nìkan* plus the focus associate corresponds to the derived scope.

⁴⁵ This is contrary to Fábùnmí's (2013) claim that there are six of them.

- (125) a. **Kíí** ʃe German_F (**nìkan**) ní John **kàn** ʃe. [neg > only]
 NEG do German only FOC John only do
 ‘It is not only German that John takes (but also French).’ (i.e., John takes German and French)
- b. German_F (**nìkan**) ní John **kò** **kàn** ʃe. [only > neg]
 German only FOC John NEG only do
 ‘It is only German that John just didn’t take.’ (i.e., John didn’t take German—he took French and Latin)

However, considering in-situ focus constructions such as the examples in (126), an important property of *kàn* is that its position is fixed in the clause. It can only follow (126-a), but can never precede the negative marker *kò*, see (126-b). It is not only with ex-situ focus, as in (125-a), the in-situ focus example in (126-a) also has a wide scope negation reading.

- (126) a. John **kò** **kàn** ʃe German_F (**nìkan**). [neg > only]
 John NEG only do German only
 ‘It is not only German that John took (but also French).’ (i.e., John took both German and French)
- b. *John **kàn** **kò** ʃe German_F (**nìkan**). [only > neg]
 John only NEG do German only
 ‘It is only German that John didn’t take.’ (i.e., he took other courses)

However, with a proper context (as in below), the narrow scope negation reading may be derived as well, see (127). This shows that it is not the structural position of *nìkan*, in relation to the negative marker, that determines the resulted scope (cf.(125)). The possibility of a narrow scope negation reading in (127), despite the rigid structural position of *kàn*, suggests that there is a higher exclusive operator which scopes above the negative marker *kò*.⁴⁶ Recall that the exclusive *Op_{only}* can occupy a higher or a lower position in the clause, depending on the scope that it takes. Therefore, in (127), I assume that while the exclusive *Op_{only}* is overtly realized with the adverbial particle *kàn* in a wide scope negation reading (i.e., [neg > *kàn*]), it is covert in the case of a narrow scope negation reading (i.e., [*Op_{only}* > neg]). The prediction would have been that *kàn* always realizes the covert exclusive *Op_{only}*. However, due to what I assume to be a language-specific/lexical restriction, *kàn* has a lower structural position below the negative marker *kò* (cf.(126)).⁴⁷ The wide scope negative reading of (127) can have a continuation such as

⁴⁶ Such a mismatch between scope and word order is not uncommon. In certain Bantu languages, there are mismatches between morphological order and semantic scope in certain constructions. The verbal morphological template is often fixed, but the semantic scope relations do not always align with the verbal template (see Wechsler, 2020; Jerro, 2016, 2023, and the references therein).

⁴⁷ Recall that such a rigid positional occurrence of *kàn* was not observed with modals. In the case of modals, *kàn* can occur below or above the modal auxiliary. I repeat the two examples in (26-c) and (27-c) below. Thus, the rigid structural position of *kàn* in negative context seems to be a peculiar case, which, of course, needs further investigation.

(i) **Yorùbá flexible exclusive occurrence with modals**

- a. Tolú **kàn** **lè** jẹ [iʃu]_F (**nìkan**).
 Tolú only can eat yam only_{adv}
 ‘Tolú can eat only YAM.’

(128).⁴⁸

[*Context*: In the school, John has the option of taking any of the three courses: German, French, and Latin. But as John’s friend, as far as I know, he didn’t take German. I don’t know about other courses. Maybe he took them or not. So the teacher asked me: Which courses did John take?]

(127) John **kò** **kàn** ʃe German_F (**nìkan**). [only > neg]
 John NEG only do German only
 ‘(As far as I know) It is only German that John didn’t take.’ (i.e., John didn’t take German— maybe he took French and Latin)

(128) ... mi ò mò bóyá ó ʃe Faransé àti Latin.
 1SG NEG know maybe 3SG do French and Latin
 ‘I do not know whether he took French and Latin.’

In English, both the wide scope negation reading and the narrow scope negation reading can be derived by re-ordering the negative marker and the exclusive adverbial particle. While example (129-a) sounds more natural, example (129-b) is a bit unnatural, but technically grammatical.⁴⁹ However, such a construction is ungrammatical in Yorùbá (see (126-b)).

(129) a. John did **not only** take German (he took something else, including German).
 b. John **only** did **not** take German (he took something else, but not German).

Furthermore, Yip & Adedéjì’s analysis did not consider the other two negative markers *kó* and *má*. I will dwell on *kó* because of its focus status. In example (130), even though the negative marker follows the adfocal particle *nìkan*, only the wide scope negation reading is available. This is not predicted by the Q-approach. The Q-approach predicts a narrow scope negation reading instead, as in (125-b) above.⁵⁰ In fact, in the absence of the adfocal

b. Tolú **lè** **kàn** jẹ [iʃu]_F (nìkan).
 Tolú can only eat yam only_{adf}
 ‘Tolú can only eat YAM.’

⁴⁸ I wish to point out here that there is a non-trivial issue here that I do not have an adequate answer to at the moment. This concerns the status and function of *kàn* in cases such as (127), where wide scope reading is available. It remains to be seen what type of analysis would account for this. One possibility might be to assume a tripartite structure at LF, where the adverbial exclusive particle *kàn* passes up its interpretable feature to the exclusive Op_{only} that is higher than NEG for a wide scope reading. In the narrow scope reading, the exclusive feature is interpreted lower, which is now on *kàn*. The accurate analysis is far from clear. I am grateful to Aron Hirsch (p.c.) for pointing this issue out to me.

(i) LF: [TP John [Op_{only}[iExcl] [NegP Neg [vP *kàn*_{only}[iExcl] [vP ʃe [DP German (nìkan)_{only}[uExcl]]]]]]]]]

⁴⁹ The Corpus of Contemporary American English (COCA) has 63 entries of sentences with the string “only did not” (accessed on 14/03/2025).

⁵⁰ A potential analysis that the Q-approach might want to adopt is to say that *kó* forms a complex constituent with the focus associate + *nìkan* (Ka-Fai Yip, p.c.). This is, more or less, treating *kó* as a constituent negator which generates a quantifier phrase, as in English ‘no book’, for example.

particle *nìkan*, the wide scope negation reading is still available, indicating that neither the position nor the overtness of *nìkan* plays a role. But based on the Op-PRT approach, the negative marker *kó* scopes above the adverbial particle *kàn*. This will always be the case because *kó* is always realized in ex-situ focus constructions. So, its structural position feeds scope (based on the Op-PRT approach).

- (130) German_F (**nìkan**) **kó** ni John **kàn** ɕe. [neg > only]
 German only NEG FOC John only do
 ‘It is not only German that John took (but also French).’ (i.e., John took German and French)

Thus, the argument with negation does not invalidate the Op-PRT approach for Yorùbá.

5.6.4 Wide scope under ellipsis

Yip & Adedjei further argue that wide scope exclusive reading under ellipsis is available in Yorùbá, and this supports their Q-approach. Recall that in Section 5.3.2, I gave an ellipsis constraints by Bassi et al. (2022), repeated below as (131). This constraint disallows the separation of the exclusive Op from its focus associate by ellipsis. Meaning that the latter must not be part of an ellipsis site, leaving the exclusive Op_{only} ‘stranded’.

- (131) **PA’s version of B&C’s Constraint:**
 Op_{only} and its associated Foc cannot be separated by a node targeted for ellipsis.

(adapted & modified from Bassi et al., 2022, p.819)

Consider the English examples in (132). Bassi et al. (2022) argue that the wide scope exclusive reading is unavailable in an elliptical construction such as (132-a). They associated to the constraint in (131). The wide scope reading is unavailable because this would violate (131); the exclusive Op_{only}, assumed to be higher than the modal, would be separated from its focus associate (132-b)-ii.

- (132) a. Adam may eat only RICE. [may > only, only > may]
 b. Sam may [~~eat only RICE~~] too. [may > only, *only > may]
 (i) [Adam [may [Op_{only} [~~eat only RICE~~]]]]
 (ii) [Adam [Op_{only} [may [~~eat only RICE~~]]]]

In contrast, Yip & Adedjei claimed that the wide scope exclusive reading is, in fact, available in Yorùbá. However, it is important to note here that the test with ellipsis is inconclusive because based on my judgment and those of others that I consulted, there are different views on whether there is a wide scope exclusive reading under ellipsis in Yorùbá, or not. So, I will not go into this further (see also the footnote in 15).

In the next section, I present an argument from split-scope possibilities which the Q-approach cannot account for, but the Op-PRT approach does.

However, Adebayo (2021) argues against such an analysis. He argues that *kó* actually negates a proposition, (by taking a FocP as its complement) and not a noun phrase (see Adebayo, 2021, §4.3). If it is a constituent negator, then it is surprising that it only occurs in ex-situ focus cases, and can never be realized in-situ.

5.6.5 Split-scope reading

The Q-approach does not predict the possibility of a split-scope reading with exclusive operator in Yorùbá. Recall that in Section 5.3.1 above, we saw the possibility of a split-scope reading in Kasem, Kusaal, and Yorùbá (cf. Hirsch, 2017). The Yorùbá data from (30-c) above is repeated in (133). Due to the presence of the quantifier *méjì* ('two'), the sentence may involve a split in the scope of the exclusive particle and the quantifier *two*. This allows the modal verb to scope between them. The sentence in (133) means that Adam does not have to slaughter more than two fowls. Notice that the reading allows her to (optionally) slaughter more fowls if she wants.

- (133) Adam ní lati pa adìḗ [méjì]_F **nìkan**. [only > require > two]
 Adam has to kill fowl two only_{adf}
 'Adam is required to slaughter only TWO fowls.'
Meaning: Adam is only required to slaughter (any) two fowls (he could slaughter more if he wishes).

In this context, the exclusive scope is dissociated from the surface position of *only*. In other words, the adfocal particle *nìkan* is syntactically below the modal verb, while the exclusive scope is wide. In Section 5.3.1, I argued that this can be accounted for under the propositional approach (a.k.a. the Op-PRT approach), where Op_{only} is realized above the modal verb, while the adfocal particle remains below it, as in the repeated structure in (134). The split-scope reading is easily accessible when Op_{only} is realized overtly with the adverbial particle *kàn*, as exemplified in (135).

- (134) **Split-scope at LF: [only > require > two fowls]**
 [TP Adam_i [ExclP **Op_{only}** [_{VP} is required [TP t_i to [_{VP} [DP *only two fowls*]_k [t_i _{VP} slaughter t_k]]]]]]]]
 (135) Adam **kàn** ní lati pa adìḗ [méjì]_F **nìkan**. [only > require > two]
 Adam only has to kill fowl two only_{adf}
 'Adam is only required to slaughter TWO fowls (but could (optionally) slaughter more if he wishes).'

Just like the quantifier approach discussed in Section 5.3, the Q-approach of Yip & Adedeji does not predict such a split in scope whereby while the exclusive takes a wide scope above the modal have, the quantifier phrase takes a narrow scope below have (cf. von Stechow and Lasnik, 2007; Hirsch, 2017). It only predicts either a narrow scope reading, where both *nìkan*+DP_F would scope below require, as in (136), or a wide scope reading, where *nìkan*+DP_F would scope above require, as in (137). Both readings are different from the split-scope reading (see (133)) in that in the narrow scope reading (136), Adam is required to slaughter no more than two fowls that are not specific. This is different from the split-scope reading because the latter reading allows Adam to slaughter more fowls if he wants. On the other hand, the wide scope reading in (137) is different from the split-scope reading because while the former restricts the fowls to be slaughtered to specific fowls, the latter does not; the fowls to be slaughtered could be any fowls. Thus, the Q-approach undergenerates the split-scope possibility of the exclusive.

- (136) **Narrow scope: [require > only > two fowls]**
 [TP Adam_i [_{VP} is required [TP t_i to [_{VP} [QP *only two fowls*]_k [t_i _{VP} slaughter t_k]]]]]]
Meaning: Adam is required to only slaughter (any) two fowls (and no more).

- (137) **Wide scope:** [only > two fowls > require]
 [TP Adam_i is [vP [QP *only **two** fowls*]_k [vP *required* [TP t_i to [t_i vP slaughter t_k]]]]]
Meaning: There are only two particular fowls that Adam is required to slaughter.

In summary, in this section, we have seen that while the Q-approach offers a novel approach to exclusive-doubling, it falls short in accounting for the phenomenon in Yorùbá. The Op-PRT approach covers a wider range of data and possibilities. Thus, Yip and Adedjeji’s (2024) claim that “there is no exclusive operator in Yorùbá, neither *kàn* nor a null one”, is not true.

5.7 Conclusion

In this chapter, I have proposed an analysis for the syntax-semantic properties of the exclusive focus sensitive particles in Kasem, Kusaal and Yorùbá. These properties include (a) the syntactic distribution of exclusive particles in the languages under investigation (i.e., the use of adfocal and adverbial exclusive particles); (b) the exclusive doubling cases; (c) backward association phenomenon (BAwF); and (d) the quantificational and evaluative property of exclusives. Using different arguments, I propose a propositional analysis (Operator-particle Approach) for exclusive properties in these languages, where the exclusive semantics is always on the adverbial operator instead of the adfocal particles. However, contrary to the earlier claim that adfocal particles are semantically vacuous (i.e., concord markers), I argued that this is not the case by showing that the adfocal particles indicate the alternative ranking that is under discussion and constrain the scale of the alternative propositions that answer a CQ. I showed that exclusives in these languages can have a complement exclusion meaning, a rank-order meaning and an evaluative meaning. Although this led us to expand our analysis, the underlying assumption, that the exclusive meaning resides in a propositional position, remains the same.

In addition, I discussed the phenomenon of BAwF in the three languages. I showed that Kasem, Kusaal and Yorùbá exhibit BAwF, and argued that BAwF in these languages can be analyzed using the propositional approach (cf. Hirsch and Wagner, 2025). In the last section, I discussed Yip and Adedjeji’s (2024) proposal on Yoruba as a language involving the quantifier approach instead of the propositional approach. I provided arguments against their proposal, and maintained that Yoruba exclusive association with focus is best analyzed using the propositional approach. Consequently, this chapter has set up the core analytical tools or approach for the remaining chapters of this dissertation.

Chapter 6

Additive focus sensitive particles

This chapter discusses additive association with focus in Yorùbá, Kusaal and Kasem respectively. I start with a short discussion on additive meaning and properties in 6.1. After the introduction, I describe additive association in these languages in Sections 6.2 (Yorùbá), 6.3 (Kusaal), and 6.4 (Kasem), respectively. I show that these languages have both adverbial and adfocal additive particles, and also allow for additive doubling (similar to the exclusive cases in the last chapter). In these sections, I further discuss the peculiarities of additive association in relation to focus marking in these languages. I provide an interim summary in 6.5. Next, in Section 6.6, I propose a propositional analysis that is analogous to the one I proposed for exclusives in the last chapter. Then, I provide some arguments and consequences for such an analysis. In Section 6.7, I discuss the phenomenon of Backward Association with Focus (BAwF) with additive particles, and show that the three languages allows for BAwF. I provide an analysis for this, which is propositional in nature. The discussion shows that additive BAwF is more common among languages than exclusive BAwF. I conclude the chapter in Section 6.8.

6.1 Introduction

Additive particles such as *also* in English asserts their prejacent and presupposes that at least one other alternative satisfies the given predication. The truth-condition of the prejacent in an additive construction is dependent on whether another alternative is true. In other words, if the alternative is false, then the prejacent of an additive clause is false as well. In (1-a), the second sentence involve additive association with the direct object. The preceding sentence explicitly shows what he ate in addition to yam. Both the assertion, which is the prejacent p , and the presupposition, which is the fact that another alternative q ($\neq p$) is true, are given in (1-b).

- (1) a. Adam ate rice. He also ate YAM.
 b. *Assertion*: Adam ate yam.
 Presupposition: Another alternative is true (i.e., Adam ate rice in this case).

Another example is given in (2-a), with the focus now on the verb. The sentence means that Adam did not only eat yam, he did something else to it. In this case, he cooked it as explicitly revealed by the context. Even without an explicit context, the additive interpretation holds. (2-b) gives the asserted prejacent and the presupposition.

- (2) a. Adam cooked yam. He also ATE yam.

- b. *Assertion*: Adam ate yam.
Presupposition: Another alternative is true (i.e., Adam cooked yam in this case).

Just like *only*, *also* is also a propositional operator. This means that *also* composes with a proposition at the semantic level. If we consider the sentences in (3), although the elements that are in focus are different, the set of alternatives similarly comprises of propositions. The only point of variation would be in the focused part, see (4). Therefore, I follow the view in the literature that focus sensitive operators such as *also* do not compose directly with the focus associate, but with the proposition containing the focus associate (Rooth, 1992; Beaver and Clark, 2008, a.o.)¹

- (3) a. Adam also ate YAM.
 b. Adam also ATE yam.
- (4) a. ALT = {Adam ate yam, Adam ate beans, Adam ate rice, ...}
 b. ALT = {Adam ate yam, Adam cooked yam, Adam bought yam, ...}

In the next three sections, I describe the properties of the additive particles in Yorùbá, Kusaal and Kasem, one after the other.

6.2 Additive association in Yorùbá

The English's *also* equivalents in Yorùbá are *pèlú* and *tún*. The difference between the two is that while *pèlú* is adfocal (i.e., adjoins directly to the focus associate), *tún* is adverbial, and can associate with the focus at a distance. Both additive particles can be doubled in a sentence, i.e., they can co-occur in a sentence and associate with a single focused constituent.

6.2.1 The adfocal additive particle *pèlú*

The adfocal additive particle *pèlú* always follows its focus associate as illustrated with the example in (5).² In addition to the two foods Tolú ate in (5-a), he also ate yam, as in (5-b).

- (5) [What did Tolú eat?]
 a. Tolù jẹ ẹwà àti ìrẹ̀sì.
 Tolù eat beans CONJ rice
 'Tolù ate beans and rice.'

¹ See the last Chapter 5.

² *Pèlú* is also used as the preposition 'with' in the language, as shown in the example below. Notice that in such usage, the tone on the final vowel is a mid tone (cf. *pèlu*), unlike the high tone that is on the final vowel of *pèlú* as an additive. Having a morpheme functioning as an additive particle, as well as a preposition 'with' is not new. This is the case in quite a number of languages (cf. König, 1991).

- (i) Tolù jẹ ị̀su pèlu síbí.
 Tolù eat yam with spoon
 'Tolù ate yam with a spoon.'

- b. Tolú jẹ́ [iṣu]_F **pèlú**.
 Tolú eat yam also
 ‘Tolú also ate YAM.’

The adfocal nature of additive *pèlú* is corroborated with the following ditransitive sentences in (6). The position of *pèlú* changes depending on the constituent that is focused. *Pèlú* follows the indirect object focus in (6-a), with the sentence having the presupposition that in addition to *Ségún*, Tolú also gave someone else money yesterday. The focus is on the direct object in (6-b), with *pèlú* immediately following it. The sentence presupposes that in addition to money, Tolú gave *Ségún* something else yesterday. Lastly, example (6-c) shows that *pèlú* does not only adjoin to arguments, it also adjoins to adverbs.

- (6) a. Tolú fún [Ségún]_F **pèlú** ní owó làná.
 Tolú give Ségún also SEC money yesterday
 ‘Tolú also gave SÉGUN money yesterday.’
 b. Tolú fún Ségún ní [owó]_F **pèlú** làná.
 Tolú give Ségún SEC money also yesterday
 ‘Tolú also gave ségun MONEY yesterday.’
 c. Tolú fún Ségún ní owó [làná]_F **pèlú**.
 Tolú give Ségún SEC money yesterday also
 ‘Tolú also gave ségun money YESTERDAY.’

A stronger evidence for *pèlú*’s adfocal status is seen when we consider ex-situ focus constructions. All the focused constituents in (6) can be fronted to the left periphery of the clause, as in (7). In each case, the additive particle *pèlú* pied-pipes with the focus to the left periphery of the clause.

- (7) a. [Ségún]_F **pèlú** ní Tolú fún ___ ní owó làná.
 Ségún also FOC Tolú give SEC money yesterday
 ‘Tolú also gave SÉGUN money yesterday.’
 b. [Owó]_F **pèlú** ní Tolú fún Ségún ___ làná.
 money also FOC Tolú give Ségún yesterday
 ‘Tolú also gave ségun MONEY yesterday.’
 c. [Àná]_F **pèlú** ní Tolú fún Ségún ní owó ____.
 yesterday also FOC Tolú give Ségún SEC money
 ‘Tolú also gave ségun money YESTERDAY.’

It is important to elaborate on the context where the ex-situ focus examples with the additive particle *pèlú* are felicitous. This is necessary because of the semantics of additives which requires that, at least, one other alternative proposition is true in a contextual world. Ex-situ focus in Yorùbá is usually exhaustive, which would mean that the examples in (7) should result in a contradiction since an exhaustive focus should not be marked for additivity. In other words, there should not be another true alternative if we are indeed dealing with exhaustive foci in the cases above. However, the sentences in (7) are all grammatical, and here is the reason. While ex-situ focus in Yorùbá can be exhaustive, they can also be contrastive. In the cases above, we are dealing with contrastive focus, rather than purely exhaustive focus. Taking the sentence in (7-a) as a representative for the other examples, this sentence is felicitous in the context given below. Going by the context, the speaker of the sentence in (8) means that in addition to the fact that Tolú gave Adam money (which is a shared knowledge), it was *Ségún* that

Tolú also gave money to, and not Adé (and Ségun). The truth of the earlier proposition that is shared between the interlocutors meets the additive requirement of the sentence in (8), while still maintaining the contrastive nature of ex-situ focus.³

[**Context:** *You and I know that Tolú gave Adam money. However, we are debating on whether Tolú also gave Ségun and Adé money as well. So I said to you that “Tolú also gave Ségun and Adé money (in addition to Adam).” The following sentence is a correction involving contrast.*]

- (8) [Ségun]_F **pèlú** ni Tolú fún ___ ní owó làná.
 Ségun also FOC Tolú give SEC money yesterday
 ‘It was SÉGUN too that Tolú gave money yesterday (i.e., in addition to Adam, and not both Ségun and Adé).’

Pèlú can also ad focally associate with the subject focus. Consider the example in (9). Based on the context in (9-a), the sentence with additive subject focus association in (9-b) is true in a context where there is at least one other person who ate yam as well. The ex-situ subject focus association context is similar to the one discussed for ex-situ object focus association above.⁴

- (9) a. Apart from Sam did Tolú and Adé also eat yam?
 b. Rara, [Tolú]_F **pèlú** ni ó jẹ iṣu.
 no Tolú also FOC 3SG eat yam
 ‘No, it was TOLÚ also who ate yam.’

6.2.2 The adverbial additive particle *tún*

Interestingly, Yorùbá also makes use of the adverbial additive particle *tún*, which can co-occur with *pèlú* in a sentence, and the result would be a single additive reading. This

³ I will return to the ex-situ focus cases in more detail in Section 6.7

⁴ There is another particle, *náà*, which may be used in additive contexts. It functions more like the English additive ‘too’. Consider the object focus association and the subject focus association in (i-a) and (i-b) respectively.

- (i) a. Adé jẹ èwa. O jẹ [iṣu]_F **náà**.
 Adé eat beans 3SG eat yam too
 ‘Adé ate beans. He ate YAM too.’
 b. Adé jẹ èwa. [Tolú]_F **náà** jẹ èwa.
 Adé eat beans Tolú too eat beans
 ‘Adé ate beans. TOLÚ too ate beans.’

However, *náà* is usually used as the definite article in Yorùbá, see (ii) (cf. Bámbóṣé, 1967; Awobuluyi, 1978). Ajiboye (2005), however, believes that *náà* is not a determiner, but a marker of salience, whose function he divided into three: *uniqueness*, *additivity*, and *identity*. The debate is still ongoing, and I will not go into the arguments here. However, it is not surprising to see *náà* function as a determiner in the language because cross-linguistic studies have shown that determiners can grammaticalize into additive (focus) particles (see, for example, Heine and Reh, 1984; König, 1991; Heine and Kuteva, 2002, a.o.).

- (ii) Okùnrin **náà** fẹ obìnrin **náà**.
 man DEF marry woman DEF
 ‘The man married the woman.’

is a case of *additive doubling* (similar to the exclusive doubling cases discussed in Chapter 4 and 5). Before I discuss the additive doubling cases, I want to briefly dissociate between the use of *tún* as an additive adverbial particle (‘also’), and as an aspectual marker (‘again’). The intransitive sentence in (10) is ambiguous between an additive interpretation or an aspectual one. For the additive interpretation, the sentence means that Tolú slept and did something else. However, for the aspectual use, the sentence means that Tolú slept before, and has slept again.

- (10) Tolú **tún** sùn.
 Tolú TÚN sleep
 (i) ‘Tolú also SLEPT.’
 (ii) ‘Tolú slept again.’

While *tún* may be confused between its use as *also* and *again* (and rightly so, since both instances share a few things in common), they usually license different word order. While the sentence with the additive *tún* retains the canonical SVO word order of the language (11-a), aspectual *tún* has an SOV word order (11-b) (cf. Parrish and Feldscher, 2019).

- (11) a. Tolú **tún** fọ [mótò náà]_F.
 Tolú also wash car DEF
 ‘Tolú also washed THE CAR.’
 b. Tolú **tún** mótò náà fọ.
 Tolú again car DEF wash
 ‘Tolú washed the car again.’

Splitting verb constructions in Yorùbá offer another clear distinction between them. According to Parrish and Feldscher (2019), when *tún* functions as an adverbial additive particle, the verb has to split (12-a). But when it functions as an aspectual marker, the verb does not split (12-b).⁵ The base order in both cases is SVO. However, it is assumed that the SOV order in aspectual *tún* cases is a derived order. Also, the split verb is also assumed to begin as a single verb before the split (see Parrish and Feldscher (2019), and some references cited, for some discussions and the analysis of splitting verbs in Yorùbá. They propose a movement-based analysis of splitting verbs in the language).

- (12) a. Adé **tún** **tàn** Akin **jẹ**.
 Adé also deceive₁ Akin deceive₂
 ‘Adé also deceived Akin.’
 b. Adé **tún** Akin **tànjẹ**.
 Adé again Akin deceive
 ‘Adé deceived Akin again’

(adapted and modified from Parrish and Feldscher, 2019, p.545)

Unlike adfocal-*pèlú*, the adverbial additive particle *tún* has a fixed position, which is preverbal. It can neither occur postverbally nor adfocally, see (13-a). In addition, *tún* can only occur below negation, as shown in (13-b). Interestingly, it can occur above or

⁵ It may well be the case that we are dealing with two independent words here. However, both words not only have the same form, they also have the same high tone. I do not engage this discussion here. The distinction between the two usage is enough for us here. I will concentrate on *tún* as an adverbial additive particle.

below a modal auxiliary (13-c). This possibility derives different scopal behaviours of the additive particle which would be important for our analysis later. Based on the structural distribution of *tún*, we can conclude that *tún* structurally occupies a position between NEG and V in the verbal domain, see (14).

- (13) a. Tolú **tún** jẹ́ (***tún**) [iṣu]_F (***tún**).
 Tolú also eat also yam also
 ‘Tolú also ate YAM.’
 b. Tolú (***tún**) ò **tún** jẹ́ [iṣu]_F.
 Tolú also NEG also eat yam
 ‘Tolú did not also eat YAM.’
 c. Tolú ò (**tún**) le (**tún**) jẹ́ [iṣu]_F.
 Tolú NEG also may also eat yam
 ‘Tolú may not also eat YAM.’

- (14) [Subject >> NEG >> (TÚN) >> MOD >> (TÚN) >> V >> O]

Based on the position of *tún*, we predict that it should be able to associate with different constituents in its scope. That is, its position should produce ambiguous association with not only the object argument, for example, but also with the verb or even with the entire VP. This prediction is borne out. Consider the example in (15). The sentence is ambiguous, and could have three interpretations. The adverbial additive particle *tún* can associate with the direct object (15)-i, the verb (15)-ii, or whole VP (15)-iii. The context (e.g. the question under discussion) can help in disambiguating the respective interpretations. For instance, the question “What else did Tolú eat?” will constrain the focus to be on the object argument. Similarly, the question “What else did Tolú do to the yam?” requires the verb to be the focus associate. Lastly, the focus associate will be the VP if the current question is “What else did Tolú do?”. Under the additive association, the different focus constituents evoke distinct contextually salient alternatives, with the additive semantics requiring that the prejacent is true iff at least one other alternative is true as well.

- (15) Tolú **tún** jẹ́ iṣu.
 Tolú also eat yam
 (i) ‘Tolú also ate YAM.’
 (ii) ‘Tolú also ATE yam.’
 (iii) ‘Tolú also ATE YAM.’

6.2.3 Additive doubling in Yorùbá

Both *tún* and *pèlú* can co-occur in the same clause. This is exemplified with the ditransitive sentences with different focused constituents in (30). Even though the additive particles are doubled, each sentence corresponds to a single additive meaning.

- (16) a. Tolú **tún** fún [Ségún]_F **pèlú** ní owó láná.
 Tolú also give Ségún also SEC money yesterday
 ‘Tolú also gave SÉGUN money yesterday.’
 b. Tolú **tún** fún Ségún ní [owó]_F **pèlú** láná.
 Tolú also give Ségún SEC money also yesterday
 ‘Tolú also gave ségun MONEY yesterday.’

- c. Tolú **tún** fún Ségún ní owó [làná]_F **pèlú**.
 Tolú also give Ségún SEC money yesterday also
 ‘Tolú also gave ségún money YESTERDAY.’

Just as we saw for the exclusive focus sensitive particle in Chapters 4 and 5, the possibility of additive doubling in Yorùbá (and as we will see, Kasem and Kusaal) presents us with interesting data that raises the question of form-meaning mismatch, where the double occurrence of additive particles (although occupying different structural position) produces a single additive meaning. This was part of the discussions in the last two chapters, where a closer look at the particles led us to discover that the previously-assumed form-meaning mismatch with exclusive doubling is illusory, as each component of the doubled exclusives play distinct roles (see Chapter 4, §4.4 and Chapter 5, §5.4).

Next, I turn to additive association with focus in Kusaal.

6.3 Additive association in Kusaal

In Kusaal, the adfocal additive particle is *mɛn*, and the adverbial additive particle is *mɛ*.

The example in (17) is from Abubakari (2018, p.130, ex.4.31), with a few modifications. *Mɛn* immediately follows the focus associate, in this case, the object argument. Both elements form a constituent. Unlike the exclusive association context (see Chapter 4 and 5), the in-situ focus marker *nɛ* is absent in additive association, according to Abubakari (2018). She connected the absence of *nɛ* in (17) to the fact that it marks exhaustivity. Since the context here is an additive context, an exhaustive interpretation will therefore be contradictory.

- (17) Pu’a la sa da’ (***nɛ**) [suma]_F **mɛn** su’os.
 woman DEF PST buy FOC groundnut also yesterday
 ‘The woman also bought GROUNDNUT yesterday (in addition to buying something else).’

On the other hand, she noted that the ex-situ focus counterpart, as in (18), is felicitous. She claimed that this is because *ka* is ‘weakly (less) exhaustive’ in comparison to *nɛ*. However, it is unclear what this means exactly. Based on my elicitation with other speakers of the language, (18) is felicitously used in a context that is similar to that of Yorùbá ex-situ additive association with focus in (8) above. I repeat a similar context below.

[**Context:** *You and I know that a particular woman bought beans yesterday. However, we are debating on whether she also bought groundnut and yam as well. So I said to you that “The woman also bought groundnut and beans yesterday.” Then you uttered the following sentence to contrast my position.*]

- (18) [Suma]_F **mɛn** *(**ka**) pu’a la sa da’ ___ su’os.
 groundnut also FOC woman DEF PST buy yesterday
 ‘It was also GROUNDNUT that the woman bought yesterday (in addition to beans, and not groundnut and yam).’

With the given context, the additive interpretation in (18) is derived. The obligatory use of the ex-situ focus marker *ka* in this case suggests that it is not an exhaustive marker.

Although one may wonder whether it is really a simple focus marker in the first place, seeing that in additive association with in-situ focus is unmarked. One probable answer to this is that, although the ins-tiu *nɛ* and the ex-situ *ka* are both contrastive/exhasutive, *ka* seems to have a secondary function of connecting or linking the left periphery to the rest of the clause in ex-situ focus contexts. The idea that *ka* is entirely not a focus marker is unlikely simply because it is associated with focus contexts. Ex-situ additive association with focus is further discussed in Section 6.7 below.

Examples showing association with adverbial additive particle *mɛ* are given below. In (19), the context is created with (19-a), and (19-b) shows *mɛ* associating with the indirect object at a distance. (20-b) shows additive association with a direct object focus, with (20-a) serving as the context. In both cases, the (a) examples produce explicit alternatives that are true in addition to the focus associates.

- (19) a. Ziema tis Napog nɛ Kojo dʊg.
 Ziema give Napog CONJ Kojo pot
 ‘Ziema gave Napog and Kojo a pot.’
 b. Ziema **mɛ** tis [Adam]_F dʊg.
 Ziema also give Adam pot
 ‘Ziema also gave ADAM a pot.’
- (20) a. Ziema tis Napog dʊg nɛ kɔp.
 Ziema give Napog pot CONJ cup
 ‘Ziema gave a pot and a cup to Napog.’
 b. Ziema **mɛ** tis Napog [diisʊŋ]_F.
 Ziema also give Napog spoon
 ‘Ziema also gave A SPOON to Napog.’

Adverbial and adfocal additive doubling is also possible in Kusaal. Consider the sentences in (21). The three sentences produce the same truth-conditional meaning. The fact that Adam also slaughtered a fowl presupposes that he slaughtered something else as well. According to the speakers, there is no meaning difference among the three constructions. The sentence with additive doubling in (21-c) only receives more emphasis on the focus associate.

- (21) [Adam slaughtered a goat.]
 a. Adam **mɛ** kɔdig [nua]_F.
 Adam also kill fowl
 ‘Adam also slaughtered A FOWL’
 b. Adam kɔdig [nua]_F **mɛn**.
 Adam kill fowl also
 ‘Adam slaughtered A FOWL too.’
 c. Adam **mɛ** kɔdig [nua]_F **mɛn**.
 Adam also kill fowl also
 ‘Adam also slaughtered A FOWL.’

6.4 Additive association in Kasem

The pattern of additive association with focus in Kasem is similar to what we saw in Yorùbá and Kusaal above. Kasem also has both an adfocal additive particle *de* and

an adverbial additive particle *ta*. Given the question context in (22-a), the example in (22-b) answers the question, and then example (22-c) gives an additional answer to the question with the adfocal additive particle *de* following the focused constituent. The adfocal particle *de* cannot precede its focus associate, as in (22-d).

- (22) a. Bε mo Adam di diim?
 what FOC Adam eat yesterday
 ‘What did Adam eat yesterday?’
 b. Adam di [mumuna]_F **mo** diim.
 Adam eat rice FOC yesterday
 ‘Adam ate RICE yesterday.’
 c. Adam di [pea]_F **de mo** diim.
 Adam eat yam also FOC yesterday
 ‘Adam also ate YAM yesterday.’
 d. #Adam di **de** [pea]_F **mo** diim.
 Adam eat also yam FOC yesterday
 ‘Adam also ate YAM yesterday.’

Association with a focused subject context is given in (23). Given the question in (23-a), the answer in (23-b) follows, and then, an additional answer is given in (23-c), where adfocal-*de* follows the subject.

- (23) a. Wɔ mo nε Sam?
 who FOC see.PFV Sam
 ‘Who saw Sam?’
 b. [Adam]_F **mo** nε Sam.
 Adam FOC see.PFV Sam
 ‘ADAM saw Sam.’
 c. [John]_F **de mo** nε Sam.
 John also FOC see Sam
 ‘JOHN too saw Sam.’

The adverbial additive particle *ta* precedes the verb, and establishes association with the focus at a distance, as in (24-c). The additive interpretation that is derived here is equivalent to the one derived in (22-c) with adfocal-*de*.

- (24) a. Bε mo Adam di-a.
 what FOC Adam eat-DISJ
 ‘What did Adam eat?’
 b. Adam di [mumuna]_F **mo**.
 Adam eat rice FOC
 ‘Adam ate RICE.’
 c. Adam **ta** di [pea]_F **mo**.
 Adam also eat yam FOC
 ‘Adam also ate YAM.’

Kasem can also double the adverbial additive particle and the adfocal additive particle, as in (32).⁶ The example produces the same additive meaning of the preadjacent as the one

⁶ It is obvious that additive doubling (just like exclusive doubling) is a common attribute of the three languages.

reported in the singleton (adfocal and adverbial) cases above. Such constructions produce cases of form-meaning mismatch where two (structurally and morphologically) distinct additive morphemes produce a single additive meaning of the prejacent.

- (25) a. Bε mo Adam di-a.
 what FOC Adam eat-DISJ
 ‘What did Adam eat?’
 b. Adam di [mumuna]_F mo.
 Adam eat rice FOC
 ‘Adam ate RICE.’
 c. Adam ta di [pea]_F de mo.
 Adam also eat yam also FOC
 ‘Adam also ate YAM.’

Lastly, a careful consideration shows that while the in-situ morphological focus marker *nε* is absent in in-situ additive association with focus in Kusaal, the in-situ focus marker *mo* is present in Kasem. My thought on this focus marking asymmetry is that while *nε* is primarily a contrastive/exhaustive focus marker, *mo* is not primarily a contrastive/exhaustive focus marker. Thus, the latter can occur in additive association with focus context without contradicting the additive interpretation.

6.5 Interim summary

The table in 6.1 summarizes the additive particles that are used in the three languages under investigation. The three languages have both adfocal additive particles and adverbial additive particles, which are realized distinctively. The sentences with either of these particles have the same additive meaning depending on the constituent that is in focus. As we have seen, these particles can also co-occur in the same sentence, and still produce a single additive meaning. This is an evidence for additive concord.

	Adverbial- <i>also</i>	Adfocal- <i>also</i>	Doubling
Kasem	<i>de</i>	<i>ta</i>	✓
Kusaal	<i>mε</i>	<i>mεn</i>	✓
Yorùbá	<i>tún</i>	<i>pèlú</i>	✓

Table 6.1: Additive focus sensitive particles in Kasem, Kusaal, & Yorùbá

6.6 Proposal

In this section, I will start by proposing a structural analysis for additive association with focus in Yorùbá, Kusaal and Kasem (cf. §6.6.1). Thereafter, I will provide supporting evidence for my structural proposal in Section 6.6.2. Then, I will conclude the proposal.

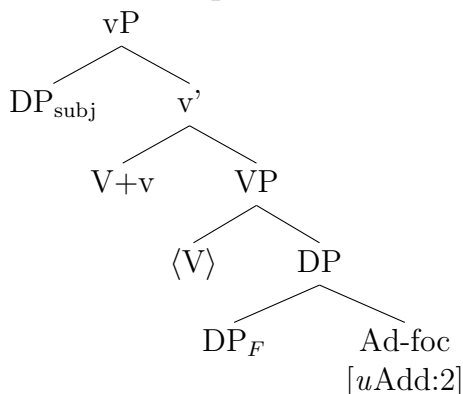
6.6.1 Structural analysis

Similar to the analysis for the exclusive particles in Chapter 5, I propose a propositional approach (Op-particle approach) analysis for additive association with focus for the three

languages. I assume that the additive semantics is not on the adfocal particles, but comes from a propositional additive operator (Op_{add}) high in the clause. The assumption is that Op_{add} can occupy different positions in the clause, provided that it scopes over a proposition.⁷ That is, Op_{add} may occupy any position between TP and vP. It may also occur above TP, as nothing stops it from doing so. Another property of Op_{add} is that it may overtly be exponed by the adverbial additive particles of these languages. On the other hand, the adfocal additive particles form a constituent with the focus association and form an agreement relation with the clausal Op_{add} . In the following discussion, I will work through the analysis step-by-step. Since the three languages are SVO, and they show a similar structural distribution of the additive particles, I will provide a simplified structure which I assume accounts for all three languages.

Using the structure in (26), first, I propose that the adfocal additive particles are specified with a valued uninterpretable feature (cf. Pesetsky and Torrego, 2007; Aremu, 2025).⁸ Their valuation is premised upon their function as syntactically delineating the constituent that is in focus in the clause. But they are uninterpretable because the additive meaning does not come from them. So, they must Agree with a corresponding Op_{add} . They adjoin to the focus associate and project an extended DP. The verb selects for this focused DP, and projects a VP. Then, the little v is merged and the external argument (i.e., the subject) as well. The structure in (26) is built until the vP level which I assume to be the minimal propositional constituent.

(26) **Structural representation until vP**

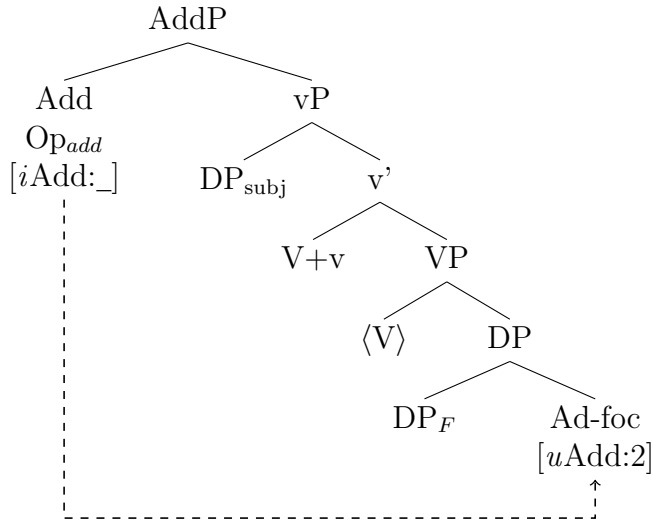


In the next step of the derivation in (27), I propose that an Additive Phrase (AddP) is merged above vP, whose head can be covert (or overtly realized with the adverbial additive particles). Op_{add} is specified with an unvalued interpretable additive feature. Op_{add} establishes an agreement relation with the adfocal particle through feature valuation. Thus, Op_{add} serves as the agreement probe, and the adfocal particle serves as the agreement goal.

⁷ Recall that the minimal propositional node that we assume for our analysis is the little vP (cf. Chomsky, 2000, 2001).

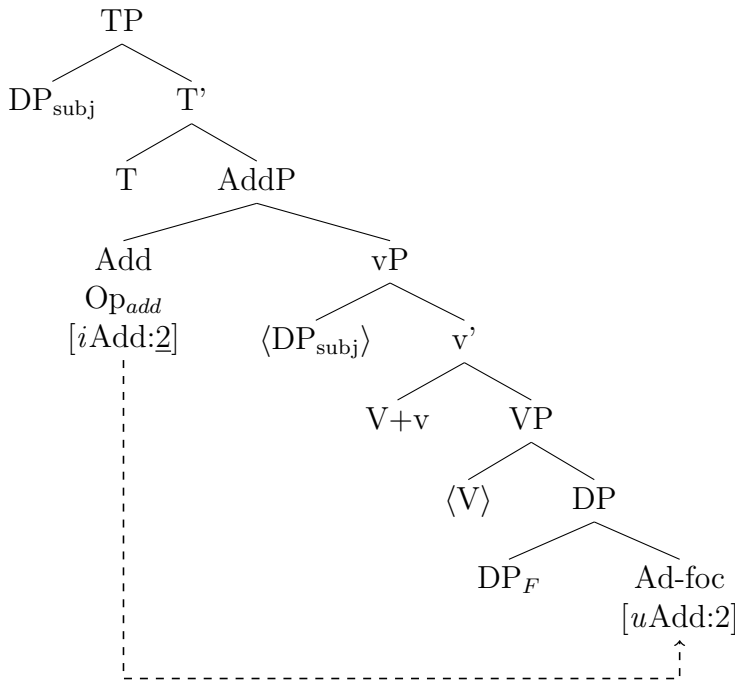
⁸ Note that nothing hinges on the valuation index [2] used here. It is just to represent a valued feature, in the spirit of Pesetsky and Torrego (2007).

(27) **Merging Op_{add} and projecting the Additive Phrase**



After agreement has taken place, the unvalued feature on Op_{add} is valued, as in (28). Unlike Chomsky's (2000; 2001) traditional agreement approach where agreement licenses feature checking and/or deletion, the approach adopted here is more of a feature sharing approach (à la Pesetsky and Torrego, 2007), where agreement creates a link between the additive operator and the adfocal particle. This establishes instances of the same feature on them.⁹ The subject of the clause then moves from Spec,vP to Spec,TP for the EPP requirement on T.¹⁰

(28) **Additive AwF structural representation**



The agreement system offers a syntactic analysis to the syntax-semantic (interface) property of association with focus. Although the focus associate can be a sub-part of

⁹ Although not shown here, the feature on the adfocal particle can project to the extended DP projection (cf. Mursell, 2021; Branan and Erlewine, 2023).

¹⁰ Again, I wish to point out here that the structure in (28) is only a simplified one for these languages. The structures for each language may be more complex depending, for instance, on whether there is v to Asp movement. For example Kusaal and Kasem has v-to Asp movement, but Yorùbá does not.

a sentence, the semantics of additive association with focus requires that the additive particles operate on a propositional level. So, while the adfocal particle identifies the sub-part of the proposition that is in focus, the position of the adverbial operator satisfies the semantic requirement of propositional association. Through feature valuation, the system connects these two positions, allowing for proper computation of the additive meaning while also maintaining the local focus marking.

At this point, we can propose a simple semantic denotation for the additive operator Op_{add} and the adverbial additive particles in Yorùbá, Kasem and Kusaal, see (29). These operators denotes a function that takes a proposition p (of type $\langle s, t \rangle$), and presupposes that p is true in world w , and then asserts that at least one distinct alternative proposition q from the set of alternatives to p is also true in w .

- (29) **Lexical denotation of the propositional additive operators**
 $\llbracket \text{Op}_{add}/\text{tún}/\text{ta}/\text{mɛ} \rrbracket = \lambda p_{\langle s, t \rangle} . \lambda w : p(w) . \exists q \in \text{ALT}(p)[q \neq p \wedge q(w)=1]$

6.6.2 Evidence for the Propositional Approach for additives

That the additive semantics is introduced by an additive operator, which is not the adfocal additive particle, is supported by some evidence, which I will discuss in this section. These arguments here are similar to the ones used in Chapter 4 to argue for a propositional approach for exclusive association with focus. They include (a) the additive doubling cases, (b) scope ambiguity, (c) additive concord, and (d) ellipsis.

Additive concord cases

A simple piece of morphological evidence for the propositional approach to additive association with focus comes from the possible co-occurrence of both the adverbial additive particle and the adfocal additive particle, which produces a single additive reading. This provides a concord phenomenon that we have already seen in the three languages above. Since the position of the adverbial additive particles is pre-verbal, I propose that they overtly realize the additive head. The examples below are for Yorùbá (30), Kusaal (31), and Kasem (32) respectively.

- | | | |
|------|--|---------------|
| (30) | Tolú tún jẹ́ [iṣu] _F pèlú .
Tolú also eat yam also
‘Tolú also ate YAM.’ | Yorùbá |
| (31) | Adam mɛ kɔ́dig [nua] _F mɛn .
Adam also kill fowl also
‘Adam also slaughtered A FOWL.’ | Kusaal |
| (32) | Adam ta di [pea] _F de mo .
Adam also eat yam also FOC
‘Adam also ate YAM.’ | Kasem |

Each example presupposes that another alternative proposition is true. The only point of variation among the contextually salient alternative propositions will be the part that is focused.

Scope ambiguity with adfocal particle

Similar to what we saw with adfocal exclusive particles in Chapter 4, adfocal additive particles may also derive ambiguous scope in the presence of other operators such as

negation and modal predicates (cf. Taglicht, 1984).

The use of the adfocal additive particle in Yorùbá in example (33) produces ambiguous scope with respect to negation. In other words, the sentence has both a narrow scope additive reading and a wide scope additive reading respectively. In the narrow scope additive reading, as in (33)-(i), the derived interpretation presupposes that Tolú ate something else (e.g. rice), and asserts that he did not eat yam in addition. For instance, this may mean that, contrary to what the other interlocutor thinks, yam is not one of the things Tolú ate. On the other hand, the wide scope additive reading is also available (33)-(ii). Assume a context where there are two foods in the room: < rice and yam >. If the additive operator scopes above negation in (33), then this means that within this specific context, the meaning derived is that Tolú neither ate yam nor rice.

- (33) **Yorùbá: scope ambiguity with negation** [neg > also, also > neg]
 Tolú ò jẹ [iṣu]_F pèlú.
 Tolú NEG eat rice also
 ‘Tolú did not also eat yam.’
 (i) Narrow scope: Tolú did not also eat yam (i.e., Tolú did not eat yam, but he ate something else, e.g. rice).
 (ii) Wide scope: Tolú also did not eat yam (i.e., Tolú did not eat yam, and he did not eat something else, e.g. rice).

This type of scope ambiguity with adfocal additives is also available in Kasem (34) and Kusaal (35) respectively. The sentences have only the adfocal additive particle, but still have not only the narrow scope additive reading, but also the wide scope additive reading.

- (34) **Kasem: scope ambiguity with negation** [neg > also, also > neg]
 Adam wo di [pea]_F de mo.
 Adam NEG eat yam also FOC
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)
- (35) **Kusaal: scope ambiguity with negation**
 Adam sa pɔ di [busa]_F mɛn.
 Adam HEST.PST NEG eat yam also
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)

Interestingly, in Kasem and Kusaal, when adverbial additive particles are inserted, the sentences have the surface scope reading (i.e, depending on the surface position of the adverbial additive particle in relation to negation). For instance, in Kasem, the adverbial additive particle *ta* can only precede the negative morpheme, but cannot follow it, see (36-a) and (36-b). This sentence can only have a wide scope additive reading over negation, in accordance to the surface position of the adverbial particle. The narrow scope additive reading is unavailable (36-a).

- (36) **Kasem: rigid scope with adverbial additive** [$*\text{neg} > \text{also}, \text{also} > \text{neg}$]
- a. Adam **ta wo** (*ta) di [pea]_F (**de**) mo.
 Adam also NEG also eat yam also FOC
 (i) *Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)
- b. *Adam **wo ta** di [pea]_F (**de**) mo.
 Adam NEG also eat yam also FOC
Int. ‘Adam did not also eat yam.’

We get the same result in Kusaal as well, see (37-a). The only available scope here is the surface scope, which in this case is the wide scope because the adverbial additive particle *mε* precedes the negative morpheme. The adverbial additive particle cannot follow the negative morpheme (37-b).

- (37) **Kusaal: rigid scope with adverbial additive**
- a. Adam **mε sa pu** (*mε) di [busa]_F (**mεn**).
 Adam also HEST.PST NEG also eat yam also
 (i) *Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)
- b. *Adam sa **pu mε** di [busa]_F (**mεn**).
 Adam HEST.PST NEG also eat yam also
Int. ‘Adam did not also eat yam.’

I wish to point out an issue here (mentioned briefly in Chapter 5), which I do not have an absolute answer for at the moment. The rigid position of the adverbial additive particles in Kasem and Kusaal above raises the question of the position of the covert additive Op_{add} . If Op_{add} is overtly realized by the adverbial additive particles, then the prediction would be that the latter should be flexible in their occurrence as well, just like what we assume for Op_{add} . Apparently, this is not the case. However, I have reasons to believe that this issue only arises with negation. In the context of other operators such as (necessity) modals (e.g., require), the adverbial additive particles can either occur below or above the respective operator. Consider the examples from Kasem (38) and Kusaal (39). In Kasem, the adverbial additive particle can occur below or above the necessity modal *jeye*. In Kusaal the adverbial additive particle can occur above or below the possibility modal *tun’e*.

- (38) **Kasem flexible adverbial additive position**
- a. Adam jege se o **ta** di mumuna mo.
 Adam require to 3SG also eat rice FOC
 ‘Adam is required to also eat rice.’
- b. Adam **ta** jege se o di mumuna mo.
 Adam also require to 3SG eat rice FOC
 ‘Adam is also required to eat rice.’

(39) **Kusaal flexible adverbial additive position**

- a. O **mɛ** tun'e di mui.
3SG also may eat rice
'He also may eat rice.'
- b. O tun'e **mɛ** di mui.
3SG may also eat rice
'He may also eat rice.'

The discussion suggests that there is an interesting structural restriction regarding adverbial additive particles and negation.¹¹ I leave the nature of this interaction with negation open for future research.

Similarly, something interesting is also going on in Yorùbá with respect to the adverbial additive particle's scopal relation with negation. Unlike Kasem and Kusaal, the adverbial additive particle *tún* in Yorùbá obligatorily occurs below the negative morpheme, see (40). Based on this position, the prediction would be that *tún* should take narrow scope below negation, as in (40-a)-i. However, the sentence also has a wide scope additive reading, as in (40-a)-ii.

(40) **Yorùbá: flexible scope with adverbial additive**

- a. Tolú ò **tún** jẹ [iʃu]_F (pẹ̀lú).
Tolú NEG also eat yam also
'Tolú did not also eat yam.'
(i) *Narrow scope*: Tolú did not also eat yam (i.e., Tolú did not eat yam, but he ate something else, e.g. rice).
(ii) *Wide scope*: Tolú also did not eat yam (i.e., Tolú did not eat yam, and he did not eat something else, e.g. rice).
- b. *Tolú **tún** ò jẹ [iʃu]_F (pẹ̀lú).
Tolú also NEG eat yam also
'Tolú also did not eat yam.'

The availability of a wide scope additive reading in (40-a) is interesting, considering the fact that I assume that the additive Op_{add} may be overtly realized with the adverbial additive particles. In contrast, similar to Kusaal and Kasem, the position of the adverbial additive particle *tún* can also be flexible in the context of modals. This is shown exemplified in (41).

(41) **Yorùbá flexible adverbial additive position**

- a. Tolú le **tún** jẹ [iʃu]_F (pẹ̀lú).
Tolú may also eat yam also
'Tolú may also eat yam.'
- b. Tolú **tún** le jẹ [iʃu]_F (pẹ̀lú).
Tolú also may eat yam also
'Tolú also may eat yam.'

While I do not have an absolute answer to these puzzles, there is, however, a potential analysis which is PF related. In Yorùbá, there seems to be a prosodic phrase boundary after negation, with the adverbial particle and the modals belonging to another prosodic

¹¹ See also Chapter 5 (esp. Section 5.3) for examples showing flexible structural positions for adverbial exclusive particles in relation to modals.

phrase.

In analyzing the unique structural position of Yorùbá clausal coordinator *sì*, as in (42), Aremu and Weisser (2024) argue that “*sì* right-adjoins to what is linearly the first prosodic phrase of its second conjunct.” The prosodic phrasing is given in (43).

- (42) Adé jẹ ẹwà. Tolú **sì** jẹ ọ̀ṣu.
 Adé eat beans Tolú CONJ eat yam
 ‘Adé ate beans and Tolú ate yam.’

(43) **Yorùbá prosodic phrasing with *sì***

{Comp < Subj < Neg/Tense/Modal } φ < **sì** < {Perf < Adv/PP-Adjuncts < Verb < Obj } φ

They propose that negation is part of a higher prosodic phrase, and the coordinator *sì* right adjoins to this prosodic phrase. We can use this phrasing to test for the position of the adverbial additive particle in relation to negation and modals. But before we do this, I will argue for a slight refinement of the prosodic phrasing in (43). I argue that modals should be part of the second prosodic phrase, and not the first. Consider the sentences in (44). The modal verb must follow the coordinator in the second conjunct, as shown in (44-a). If the modal precedes the coordinator, then the second conjunct becomes ungrammatical, see (44-b).

- (44) a. Adé le jẹ ẹwà. [Tolú **sì** **le** jẹ ọ̀ṣu].
 Adé may eat beans Tolú CONJ may eat yam
 ‘Adé may eat beans and Tolú may eat yam.’
 b. Adé le jẹ ẹwà. *[Tolú **le** **sì** jẹ ọ̀ṣu].
 Adé may eat beans Tolú may CONJ eat yam
 ‘Adé may eat beans and Tolú may eat yam.’

The same result is observed with the modal *gbọ̀dọ̀* (‘must’) in (45). The modal auxiliary cannot precede the coordinator.¹²

- (45) a. Adé gbọ̀dọ̀ jẹ ẹwà. [Tolú **sì** **gbọ̀dọ̀** jẹ ọ̀ṣu].
 Adé must eat beans Tolú CONJ must eat yam
 ‘Adé must eat beans and Tolú must eat yam.’
 b. Adé **gbọ̀dọ̀** jẹ ẹwà. *[Tolú gbọ̀dọ̀ **sì** jẹ ọ̀ṣu].
 Adé must eat beans Tolú must CONJ eat yam
 ‘Adé must eat beans and Tolú must eat yam.’

Based on this empirical evidence, I propose a revised version of the prosodic phrasing in (43) below. The modal is now part of the second prosodic phrase. With this background, we can now return to our discussion with adverbial additives.

(46) **Yorùbá prosodic phrasing with *sì* (revised)**

{Comp < Subj < Neg/Tense } φ < **sì** < {**Modal**/Perf < Adv/PP-Adjuncts < Verb < Obj } φ

¹² It is important to point out here that this may depend on the type of modal involved, as modals are known for their different scopal behaviour. For instance, while epistemic modals are usually assumed to take high scope (e.g., over negation), deontic modals are said to take lower scope. This in turn may predict their structural position (at LF) (see Jackendoff, 1972; Cinque, 1999; Portner, 2009, a.o.).

Adopting the prosodic phrasing in (46), we may be able to account for what is going on in Yorùbá, where the adverbial particle cannot precede negation. However, there is no such restriction with modals. The assumption is that the adverbial additive particle is part of the second prosodic phrase in (46). This is supported with (47). The second conjunct in (47-a) shows that indeed *tún* occurs below the coordinator *sì*, which means that it belongs to the second prosodic phrase. *Tún* cannot precede *sì*, as in (47-b) and (47-c), which explains why it cannot occur above negation (see (40) above).

- (47) a. Tolú le jẹ ẹwà. [Tolú **sì** **tún le** jẹ ịṣu].
 Tolú may eat beans Tolú CONJ also may eat yam
 ‘Tolú may eat beans and Tolú also may eat yam.’
 b. Tolú le jẹ ẹwà. *[Tolú **tún le** **sì** jẹ ịṣu].
 Tolú may eat beans Tolú also may CONJ eat yam
 Int. ‘Tolú may eat beans and Tolú also may eat yam.’
 c. Tolú le jẹ ẹwà. *[Tolú **tún sì** **le** jẹ ịṣu].
 Tolú may eat beans Tolú also CONJ may eat yam
 Int. ‘Tolú may eat beans and Tolú also may eat yam.’

The adverbial additive seems to have a fixed linear position, which may be related to these requirements on prosodic phrasing, which sometimes might not reflect the semantic scope of the additive Op_{add} .

Note that the scope ambiguity test, that we discussed here, is only meant to show the possibility of having a high additive operator, as in the propositional approach, other than the adfocal additive particles. This does not preclude the quantifier approach, which may also account for such an ambiguity. However, the next two pieces of evidence argue for the advantage of the propositional approach to additive association with focus.

Multiple focus association and concord

If the additive semantics is on a clausal additive Op_{add} , and not the adfocal additive particles, then a multiple adfocally marked focus construction should be able to result in a single concord interpretation. In other words, if multiple focus association with adfocal additive particles results in a single additive concord reading, then the additive meaning cannot be on the adfocal additive particles. Rather, the additive meaning has to be on a propositional operator. This prediction is borne out.

Consider the context in (48), which involves two individuals (Adam and Tolú) who ate one food each, with no other eating taken place. Specifically, Adam ate rice, and Tolú ate beans. Taking into consideration this context, the sentence in (48-a) is felicitous, and reflects a concord reading of the two adfocal additive particles, where Adam ate rice, and the pair <Tolú, ẹwà>, satisfies the other eating relation, and no other. This reading is derived with multiple adfocal additive association on the subject and the object.¹³

(48) Yorùbá additive concord

[**Context:** Adam and Tolú are the only ones in the room, and there is only one portion each of two foods, rice and beans, in the room for them to eat. While Adam ate rice, Tolú ate beans, and no other eating relation happened.]

- a. Adam jẹ ịrẹsì. [Tolú **pẹlú sì** jẹ ẹwà **pẹlú**].
 Adam eat rice Tolú also CONJ eat beans also
 Lit. ‘Adam ate rice and Tolú also ate beans also.’ (i.e., the pair <Tolú,

¹³ Note that English does not have this concord reading.

- èwà> satisfy the other eating relation)
- b. #Adam jẹ ìrèsì. Tolú sì jẹ [èwà]_F pèlú.
Adam eat rice Tolú CONJ eat beans also
‘Adam ate rice and Tolú ate beans also.’
- c. #Adam jẹ ìrèsì. [Tolú]_F pèlú sì jẹ èwà.
Adam eat rice Tolú also CONJ eat beans
‘Adam ate rice and Tolú also ate beans.’

The context predicts that additive association with each of the individual constituents in focus in (48-a) (i.e., *Tolú* and *èwà*) should be infelicitous. This prediction is borne out. If all that happened was that Adam ate rice, and Tolú ate beans, then it is infelicitous to say that Tolú also ate [beans]_F, i.e., in addition to something else that he ate. This is why (48-b) is infelicitous, based on the given context. Similarly, additive association on the subject is also infelicitous in this context as shown in (48-c) because, apart from Tolú, no one else ate beans. The consequence of our discussion is that the naturalness of the sentence in (48-a) must reflect the possibility of a concord reading, where the pair <Tolú, èwà> satisfies the second eating relation out of two.¹⁴ The set of alternatives is contextually constrained, and is given in (49)

(49) ALT: {Adam ate rice, Tolú ate beans}

The possibility of a concord reading with multiple adfocal association supports the propositional approach because the additive semantics must reflect a single additive operator, with the multiple adfocal particles functioning only as concord markers. The structural representation is given in (50). The additive operator *Op_{add}* Agrees with the two adfocal additive particles, which in turn are local to the focus constituents. This produces the concord meaning that is available in (48-a).¹⁵

(50) [AddP Op_{add} [TP [DP Tolú_F pèlú] [T' [vP jẹ [DP èwà_F pèlú]]]]]

Kusaal also allows for an additive concord construction. Based on a similar context to that of Yorùbá in (51), the sentence with multiple adfocal association in (51-a) is felicitous. This means that, apart from the eating relation, that involves Sam and rice, there is also another eating relation involving Adam and yam. This produces a concord reading. Given the context, the sentence cannot mean that in addition to Adam who ate yam and something else, someone else also ate yam and something else. The reading where, in addition to Adam, someone else ate yam is also not available, as shown by the infelicity of the sentence in (51-b). The other reading involving the object, which would mean that Adam ate something else in addition to yam, is also unavailable, see (51-c).

¹⁴ The first eating relation is satisfied by the pair <Adam, ìrèsì>.

¹⁵ On the other hand, the structure in (i) represents a context where there are multiple additive meanings. Based on the propositional approach, this will result in multiple *Op_{add}*s agreeing with individual adfocal additive particles. The set of alternatives would also be separate. However, as we have seen already, such an interpretation is not available based on the present context.

(i) [AddP Op_{add} [TP [DP Tolú_F pèlú] [T' [AddP Op_{add} [vP jẹ [DP èwà_F pèlú]]]]]]]

(51) **Kusaal additive concord**

[**Context:** Only Adam and Sam are in the room, and there are only two dishes for them to eat. While Sam ate rice, Adam ate yam]

- a. Sam di mui ka [Adam **mɛn** di busa **mɛn**].
Sam eat rice CONJ Adam also eat yam also
Lit. ‘Sam ate rice and Adam also ate yam also.’
- b. #Sam di mui ka [Adam]_F **mɛn** di busa.
Sam eat rice CONJ Adam also eat yam
‘Sam ate rice and Adam also ate yam.’
- c. #Sam di mui ka Adam di [busa]_F **mɛn**.
Sam eat rice CONJ Adam eat yam also
‘Sam ate rice and Adam ate yam also.’

In summary, what we see here is that despite multiple morphological expressions of the additive particles, there is only one semantic contribution of additivity at LF.

Ellipsis test

The next evidence for a propositional approach comes from ellipsis. Recall that in Chapter 5 (esp. §5.3.2), I argued that the observed scopal behaviour of focus association in elliptical constructions is predicted by the propositional approach. However, the quantifier approach does not predict such a behaviour. In Chapter 5, we saw the constraint by Bassi et al. (2022), repeated below in (52). This constraint was originally proposed for exclusives, and it disallows a propositional exclusive operator from associating with an adfocal exclusive particle (plus the focus).

(52) **Constraint on ellipsis and focus association:**

Op_{only} and its associated Foc cannot be separated by a node targeted for ellipsis.

(adapted & modified from Bassi et al., 2022, p.819)

However, since we assume that the additive particle (as well as the scalar-additive particle) is also a propositional operator, I argue that the constraint in (52) is true for additive *Op_{add}* as well.

We can apply the constraint on ellipsis and focus association on the following examples involving additive scope with a control predicate *sɛ̀lé́rí* ‘promise’. The sentence in (53-a) is ambiguous between a narrow scope additive reading (see (53-a)-i) and a wide scope additive reading (see (53-a)-ii) respectively. However, when the complement of the raising predicate, containing the focus associate and the adfocal additive particle, is elided, as in (53-b), only the narrow scope additive reading (see (53-b)-i), and not the wide scope additive reading (see (53-b)-ii) is available.

(53) **Yorùbá: No wide additive scope in ellipsis**

- a. Tolú sɛ̀lé́rí láti jẹ́ [iṣu]_F **pɛ̀lú**. [prom. > also, also > prom.]
Tolú promise to eat yam also
‘Tolú promised to eat YAM also.’
 - (i) Narrow scope: Tolú promised to also eat yam (i.e., the content of the promise is to eat yam in addition to what had already been eaten).
 - (ii) Wide scope: Tolú also promised to eat yam (i.e., in addition to his promise to eat something else).

- b. Dàpò náà sèlérí [_{TP} látì jẹ́ [_{isu} ~~pèlú~~].
 Dàpò too promise
 ‘Dàpò promised too.’ [prom. > also, also > prom.]
 (i) Narrow scope: Dàpò promised to also eat yam (i.e., the content of the promise is to eat yam in addition to what had already been eaten).
 (ii) *Wide scope: Dàpò also promised to eat yam (i.e., in addition to his promise to eat something else).

We can account for this behaviour under ellipsis with the propositional approach by assuming an additive version of the constraint in (52), and the structures in (54). The structure for the narrow scope additive reading is given in (54-a). Here, both the additive Op_{only} and the adfocal additive particle are inside the ellipsis site. So, the constraint in (52) (additive version) is not violated, and we get the narrow scope additive reading. In other words, the additive Op_{only} and the associated Foc (adfocal additive particle) are not separated by ellipsis. On the other hand, based on the structure in (54-b), the wide scope additive reading is not available because the constraint on ellipsis and focus association is now violated. In order to take wide scope, the additive Op_{only} has to occur above the raising predicate. At this position, it is separated from the adfocal particle by ellipsis, and so the wide scope additive reading is not available. The point of emphasis here is that assuming a propositional approach to additive association with focus predicts the behaviour of additives under ellipsis.

- (54) a. Narrow scope LF: [_{TP} Dàpò náà [_{VP} sèlérí pèlú [_{TP} látì [_{ExclP} Op_{only} jẹ́ [_{isu} ~~pèlú~~]]]]]
 b. Wide scope LF: *[[_{TP} Dàpò náà [_{ExclP} Op_{only} [_{VP} sèlérí pèlú [_{TP} látì jẹ́ [_{isu} ~~pèlú~~]]]]]]]

On the other hand, if we assume that the additive semantics is on the adfocal additive particle instead, then we will over-generate the availability of wide scope in elliptical constructions, contrary to fact. This is what the Quantifier Approach (QA) that I introduced in Chapter 5 predicts. It predicts the availability of the wide scope additive reading under ellipsis (cf. (53-b)-ii) (cf. Bassi et al., 2022; Beaver and Clark, 2008).

In the next section, I discuss the phenomenon of BAwF with respect to additives. This will also constitute a discussion on the ex-situ focus association, since we have been considering association with in-situ focus hitherto.

6.7 Additive Backward Association with Focus

The phenomenon of Backward Association with Focus (BAwF) was introduced in Section 5.5 of Chapter 5. This is summarized again with the repeated examples below. The examples in (55-a) and (56-a) satisfy the c-command requirement between the focus sensitive particles and their focus associate (cf. PLA in Ch. 2). In (55-b), although the object argument is fronted, the scalar additive particle is still able to associate (backward based on syntactic hierarchy) with it in the surface syntax. However, the same backward association is not available with exclusives, as in (56-b) in English and some other languages that were mentioned in 4 (cf. Jackendoff, 1972; Tancredi, 1990; Beaver and Clark, 2008; Erlewine, 2014a,b, a.o.).

- (55) a. Adam even ate RICE
 b. RICE, Adam even ate ____.
- (56) a. Adam only ate RICE.
 b. *RICE, Adam only ate ____.

This led to the constraint in (57) which bans the exclusive operator from associating backward with a fronted constituent. This constraint takes inspiration from the Principle of Lexical Association (PLA), which requires that focus sensitive particles such as *only* must c-command their focus associates (see chapter 5).

- (57) **No Backwards Association**
 ONLY must c-command F in the surface syntax.

(adapted from Hirsch and Wagner, 2025, p.14)

However, in Section 5.5 of the last chapter, I argue that unlike English, BA_wF is available in Yorùbá, Kasem and Kusaal respectively. This means that these languages violate the constraint in (57). In this section, I will show that not only do these languages have BA_wF with exclusives, but they also have BA_wF with additive focus sensitive particles. Unlike exclusive BA_wF, less has been said about additive BA_wF, but see Erlewine (2014b). Similar to the presence of BA_wF with *even* above, BA_wF with *also* is also available in English, as in (58-b). This suggests that the constraint in (57) is peculiar to BA_wF with exclusives.

- (58) a. Adam also ate RICE
 b. RICE, Adam also ate ____.

I already showed in Chapter 3 that the three languages employ focus fronting as one of its focus marking strategies. Thus, it is interesting to see what happens when a focus associate is fronted out of the c-command domain of an additive operator. The question is whether the additive association is still maintained in such a context. The answer is yes, and this is the case for all three languages under investigation. After describing the data, I will show that the propositional approach also accounts for additive BA_wF in these languages.

6.7.1 Additive BA_wF in Yorùbá, Kasem and Kusaal

Because of the nature of ex-situ focus in Yorùbá, Kasem and Kusaal, additive BA_wF would have to be properly contextualized to avoid any contradiction. As I mention earlier, ex-situ focus in these languages may mark exhaustive focus, but may also be simply contrastive. I argue that in additive BA_wF, the ex-situ foci are contrastive in nature rather than exhaustive. This is because additive semantics should be incompatible with exhaustivity.

The sentence in (59-b) shows BA_wF with an adverbial additive particle where the focus associate is not within the c-command domain of the particle in the surface syntax. For this sentence to be felicitous, a context which includes the sentence in (59-a) has to be assumed. Uttering (59-a) itself already presupposes that Tolú ate something else. So, assume a context, where speaker A utters the sentence in (59-a). Then, speaker B corrects that assertion by uttering the sentence in (59-b). This means that Tolú did not eat both yam and beans, in addition to what was eaten already in the background. Rather, he ate only yam in addition to what had been eaten. This context maintains

the contrastive nature of ex-situ foci in the language, while also producing an additive interpretation of the fronted focus. Notice that the occurrence of the adfocal additive particle *pèlú* is optional. So, the additive meaning is not dependent on it. Therefore, the adverbial additive particle *tún* associates backward with the focus.

(59) **Yorùbá additive BA_wF**

- a. Tolú **tún** jẹ [iṣu àti ẹwà]_F (pèlú).
Tolú also eat yam CONJ beans also
'Tolú also ate YAM AND BEANS.'
- b. Rárá, [iṣu]_F (pèlú) ni Tolú **tún** jẹ ____.
no yam also FOC Tolú also eat
'No, it was yam also that Tolú ate (he did not also eat beans).'

The same BA_wF with adverbial additive particles is also available in Kasem (60-b) and Kusaal (61-b) respectively. The same context as in the Yorùbá example above applies here. In both languages, the adfocal additive particles are optional, which means that in the surface syntax, the adverbial additive particles associate backward with the fronted foci.

(60) **Kasem additive BA_wF**

- a. Adam **ta** di [pea de swoona]_F (de).
Adam also eat yam CONJ beans also
'Adam also ate yam and beans.'
- b. Aawo, [pea]_F (de) mo Adam **ta** di-a ____.
no yam also FOC Adam also eat-DISJ
'No, it was yam also that Adam ate (he did not also eat beans).'

(61) **Kusaal additive BA_wF**

- a. Adam **mɛ** di [busa nɛ bɛŋa]_F (mɛn).
Adam also eat yam CONJ beans also
'Adam also ate yam and beans.'
- b. Ayei, [busa]_F (mɛn) ka Adam **mɛ** di ____.
no yam also FOC Adam also eat
'No, it was yam also that Adam ate (he did not also eat beans).'

Since the original formulation of PLA does not preclude other focus sensitive particles, such as *also* and *even*, we may also assume a constraint for additive association which is similar to the one in (57), as in (62) (cf. Tancredi, 1990; Beaver and Clark, 2008).¹⁶ The constraint prohibits additive BA_wF. However, as we have seen in this section so far, the Yorùbá, Kasem and Kusaal have additive BA_wF, thereby violating (62).

(62) **No Backwards Association (additive version)**

ALSO/Op_{add} must c-command its focus associate in the surface syntax.

Interestingly, the availability of additive BA_wF is not peculiar to these three languages. Additive BA_wF is also available in languages which do not have exclusive BA_wF: Mandarin, Vietnamese and Cantonese.¹⁷ The following examples are from Mandarin (63),

¹⁶ Tancredi (1990) actually treats *only* and *even* as propositional operators, while Beaver and Clark (2008) categorizes the three (*only*, *even*, and *also*) as conventional (propositional) operators.

¹⁷ As shown in example (58-b), additive BA_wF is also available in English.

Vietnamese (64), and Cantonese (65).¹⁸ While the (a) examples show the normal association with the focus in the c-command domain of the additives, the (b) examples show additive backward association with fronted topics.

- (63) **Mandarin BA_wF** (Ka-Fai Yip, p.c.)
- a. Zhangsan kan-le zhe-ben shu. Ta **ye** kan-le [na-ben shu]_F.
 Zhangsan read-PFV this-CL book 3SG also read-PFV that-CL book
 ‘Zhangsan read this book. He also read that book.’
- b. Zhangsan kan-le zhe-be shu. [Na-ben shu]_F ne, ta **ye** kan-le
 Zhangsan read-PFV this-CL book that-CL book TOP 3SG also read-PFV
 —.
 ‘Zhangsan read this book. That book, he also read.’
- (64) **Vietnamese BA_wF** (Ka-Fai Yip, p.c.)
- a. Minh đọc quyển sách này. Minh **cũng** đọc [quyển sách kia]_F.
 Minh read CL book this Minh also read CL book that
 ‘Minh read this book. Minh also read that book.’
- b. Minh đọc quyển sách này. [Quyển sách kia]_F, Minh **cũng** đọc —.
 Minh read CL book this CL book that Minh also read
 ‘Minh read this book. That book, Minh also read.’
- (65) **Cantonese BA_wF** (Ka-Fai Yip, p.c.)
- a. Aaming tai-zo ni-bun syu. Keoi **dou** tai-zo [go-bun syu]_F.
 Ming read-PFV this-CL book 3SG also read-PFV that-CL book
 ‘Ming read this book. He also read that book.’
- b. Aaming tai-zo ni-bun syu. [Go-bun syu]_F ne, keoi **dou** tai-zo —.
 Ming read-PFV this-CL book that-CL book TOP 3SG also read-PFV
 ‘Ming read this book. That book, he also read.’

The implication is that all these languages seem to violate the constraint in (62) because they allow BA_wF. However, in the next section, I will show, following Erlewine (2014b), that *also*’s association with a fronted constituent is an illusion as it can associate with a lower copy of the moved constituent. All that matters is that the fronted constituent originates from its scope.

6.7.2 Analyzing additive BA_wF in Yorùbá, Kasem and Kusaal

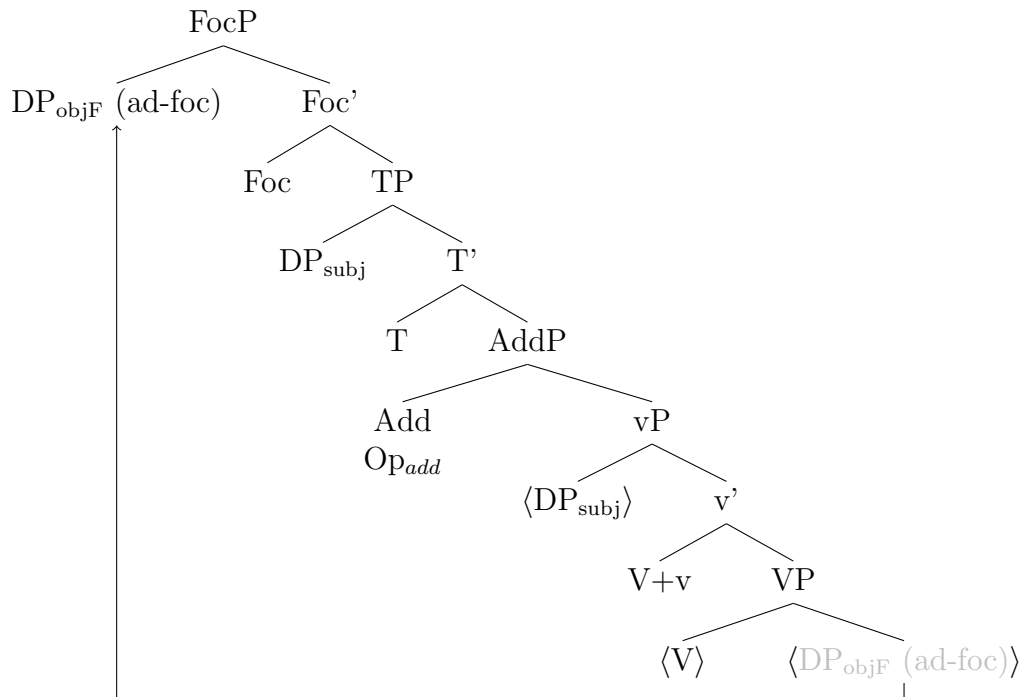
Additive BA_wF in Yorùbá, Kasem and Kusaal is accounted for by the propositional approach. I assume that backward association with *also* does not require reconstruction like the one that was argued for exclusive backward association in the last chapter. The reason for this is attributed to the semantics of *only* on the one hand, and that of *also* and *even* on the other. While *only* affects the truth-condition of its assertion (i.e., its prejacent), *also* and *even* do not affect the truth-condition of their assertion. They introduce a not-at-issue meaning in the form of presuppositions. So, while *only* asserts that its prejacent is true iff all other propositional alternatives are false, *also* and *even* presuppose that other alternatives are true in addition to their prejacent. The latter goes further to introduce a scalar presupposition, which indicates that the prejacent is less likely on a likelihood or

¹⁸ I am grateful to Ka-Fai Yip for providing these data for me.

expectation scale.¹⁹ The fact that *only* asserts the truth-condition of its prejacent makes it rigid when it comes to the structure of its prejacent. So, in order not to end up with contradictions, all the constituents of its prejacent would have to be interpreted in their base positions. This is not the case with *also* (and *even*), which allows for more flexibility with respect to its structural requirement Erlewine (2014a,b).

Based on this idea, it is enough for the adverbial additive particles in these languages to associate with the trace of their focus associate. Adopting the copy theory of movement, this trace would be an unpronounced copy of the moved constituent (cf. Chomsky, 1993; Sauerland, 1998; Fox, 1999, 2002; Bošković and Nunes, 2007, a.o.).²⁰ So, based on the structure in (66), the assumption is that at PF, the higher copy of the moved constituent is pronounced. However, at LF, it is the lower copy that is interpreted. This simply allows the adverbial additive Op_{add} to associate with the focus.

(66) **Additive BA_wF Structure**



In the last chapter, we saw that the fronted focus constituents under the exclusive association were able to take not only wide scope, but also narrow scope. The prediction is that the same should be observable with additive backward association. This prediction is borne out. Recall that in Section 6.6.2 above, I showed that we can have a narrow scope additive reading or a wide scope additive reading in relation to negation. The examples given above involve in-situ focus constructions, where the focus associate and the adfocal particle are in the c-command domain of the additive Op_{add} . The Yorùbá example is repeated in (67-a) below. The sentence has both the narrow scope additive meaning (67-a)-i and the wide scope additive meaning (67-a)-ii respectively. Interestingly, if the focus is fronted, as in (67-b), the scope ambiguity is still available. Both the wide scope additive reading and the narrow scope additive reading are available, see (67-b)-i and

¹⁹ I will return to the scalar-additives in the next Chapter.

²⁰ The compositional semantic implication of having multiple instances of a constituent at LF has been up for debate. A proposed solution for this has been a process called *Trace Conversion*, which involves converting the lower copy of the moved constituent into a definite description. I will not go into the intricacies of this process here, but see Rullmann and Beck (1998); Erlewine (2014b).

(67-b)-ii. The fact that the narrow scope additive reading is available under negation supports our assumption that the fronted constituent started out from a lower position.

(67) **Yorùbá: preservation of scope ambiguity with negation**

- a. Tolú ò jẹ [iṣu]_F pẹ̀lú. [neg > also, also > neg]
 Tolú NEG eat yam also
 (i) Narrow scope: Tolú did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Tolú also did not eat yam (i.e., he did not eat something else as well)
- b. [Iṣu]_F pẹ̀lú ni Tolú ò jẹ _____. [neg > also, also > neg]
 yam also FOC Tolú NEG eat
 (i) Narrow scope: Tolú did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Tolú also did not eat yam (i.e., he did not eat something else as well)

The same result is found in Kasem (68) and Kusaal (69) respectively. There is additive scope ambiguity with ex-situ focus in relation to negation, see (68-b) and (69-b) below. Importantly, the availability of narrow scope in ex-situ focus supports our propositional approach because if the additive semantics is on the adfocal additive particles, then we would expect to get only a wide scope additive reading over negation. However, this is not the case.

(68) **Kasem: scope preservation with negation**

- a. Adam wo di [pea]_F de mo. [neg > also, also > neg]
 Adam NEG eat yam also FOC
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)
- b. [Pea]_F de mo Adam wo di _____. [neg > also, also > neg]
 yam also FOC Adam NEG eat
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)

(69) **Kusaal: scope preservation with negation**

- a. Adam sa pu di [busa]_F men.
 Adam HEST.PST NEG eat yam also
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)
 (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)
- b. [Busa]_F men ka Adam sa pu di _____.
 yam also FOC Adam HEST.PST NEG eat
 (i) Narrow scope: Adam did not also eat yam (i.e., he ate something else excluding yam)

- (ii) Wide scope: Adam also did not eat yam (i.e., he did not eat something else as well)

The assumption is that depending on whether we have a narrow scope additive reading or a wide scope additive reading, the propositional additive Op_{add} occurs below or above the negative marker. So, in both readings, the additive association is with the lower copy of the moved element, but the different scope is derived based on the position of Op_{add} . The structures are presented below. Starting with the wide scope additive reading, (70) is the surface structure, and Op_{add} occurs above NEG.

- (70) **Wide scope additive reading with negation** [also > neg]
 [FocP [DP_{objF} (ad-foc)] Foc [TP DP_{subj} [**Op_{add}** [**Neg** [vP V [DP_{objF} (ad-foc)]]]]]]]

For the narrow scope reading on the other hand, Op_{add} occurs below NEG, see (71).

- (71) **Narrow scope additive reading with negation** [neg > also]
 [FocP [DP_{objF} (ad-foc)] Foc [TP DP_{subj} [**Neg** [**Op_{add}** [vP V [DP_{objF} (ad-foc)]]]]]]]

The discussion in this section has shown that the treatment of exclusive BAwF (see Section 5.5 of Chapter 5) is different from that of additive BAwF. The reason for this stems from their individual semantic contribution in association with focus contexts.

6.8 Conclusion

In this chapter, I have argued for a propositional approach for additive association in Kasem, Kusaal and Yorùbá. The implication is that the syntactic distribution of both adfocal additives and adverbial additives can uniformly be accommodated under the propositional approach, where the additive interpretation is located on a clausal additive operator, which may be overtly realized by the adverbial additive particles of the three languages. However, the covert additive operator has a wider scopal distribution than the adverbial additive particles. Evidence for my proposal comes from (a) the availability of not only adfocal and adverbial additive particles in all the three languages, but also additive doubling/concord, which derive a single additive meaning, (b) the scopal ambiguity with other quantificational operators such as negation, (c) multiple additive association with a concord reading, and (d) additive association under ellipsis.

I also showed that Kasem, Kusaal and Yorùbá allow for additive BAwF, which is interestingly also available in other languages that do not allow exclusive BAwF. I argued that the constraint on additive BAwF in (62) is not active in these languages, and that our approach accounts for the observed phenomenon. Unlike exclusive BAwF, additive BAwF does not require reconstruction. The only requirement is for the fronted constituent to start out within the scope of the additive operator. Adopting a copy theory of movement, I assume that the trace of the moved constituent is realized as a lower copy of the constituent. It is this lower copy that the additive operator associates with at LF. Therefore, additive BAwF is merely an illusion. The implication here is that focus movement does not bleed additive association with focus in the languages under investigation.

Chapter 7

Scalar-Additive focus sensitive particles

This chapter discusses the third focus sensitive particle that this dissertation investigates. This is the scalar-additive particle *even*. I will begin by describing the associative properties of *even* in these languages. Then, I will propose a structural analysis for them. I will argue that the minor difference in the distribution of *even* in these languages can uniformly be accounted for by the propositional approach that I have assumed all through this dissertation. I argue that only two of the three languages (Kasem and Kusaal) allows scalar-additive backward association, and that this is simply accounted for by assuming a lower copy of the moved element, instead of reconstruction. In short, the analysis provided for scalar-additive backward association here is the same as the one proposed for additives backward association in the last chapter. Lastly, I argue that the three focus sensitive particles investigated in this study all involve conventional association with focus.

7.1 Introduction

Traditionally, scalar-additive particles like *even* are said to assert their prejacent and produce two presuppositions: an additive presupposition, and a scalar presupposition which is associated with the ‘unlikelihood’ or ‘noteworthiness’ or ‘unexpectedness’ of the focus associate (Horn, 1969; Karttunen and Peters, 1979; Kay, 1990; Krifka, 1998). Consider the context below and the example that follows it in (1). Based on our knowledge of Adam, it is more unlikely for him to get an F, than to get other grades. In addition, the sentence presupposes that someone else also got an F.¹ The scalar presupposition of *even* means that the alternative propositions are ordered on a likelihood scale.

- (1) [Adam is known to be the best student in his class. But after a particular math exam, many students failed, including Adam.]
Even [Adam]_F got an F.
(i) Assertion: Adam got an F.
(ii) Additive presupposition: Someone else got an F as well.

¹ The discussion on whether these two presuppositions are always available in sentences with *even* is still active in the literature (cf. Greenberg, 2016, 2018, 2022; Zhang, 2022; Liao and Jheng, 2025, a.o.). There is also a discussion on whether they should be called presuppositions or just conventional implicatures (see Horn, 1969; Karttunen and Peters, 1979, for example). In this study, I take the position that *even* has scalar presuppositions in the languages under investigation.

(iii) Scalar presupposition: Compared to other students, Adam is the most unlikely student to get an F.

Further examples are given in (2) and (3), taken from the Bible (New International Version). The prejacent assertion in (2), for example, is that the dogs under the table eat children’s food, and this sentence presupposes that other animals do as well, while dogs are more unlikely to eat children’s food.

- (2) Lord, she replied, “**even** THE DOGS under the table eat the children’s crumbs.” (Mark 7:28, NIV)
Meaning: “dogs” are unlikely animals to eat children’s food.
- (3) “What is this? ... He (Jesus) **even** gives orders to IMPURE SPIRITS and they obey Him.” (Mark 1:27, NIV)
Meaning: “impure spirits” are unlikely entities to be given orders.

The presuppositions of *even* are formally represented in (4) (cf. Horn, 1969; Karttunen and Peters, 1979; Zhang, 2022; Liao and Jheng, 2025). (4-a) means that there is at least one member of a contextually relevant set of alternatives, apart from the prejacent, that is true. The likelihood denotation, (4-b), shows that, given all the alternatives of a contextual set, the prejacent is more unlikely than all the other contextual alternatives.

- (4) a. Additive presupposition of *even*:

$$\llbracket \text{even} \rrbracket^w(p) = \exists q \in C[q \neq p \wedge q=1]$$
b. Likelihood presupposition of *even*:

$$\llbracket \text{even} \rrbracket^w(p) = \forall q \in C[q \neq p \rightarrow p >_{\text{Unlikely}} q]$$

In the next section, I discuss scalar-additive association with focus in the three languages.

7.2 Scalar-additive particles in Yorùbá, Kusaal and Kasem

So far, in the three languages, we have consistently seen that they have distinct (a) adverbial and adfocal exclusive particles, and (b) adverbial and adfocal additive particles. The three languages also allow for exclusive and additive doubling of these particles. However, as I will show in this section, in scalar-additive association with focus, only Yorùbá has both adverbial and adfocal scalar-additive particles. The other two languages, Kusaal and Kasem, only have adverbial scalar-additive particles. They lack adfocal scalar-additives. The fact that the three languages have adverbial scalar-additive particles supports the propositional approach that we adopt in this study. In the next three subsections, I will describe scalar-additive association with focus in the three languages one after the other, and then provide a proposal in the Section 7.3.

7.2.1 Scalar-additive particles in Yorùbá

Yorùbá has both an adfocal and an adverbial scalar-additive particles. The morpheme *pàápàá* adjoins directly to the focus associate, while the morpheme *kódà* associates with the focus at a distance. Unlike what we have seen so far with the adverbial exclusive

and additive particles, the adverbial scalar-additive particle *kódà* occupies the sentence initial position in Yorùbá.² Consider the following context and the examples that follow. In (5-a), the adfocal scalar-additive particle *pàápàá* follows the focus associate. In (5-b), we have the adverbial scalar-additive particle *kódà* associating with the focus constituent at a distance. Both particles can also be doubled as shown in (5-c). The three constructions share the same scalar-additive interpretation, which means that although Tolú ate something else at the party, the fact that Tolú ate pork as well is unexpected. Based on our knowledge of Tolú, the likelihood of him eating pork is low.³

- (5) [We all went for a birthday party, and we know that Tolú dislikes pork. But surprisingly when the food was served, Tolú reached out for a pork and ate it.]
- a. Tolú jẹ [ẹran ẹlédè]_F **pàápàá**.
Tolú eat meat pig even
'Tolú even ate PORK.'
- b. **Kódà** Tolú jẹ [ẹran ẹlédè]_F.
even Tolú eat meat pig
'Tolú even ate PORK.'
- c. **Kódà** Tolú jẹ [ẹran ẹlédè]_F **pàápàá**.
even Tolú eat meat pig even
'Tolú even ate PORK.'

The fact that *pàápàá* is adfocal is supported with the ditransitive sentences in (6). The context creates a situation where Sam is an unlikely person to be given a gift by Sandra. In (6-a), *pàápàá* follows the indirect object focus associate. The second sentence (6-b) involves the adverbial particle *kódà*, with the possibility of doubling as well.

- (6) [Sam broke up with Sandra last December, which caused her a lot of pain. Now this December, Sandra gave out gift on Boxing Day, and gave one to Sam.]
- a. Sandra fún [Sam]_F **pàápàá** ní ẹbùn.
Sandra give Sam even SEC gift
'Sandra even gave SAM a gift.'
- b. **Kódà** Sandra fún [Sam]_F (**pàápàá**) ní ẹbùn.
even Sandra give Sam even SEC gift
'Sandra even gave SAM a gift.'

If the focus shifts to the direct object, *pàápàá* would have to follow the direct object, see (7). This would mean that Sandra gave something else to Sam. However, it is unlikely that Sandra would give a gift to Sam. Perhaps, Sandra is a stingy person.

- (7) **Kódà** Sandra fún Sam ní [ẹbùn]_F (**pàápàá**).
even Sandra give Sam SEC gift even
'Sandra even gave Sam A GIFT.'

Association with subject focus is given in (8), taken from the Yoruba Bible.

² In some context, the word *kódà* may be used as the phrase "in fact". However, whenever it is used as such, there is always a prosodic break before the next constituent. In the case of the scalar-additive use, there is no prosodic break. The use of *kódà* for both expressions is unsurprising because both expressions certainly share some scalar properties.

³ Of course, without appropriate context, the sentence in (5-b) (and others like it, i.e., with the scalar-additive adverbial particle) can be ambiguous with other foci such as subject focus, VP focus, or even TP focus.

- (8) **Kódà** [alàìgbón]_F maa ní dàbí ọlọgbón bí ó bá dáké ...
 even fool FUT PROG be.like wise if 3SG BA quiet
 ‘Even a fool is considered wise if he is quiet ...’
Meaning: While other people are considered wise, a fool is the most unlikely person to be wise. In fact, it appears ironic

From Òwe (Proverbs) 17:28 (YCB)

Although it is not used in that verse of the Bible, the adfocal scalar-additive particle can as well be felicitously used, as in (9). The position of the adverbial scalar-additive particle, which is clause-initial, makes the association with the subject focus clearly realized.

- (9) (**Kódà**) [alàìgbón]_F **pàápàá** maa ní dàbí ọlọgbón bí ó bá dáké ...
 even fool even FUT PROG be.like wise if 3SG BA quiet
 ‘Even a fool is considered wise if he is quiet ...’
Meaning: While other people are considered wise, a fool is the most unlikely person to be wise.

An interesting property of the adverbial scalar-additive particle *kódà* is that it has a flexible distribution, i.e., it can occur in different positions outside the clause that contains the focus associate. The only condition is that it must precede a clause which is as big as a TP. Consider the sentence in (10). The focus is on the embedded subject *Tolú*, and the sentence can be interpreted to mean that *Tolú* is an unlikely person to dance, but the party was so sweet that he danced. *Kódà* can occur above the embedded subject, or above the entire clause. It associates with the focused constituent at a distance. The adfocal particle can optionally occur.

- (10) [How sweet was the party?]
 (Kódà) ọ̀de nàà dùn dé ibi wípé (kódà) [Tolú]_F (pàápàá) jó.
 even party DEF sweet to place COMP even Tolú even dance
 ‘The party was so sweet that even TOLÚ danced.’

7.2.2 Scalar-additive particles in Kusaal

In Kusaal, scalar-additivity is marked with the adverbial particle *la’am* and sometime *hali*. In my elicitation, I did not find an adfocal scalar-additive particle in Kusaal. Although both *la’am* and *hali* are adverbial, while the former occurs in a pre-verbal position inside the clause, *hali* occur in a sentence-initial position. Consider the following examples. Based on the context in (11), the sentences marks scalar-additivity on the object argument. In (11-a), *la’am* occurs pre-verbally and associates at a distance with the object focus. In (11-b), *hali* on the other hand, occurs before the subject, in a sentence-initial position. Both sentence produce a scalar-additive meaning, which means that Adam ate something else apart from pork. However, pork is the most unlikely thing for Adam to eat. The example in (11-c) shows that both *hali* and *la’am* cannot co-occur.⁴ The position of *hali* in Kusaal is similar to the position of *kódà* in Yorùba.

⁴ In Section 7.3, I will propose that both particles can overtly realize the different positions that the scalar-additive operator can occupy.

- (11) [There were different meats at the party. We know that Adam loves meat, but he never eats pork. But at the party, he did not only eat chicken, he even ate pork.]
- a. Adam **la'am** ɔnb nɛ [kukur ni'im]_F.
Adam even chew FOC pig meat
'Adam even ate PORK.'
 - b. **Hali** Adam ɔnb nɛ [kukur ni'im]_F.
even Adam chew FOC pig meat
'Adam even ate PORK.'
 - c. ***Hali** Adam **la'am** ɔnb nɛ [kukur ni'im]_F.
even Adam even chew FOC pig meat
'Adam even ate PORK.'

Further examples are given with the ditransitive examples below. The focus in (12) is on the direct object *lɔr* ('car'). Again, both *la'am* and *hali* can be used, see (12-b) and (12-c) respectively. The meaning of the examples is that Ziema gave many things to Napog, including a car, which is an unlikely thing to give to Napog.

- (12) [Ziema and Napog were neighbors, but were not that close. Ziema was relocating to another city. So he decided to give Napog some of his things. Surprisingly, Ziema also gave Napog his car.]
- a. Ziema tis Napog la'ad bɛdigu.
Ziema give Napog thing many
'Ziema gave many things to Napog.'
 - b. Ziema **la'am** tis Napog nɛ [lɔr]_F.
Ziema even give Napog FOC car
'Ziema even gave A CAR to Napog.'
 - c. **Hali** Ziema tis Napog nɛ [lɔr]_F.
even Ziema give Napog FOC car
'Ziema even gave A CAR to Napog.'

In (13), the focus is now on the indirect object *Napog*. The two scalar-additive particles can associate with the focus at a distance, and produce the meaning that, out of the people who received a car from Ziema last year, Napog was the most unlikely person to receive a car.

- (13)
- a. Ziema da tis nidib bɛdigu. lɔr yu'um.
Ziema PST give people many car year
'Ziema gave cars to many people last year.'
 - b. Ziema da **la'am** tis nɛ [Napog]_F lɔr yu'um.
Ziema PST even give FOC Napog car year
'Ziema even gave NAPOG a car last year'
 - c. **Hali** Ziema da tis nɛ [Napog]_F lɔr yu'um.
even Ziema PST give FOC Napog car year
'Ziema even gave NAPOG a car last year'

Scalar-additive association with verb focus is also possible. Consider the intransitive construction in (14). The derived interpretation is that singing is the most unlikely thing for Adam to participate in, maybe because he does not like singing, or he is not good at it.

(14) [How well did Adam participate?]

- a. Adam **la'am** yu'umne.
Adam even sing
'Adam even SANG.'
- b. **Hali** Adam yu'umne.
even Adam sing
'Adam even SANG.'

A scalar-additive association with a subject focus in Kusaal can be realized in two ways. The first is by the use of the adverbial scalar-additive *la'am*, which occurs pre-verbally, see (15). In this case, the subject focus is adfocally marked by the adfocal additive particle, which is obligatory in this case. The second strategy has to do with the clause-initial additive particle *hali*, which can precede the subject, and also associate with it, see (15-b). In this case, the adfocal additive particle is optional. In both cases, the reflected meaning is that the economy was so good that Adam, who is more unlikely to buy a car, bought one.

(15) [The economy of the city was so good that even Adam bought a car.]

- a. [Adam]_F *(me) **la'am** da' lor.
Adam also even buy car
'Even ADAM bought a car.'
- b. **Hali** [Adam]_F (me) da' lor.
even Adam also buy car
'Even ADAM bought a car.'

A further example on scalar-additive association with subject focus is given in (16). This example was taken from the Kusaal dictionary. It introduces an interesting discussion in the literature about the behaviour of *even* in the presence of negation, i.e., in a downward-entailing context. In a downward-entailing context, the scalar presupposition of *even* is said to be reversed (Karttunen and Peters, 1979). So, the sentence would mean that my parents are the most likely people that should be informed about a particular matter. Compare this with the example in (17), where the reverse is the case. The scalar inference here is that my parents are less likely to know about a matter in comparison to other people.⁵

(16) M da pu yel so'oso, **hali** [m du'adib]_F da zi'.
1SG PST NEG say anyone even 1SG.POSS parent NEG know
'I did not tell anyone, even MY PARENTS did not know.'
↔ (My parents know) >_{likely} (<ALT> know)

(17) **Hali** [m du'adib]_F zi'.
even 1SG.POSS parent know
'Even MY PARENTS know.'
↔ (My parents know) <_{likely} (<ALT> know)

⁵ This kind of observation has led to the proposal of two variants of *even*, the positive polarity *even* (*even_{PP1}*) and the negative polarity *even* (*even_{NP1}*). See Rooth (1985).

7.2.3 Scalar-additive particles in Kasem

In Kasem, the scalar-additive particle is made up of the adverbial additive particle *ta*, discussed in the last chapter (cf. §6.4), and the morpheme *maje*.⁶ Both particles form a compound *ta-maje* in order to produce a scalar-additive interpretation. I will gloss the compound word as *even*. With the context in (18), snail is the most unlikely thing that Adam would eat. So, in addition to eating rice, see (18-b), Adam even ate a snail, (18-c). There are two more things to note about the example in (18-c) (and scalar-additive constructions in general): (a) similar to the adverbial exclusive particle *weeni* (see Ch.4), scalar-additive constructions also have a complex structure with a copy of the subject following the particle; and (b) the adfocal additive particle *de* may optionally follow the focus associate.

- (18) [We went for a party with Adam, and we know that he does not like eating snail. But Adam ended up eating snail. Now Adam's wife wants to know what Adam ate.]

- a. Bε mo Adam di-a.
what FOC Adam eat-DISJ
'What did Adam eat?'
- b. Adam di [mumuna]_F mo.
Adam eat rice FOC
'Adam ate RICE.'
- c. Adam **ta-maje** o di [kunkwoara]_F (de).
Adam even 3SG eat snail also
'Adam even ate SNAIL.'

(19) gives another example. The focus here is on the adjective *gonja*, which is the name of a place in Northern Ghana. The sentence in (19-b) means that a Gonja woman is the most unlikely person to get married to.⁷

- (19) [Due to an unpleasant experience, Napog swore not to ever marry a Gonja woman. Surprisingly, he married four wives, and a Gonja woman is one of them.]

- a. Napog di kaana bana.
Napog marry woman four
'Napog married four wives.'
- b. O **ta-maje** o di [Gonja]_F bukɔ (de).
3SG even 3SG marry Gonja woman also
'He even married a GONJA woman.'

Another example is given in (20), where (20-b) means that although Napog saw many animals, a pig is more unlikely to be seen.

⁶ According to the language consultants, *maje* is used as a temporal adverb 'already' in the language, see (i). But when it forms a compound with *ta*, resulting in *ta-maje*, then the scalar-additive meaning is derived.

- (i) a. Someone says: Get married to a Gonja woman.
- b. O maje o di Gonja bukɔ.
3.SG even 3.SG marry Gonja woman
'I am already married to a Gonja woman.'

⁷ This is a cultural thing. It may be that there are some unpleasant things about Gonja women.

- (20) [Napog is a Muslim, and lives in a Muslim community where pigs are prohibited. So, on his way back home yesterday, he saw different animals, including a pig.]
- a. Napog nɛ vara dedɛ diim.
 Napog see.PFV animal many yesterday
 ‘Napog saw many animals yesterday.’
- b. O **ta-maje** o na [teiri]_F (de).
 3SG even 3SG see pig also
 ‘He even saw A PIG.’

Association with subject focus shows a backward association context, which I will return to in Section 7.4. The subject focus precedes *ta-maje* in (21-b), but the sentence can be interpreted based on focus on the subject. This is supported by the context in (21-a). Assuming that Napog is a stingy fellow, then he is more unlikely to give the king anything, let alone a goat. An interesting property that is peculiar to scalar-additive association with subject focus is the obligatory use of the adfocal additive particle *de*, see (21-b) again. If *de* is absent in this context, then the scalar-additive association cannot be with the subject. Rather, it would be with any of the constituents below *ta-maje*.

- (21) [Napog is a very stingy person. He hardly gives anyone anything.]
- a. Kateiri dem nɔɔna pɛ pɛ wom bone dedɛ diim.
 community DEF people give.PFV chief DEF goat.PL many yesterday
 ‘The villagers gave the king many goats yesterday.’
- b. [Napog]_F *(de) **ta-maje** o pa pɛ wom boɔo diim.
 Napog too even 3.SG give chief DEF goat yesterday
 ‘Even NAPOG too gave the king a goat yesterday.’

Lastly, unlike Yorùba and Kusaal, the adverbial scalar-additive *ta-maje* in Kasem cannot occur in the clause-initial position, see (22) and (23).

- (22) ***Ta-maje** [Napog]_F (de) pa pɛ wom boɔo diim.
 even Napog also give chief DEF goat yesterday
 ‘Even NAPOG too gave the king a goat yesterday.’
- (23) ***Ta-maje** Adam na [teiri]_F (de).
 even 3SG see pig also
 ‘He even saw A PIG (too/in addition).’

To summarize, the discussion in this section has shown that while Yorùbá has both adverbial and adfocal scalar-additive particles, Kasem and Kusaal only have adverbial scalar-additive particles. In Yorùbá, both particles can be doubled or may occur individually. The interpretation remains the same. In contrast, since Kasem and Kusaal do not have adfocal scalar-additive particles, then particle doubling does not apply to them. This is summarized in Table 7.1 below.

	Adverbial- <i>also</i>	Adfocal- <i>also</i>	Doubling
Kasem	<i>ta-maje</i>	-	N/A
Kusaal	<i>la'am/hali</i>	-	N/A
Yorùbá	<i>kódà</i>	<i>pàápàá</i>	✓

Table 7.1: Scalar-additive focus sensitive particles in Kasem, Kusaal, & Yorùbá

7.3 Proposal for the syntax of scalar-additives

The proposal for scalar-additive association is similar to what we have seen already for both exclusive and additive association. I assume that there is a scalar-additive operator (Op_{even}) which projects a Scalarity Phrase (ScalP), and may be overtly realized with the adverbial scalar-additive particles in the three languages. I will argue that ScalP can be projected high in the clause, i.e., as high as the clause-initial position. Evidence for this assumption comes from the two languages (Yorùbá and Kasem), which have adverbial scalar-additive particles occupying a clause-initial position. Taking into consideration all three languages, they show overt morphology for both a high ScalP position and a low preverbal ScalP position. Although, adfocal scalar-additive particles are not present in all three languages, except Yorùbá, I assume that they all have an adfocal projection adjoined to the the focus constituent. While, it can be overtly realized in Yorùbá, it is covertly realized in Kasem and Kusaal respectively. One piece of evidence comes from the other dialect of Kusaal, the Toende dialect, where there is a morpheme *gba*, which can be considered as an adfocal scalar-additive particle, see (24).⁸

(24) **Adfocal scalar-additive in Toende dialect of Kusaal**

Hali Adam ɔnb ne [kukur ni'im]_F **gba** su'os.
 even Adam chew FOC pig meat even yesterday
 'Adam even ate A PORK yesterday.'

Furthermore, I propose an agreement relationship between the scalar-additive Op_{even} and the adfocal particle, which must be local to the focus associate. For this agreement process, I argue that there are two features involved: a scalar feature [Scal] and an additive feature [Add]. Both features are on the constituents involved in agreement. The scalar-additive Op_{even} is specified with a feature bundle, which includes both an unvalued interpretable scalar feature and an unvalued interpretable additive feature: [iScal:__, iAdd:__]. The adfocal constituent, on the other hand, is specified with the uninterpretable but valued counterparts of these features: [uScal:2, uAdd:2].⁹ The assumption is that the agreement between Op_{even} and the adfocal particle involves a single probe with multiple feature specifications. Just like we saw in the previous chapters on exclusives and additives, I want to mention here as well that the features of the adfocal particle are shared with the focus associate, since both constituents share the same extended projection, which in turn bears all the features of the adjoined constituents (i.e., focus feature and scalar-additive features).

Due to the peculiarities that exist with respect to the structural positions of the adverbial scalar-additive particles in the three languages, I will discuss the structures for each language separately. However, this does not affect the general proposal, which is the fact that the scalar-additive meaning is on a scalar-additive Op_{even} , which may be overtly realized as the adverbial scalar-additive particles. This means that the propositional approach also accounts for scalar-additive association with focus in these languages. Empirically, Op_{even} will always be overtly realized by the adverbial scalar-additive particles in Kasem and Kusaal because these languages do not have an adfocal scalar-additive particles. However, in Yorùbá, Op_{even} can either be covertly realized or overtly realized

⁸ Recall that this study is based on the Agole dialect of Kusaal. The provided data was discovered during one of the elicitation sessions.

⁹ Recall that I use this number as a simple index for valuation. It does not carry any semantic import.

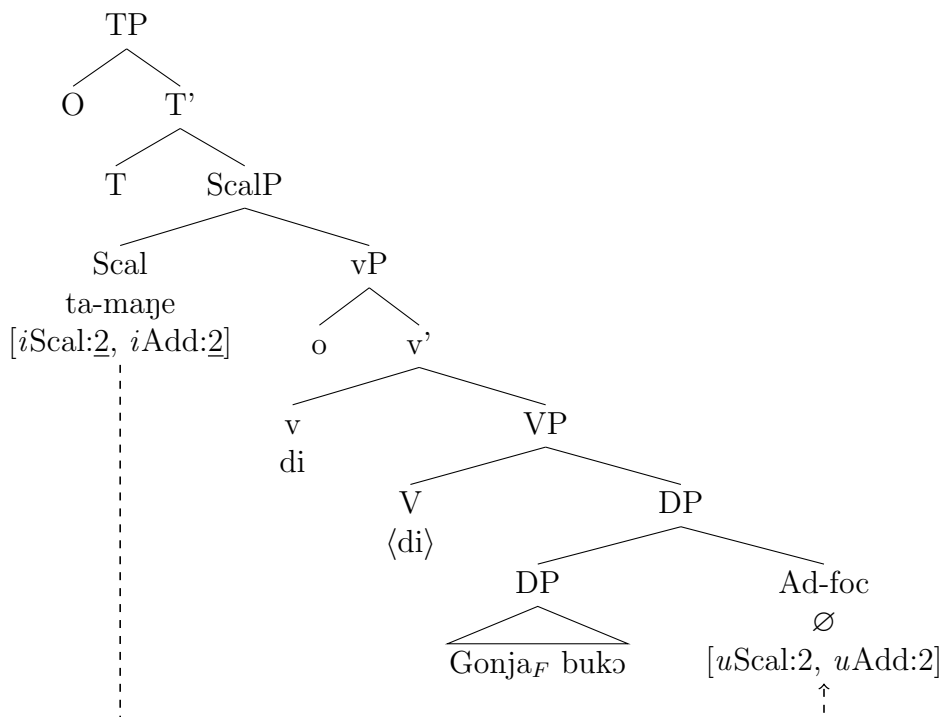
in the narrow syntax.

Kasem

I will begin with Kasem. I use the example in (19-b), which is repeated in (25) to propose the structure in (26) for scalar-additive association in Kasem. The scalar-additive Op_{even} , which is always overtly realized as *ta-maje* in Kasem, probes for feature valuation of its feature bundle. It finds the goal, which in this case is covertly realized. The goal provides feature valuation for the collective probe.

- (25) O **ta-maje** o di [Gonja]_F bukɔ (de).
 3SG even 3SG marry Gonja woman also
 ‘He even married a GONJA woman.’

(26) Scalar-additive AwF structure for Kasem



There are a few notes to make here. The first is with respect to the covert realization of the adfocal scalar-additive particle in Kasem. There are two pieces of evidence that justify its existence in this syntactic structure. The first is a cross-linguistic one. Although not marked in Kasem, we have already seen above that Yorùbá has a morphological realization of the adfocal scalar-additive particle. Similarly, we also saw that the Toende dialect of Kusaal, a closer language to Kasem, also has an adfocal scalar-additive particle, see (24). Another indirect piece of evidence comes from the fact that the other two focus sensitive associations in Kasem (*only* and *also*) involve not only adverbial particles, but also adfocal particles. In fact, the example in (25) shows that the adfocal additive particle *de* can optionally occur in scalar-additive context. This is in principle not surprising, since one of the presuppositions of scalar-additive constructions is the additive inference. If it occurs, it simply expone the additive feature on Ad-foc. Importantly, having a covert element valuing a feature is not new. This is similar to what has been argued for in pro-drop languages, such as Spanish and Italian, where phi-feature agreement may involve a null pro (cf. Rizzi, 1982; Müller, 2005a; Barbosa, 2011; Roberts, 2019, a.o.).

A second point to note is in relation to the composition of the adverbial scalar-additive

particle itself. It involves the adverbial additive particle *ta* and another morpheme *maje*, which is independently translated to mean *already* in the language. The composition morphologically supports the additive feature that we assume here. In isolation *ta* does not give any scalar-additive reading. But when it is combined with *maje*, the scalar-additive reading is derived.

Lastly, recall that the scalar-additive construction is one of the constructions discussed in Section 4.1.4 of Chapter 4, which licenses the use of a subject copy lower in the clause. Pending further investigation, I simply assume that the subject copy in (25) realizes the trace of the moved subject in Spec,vP of the structure in (26).¹⁰

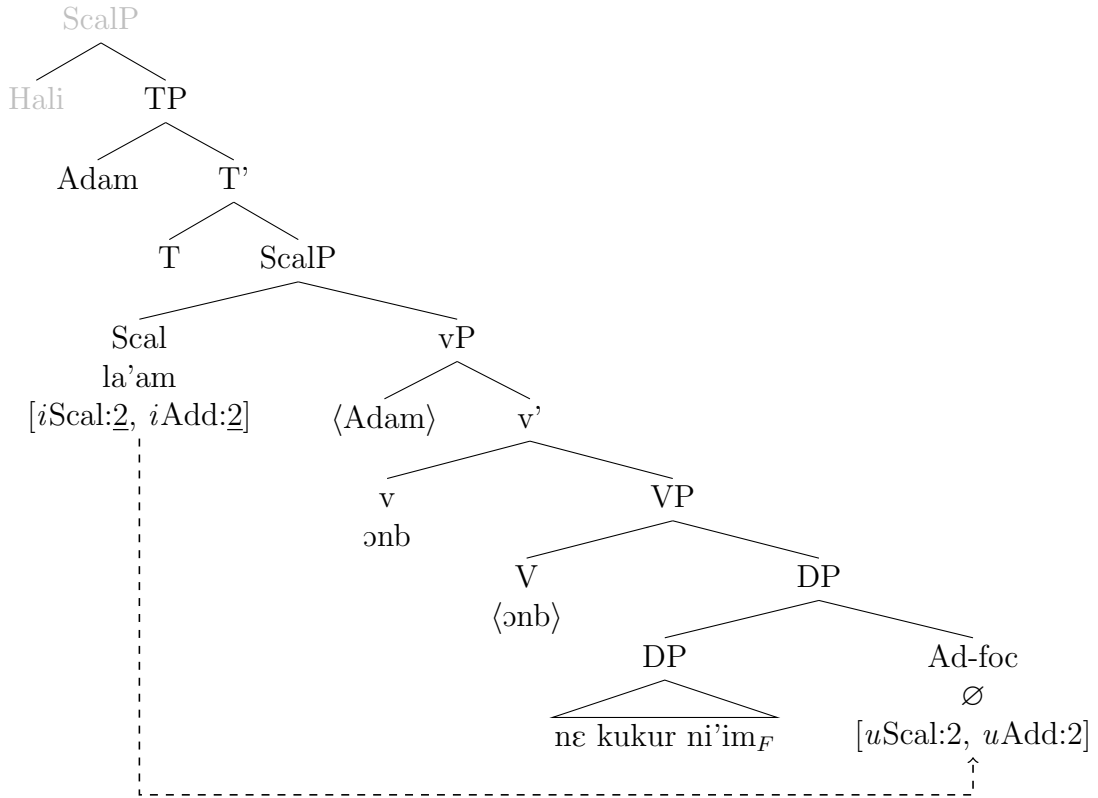
Kusaal

Recall that in Kusaal there are two adverbial scalar-additive particles, which occupy different structural position in the syntax, and cannot co-occur. While *la'am* occurs below the pre-verbally, *hali* occurs clause-initially, see (27). I assume that their distribution reflects the potential positions that the scalar-additive Op_{even} may occupy. These positions are consistently propositional in nature. I propose the structure in (28) for scalar-additive association in Kusaal, where *hali* would be the head of a projected ScalP above TP, whenever it is used. In order to avoid repetition, I will only discuss the structure with *la'am*. Notice that the two ScalP cannot be present at the same time. This is why the higher ScalP is in gray. The agreement mechanism follows the same processes as discussed for Kasem above. The adverbial scalar-additive particle Agrees with the covert adfocal scalar-additive, which adjoins to the focused constituent, and project an extended DP. The covert adfocal particle creates a local link with the focus associate, thereby allowing a direct access to the focus associate by the scalar head. This is realized through syntactic agreement with the covert adfocal particle.

- (27) [There were different meats at the party. We know that Adam loves meat, but he never eats pork. But at the party, he did not only eat chicken, he even ate pork.]
- a. Adam **la'am** ɔnb nɛ [kukur ni'im]_F.
 Adam even chew FOC pig meat
 'Adam even ate PORK.'
- b. **Hali** Adam ɔnb nɛ [kukur ni'im]_F.
 even Adam chew FOC pig meat
 'Adam even ate PORK.'

¹⁰ Of course, this structure might be more complex than this. It might even be a bi-clausal structure as I have discussed in Chapter 4. However, the overall assumption and argument here does not depend on this. So, I leave this for a future research.

(28) **Scalar-additive AwF structure for Kusaal**



For both Kusaal and Kasem, in association with subject focus, the agreement takes place before the subject moves to its surface position before ScalP. This is possible because the subject originates from Spec,vP before movement. As I have discussed above, merging ScalP above vP already provides us with a proposition, which is the main requirement for the approach. So, the Scal head can already establish agreement with any of the constituents in its c-command domain at that point, subject focus inclusive.

Yorùbá

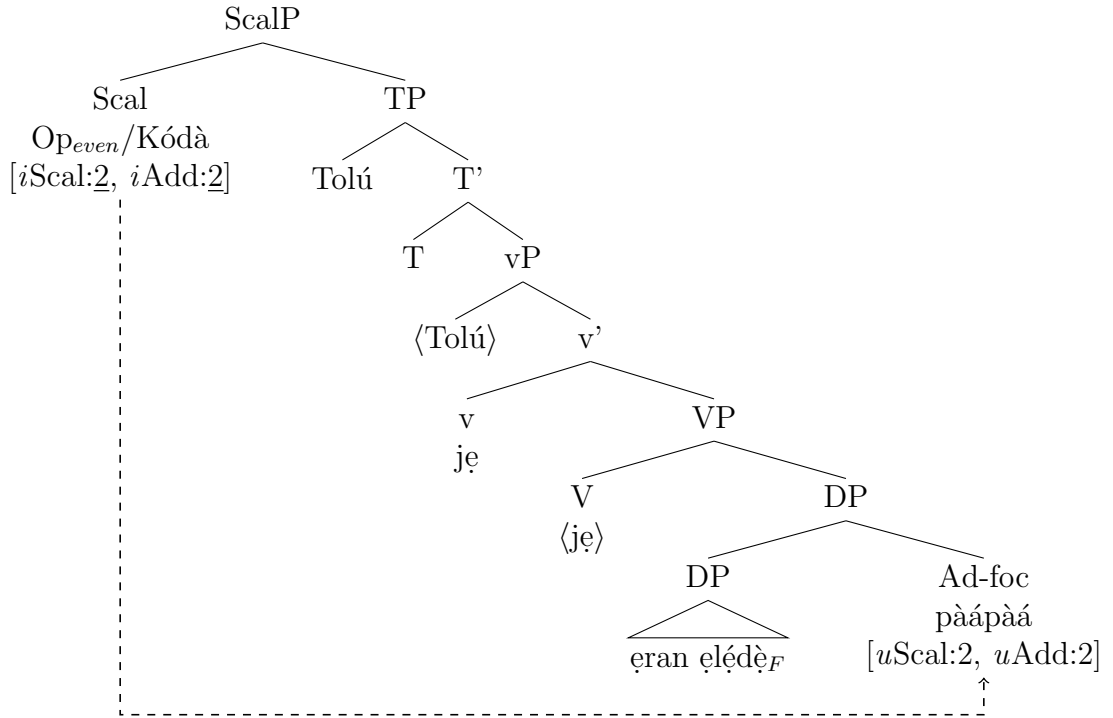
For Yorùbá, consider the structure in (30), which represents the example in (5-c), repeated in (29). I assume that ScalP is projected in the left periphery. This is because the adverbial scalar-additive particle *kódà* occupies such a high position. It is from this position that the Scal head creates an agreement relation with the adfocal scalar-additive particle *pàápàá*, which adjoins with the focus associate.¹¹

Recall that it is only Yorùbá that has the adfocal scalar-additive particle overtly realized. The agreement follows as assumed for the other languages above.

- (29) **Kódà** Tolú jẹ [ẹran ẹlédè]_F **pàápàá**.
 even Tolú eat meat pig even
 ‘Tolú even ate PORK.’

¹¹ Recall that, out of the three languages, only Yorùbá has an overt adfocal scalar-additive particle.

(30) **Scalar-additive AwF structure for Yorùbá**



Since the scalar-additive meaning is argued to be on the adverbial operator/particle, the agreement mechanism employed here provides a syntactic means of connecting this head with the focus associate through the adfocal particle. So, the value on the adfocal particle is copied onto the adverbial operator after agreement has taken place.

7.4 Scalar-additive Backward Association with Focus

That scalar-additive particles can associate with a fronted focus constituent to their left is not peculiar to the languages in the current study. As we have seen above, English also allows for a backward association with focus in scalar-additive contexts (see Taglicht, 1984; Erlewine, 2014a,b, 2018a). In the discussions that follow, I show that while scalar-additive BAwF is available in Kasem and Kusaal, it is not available in Yorùbá. Recall from above that Kusaal has two adverbial scalar-additives, *la'am* and *hali*, while the former allows BAwF, the latter does not (similar to Yorùbá's *kódà*).

7.4.1 Scalar-additive BAwF in Kasem

Recall that the scalar-additive in Kasem is *ta-maje*. *Ta-maje* can associate backward with the fronted constituent in (31-a). *Ta-maje* cannot occur immediately after the fronted constituent, see (31-b). This shows that *ta-maje* indeed associates with a fronted focus to its left.

(31) **Scalar-additive BAwF in Kasem**

[We all know how Adam dislikes Gonja women, but to our surprise, he married one.]

- a. Gonja bukɔ de *(mo) o **ta-maje** *(o) di ____.
 Gonja woman also FOC 3SG even 3SG marry
 'He even married a GONJA woman.'

- b. *[Gonja]_F bukɔ ta-maje *(mo) o (o) di ____.
 Gonja woman even FOC 3SG 3SG marry
 ‘He even married a GONJA woman.’

Ta-maje can also associate with a fronted subject focus, as in (32-b). This means that Napog is unlikely to get a car, due to his financial status. The use of the adfocal additive particle is obligatory here in order to avoid ambiguity with the constituents below *ta-maje*. In other words, in a subject focus association, without the use of the adfocal additive particle, the sentence could mean that “a car is the most unlikely thing that Adam is expected to buy”.

(32) **Scalar-additive BA_wF in Kasem with subject focus**

[Napog is a worker, whose salary is not enough for many basic things. So while others buy a car, Napog is not expected to.]

- a. Nɔɔna dedɛ yeigi lɔɔre deem.
 people many buy.PFV car last.year
 ‘Many people bought a car last year.’
 b. Napog *(de) mo **ta-maje** o yeigi lɔɔre.
 Napog also FOC even 3SG buy.PFV car
 ‘Even NAPOG bought a car.’

A further example is given in (33), where *ta-maje* associates with the fronted object focus. This means that although Napog saw many animals yesterday, a pig was more unlikely to be seen based on the given context.

(33) **Scalar-additive BA_wF in Kasem with direct object focus**

[Napog is a Muslim, and lives in a Muslim community where pigs are prohibited. So, on his way back home yesterday, he saw different animals, including a pig.]

- a. Napog nɛ vara dedɛ diim.
 Napog see.PFV animal many yesterday
 ‘Napog saw many animals yesterday.’
 b. [Teiri]_F *(mo) o **ta-maje** o na ____.
 pig FOC 3SG even 3SG see
 ‘He even saw A PIG.’

7.4.2 Scalar-additive BA_wF in Kusaal

Kusaal also exhibit the same scalar-additive backward association with focus. This is only realized with the adverbial scalar-additive particle *la'am*. The other adverbial scalar additive particle *hali*, that we discussed above, do not associate backward, since they are always in the clause-initial position. In (34), the focus is in-situ in (34-b). However, the same focus can be fronted to the left periphery of the clause (34-c). Despite the fronting of the focus, the scalar-additive reading is still available. Both sentences in (34-b) and (34-c) have the interpretation that Gonja is the most unlikely place to marry from.

(34) **Scalar-additive BA_wF in Kusaal**

- a. Napog disne pu'ab anaasi.
 Napog eat.PFV woman four
 ‘Napog married four wives.’

- b. O **la'am** disne Gonja pu'a.
 3SG even eat.PFV Gonja woman
 'He even married a GONJA woman.'
- c. [Gonja]_F pu'a **ka** o **la'am** di ____.
 Gonja woman FOC 3.SG even eat.PFV
 'He even married a GONJA woman.'

The same ex-situ example can be used with *hali*, but now *hali* has to occur higher than the fronted focus, as in (35). Also, there cannot be doubling with *la'am*. This example shows that *hali* occupies a very high position, at least, above the left peripheral FocP.

(35) **No Scalar-additive BA_wF with *hali* in Kusaal**

Hali [Gonja]_F pu'a **ka** o (***la'am**) di ____.
 even Gonja woman FOC 3SG even eat.PFV
 'He even married a GONJA woman.'

The example in (36) involves a fronted adverbial constituent, which the *la'am* associates with backwardly.

(36) **Scalar-additive BA_wF in Kusaal = adjunct**

[Yesterday was a public holiday. So, everyone was with their family. But Adam, as in other days, worked yesterday.]

[Su'os]_F Adam sa **la'am** tumya.
 yesterday A. PST even work
 'Adam even worked YESTERDAY.'

I repeat the association with fronted subject focus in Kusaal here, since we can classify it as backward association with focus as well.

(37) **Scalar-additive BA_wF in Kusaal = subject**

[The economy of the city was so good that even Adam bought a car.]

[Adam]_F *(me) **la'am** da' lor.
 Adam also even buy car
 'Even ADAM bought a car.'

7.4.3 No scalar-additive BA_wF in Yorùbá

While there are instances of scalar-additive BA_wF in Kasem and Kusaal, there are no such cases in Yorùbá. Consider the root clause examples in (38). Because *kódà* occurs very high in the clause (presumably higher than the CP, see ex. (39)), every fronted constituent in a root clause still ends up in its scope, as in (38-b).

(38) **No scalar-additive BA_wF in simple clauses in Yorùbá**

[The whether was so hot that Tolù had his bath six times today.]

- a. **Kódà** Tolù wè ní [èmẹ̀ẹ̀fà]_F.
 even Tolù bathe LOC six.time
 'Tolù even bathe SIX TIMES.'
- b. **Kódà** [èmẹ̀ẹ̀fà]_F ni Tolù wè ____.
 even six.time FOC Tolù bathe
 'Tolù even bathe SIX TIMES.'

Similarly, if *kódà* appears at the lower clause edge of an embedded clause, the sentence is ungrammatical (39-a). In long distance extraction, *kódà* must precede the fronted constituent, as in (39-b). This suggests the unavailability of scalar-additive BAwF in Yorùbá.

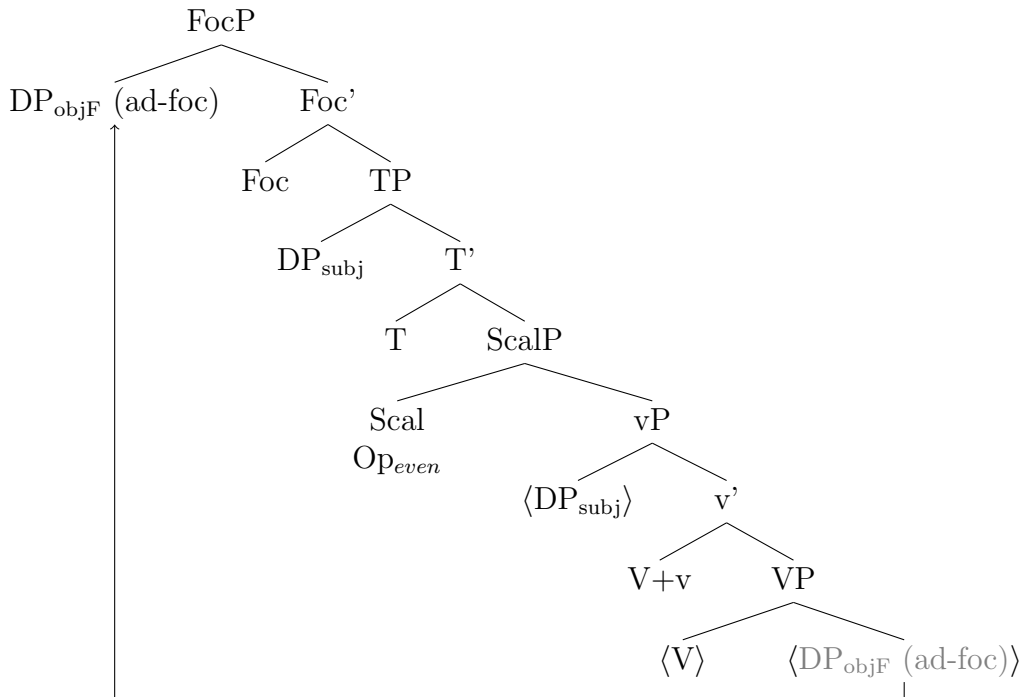
(39) **No scalar-additive BAwF in embedded clauses in Yorùbá**

- a. *[Èmẹ̀ẹ̀fà]_F ni mo mò wípé **kódà** Adé wẹ ____.
 six.time FOC 1SG know COMP even Adé bathe
 ‘It was six times that I know that Adé even bathe.’
- b. **Kódà** [èmẹ̀ẹ̀fà]_F ni mo mò wípé Adé wẹ ____.
 even six.time FOC 1SG know COMP Adé bathe
 ‘It was six times that I know that Adé even bathe.’

7.4.4 Proposal for Scalar-additive BAwF

In Chapter 6, I argued that additive BAwF does not require reconstruction of the fronted constituent, unlike exclusive BAwF, which does. Here, I argue that the scalar-additive BAwF does not require reconstruction as well. This claim has already been made by Erlewine (2014b), who shows that in English, *even*, and not *only*, allows for what might be called backward association (cf. Jackendoff, 1972). I already showed this above. However, Erlewine argues that the backward association with *even* in English is illusory. This is because *even* associates with a lower copy of the fronted constituent instead. I make a similar claim for the languages under investigation here. Although the focus associate is fronted to the left periphery, scalar-additive association is derived when the scalar-additive particle c-commands the lower copy of the fronted constituent. A simplified structure is provided in (40). I assume that ex-situ focus movement is triggered independently of the additive association. Only Kusaal and Kasem would require such a structure since there is no scalar-additive BAwF in Yorùbá. As I have shown above, this is not because the language does not allow scalar-additive BAwF. Rather, it is because scalar-additive BAwF is empirically not applicable in the language due to the structural position of the adverbial scalar-additive particle *kódà*, which precedes fronted focus constituents. Assuming a lower copy for the observed BAwF is enough for the semantic computation of the scalar-additives in these languages. The reason is that, similar to *also*, *even* semantics does not modify the truth-condition of its assertion. Therefore, it does not require a strict structural compliance in its scope.

(40) **Scalar-additive BA_wF Structure**



The structure above can also account for backward association with subject focus. This is derived based on the fact that the subject originates in the scope of the scalar-additive particle (specifically, Spec,vP) before movement to Spec,TP. This is anchored on the VP-Internal Subject Hypothesis by Koopman and Sportiche (1991). The lower copy of the subject is available in Spec,vP, and this is in the scope of the scalar-additive particle. At this point, the scalar-additive semantics can be derived since vP is the minimal propositional node.¹²

7.5 Conventional focus association

In Chapter 2, I introduced the QFC model of association with focus by Beaver and Clark (2008). The three focus sensitive particles (*only*, *even*, and *also*) that we have considered in this study, so far, have been classified as particles that exhibit conventional association. This means that they have focus sensitivity built into their lexical semantics. Thus, they are able to reference focus and its evoked alternatives directly. For Beaver and Clark (2008), a major requirement for association with focus is c-command. The conventional focus sensitive particles must c-command their focus associate. They adopt Tancredi's (1990) Principle of Lexical Association (PLA) which has been referenced a number of times in this study, see (41). This simply means that syntax plays a role in focus sensitivity

(41) **Principle of Lexical Association**

An operator like *only* must be associated with a lexical constituent in its c-command domain. (Tancredi, 1990, p.30).

In contrast, Beaver and Clark (2008) argue that operators which undergo free association, such as *always*, do not directly reference focus in their semantics. Rather, they associate with focus based on more general pragmatic principles, and not based on a direct

¹² See Erlewine (2014a,b) for a more comprehensive semantic analysis of scalar-additive BA_wF.

semantic sensitivity to focus alternatives. Therefore, for such operators, the prediction is that the focus does not need to be conventionally associated, i.e., in the c-command position of the free association operator. According to Beaver and Clark (2008), this prediction is borne out. While operators such as *only* associate conventionally with their focus, operators like *always* associate freely, i.e., with no syntactic (c-command) requirement.

Consider the examples in (42-a) and (42-b), which compares association with *always* and association with *only*. The two sentences can have different interpretations, as in (42-a)-i & ii, and (42-b)-i & ii.

- (42) a. Sam *always* buys mango from Adam.
 (i) Whenever Sam buys something from Adam, it is mango.
 (ii) Sam buys mango from no one else but Adam.
 b. Sam *only* buys mango from Adam.
 (i) Sam buys nothing but mango from Adam.
 (ii) Sam buys mango from no one else but Adam.

However, everything changes if, for example, the direct object is fronted, see (43). Here, *always* can still have the interpretation in (43-a), which appears to show that it associates with the extracted element. However, *only* does not have such an interpretation, see (43-b). In other words, *only* fails to associate leftward with an extracted element.

- (43) a. Mango, Sam *always* buys from Adam.
 Means: Whenever Sam buys something from Adam, it is mango.
 b. Mango, Sam *only* buys from Adam.
 Cannot mean: Sam buys nothing but mango from Adam.

Based on this association and the interpretation asymmetry, Beaver and Clark (2008) argued that while *always* is a free focus sensitive operator (thus allowing leftward association), *only* is a conventional focus sensitive operator. Although Beaver and Clark's (2008) discussion focused more on *only*, they assumed that the same holds for *even* and *also*, since they classify all three operators as conventional operators.

Interestingly, further cross-linguistic studies have cast doubt on the universality of the proposal that *only*, *also*, and *even* are conventional operators. This was briefly discussed in Section 2.2.3 of Chapter 2. In this section, I will argue that these three focus sensitive operators, as discussed extensively all through this dissertation, are indeed conventional focus sensitive operators in Yorùbá, Kusaal and Kasem. I will argue for this position by discussing two of the arguments that Beaver and Clark (2008) used in their study: (a) extraction test, and (b) ellipsis test.

7.5.1 Extraction

I have already introduced the argument from extraction by Beaver and Clark (2008) above. Going by Beaver and Clark, the prediction is that while free association operators (e.g., *always*) should allow extraction of the focus associate, conventional focus sensitive operators should not.

However, as we have seen so far, all three focus sensitive particles (*only*, *also*, and *even*) that we have discussed in this study allows for the extraction of their focus associates,

see (44), (45), and (46) below.¹³ These are particles that Beaver and Clark classified as involving conventional association. Based on the discussion given above, we might want to classify all the three focus sensitive particles as involving a free association. However, we have reasons to assume otherwise. In addition to the explanation that would be provided in this section, further arguments for a conventional association of these particles in the languages under investigation would be discussed in the next section.

- (44) **Extraction in the context of *only***
- a. [Chworo]_F (yerane) *(**mo**) Adam **weeni** o go-a _____. **Kasem**
 fowl only FOC Adam only 3SG kill-DISJ
 ‘Adam slaughtered only A FOWL.’
- b. [Nua]_F (ma’aa) *(**ka**) Adam **kudim** kɔdig _____. **Kusaal**
 fowl only FOC Adam only kill
 ‘Adam slaughtered only A FOWL.’
- c. [Ìrẹ̀sì]_F (nìkan) *(**ni**) Tolú **kàn** jẹ _____. **Yorùbá**
 rice only FOC Tolú only eat
 ‘Tolú ate only RICE.’
- (45) **Extraction in the context of *also***
- a. [Pea]_F (de) mo Adam **ta** di-a _____. **Kasem**
 yam also FOC Adam also eat-DISJ
 ‘It was yam also that Adam ate.’
- b. [Busa]_F (mɛn) ka Adam **mɛ** di _____. **Kusaal**
 yam also FOC Adam also eat
 ‘It was yam also that Adam ate.’
- c. [Iṣu]_F (pẹ̀lú) ni Tolú **tún** jẹ _____. **Yorùbá**
 yam also FOC Tolú also eat
 ‘It was yam also that Tolú ate.’
- (46) **Extraction in the context of *even***
- a. [Teiri]_F mo o **ta-maje** o na _____. **Kasem**
 pig FOC 3.SG even 3SG see
 ‘He even saw A PIG.’
- b. [Kukur]_F ka o sa **la’am** nyɛ _____. **Kusaal**
 pig FOC 3SG PST even see
 ‘He even saw A PIG.’

The data presented above do not invalidate the ‘conventional association’ claim of these particles because all the fronted constituents started out in the scope of the focus sensitive particles. This predicts that cases where the focus is base-generated outside of the scope of the focus operators (adverbs) (subject focus in Yorùbá for instance), there

¹³ For the scalar-additive association with focus, recall that Yorùbá’s adverbial scalar-additive particle occurs very high in the clause, whereby it precedes fronted focus constituents. This shows that there is no backward association with scalar-additive in Yorùbá. Similarly, recall that in Kusaal, apart from *la’am*, which is used in (46-b), there is another scalar-additive particle *hali*, which occur in a similarly high position, as in the Yorùbá case, see (i).

- (i) **Hali** [Kukur]_F ka o sa nyɛ _____. **Kusaal**
 even pig FOC 3SG PST see
 ‘He even saw A PIG.’

should not be backward association. This prediction is borne out, as shown in the Yorùbá examples in (47).

(47) **No BA_wF for base-generates local subject focus in Yorùbá**

- a. [Tolú]_F *(ni) ó kàn jẹ ìrẹ̀sì.
Tolú FOC 3SG only eat rice
'TOLÚ only ate rice.'
- b. [Tolú]_F *(ni) ó tún jẹ ìrẹ̀sì.
Tolú FOC 3SG also eat rice
'TOLÚ also ate rice.'

One of the major difference between conventional association and free association is that the focus in free association context does not need to start out within the scope of a free association particle, such as *always*, whereas the focus in conventional association context must start in the scope of conventional focus sensitive particles, such as *only*, *also*, and *even* (cf. Beaver and Clark, 2008). For example, while *only* requires that an extracted focus material should reconstruct to its base position for the purpose of semantic composition, the same is not required of *always*. As we have seen, the lexical meaning of the conventional focus sensitive particles compose directly with the focus associate, and they need the set of alternatives of the focus in order to be interpreted. This is not the case with free association, which have been analyzed with purely pragmatic accounts.

We have seen that all three focus sensitive particles discussed in this study can be interpreted based on which constituent is in focus. Their interpretations change with the change of focus. This shows their direct dependence on focus. Therefore, the extraction cases above do not constitute any challenge to the fact that all the focus sensitive particles discussed in this study (cf. (44) - (46)) are conventional particles.

7.5.2 Ellipsis

The next argument for a conventional association status for all three focus sensitive particles in these languages comes from VP ellipsis. According to Beaver and Clark, ellipsis must not separate a conventional focus sensitive operator from its focus associate. This idea has been used extensively in this dissertation, especially in arguing for a propositional approach for *only*, *also*, and *even* in the languages of this study. Consider the examples in (48) from English.¹⁴ Beaver and Clark claimed that the sentence in (48-b) with *only* does not have the reading below it because the focus associate (i.e., the verb) has been elided. The reverse is the case with *always* in (48-a), where even with ellipsis, the sentence still means that Kim salutes at every ceremony because Sandy salutes at every ceremony.

- (48) a. Kim *always* SALUTES because Sandy *always* does ~~salute~~.
[can mean: 'Kim salutes at every ceremony because Sandy salutes at every ceremony']
- b. *Kim *only* SALUTES because Sandy *only* does ~~salute~~.
[cannot mean: 'Kim salutes (and does nothing else) because Sandy salutes (and does nothing else).']

(adapted from Beaver and Clark, 2008, p.177)

¹⁴ This example was first given and discussed in Section 2.2.2 of Chapter 2. It has subsequently been used in other places. But I chose to repeat it here for easy access and reference.

Turning to the languages in this study. I have already showed how ellipsis data are instrumental to our discussion in this study. In this section, I will provide ellipsis data, which involve the adverbial focus sensitive particles, with the optional occurrence of the adfocal particles, where one is available.

The first set of data in (49)-(51) supports the view that the exclusive particles in these languages are conventional focus sensitive particles. In each cases (the (b) examples in particular), the elision of the verb phrases (or TP, as in the Yorùbá case) precludes the expected interpretation. Take Kasem for instance, the example with VP ellipsis in (49) cannot mean that it is only bread that John is allowed to eat (with no other permission). In all three languages, the separation of the adverbial exclusive particles from their focus associate, through ellipsis, prevents the desired meaning of the sentences involved.

(49) **Ellipsis and exclusive AwF in Kasem**

- a. Adam **weeni** o **wae** o yeigi [paano]_F (yerane) **mo**.
Adam only 3SG can 3SG buy bread only_{adf} FOC
‘Adam only may buy BREAD.’
- b. *John de **weeni** o wae [~~VP o yeigi paano (yerane) mo~~].
John too only 3SG can
Intended: It is only bread *x*, such that John is allowed to buy *x* (and no other permission).

(50) **Ellipsis and exclusive AwF in Kusaal**

- a. Adam **kudim tun’e** da’ad **ne** [bodobodo]_F (ma’a).
Adam only may buy.IPFV FOC bread only_{adf}
‘Adam only may buy BREAD.’
- b. *John me **kudim tun’e** [~~VP da’ad ne bodobodo ma’a~~].
John also only may
Intended: It is only bread *x*, such that John is allowed to buy *x* (and no other permission).

(51) **Ellipsis and exclusive AwF in Yorùbá**

- a. Tolú **kàn** sèlérí látí wá [ní òlǎ]_F (nìkan).
Tolú only promise to come LOC tomorrow only
‘Tolú only promised to come TOMORROW.’
- b. *Dàpò nàà **kàn** sèlérí pèlú [~~TP látí wá [ní òlǎ]_F (nìkan)~~].
Dàpò too only promise also
Intended: Only tomorrow is such that Dàpò promised to come, i.e., he made no other promise about another day.

Additive particles also show a similar result like the one we saw above with exclusives. The adverbial additive particles in Yorùbá, Kasem and Kusaal cannot associate with an elided constituent. The is presented in (52)-(54).

(52) **Ellipsis and additive AwF in Yorùbá**

- a. Tolú **tún** sèlérí látí jẹ [iṣu]_F (pèlú).
Tolú also promise to eat yam also
‘Tolú also promised to eat YAM.’
- b. *Dàpò nàà **tún** sèlérí pèlú [~~TP látí jẹ [iṣu]_F (pèlú)~~].
Dàpò too also promise also
‘Dàpò also promised.’

Intended: Dàpò also promised to eat yam (i.e., in addition to his promise to eat something else).

(53) **Ellipsis and additive AwF in Kasem**

- a. Adam **ta** sɛ se o ba [jwa]_F ne ye ...
 Adam also promise to 3SG come tomorrow LOC and
 ‘Adam also promised to come tomorrow and...’
- b. *Sam de **ta** sɛ [~~se o ba~~ [jwa]_F ne].
 Sam too also promise
 ‘Sam also promised.’
Intended: Sam also promised to come tomorrow (i.e., in addition to his promise to come another day).

(54) **Ellipsis and additive AwF in Kusaal**

- a. Adam mɛ tun’e da’ bodobodo.
 Adam also may buy bread
 ‘Adam also may buy BREAD.’
- b. *Sam mɛ tun’e [~~da’ bodobodo~~].
 Sam also may
Intended: It is bread *x*, such that Sam is allowed to buy *x* (and is allowed to buy something else).

Similarly, the Scalar-additive particles are also conventional particles in these languages, based on their behaviour in elliptical contexts. They cannot associate with an elided material, see (55) and (56).¹⁵

(55) **Ellipsis and scalar-additive AwF in Kasem**

[We know that Adam and Sam work very hard, but tomorrow is a public holiday, so we expected them not to come to the office. However, they promised to come tomorrow]

- a. Adam ta-mange o sɛ se o ba jwa ne (mo) ye ...
 Adam even 3SG promise to 3SG come tomorrow LOC FOC and
 ‘Adam even promised to come tomorrow and ...’
- b. *Sam de ta-mange o sɛ [~~se o ba jwa ne (mo)~~]
 Sam too even 3SG promise
Intended: San even promised to come tomorrow (i.e., in addition to his promise to come another day, the promise that he would come tomorrow is more unlikely).

(56) **Ellipsis and scalar-additive AwF in Yorùbá**

[We know that Tolú and Dàpò work very hard, but tomorrow is a public holiday, so we expected them not to come to the office. However, they promised to come tomorrow]

- a. Kódà Tolú sèlérí láti wá ní òlà.
 even Tolú promise to come LOC tomorrow
 ‘Tolu even promised to come TOMORROW.’
- b. *Kódà Dàpò nàà sèlérí pèlú [~~láti wá ní òlà~~].
 even Dàpò too promise also

¹⁵ The Kusaal data for this is not available. However, I strongly believe that it shows a similar pattern with the others.

Intended: Dàpò even promised to come tomorrow (i.e., in addition to his promise to come another day, the promise that he would come tomorrow is more unlikely).

What I have argued for in this section is that all the three focus sensitive particles in these languages involve conventional association with focus. Beaver and Clark (2008) gave further tests, such as their behaviour in second occurrence focus contexts, and their behaviour with reduced materials. All these forms areas for further investigation, which I highlight in the next chapter on conclusion.

7.6 Conclusion

Similar to the previous chapters, I have argued for a propositional account for scalar-additive association in this chapter. I showed that all three languages have adverbial scalar-additives, but only Yorùbá has adfocal scalar-additives. This morphological evidence is enough for us to assume a ScalP in the regions above vP. I show that each language has peculiar structural properties for these scalar-additives, which licenses the position of ScalP. I assume an agreement which involves a feature bundle consisting of a scalar feature and an additive feature, which correspond to the scalar-additive presuppositions of *even*. In addition, I provided evidence for scalar-additive backward association with focus, which is handled by assuming a lower copy of the moved element, instead of reconstruction. Lastly, I provided evidence for conventional association for all three focus sensitive particles in all three languages.

Chapter 8

Conclusion and further research

8.1 Summary and major findings

In this dissertation, I have investigated the syntax-semantics properties of *only*, *also*, and *even* from a cross-linguistic point of view. I investigated these particles in Kasem, Kusaal and Yorùbá. Each chapter of the dissertation was dedicated to one of the focus sensitive particles. All through the Chapters, one theme re-occurred: the syntactic and semantic properties of these particles can uniformly be accounted for using the Propositional Approach. This means that the propositional semantic requirement of these focus sensitive operators can be mapped onto the syntax as well. I argued that regardless of whether the particles associate locally or from a distance, the propositional approach provides a uniform account for all three particles. The thesis provides morphological, syntactic, and semantic evidence for this claim. I showed that assuming that the individual association meaning comes from an adfocal particle does not provide us with an adequate approach to account for the observed properties of these particles in these languages. I provided evidence mainly from concord phenomenon, which is associated with multiple focus association, split-scope possibilities, and ellipsis.

As a way of setting the stage, I described focus constructions in these languages, and I provided syntactic analysis for them. I argued that all three languages project FocP in the left periphery, which host fronted constituents after agreement relation has taken place. Each of the languages have their peculiarities. For instance, while Kasem and Kusaal have both in-situ and ex-situ morphologically focus markers, Yorùbá only has an ex-situ morphological focus marker. In-situ focus is not morphologically marked. The distribution of these morphemes also differ. I show that while Kasem's in-situ focus marker follows the focus constituent, Kusaal's in-situ focus marker precedes the constituent. However, in both cases, I argued that the in-situ focus markers form a constituent with the focus element. I provided a syntactic agreement analysis, where the in-situ focus markers expone the focus features on the focus constituent and Agree with a high Foc head in the left periphery. I also discussed the phenomenon of anti-pied-piping, which is present in all three languages.

Next, I discussed exclusive association with focus in Kasem, Kusaal and Yorùbá. Using empirical evidence, I argued that exclusives in these languages allow for exclusive doubling. This description was important for my analysis because it constitute an argument for the presence of a clausal exclusive operator, which is responsible for the exclusive semantics. I assumed an agreement mechanism, which connects the adfocal exclusive particle and the adverbial or clausal exclusive Op. The agreement with the adfocal particle

also has a consequence on the focus associate because both the latter and the former form a single constituent and have the same extended project. Furthermore, I have shown that contrary to the earlier assumption that the adfocal particles are semantically vacuous elements, they perform the function of constraining the scale of the alternative propositions. The exclusive association can result in a (a) complement exclusion interpretation, (b) rank-order interpretation, and (c) an evaluative interpretations, where the proposition is considered little/insufficient or bad. Of the three languages, Yorùbá provided a more robust distinct morphology for these different exclusive contributions. While the other two languages do not show robust morphology, they show consistency. In other words, the only adfocal exclusive particles in Kasem and Kusaal are only felicitous in complement exclusion contexts. They are infelicitous in rank-order and evaluation of proposition contexts. What is common to all three languages is that the adverbial exclusive particles can independently occur in all three contexts. This solidifies my claim for a propositional approach. The discussion here forms a strong evidence against the ‘vacuous claim’ (at least for exclusives), by showing that the adfocal exclusive particles are more than vacuous semantic elements. In addition, the three languages allow exclusive backward association, a phenomenon which is not allowed in English (cf. Jackendoff, 1972; Erlewine, 2014b). The argument provided for such cases is that the fronted constituent is required to reconstruct back to its based position at LF. Evidence for such reconstruction effect comes from the possibility of the fronted constituent taking a narrow scope in addition to the expected surface (wide) scope. This can only be explained if it reconstructs.

I extended the propositional approach to both additive and scalar-additive particles as well. I showed that the three languages have additive doubling cases. I argued in Chapters 6 and 7 that a high additive Op and a high scalar-additive Op carries the additive and scalar additive meaning respectively. I propose an agreement system, which connects these operators, that can be overtly realized by the adverbial focus sensitive particles, with the adfocal particles, who in turn must be local to the focus associates. (Scalar-)additive backward association with focus is also allowed across the three languages. I argued that, in order to account for this, reconstruction is not required, unlike the *only* cases. All that matters is that the fronted constituent originates from the scope of the (scalar-)additive operators. There is lower copy of the moved constituent, which allows for (scalar-)additive associate. Since the fronted foci in exclusive backward association context are able to take both wide scope and narrow scope, which create a scope ambiguity, the prediction is that the same should be available with the other focus sensitive particles as well. I showed that this prediction is borne out with respect to additives. The fronted constituent is able to take narrow scope with additives as well, and this is present in all three languages.

In summary, I provided a unified treatment of the conventional focus sensitive particles as propositional operators, and argued that instead of assuming a cross-categorical analysis for each particles, depending on the categories that they associate with, a uniform propositional analysis for both adfocal and adverbial conventional focus sensitive particles is better.

8.2 Open issues and areas for further research

The study here does not in any way assume an absolute coverage of association with focus phenomenon in the three languages investigated, which are representatives of the language groups they belong to (Mabia and Yoruboid). There are still quite a number of open issues, which I am going to discuss briefly in this section. In addition, I will also

highlight areas for future studies.

8.2.1 Some open issues

There are several points that were left open at different point of the dissertation. For instance, what is the exact structure for the exclusive and scalar-additive constructions in Kasem? We saw that these constructions involve a subject copy lower in the clause. Further investigation is needed to determine the exact structures.

There is another important issue bearing on my proposal, which I do not have a definite answer to. This has to do with the overt realization of the covert operators. Although I argued that the adverbial focus sensitive particles overtly realize these covert operators, we saw that these operators have more flexible distribution than the adverbial focus sensitive particles. This is an issue that would need a proper analysis on the constraint that might be at play here.

Another notable open issue has to do with the fact that while morphological focus marking seems to be obligatory in exclusive association with focus, the same is not true for additive and scalar-additive association with focus. Although previous analysis (e.g., Grubic and Zimmermann, 2011; Duah, 2015) have used this focus marking asymmetry to argue that only *only*, but not *also* and *even*, is a conventional focus sensitive particle, I, however, do not hold the same view. This may be related to focus as a contrastive notion or as new information. The assumption is that in languages where the (in-situ) morphological focus markers are absent with *also* and/or *even*, these markers seem to be contrastive/exhaustive markers, and so they contradict the meaning of *also* and *even*, but not *only*. This, therefore, leads to their omission in (scalar-)additive contexts. This is an interesting point as it has the potential to contribute to the ongoing debate as to whether grammatical focus marking are purely contrastive on the one hand, or involve both new information and contrast on the other (Zimmermann and Onea, 2011; Rochemont, 2013a,b; Kratzer and Selkirk, 2020).

8.2.2 Areas for further research

There are also some interesting areas that this dissertation did not investigate. One of such is with respect to the combination of the investigated particles. Although I investigated multiple focus association with the same focus sensitive particle (e.g., exclusives), it would be interesting to investigate multiple association with focus involving (a) different focus sensitive particles associating with a single focus associate, see (1), or (b) different focus sensitive particles associating with distinct focus associates, see (2).

- (1) Adam *even only* ate [rice]_F.
- (2) *Even* [Adam]_F *only* ate [rice]_F

Another potential area for further research, especially with respect to African languages, is a wider typological investigation of these focus sensitive particles. As I have shown in this study, African languages provide robust morphology for the focus sensitive particles. Unlike in English, and many other languages, they distinguish between the adfocal and the adverbial particles. This helps in avoiding controversies that surrounds their distribution or use. For instance, as I have shown above, there is a continuous debate on whether German *nur* should be analyzed as only an adverbial particle or as both

adverbial and adfocal. The languages discussed here show clear morphological distinction between the two. Another reason why such a typological investigation is necessary is because it can provide answers to long-standing issues in information structure, like the notion of Second Occurrence Focus (SOF). There is always the debate on whether SOF is focally marked or not. Prosodic studies have shown different variations. However, carrying out studies on morphosyntactic focus marking languages may throw some light to this discussion.

Another important area, which I did not investigate in this study has to do with the presuppositions that scalar-additives introduce. For want of time, in this study, I assume the traditional view that the scalar-additive particles in these languages have both additive presupposition and likelihood presupposition. Some studies, however, have shown that both the additive and the scalar presuppositions are not always available in a sentence with *even* (Rullmann, 1997; Herburger, 2000; Gast and der Auwera, 2011; Greenberg, 2016, 2018; Zhang, 2022; Liao and Jheng, 2025). For instance the sentence in (3), from (Szabolcsi, 2017, p.458), does not have an additive presupposition. This is because, given the context, only Eeyore spat the thistles out. No other person did. However, the scalar, (i.e., unlikely) presupposition is available.

- (3) [Context: Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known for favoring thistles) takes a bite but spits it out.]
Those thistles must be really prickly! Even Eeyore spat them out!

Similarly, it has been argued that the unlikely presupposition is not always necessary in sentences with *even* either. Consider the following example from Liao and Jheng (2025) citing Kay (1990). The two sentences do not involve an unlikely scale. The use of *even* here is more of an additive use. This has prompted the discussion on the appropriate scale that should be associated with *even*. In response to this, Greenberg (2016, 2018) proposed alternative scale for *even*, which is a gradability-based account.

- (4) a. George drank a little wine, a little brandy, a little rum, a little calvados, and even a little Armagnac.
b. George drank a little wine, a little rum, a little Armagnac, a little calvados, and even a little brandy.

The reason for bringing up this discussion is to highlight that the languages under investigation would definitely provide a good testing ground for these properties of *even*. We saw that there are different adverbs which are used in scalar-additive contexts. It might be the case that these different adverbial particles, or even the adfocal particles may instantiate different scalar presuppositions. This is what Liao and Jheng (2025) recently argued for in Mandarin Chinese *shenzhi* and *lian.dou*.

The goal is that this study will motivate further research on association with focus among the African languages, which will in no doubt contribute immensely to our understanding of the phenomenon. Such endeavours may also motivate a re-think or a reanalysis of the theory of focus in general.

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